A hitch-hiker's guide to the metaplectic trace formula for $SL(2)$: basic theory*

Draft 1. Corrected and expanded version

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Abstract

This revision was written with the help of AI systems: Claude (Anthropic) refereed the 1995 notes and drafted the revision as a co-author, and ChatGPT (OpenAI) refereed the revised draft. See the Acknowledgments for details. We write out the Arthur–Selberg trace formula, in its non-invariant and in its invariant form, for the metaplectic double cover of $G = \mathrm{Res}_{k/\mathbb{Q}} SL(2)$ over a number field k, and take the first steps towards its stable form. The general theory is due to Arthur and, for covering groups, to W.-W. Li; here every step is specialized to this group, and the terms are made explicit. The coarse geometric expansion is proved directly from one inequality between heights. On the spectral side we determine the classes fixed by the Weyl group (Weil's characters χ_{ψ_b}), the residual spectrum (the even Weil representations $r^+_{\psi_{-b}}$), the normalizing factor $L(2\lambda,\chi^2)/L(2\lambda+1,\chi^2)$ of the intertwining operator, and the value $M(w,0) = -1$. The invariant distributions are constructed by Fourier inversion on the abelian cover of the diagonal torus, without appeal to a trace Paley–Wiener theorem, and the unipotent coefficients of the fine geometric expansion are computed in terms of the values at 1 of quadratic Hecke L-functions. In the last section we determine the action of $GL(2)$ on the cover: conjugation by $GL(2)$ can exchange the two lifts of an element, and stable conjugacy in the sense of Adams and Li is conjugation by $GL(2)$ twisted by the sign $(\det h, \det(1+\gamma))$. The regular elliptic terms of the trace formula then take the form $S(f) + S(f_-)$, with S a stable distribution and f_- the translate of f by the central element -1; the theta functions contribute to the spectral side in the same form. The stabilization of the remaining terms, which is due to Li, is not reproduced. One ingredient is not computed here but taken from the work of Szpruch: the explicit form of the local normalizing factors at the dyadic and ramified places. This is a revision, restricted to the double cover, of notes of the first author from 1995; the explicit formulas are checked by accompanying scripts.

*This is a revision of the unpublished notes of the same title by D. Joyner (January 1995, slightly revised June 2002). The revision was made in November 2026. It is restricted to the double cover of $SL(2)$, while the 1995 notes treated the n-fold cover.

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‡Claude is an AI model made by Anthropic. See the Acknowledgments for the division of work.

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1 Introduction

Let k be a number field, $G = \mathrm{Res}_{k/\mathbb{Q}} SL(2)$, and $\overline{G(\mathbb{A})}$ the metaplectic double cover of $G(\mathbb{A}) = SL(2, \mathbb{A}_k)$. This paper writes out the Arthur–Selberg trace formula for $\overline{G(\mathbb{A})}$, first in its non-invariant form and then in its invariant form, with every step specialized to this group, and then begins its stabilization. The general theory exists: Arthur's trace formula for connected reductive groups, and its extension to covering groups by W.-W. Li. The aim here is the opposite of generality. The group has rank one, its only proper Levi subgroup is the diagonal torus A, and the cover of A is abelian. As a result most of the general machinery reduces to statements that can be proved in a few lines and checked numerically, and the terms of the formula can be written down.

1.1 Overview

The paper has six parts, each resting on the ones before it. They are described here in order, with the main result after them.

The cover (Section 2). The double cover is defined by Kubota's cocycle, place by place, and assembled into an adelic group that splits over $G(\mathbb{Q})$. The cocycle is not continuous, so the topology of the cover and the spaces of test functions are defined intrinsically, and the coordinates (g, ζ) are used only as Borel coordinates. Test functions are *anti-genuine*: they transform by ζ^{-1} under μ_2.

Rank 0 **(Section 3).** For the cover of a torus the trace formula is Poisson summation, twisted by a genuine automorphic character. For the split torus Weil's character χ_ψ is a canonical choice. This case is used twice later: to describe the cuspidal data of the Borel subgroup, and to match the correction terms on the two sides of the invariant formula.

The geometric side (Section 4). The kernel of $R(f)$ is truncated, and the integral of the truncated kernel is computed class by class: orbital integrals for the elliptic classes, weighted orbital integrals for the hyperbolic classes, and the finite part of a zeta integral for the two central classes.

The spectral side (Section 5). The same truncated integral is computed from the spectral decomposition. It gives the traces on the cuspidal spectrum and on the residual spectrum (even Weil representations), an integral of the logarithmic derivative of the intertwining operator $M(w,\lambda)$ along the unitary principal series, and one extra term $\frac{1}{4}\operatorname{tr}(M(w,0)I(\mu_0,f))$ for each character μ fixed by the Weyl group.

The invariant form (Section 6). The weighted orbital integrals and the term with $M^{-1}M'$ are not invariant under conjugation. Both fail in the same way, and subtracting a common correction, built from the normalized local intertwining operators by Fourier inversion on the cover of A, makes both sides invariant.

The stable(r) form (Section 7). $GL(2)$ acts on the cover, but conjugation by $GL(2)$ is not stable conjugacy there: it can exchange the two lifts of an element. Stable conjugacy is conjugation by $GL(2)$ twisted by a sign. With it, the regular elliptic terms are the sum of a stable distribution evaluated at f and at the translate f_- of f by the central element -1, and the theta functions contribute to the spectral side in the same form. The behavior of the other terms under $GL(2)$ is determined, and the unipotent coefficients are computed. The terms are not transferred to $SO(3)$; that is Li's theorem.

The result is Theorem 6.11 with Corollary 6.14. For $f = f_S \otimes e^S$, with S a large enough finite set of places,

$$\begin{aligned}
&\sum_{z=\pm 1}\Big(\operatorname{vol}\cdot f_S(z)+\sum_{\xi} a^G(S,u_\xi)\,J_G(zu_\xi,f_S)+\tfrac{1}{2}c_k\,I_A(z,f_S)\Big)\\
&\qquad+\sum_{\mathscr{O}\ \text{elliptic}}\operatorname{vol}(T_\mu)\,J_G(\mu_S,f_S)+\sum_{\mathscr{O}\ \text{hyperbolic}} c_k\,I_A(\gamma_S,f_S)\\
&=\sum_{\pi} m_{\text{disc}}(\pi)\operatorname{tr}\pi(f)-\frac{1}{4}\sum_{b\in k^\times/k^{\times 2}}\operatorname{tr} I(\chi_{\psi_b,0},f)\\
&\qquad-\frac{1}{2}\sum_{\mu}\frac{1}{2\pi}\int_{\mathbb{R}}\frac{m'}{m}(\mu,it)\operatorname{tr} I(\mu_{it},f)\,dt.
\end{aligned}$$

Here J_G are orbital integrals, I_A the invariant distributions that replace the weighted orbital integrals, π runs over the discrete spectrum (cuspidal and residual), μ over the genuine automorphic characters of the cover of A, and m is the global normalizing factor of the intertwining operator, essentially $L(2\lambda,\chi^2)/L(2\lambda+1,\chi^2)$. The notation is collected in the Glossary at the end.

Three features distinguish the double cover from $SL(2)$ itself, and they are the main mathematical content of Sections 5 and 6. First, the Weyl group does not act on the cover of A by inversion; the characters it fixes are the Weil characters χ_{ψ_b}, one for each square class b (Lemma 5.3). Second, the residual spectrum is not the trivial representation but the sum of the even Weil representations $r^+_{\psi_{-b}}$, which come from the pole of $M(w,\lambda)$ at $\lambda = \frac{1}{2}$ (Theorem 5.8). Third, $M(w,0) = -1$ for every one of these characters (Proposition 6.13), whereas for $SL(2)$ and a nontrivial quadratic character the corresponding operator is not a scalar and is the source of the multiplicity formula of Labesse and Langlands [LL79].

Section 7 adds a fourth. For $SL(2)$, conjugation by $GL(2)$ is stable conjugacy, and the unstable part of the trace formula is carried by the elliptic tori. For the double cover, an element h of $GL(2)$ that commutes with γ multiplies the lifts of γ by $(\det h, \det(1+\gamma))_v$ (Lemma 7.2); stable conjugacy is the action $h \star x = (\det h, \det(1+x))_v \operatorname{Ad}(h)x$, whose orbits are the level sets of the character of $r^+_\psi - r^-_\psi$ (Proposition 7.5); and the κ-orbital integral of f at γ is the stable orbital integral of f_- at $-\gamma$ (Lemma 7.6). So the elliptic regular terms are $S_{\mathrm{ell}}(f) + S_{\mathrm{ell}}(f_-)$ (Theorem 7.7): both endoscopic groups are $SO(3)$, and no tori occur. The unipotent coefficients in the first line of the display are $a^G(S,u_\xi) = \frac{1}{2}\big(c_S + \sum_{\kappa\neq 1} \kappa_S(\xi) L^S(1,\kappa)\big)$ (Proposition 7.10).

1.2 What is proved here and what is cited

Each section ends with a subsection "Status of the results", which lists the statements taken from the literature and the statements checked numerically. In summary:

Section	Proved in the text	Taken from the literature
2	the splittings, the center, commutativity of preimages of tori, the local triviality of the cocycle, the adelic cover and its splitting over $G(\mathbb{Q})$	Kubota's cocycle and the existence of the topological cover [Ku67, Mo68]; the splitting over K_v [Ku69]; strong approximation
3	the trace formula for covers of tori and its reduction to Poisson summation	the properties of the Weil index [We64]
4	the coarse geometric expansion and the polynomials $J^T_{\mathcal{O}}$; the local distribution $J_A(z,f)$	the fine expansion at the central classes [Ar86, Li14a] (rederived, with its coefficients, in Section 7.6)
5	cuspidal data and the Weyl action; the unramified intertwining operator; the terms J^T_χ from the Maass–Selberg relations; the matching $V_b \cong r^+_{\psi_{-b}}$	the spectral decomposition and Eisenstein series [La76, MW95]; the convergence and polynomiality of the spectral side [Ar80, Li13]; the residual spectrum as theta functions [Ge76, GPS80, Wa80]
6	the variance formulas; the real normalizing factors; the invariance of I_A; the invariant trace formula; $M(w,0) = -1$	the existence of normalizing factors [Ar89, Li12a], which are not explicit at the dyadic and ramified places
7	the action of $GL(2)$; stable conjugacy and the function Δ_ψ; the elliptic regular terms as $S_{\mathrm{ell}}(f) + S_{\mathrm{ell}}(f_-)$; the equivariance of the weighted terms, for normalizing factors with the property (N5), and the existence of such factors; the unipotent coefficients; the character of $r^+_\psi - r^-_\psi$ on the regular set; the trace on the theta functions	the regularity of the character of the Weil representation [Ma99, Th08]; the Hasse norm theorem; everything about the transfer to $SO(3)$ [Li11, Li15, Li21], which is not used

One ingredient of the invariant formula is therefore defined but not evaluated: the local normalizing factors r_v at the places where the additive character or the inducing character is ramified, and at the dyadic places. (The unipotent coefficients $a^G(S, u_\xi)$, a second such ingredient in Sections 4–6, are computed in Proposition 7.10.) Remarks numbered "2026.n" record these and other points that rest on citation or on numerical checks only. The explicit formulas are tested by the scripts in `checks/`; they are evidence, not proofs.

The stable form of the trace formula is only begun. Section 7 shows that stable conjugacy on the cover is conjugation by $GL(2)$ twisted by a sign, relates the twisted notion to that of Adams and Li [Ad98, Li11], and stabilizes the regular elliptic terms. The hyperbolic and unipotent terms and the cuspidal spectrum are not stabilized; Section 7.8 lists what is missing. The statements of Section 7 about the weighted terms and about the spectral terms I_χ use the property (N5) of the normalizing factors. It holds for the explicit factors, and Proposition 7.8′ shows that factors with this property exist at every place; at the dyadic and ramified places their explicit form is taken from Szpruch [Sz09, Sz16] (Remark 2026.18).

1.3 A running example

The case $k=\mathbb{Q}$, $S=\{\infty,2\}$ is the smallest one in which every feature of the formula appears, and it is followed through the paper. We collect it here; ψ is the character of $\mathbb{A}/\mathbb{Q}$ with $\psi_\infty(x)=e^{-2\pi ix}$.

- *The cover.* At the real place the cover is $Mp(2,\mathbb{R})$; its center is cyclic of order 4, and so is the center at the place 2 (Lemma 2.2(e), Example 2.B). At an odd prime p the cover splits over $SL(2,\mathbb{Z}_p)$ by $k\mapsto(k,\kappa_p(k))$, and on $SL(2,\mathbb{Z})$ the product of the κ_p is a Jacobi symbol (Example 2.C). At 2 we use a splitting over a congruence subgroup (Lemma 2.2(c), Remark 2026.1). Test functions are $f=f_\infty\otimes f_2\otimes\bigotimes_{p\text{ odd}}e_p$.

- *Rank* 0. $\chi_{\psi,\infty}(a(-1),1)=i$ and $\chi_{\psi,p}(a(p),1)\in\{1,i\}$ for odd p (Examples 3.A and 3.B): a genuine character of the cover of A is not an ordinary character times a sign.

- *Geometric side.* $c_\mathbb{Q}=1$. Besides the elliptic classes there are the two central classes ±1 and the hyperbolic classes of $a(t)$, $t=\pm2^j$, $j\geq1$. The unipotent terms are indexed by $\mathbb{Q}_S^\times/\mathbb{Q}_S^{\times2}$, which has $2\cdot8=16$ elements. The constant term of the unipotent contribution is computed for a Gaussian in Example 4.B, and the 16 coefficients in Example 7.G: they involve γ_E, $\log2$, $L(1,\chi_{-4})$, $L(1,\chi_8)$ and $L(1,\chi_{-8})$.

- *Spectral side.* The characters fixed by the Weyl group that are unramified outside S are the four χ_{ψ_b}, $b\in\{\pm1,\pm2\}$ (Example 5.A). Each carries a residual representation $V_b\cong r^+_{\psi_{-b}}$; V_1 contains the classical theta function $\sum_n e^{2\pi in^2z}$ (Example 5.B). The partial normalizing factor is $(1-2^{-2\lambda})\zeta(2\lambda)/((1-2^{-1-2\lambda})\zeta(2\lambda+1))$, with residue $2/\pi^2$ at $\lambda=\frac12$ (Examples 5.B and 5.C).

- *Invariant form.* At the real place the normalized intertwining operator is 1 on the K-type of weight $\pm\frac12$ and a product of factors $(c-\lambda)/(c+\lambda)$ on the others (Example 6.B), and the correction Φ_A is an Abel transform smoothed by an exponentially decaying kernel (Example 6.C). The only bad place is 2, where the normalizing factor is not computed in Section 6; by Proposition 7.8′(c) it is a constant times $L_2(2\lambda,\chi_2^2)/L_2(2\lambda+1,\chi_2^2)$. The discrete correction is $-\frac14\sum_{b=\pm1,\pm2}\operatorname{tr}I(\chi_{\psi_b,0},f)$ (Example 6.D).

- *Stable(r) form.* At the real place the outer automorphism $\mathrm{Ad}(\mathrm{diag}(-1,1))$ of $Mp(2,\mathbb{R})$ sends $\tilde k_\theta$ to $\tilde k_{-\theta}$, a stable conjugate, and exchanges the two lifts of $a(t)$, $t<0$; the stable character $\Theta^+-\Theta^-$ is $1/(2\cos\frac\theta2)$ (Examples 7.A and 7.B). The classes of the elements of order 4 are treated in Example 7.E. The theta functions of ψ_{-b}, $b\in\{\pm1,\pm2\}$, span four spaces $\Theta_b\supset V_b$ whose traces are $\frac12\Theta_b^{\mathrm{st}}(f)+\frac12\Theta_b^{\mathrm{st}}(f_-)$ (Proposition 7.12).

1.4 Conventions

This subsection fixes the scope of the paper and says where the reference material is. The revision treats the double cover only; the 1995 notes treated the n-fold cover, and remarks in Sections 2 and 3 say what changes for $n>2$. Comparisons with the 1995 notes and with the 1997 book manuscript, including the corrections made, are collected in the revision log that accompanies the source and are not needed to read the paper. The literature review follows this section. The main results and the open questions are collected in Section 8, and notation and terms are listed in the Glossary at the end.

Literature review

This section summarizes the work on which the paper rests and says how the present results fit in. The first part covers work up to 1995, the date of the original notes. The second part covers later work, most of which the revision uses.

Work up to 1995

Covering groups and the Weil representation. The double cover of $SL(2)$ over a local field and over the adeles first appeared in Weil's work on the representation that now carries his name [We64]. Kubota gave the explicit cocycle used here, for the n-fold cover of $SL(2)$ over a local field containing the n-th roots of unity, and the splitting over the rational points [Ku67, Ku69]. Moore [Mo68] and Matsumoto [Ma69] treated the central extensions of split groups over local fields and adele rings in general. Kazhdan and Patterson [KP84] studied the covers of $GL(r)$ and their automorphic forms. Gelbart's lecture notes [Ge76] remain the standard account of the double cover of $SL(2)$ and $GL(2)$, the Weil representation and theta functions. Covering groups are not algebraic groups, and the Borel cocycles that describe them are not continuous; the relevant general theorem on the topology of such extensions is Mackey's [Ma57].

Half-integral weight and the Shimura correspondence. Shimura [Sh73] attached to a cusp form of weight $k/2$ a form of even weight $k-1$. Niwa [Ni75] and Shintani [Sh75] realized the correspondence and its inverse by theta series. Gelbart and Piatetski-Shapiro [GPS80, GPS83] and Waldspurger [Wa80, Wa81, Wa91] recast it in terms of automorphic representations of the double cover. Waldspurger's description of the discrete spectrum of the double cover of $SL(2)$, through the theta correspondence with $PGL(2)$ and its inner forms, is the definitive result for the group studied here. The identification of the residual spectrum with Weil representations, used in Theorem 5.8, belongs to this circle of results [Ge76, GPS80, Wa80].

The trace formula. The trace formula goes back to Selberg [Se56]. For $GL(2)$ it was put in adelic form by Jacquet and Langlands [JL70]; the treatment of Gelbart and Jacquet [GJ79] is the closest in spirit to the present paper, and we compare signs and constants with it in Section 5. The classical Selberg trace formula with multiplier systems and real weight, as in Hejhal [He83], covers forms of half-integral weight on congruence subgroups of $SL(2,\mathbb{Z})$. For a general connected reductive group the formula is Arthur's: the coarse expansions [Ar78, Ar80], the invariant form [Ar81], the fine expansions [Ar86, Ar82], the local theory of weighted orbital integrals and normalized intertwining operators [Ar88a, Ar88b, Ar89], and the invariant trace formula [Ar88c]. The spectral decomposition on which the spectral side rests is due to Langlands [La76], and Mœglin and Waldspurger [MW95] wrote it in a form that includes finite central covers. For $SL(2)$ the stable trace formula is the subject of Labesse and Langlands [LL79].

The trace formula for covering groups. Flicker [Fl80] used a trace formula for the n-fold covers of $GL(2)$ to prove the metaplectic correspondence with $GL(2)$, and Flicker and Kazhdan [FK86] did the same for $GL(r)$; see also Kazhdan and Patterson [KP86]. The first author studied a meta-

plectic analog of Kazhdan's endoscopic lifting in [Jo88]. The 1995 version of the present notes wrote out Arthur's formula for the n-fold cover of $SL(2)$, with several steps conjectural.

Local harmonic analysis on the double cover of $SL(2)$. The irreducibility of the unitary genuine principal series is due to Moen [Mo88]. The unramified computation of the intertwining operator (the Gindikin–Karpelevich formula) for general covers is in [KP84]. See also Savin [Sa88] for the Hecke algebra correspondence.

Work since 1995

Covering groups. Brylinski and Deligne [BD01] placed all these covers in one algebraic framework, as extensions by K_2. For the Langlands program for covering groups, and its history, see Gan, Gao and Weissman [GGW18] and Weissman [We18].

Half-integral weight. Gan and Savin [GS12] obtained the local Langlands correspondence for the double cover of $Sp(2n)$ from the theta correspondence, and Gan and Ichino [GI18] extended Waldspurger's global description to the generic part of the discrete spectrum of the double cover of $Sp(2n)$.

The trace formula. Arthur's Clay lectures [Ar05] are the best introduction to his trace formula.

The trace formula for covering groups. Mezo [Me02, Me01] prepared the comparison of the full invariant trace formulas of $GL(r)$ and its covers. The conjectural steps of the 1995 notes are now theorems of Li, who extended Arthur's trace formula to covering groups of connected reductive groups in the four papers [Li14a, Li12a, Li13, Li14b]: the fine geometric expansion, the local harmonic analysis including normalized intertwining operators, the fine spectral expansion, and the invariant distributions (the invariant formula under a hypothesis at the real places, which he verifies for the double cover of $Sp(2n)$). For the double cover of $Sp(2n)$ Li went on to develop endoscopy, building on Adams [Ad98] and Renard [Re99] at the real place: the transfer of orbital integrals [Li11], the weighted fundamental lemma [Li12b], the stabilization of the elliptic terms [Li15], local character relations [Li19], the stabilization of the full trace formula [Li21], and a multiplicity formula for the discrete spectrum in terms of Arthur parameters [Li24]. For $n = 1$ these results apply to the group of this paper (Li works with an eight-fold cover that contains the double cover). A p-adic character lifting between the double cover of $SL(2)$ and $SO(1,2)$ is in Schultz's thesis [Sc98], and for $Sp(2n)$ in Howard [Ho10]. Stable conjugacy on these covers is defined through the character of the Weil representation, which was computed by Maktouf [Ma99] and Thomas [Th08]; Thomas [Th13] also treats the action of the similitude group on the cover and the character of the difference of the two halves of the Weil representation. According to [Li15], Hiraga and Ikeda have considered the stabilization of the trace formula for covers of $SL(2)$ (unpublished); see [HI13] for an application of the trace formula of the double cover to the Kohnen plus space.

Local harmonic analysis on the double cover of $SL(2)$. Szpruch [Sz09] computed the local coefficients of the genuine principal series of the double cover of $SL(2,F)$, and Goldberg and Szpruch [GS16] the Plancherel measures for the n-fold covers; the Plancherel measure of the double cover in all residue characteristics is also in [Sz16]. These results give the normalizing factors at the places where our Section 6.2 is not explicit (Proposition 7.8′). The unramified computation of the intertwining operator for general covers is also in [Mc12, Ga14].

The place of this paper

Nothing in Sections 2–6 goes beyond what the general theorems of Arthur and Li give. What the paper adds is an account in which the group is fixed, the constants are determined, the proofs that are short for this group are given in full, and the remaining appeals to the literature are listed. Section 7 is likewise a special case of Li's endoscopy for the double cover of $Sp(2n)$, reached by a different route: there the elementary description of stable conjugacy through $GL(2)$, the identity between κ-orbital integrals and stable orbital integrals of f_-, and the formula for the unipotent coefficients are worked out directly. We hope that the paper is useful as a worked example of Li's theory and as a companion to the classical treatments of half-integral weight.

2 Background on the metaplectic double cover

This section fixes the double cover of $G = \mathrm{Res}_{k/\mathbb{Q}} SL(2)$ and the conventions used in the rest of the paper. Section 2.1 treats a single place: Kubota's cocycle, the splittings over unipotent and compact subgroups, the center, and the preimages of tori. These are then assembled into the cover of $G(\mathbb{Q}_p)$. Section 2.2 builds the adelic cover and its splitting over $G(\mathbb{Q})$, and Section 2.3 fixes a double cover of $GL(2)$ and the resulting $GL(2)$-action on the cover of $SL(2)$, which is what Section 7 uses.

One point needs care throughout. The cover is described by an explicit cocycle, in coordinates (g,ζ), but the cocycle is only a Borel function: it is not continuous, and the section $g \mapsto (g,1)$ is not continuous either. The topology of the cover is therefore not the product topology in these coordinates. It is fixed in Lemma 2.1(c), smooth functions are defined with respect to it (Section 2.1.4), and Lemma 2.2(i) lists the sections that are continuous. The coordinates are used only for algebraic computations and for integration.

From now on $n = 2$. The 1995 notes treated the n-fold cover; for $n = 2$ several statements become simpler, and one, the commutativity of preimages of abelian subgroups (Lemma 2.2(g)), becomes a two-line argument. What changes for general n is listed in Remark 2026.3.

Notation

This subsection fixes the notation for number fields, places and the quadratic Hilbert symbol that is used throughout the paper.

Let k be a number field. (For $n = 2$ the hypothesis $\mu_n \subset k$ of the 1995 notes always holds, and k may have real places.) Let Σ_k be the set of places of k, Σ_k^∞ the archimedean places and Σ_k^f the finite places. Let $\mathbb{A}$ be the adele ring of $\mathbb{Q}$ and $\mathbb{A}_k$ that of k. For $v \in \Sigma_k$ let k_v be the completion. For $F = k_v$ non-archimedean let $\mathcal{O} = \mathcal{O}_F$ be its ring of integers, $\varpi = \varpi_F$ a uniformizer, q the order of

the residue field and $\mathrm{ord} = \mathrm{ord}_F : F^\times \to \mathbb{Z}$ the normalized valuation, so $\mathrm{ord}(\varpi) = 1$ and $|\varpi| = q^{-1}$. We call v *dyadic* if $v \mid 2$, and put $e_v = \mathrm{ord}_v(2)$. Absolute values are normalized; at a complex place $|z|_v = z\bar{z}$.

Let $(a,b) = (a,b)_v$ be the quadratic Hilbert symbol of $F = k_v$ [We74, Se73]. We use only the following properties:

(H1) $(\cdot,\cdot)$ is bimultiplicative and symmetric, with values ± 1; hence $(a^{-1},b) = (a,b)$.

(H2) $(a,b) = 1$ if a or b lies in $F^{\times 2}$; $(a,-a) = 1$; and $(a,1-a) = 1$ for $a \neq 0,1$ (the Steinberg relation).

(H3) If v is finite and not dyadic and $a,b \in \mathscr{O}^\times$, then $(a,b) = 1$. If v is complex, $(\cdot,\cdot) \equiv 1$. If v is real, $(a,b) = -1$ exactly when $a < 0$ and $b < 0$.

(H4) (Product formula) For $a,b \in k^\times$, $(a,b)_v = 1$ for almost all v and $\prod_v (a,b)_v = 1$.

By (H3), $(-1,-1)_v = 1$ at non-dyadic finite places and at complex places, and $(-1,-1)_v = -1$ at real places. At a dyadic place both values occur; $(-1,-1)_{\mathbb{Q}_2} = -1$.

Let G be the group over $\mathbb{Q}$ obtained from $SL(2)$ by restriction of scalars from k to $\mathbb{Q}$, so $G(\mathbb{Q}) = SL(2,k)$ and $G(\mathbb{Q}_p) = \prod_{v|p} SL(2,k_v)$.

2.1 Local aspects

Here $F = k_v$ for one place v. We define the cover, compute it on the subgroups that matter later (unipotent, compact, diagonal, central, anisotropic), and then pass to $G(\mathbb{Q}_p)$.

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,F)$ put $x(g) = c$ if $c \neq 0$ and $x(g) = d$ if $c = 0$, and define $\beta = \beta_v : SL(2,F) \times SL(2,F) \to \mu_2$ by

$$\beta(g_1,g_2) = \Big(\frac{x(g_1g_2)}{x(g_1)}, \frac{x(g_1g_2)}{x(g_2)}\Big) = \big(x(g_1),x(g_2)\big)\big(-x(g_1)^{-1}x(g_2), x(g_1g_2)\big). \tag{2.1}$$

At a complex place $\beta \equiv 1$ by (H3).

Lemma 2.1 (Kubota [Ku67], Moore [Mo68])**.** *(a) β is a Borel 2-cocycle on $SL(2,F)$ with values in μ_2. With the group law (2.2) below it defines a central extension of groups*

$$1 \to \mu_2 \to \overline{SL(2,F)} \xrightarrow{p} SL(2,F) \to 1,$$

which is nontrivial unless $F = \mathbb{C}$.

(b) β is not continuous unless $F = \mathbb{C}$ (Example 2.A). There is, however, an open neighborhood U of 1 in $SL(2,F)$ with $\beta \equiv 1$ on $U \times U$: for $F = \mathbb{R}$ one may take $U = \{g \mid a > 0,\ d > 0\}$, and for F non-archimedean $U = K_m$ with m as in Lemma 2.2(c).

(c) There is exactly one topology on $\overline{SL(2,F)}$ for which it is a topological group and the sets $\{(g,1) \mid g \in V\}$, with $V \subset U$ an open neighborhood of 1 in $SL(2,F)$, form a base of neighborhoods of the identity. With this topology $\overline{SL(2,F)}$ is a locally compact group, p is a two-fold covering map, and $g \mapsto (g,1)$ is a homeomorphism of U onto an open subset. Unless $F = \mathbb{C}$ it is not the product topology of $SL(2,F) \times \mu_2$, but it has the same Borel sets.

From now on $\overline{SL(2,F)}$ carries the topology of (c). For $F=\mathbb{R}$ it is the metaplectic group $Mp(2,\mathbb{R})$, a connected Lie group. The proof is given after Lemma 2.2. The two expressions in (2.1) agree, and the cocycle identity holds; both are also checked on random elements over $\mathbb{R}$, $\mathbb{Q}_2$, $\mathbb{Q}_3$, $\mathbb{Q}_5$ and $\mathbb{Q}_7$ in `checks/sec2_n2_check.py`.

For a subgroup $H\subset SL(2,F)$ put $\overline{H}=p^{-1}(H)$. Elements of $\overline{SL(2,F)}$ are written (g,ζ) with $g\in SL(2,F)$, $\zeta\in\mu_2$, and multiplied by

$$(g_1,\zeta_1)(g_2,\zeta_2)=(g_1g_2,\ \beta(g_1,g_2)\zeta_1\zeta_2). \tag{2.2}$$

Since $\zeta^{-1}=\zeta$, inverses and conjugates are

$$(g,\zeta)^{-1}=(g^{-1},\beta(g,g^{-1})\zeta),\quad (g,\zeta)(h,\zeta')(g,\zeta)^{-1}=\bigl(ghg^{-1},\ \beta(g,h)\beta(gh,g^{-1})\beta(g,g^{-1})\zeta'\bigr). \tag{2.3}$$

The pairs (g,ζ) are Borel coordinates on $\overline{SL(2,F)}$, not topological ones. A statement about continuity or smoothness on the cover always refers to the topology of Lemma 2.1(c). A function ϕ on the cover that is invariant under μ_2 is a function on $SL(2,F)$, and for the measures of Section 2.1.5 its integral over the cover is $\int\phi(g,1)\,dg$.

Let $B\subset SL(2)$ be the upper triangular Borel subgroup, N its unipotent radical and A the diagonal torus. Write $a(t)=\mathrm{diag}(t,t^{-1})$ and $n(u)=\begin{pmatrix}1&u\\0&1\end{pmatrix}$. For $\theta\in F^\times\smallsetminus F^{\times2}$ let

$$T_\theta=\left\{\begin{pmatrix}a&b\theta\\b&a\end{pmatrix}\ \middle|\ a,b\in F,\ a^2-\theta b^2=1\right\}\cong E^1,\qquad E=F(\sqrt{\theta}),$$

the norm-one group of E. Every anisotropic maximal torus of $SL(2,F)$ is $GL(2,F)$-conjugate to some T_θ.

If v is finite let $K=K_0=SL(2,\mathcal{O})$ and, for $m\geq1$,

$$K_m=\left\{\begin{pmatrix}a&b\\c&d\end{pmatrix}\in SL(2,\mathcal{O})\ \middle|\ a-1,\ d-1,\ b,\ c\in\varpi^m\mathcal{O}\right\}.$$

If v is non-dyadic, define $\kappa=\kappa_v\colon K\to\mu_2$ by

$$\kappa\begin{pmatrix}a&b\\c&d\end{pmatrix}=\begin{cases}(c,d)_v, & 0<|c|_v<1,\\ 1, & \text{otherwise.}\end{cases} \tag{2.4}$$

If v is archimedean let $K=SO(2)$ (v real) or $K=SU(2)$ (v complex).

Lemma 2.2. *Let $F=k_v$.*

(a) $\beta\equiv1$ *on $N(F)\times N(F)$, so $n\mapsto(n,1)$ is a homomorphism and $\overline{N(F)}=N(F)\times\mu_2$.*

(b) Let v be finite and non-dyadic. Then $\beta(k_1,k_2)=\kappa(k_1)\kappa(k_2)\kappa(k_1k_2)$ for $k_1,k_2\in K$, so $k\mapsto(k,\kappa(k))$ is a homomorphism and $\overline{K}=K^*\times\mu_2$ with $K^*=\{(k,\kappa(k))\}$. The restriction of β to $K\times K$ is not identically 1*, so the set $\{(k,1)\mid k\in K\}$ is not a subgroup.*

(c) Let v be finite and $m\geq1$ with $1+\varpi^m\mathcal{O}\subset F^{\times2}$. Then $\beta\equiv1$ on $K_m\times K_m$, so $k\mapsto(k,1)$ splits $\overline{K_m}$. The hypothesis holds for every $m\geq1$ if v is non-dyadic, and for $m\geq2e_v+1$ if v is dyadic.

(d) *Let v be archimedean. If v is complex, $\overline{SL(2,F)} = SL(2,F)\times\mu_2$. If v is real, $\beta\equiv 1$ on $A(\mathbb{R})^0\times A(\mathbb{R})^0$, so $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0\times\mu_2$, while $\overline{SO(2)}$ is connected, the nontrivial double cover of the circle [Ge76].*

(e) *The center of $\overline{SL(2,F)}$ is $\overline{\{\pm 1\}}$, a group of order 4 in which $(-1,\zeta)^2 = (1,(-1,-1)_v)$. It is cyclic when $(-1,-1)_v = -1$ (for example $F=\mathbb{R}$ or $F=\mathbb{Q}_2$) and is $\{\pm1\}\times\mu_2$ otherwise.*

(f) *On $A(F)$, $\beta(a(t_1),a(t_2)) = (t_1,t_2)_v$. Hence $\overline{A(F)}$ is abelian, and $(a(t),\zeta)^2 = (a(t^2),(t,-1)_v)$.*

(g) *If $g_1,g_2\in SL(2,F)$ commute, then $\beta(g_1,g_2)=\beta(g_2,g_1)$. Consequently $\overline{H}$ is abelian for every abelian subgroup $H\subset SL(2,F)$; in particular $\overline{A(F)}$ and every $\overline{T_\theta}$ are abelian.*

(h) *Let $U = gNg^{-1}$ with $g\in SL(2,F)$. Then $u\mapsto(u,1)$ is the unique homomorphic section of $\overline{U}\to U$. Consequently, for every $(g,\zeta)\in\overline{SL(2,F)}$ and $u\in N(F)$,*

$$(g,\zeta)(u,1)(g,\zeta)^{-1} = (gug^{-1},1).$$

(i) *(Continuity.) The following maps are continuous for the topology of Lemma 2.1(c): $n\mapsto(n,1)$ on $N(F)$ and on every conjugate $gN(F)g^{-1}$; $a\mapsto(a,1)$ on $A(F)$; $k\mapsto(k,1)$ on K_m, for m as in (c); and, for v finite and non-dyadic, $k\mapsto(k,\kappa(k))$ on K. In particular K^* and the image of K_m are compact open subgroups of $\overline{SL(2,F)}$. The map $g\mapsto(g,1)$ is not continuous on $SL(2,F)$ unless $F=\mathbb{C}$; in general it is not continuous on K, nor on an anisotropic torus T_θ.*

Proof. (a) For $n_1,n_2\in N(F)$ all three values of x equal 1.

(b) This is Kubota's computation [Ku69]; see also [Ge76]. The identity is also checked numerically for $F=\mathbb{Q}_3,\mathbb{Q}_5,\mathbb{Q}_7$, together with the existence of pairs in $K\times K$ with $\beta=-1$.

(c) Let $k_i=\begin{pmatrix}a_i & b_i\\ c_i & d_i\end{pmatrix}\in K_m$ for $i=1,2$, and $k_1k_2=\begin{pmatrix}a_3 & b_3\\ c_3 & d_3\end{pmatrix}$. All a_i and d_i lie in $1+\varpi^m\mathscr{O}\subset F^{\times 2}$, and $c_3 = c_1a_2+d_1c_2$. If $c_1,c_2,c_3\neq 0$, then $c_1a_2/c_3 + d_1c_2/c_3 = 1$, so by (H2) and (H1)

$$\beta(k_1,k_2) = \left(\frac{c_3}{c_1},\frac{c_3}{c_2}\right) = \left(\frac{c_1a_2}{c_3},\frac{d_1c_2}{c_3}\right) = 1,$$

because a_2 and d_1 are squares. If $c_3=0\neq c_1c_2$, then $x(k_1k_2)=d_3$ is a square and $c_2=-c_1a_2/d_1$, so $\beta=(c_1,c_2)=(c_1,-c_1)(c_1,a_2/d_1)=1$. If $c_1=0\neq c_2$, then $x(k_1)=d_1$ is a square and $c_3=d_1c_2$, so $\beta=(c_3/d_1,d_1)=1$; the case $c_2=0\neq c_1$ is the same with a_2 in place of d_1. If $c_1=c_2=0$, all x-values are squares. Finally, $1+\varpi\mathscr{O}\subset F^{\times 2}$ for non-dyadic F by Hensel's lemma, and $1+4\varpi\mathscr{O}\subset F^{\times 2}$ for dyadic F by the binomial series for $(1+4\varpi y)^{1/2}$; and $4\varpi\mathscr{O}=\varpi^{2e_v+1}\mathscr{O}$.

(d) At a complex place $\beta\equiv 1$ by (H3). For $t_1,t_2>0$, $(t_1,t_2)_{\mathbb{R}}=1$ by (H3) and (f). The statement about $SO(2)$ is standard [Ge76].

(e) Commuting elements give commuting lifts by (g), so $\overline{\{\pm1\}}$ is central; conversely, the image of a central element is central in $SL(2,F)$. Since $x(-1)=-1$ and $x(1)=1$, $(-1,\zeta)^2 = (1,\beta(-1,-1)) = (1,(-1,-1)_v)$ by (2.1). A group of order 4 with an element of order 4 is cyclic, and one of exponent 2 is $\{\pm1\}\times\mu_2$.

(f) Here $x(a(t))=t^{-1}$, so (2.1) gives $\beta(a(t_1),a(t_2)) = (t_2^{-1},t_1^{-1}) = (t_1,t_2)$ by (H1). The square is $(a(t^2),(t,t))$, and $(t,t)=(t,-1)$ by (H2).

(g) If $g_1g_2 = g_2g_1$, then $x(g_1g_2) = x(g_2g_1)$, and the first expression in (2.1) is symmetric in g_1, g_2 because the quadratic Hilbert symbol is symmetric (H1). By (2.2), lifts (g_1, ζ_1) and (g_2, ζ_2) commute if and only if $\beta(g_1, g_2) = \beta(g_2, g_1)$.

(h) If $g \in B(F)$, then $U = N$ and the claim is (a). Otherwise g has lower left entry $\gamma \neq 0$, and $u_s = g\,n(s)\,g^{-1}$ has lower left entry $-\gamma^2 s$. So for $s_1, s_2 \neq 0$ with $s_1 + s_2 \neq 0$,

$$\beta(u_{s_1}, u_{s_2}) = \left(\frac{s_1+s_2}{s_1}, \frac{s_1+s_2}{s_2}\right) = \left(\frac{s_1}{s_1+s_2}, \frac{s_2}{s_1+s_2}\right) = 1$$

by (H1) and the Steinberg relation (H2). If $s_1 + s_2 = 0$, then $x(u_{s_1}u_{s_2}) = 1$ and $\beta = (1/s_1, -1/s_1) = 1$. So $u \mapsto (u, 1)$ is a homomorphism on U. Two homomorphic sections differ by a homomorphism $U \cong F \to \mu_2$, which is trivial because $F = 2F$. Conjugation by (g, ζ) carries the section $n \mapsto (n, 1)$ of $\overline{N}$ to a homomorphic section of $\overline{gNg^{-1}}$, which is therefore $u \mapsto (u, 1)$.

(i) Let $X \subset SL(2, F)$ and $\varepsilon\colon X \to \mu_2$. For $g_0 \in X$ and u near 1 with $g_0 u \in X$,

$$(g_0 u, \varepsilon(g_0 u)) = (g_0, \varepsilon(g_0))\,(u, 1) \cdot \beta(g_0, u)\,\varepsilon(g_0)\,\varepsilon(g_0 u),$$

and $u \mapsto (u, 1)$ is continuous at 1 by Lemma 2.1(c). So $g \mapsto (g, \varepsilon(g))$ is continuous at g_0 if $\varepsilon(g_0 u) = \varepsilon(g_0)\beta(g_0, u)$ for all such u close enough to 1. For $X = N(F)$ and $\varepsilon = 1$ this holds by (a), and for $gN(F)g^{-1}$ by (h). For $X = A(F)$ and $\varepsilon = 1$: $\beta(a(t_0), a(s)) = (t_0, s)_v = 1$ when $s \in F^{\times 2}$, and $F^{\times 2}$ is a neighborhood of 1 in $F^\times$. For $X = K_m$ it is part of Lemma 2.1(c). For $X = K$ and $\varepsilon = \kappa$: $k \mapsto (k, \kappa(k))$ is a homomorphism by (b), and on K_1 it is $k \mapsto (k, 1)$, because $\kappa(k) = (c, d)_v$ with $d \in 1 + \varpi\mathcal{O} \subset F^{\times 2}$; a homomorphism that is continuous at 1 is continuous. The image of K_m is compact, and open by Lemma 2.1(c); K^* is compact and contains the image of K_1, so it is open. If $g \mapsto (g, 1)$ were continuous on $SL(2, F)$, then β would be continuous. For K and T_θ see Example 2.A and (d). □

Proof of Lemma 2.1. (a) is due to Kubota [Ku67]; see also [Ku69, Ge76]. β is a Borel function because $x(g)$ is continuous on each of the Borel sets $\{c \neq 0\}$ and $\{c = 0\}$.

(b) For F non-archimedean this is Lemma 2.2(c). Let $F = \mathbb{R}$ and $g_1, g_2 \in U$, with entries a_i, b_i, c_i, d_i, and let $g_3 = g_1g_2$, so $c_3 = c_1a_2 + d_1c_2$. By (H3), a real Hilbert symbol is 1 as soon as one of its entries is positive. If $c_1c_2c_3 \neq 0$, then $\beta(g_1, g_2) = (c_3/c_1, c_3/c_2)$: when c_1 and c_2 have the same sign, c_3 has that sign too, since $a_2, d_1 > 0$; when they have opposite signs, so do the two entries. If $c_3 = 0 \neq c_1c_2$, then $c_1a_2 = -d_1c_2$, so c_1 and c_2 have opposite signs, and $\beta = (d_3/c_1, d_3/c_2) = 1$. If $c_1 = 0 \neq c_2$, then $x(g_1) = d_1$, $c_3 = d_1c_2$ and $\beta = (c_2, d_1) = 1$; if $c_2 = 0 \neq c_1$, then $\beta = (a_2, c_1a_2/d_2) = 1$. If $c_1 = c_2 = 0$, all three values of x are positive.

(c) A group topology is determined by its neighborhoods of the identity, so there is at most one such topology. By Kubota [Ku67] and Moore [Mo68] there is a topology $\mathcal{T}$ on $\overline{SL(2, F)}$ for which it is a locally compact group and p is a covering map. (For an arbitrary Borel cocycle on a locally compact second countable group this is a theorem of Mackey [Ma57].) We show that $\mathcal{T}$ has the stated base of neighborhoods. Since p is a covering, there are a $\mathcal{T}$-open neighborhood W of the identity, with $W \cap \mu_2 = \{1\}$, and an open neighborhood $V \subset U$ of 1 such that p maps W homeomorphically onto V. Write the inverse map as $g \mapsto (g, \chi(g))$, $\chi\colon V \to \mu_2$. Let $W' \subset W$ be an open neighborhood of the identity with $W'W' \subset W$, and $V' = p(W')$. For $g, h \in V'$ the product $(g, \chi(g))(h, \chi(h))$ lies in W and maps to gh, so it equals $(gh, \chi(gh))$; since $\beta(g, h) = 1$ on $U \times U$, this gives $\chi(gh) = \chi(g)\chi(h)$. Now every g in a small enough neighborhood $V'' \subset V'$ of 1 is a

square $g = h^2$ with $h \in V'$: take $g = \exp X$ and $h = \exp(X/2)$, for X in a small ball in the Lie algebra if $F = \mathbb{R}$, and in a lattice $\varpi^M \mathfrak{sl}_2(\mathscr{O})$ with M large if F is non-archimedean. Hence $\chi = 1$ on V'', the set $\{(g,1) \mid g \in V''\}$ is $\mathscr{T}$-open, and $g \mapsto (g,1)$ is a homeomorphism of V'' onto it. So the sets in (c) form a base of neighborhoods of the identity for $\mathscr{T}$. For $g_0 \in U$ and $u \in U$ we have $(g_0 u, 1) = (g_0, 1)(u, 1)$ by (b), so $g \mapsto (g,1)$ is a local homeomorphism on U; being injective, it is a homeomorphism of U onto an open subset. The topology is not the product topology, because $g \mapsto (g,1)$ is not continuous (Lemma 2.2(i)). The map $(g,\zeta) \mapsto (g,\zeta)$ from the product to the cover is a Borel isomorphism: on each translate $g_0 U$ it is $(g_0 u, \zeta) \mapsto (g_0,1)(u,1) \cdot \beta(g_0,u)\zeta$, countably many translates cover $SL(2,F)$, and β is Borel. □

Example 2.A. *The cocycle and the section are not continuous.* Let $\bar{n}(c) = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$. For $t \in F^\times$ and $c \neq 0$, formula (2.1) gives $\beta(a(t), \bar{n}(c)) = (c,t)_v$. If t is not a square, there are c arbitrarily close to 0 with $(c,t)_v = -1$, and others with $(c,t)_v = 1$; but $\bar{n}(c) \to 1$ and $\beta(a(t),1) = 1$. For $F = \mathbb{R}$ and $t = -1$ this is $\beta(-1, \bar{n}(c)) = \operatorname{sgn}(c)$. So β is discontinuous at $(a(t),1)$, and

$$(a(t),1)\,(\bar{n}(c),1) = \big(a(t)\bar{n}(c),\,(c,t)_v\big)$$

shows that $g \mapsto (g,1)$ is discontinuous at $a(t)$.

Now let v be finite and non-dyadic, $d \in \mathscr{O}^\times$ a non-square, and $k_j = \begin{pmatrix} d^{-1} & 0 \\ \varpi^j & d \end{pmatrix} \in K$. Then $k_j \to a(d^{-1})$ as $j \to \infty$, while $\kappa(k_j) = (\varpi^j, d)_v = (-1)^j$. So κ is not continuous on K. For $F = \mathbb{Q}_3$ and $d = -1$: $k_j = \begin{pmatrix} -1 & 0 \\ 3^j & -1 \end{pmatrix} \to -1$ and $\kappa(k_j) = (-1)^j$. Since $k \mapsto (k, \kappa(k))$ is continuous (Lemma 2.2(i)), $k \mapsto (k,1)$ is not continuous on K.

Example 2.B. Take $F = \mathbb{R}$. By (e) and (f), the element $(a(-1),1) = (-1,1)$ has order 4, so $\overline{\{\pm 1\}} \cong \mathbb{Z}/4$ and $\overline{A(\mathbb{R})}$ is an abelian group which is not $A(\mathbb{R}) \times \mu_2$. By (d), $\overline{A(\mathbb{R})}$ is generated by $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$ and $(-1,1)$. At a non-dyadic place, on the other hand, $(-1,-1)_v = 1$, so the center is $\{\pm 1\} \times \mu_2$; and $\overline{A(\mathscr{O}^\times)}$ is split by $a \mapsto (a,1)$, because $(t_1,t_2)_v = 1$ for units by (H3). This is consistent with (b), since $\kappa \equiv 1$ on $A(\mathscr{O}^\times)$.

Remark 2026.1. For $F = \mathbb{Q}_2$, Lemma 2.2(c) gives the splitting $k \mapsto (k,1)$ over K_3, and β is not identically 1 on $K_2 \times K_2$. The cover nevertheless splits over K_2, by $k \mapsto (k, (c,d)_2)$ for $c \neq 0$ and $k \mapsto (k,1)$ for $c = 0$. This is the local form at 2 of the classical splitting over $\Gamma_1(4)$ [Ku69, Ge76]. We have verified it numerically only (`checks/sec2_n2_check.py`), and the same section fails on K_1. Nothing below depends on it: the rest of the paper uses only that some K_m splits at each dyadic place.

The cover of $G(\mathbb{Q}_p)$. Let $\overline{G(\mathbb{Q}_p)}$ be the extension of $G(\mathbb{Q}_p) = \prod_{v|p} SL(2,k_v)$ by μ_2 defined by the cocycle

$$\beta_p(g_1,g_2) = \prod_{v|p} \beta_v(g_{1,v}, g_{2,v}).$$

It is the *quotient* of $\prod_{v|p} \overline{SL(2,k_v)}$ by the kernel $\{(\zeta_v) \mid \prod_v \zeta_v = 1\}$ of the product map $\prod_{v|p} \mu_2 \to \mu_2$, and it carries the quotient topology. For $H \subset G(\mathbb{Q}_p)$ put $\overline{H} = p^{-1}(H)$. Let B, N, A be the $\mathbb{Q}$-subgroups of G obtained from those of $SL(2)$, and let $T = \prod_{v|p} T_v$ with each T_v a maximal torus

of $SL(2,k_v)$. Let $K_p = \prod_{v|p} K_v$. For p odd, $K_p^* = \prod_{v|p} K_v^*$, with K_v^* as in Lemma 2.2(b), maps to a subgroup of $\overline{G(\mathbb{Q}_p)}$ which we also denote by K_p^*.

Lemma 2.3. *(a) $n \mapsto (n,1)$ splits $\overline{N(\mathbb{Q}_p)}$.*

(b) If p is odd, $\overline{K_p} = K_p^* \times \mu_2$.

(c) For every p, $k \mapsto (k,1)$ splits $\overline{K_{m,p}}$, where $K_{m,p} = \prod_{v|p} K_{m,v}$, as soon as $m \geq 2e_v + 1$ for every $v \mid p$; if p is odd, $m \geq 1$ suffices.

(d) If $p = \infty$, $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$, where $A(\mathbb{R})^0$ is the identity component of $A(k \otimes \mathbb{R})$.

(e) The center of $\overline{G(\mathbb{Q}_p)}$ is $\overline{\{\pm 1\}^{S_p}}$, where $S_p = \{v \mid p\}$. It is abelian, and the square of a lift of $\varepsilon = (\varepsilon_v)$ is $\prod_{v|p,\ \varepsilon_v = -1} (-1,-1)_v$.

(f) $\beta_p(a_1,a_2) = \prod_{v|p} (t_{1,v}, t_{2,v})_v$ for $a_i = a(t_i) \in A(\mathbb{Q}_p)$, and $\overline{A(\mathbb{Q}_p)}$ is abelian.

(g) The preimage of any abelian subgroup of $G(\mathbb{Q}_p)$, in particular $\overline{T}$, is abelian, and Lemma 2.2(h) holds for $G(\mathbb{Q}_p)$.

Proof. Each statement holds componentwise by Lemma 2.2, and the quotient map is compatible with all of them. □

2.1.1 *NAK* decompositions

Every element of the local cover factors as a unipotent element, a lift of a diagonal element and a lift of an element of the maximal compact subgroup. This factorization defines the height in the next subsection.

Lemma 2.4. *$\overline{G(\mathbb{Q}_p)} = N(\mathbb{Q}_p) \cdot \overline{A(\mathbb{Q}_p)} \cdot \overline{K_p}$, with $N(\mathbb{Q}_p)$ embedded by $n \mapsto (n,1)$. If $(n,1)\,\tilde{a}\,\tilde{k} = (n',1)\,\tilde{a}'\,\tilde{k}'$ are two such decompositions, then $p(\tilde{a}') \in p(\tilde{a})\,(A(\mathbb{Q}_p) \cap K_p)$. In particular $|t|_p := \prod_{v|p} |t_v|_v$ is well defined for $p(\tilde{a}) = a(t)$, and depends only on (g,ζ). For p odd one may take $\tilde{k} \in K_p^*$.*

Proof. The Iwasawa decomposition $G(\mathbb{Q}_p) = N(\mathbb{Q}_p)A(\mathbb{Q}_p)K_p$ holds, with a determined modulo $A(\mathbb{Q}_p) \cap K_p$, which is $\prod_{v|p} A(\mathcal{O}_v^\times)$ for $p < \infty$ and a product of copies of $\{\pm 1\}$ and of $\{\mathrm{diag}(z,\bar{z}) : |z| = 1\}$ for $p = \infty$. Since μ_2 is central and lies in $\overline{A(\mathbb{Q}_p)}$, this lifts to $\overline{G(\mathbb{Q}_p)}$. For p odd, $\overline{K_p} = K_p^* \mu_2$ and the μ_2 factor can be moved into $\tilde{a}$. The absolute value $|t|_p$ is trivial on $A(\mathbb{Q}_p) \cap K_p$. □

2.1.2 Heights

Heights measure how far an element lies from the maximal compact subgroup. They are used to truncate kernels in Section 4.2.

For $g = (g_v) \in G(\mathbb{Q}_p)$ put $\|g\|_p = \prod_{v|p} \|g_v\|_v$, where $\|g_v\|_v = \max(|a|_v, |b|_v, |c|_v, |d|_v)$ for v finite and $\|g_v\|_v = (|a|^2 + |b|^2 + |c|^2 + |d|^2)^{1/2}$, raised to the power $[k_v : \mathbb{R}]$, for v archimedean (here $|\cdot|$ is the usual absolute value). Extend to the cover by $\|(g,\zeta)\|_p = \|g\|_p$.

Define $H = H_B \colon \overline{G(\mathbb{Q}_p)} \to \mathbb{R}$ by

$$H(g,\zeta) = \log|t|_p = \sum_{v|p} \log|t_v|_v, \tag{2.5}$$

where $(g,\zeta) = (n,1)\,\tilde{a}\,\tilde{k}$ as in Lemma 2.4 and $p(\tilde{a}) = a(t)$. This is the product of the normalized absolute values over $v \mid p$, that is $|N_{k\otimes\mathbb{Q}_p/\mathbb{Q}_p}(t)|_p$; the reason is given in Section 2.1.3.

Lemma 2.5. *(a)* $H(\tilde{a}\tilde{a}') = H(\tilde{a}) + H(\tilde{a}')$ *for* $\tilde{a},\tilde{a}' \in \overline{A(\mathbb{Q}_p)}$.

(b) $H(\tilde{n}\,x\,\tilde{k}) = H(x)$ *for* $\tilde{n} \in \overline{N(\mathbb{Q}_p)}$, $x \in \overline{G(\mathbb{Q}_p)}$ *and* $\tilde{k} \in \overline{K_p}$.

(c) $|H(x)| \ll 1 + \log\|x\|_p$ *for* $x \in \overline{G(\mathbb{Q}_p)}$ *[Ar78].*

The global analog, including invariance under $B(\mathbb{Q})$ on the left, is Lemma 2.8.

2.1.3 Characteristic functions of Weyl chambers

This subsection identifies the space $\mathfrak{a}$ with $\mathbb{R}$ and records the characteristic functions of Arthur that enter the truncation. In rank one they are elementary.

Let $P_0 = B$; the standard parabolic subgroups are B and G. For a standard $P = MN_P$ let A_P be the split component of the center of M over $\mathbb{Q}$. Thus $A_G = 1$, because the center $\{\pm 1\}$ of $SL(2)$ is finite, and $A_B \cong \mathbb{G}_m$ is the image of $t \mapsto a(t)$, $t \in \mathbb{Q}^\times \subset k^\times$: the $\mathbb{Q}$-split part of the torus $A \cong \mathrm{Res}_{k/\mathbb{Q}}\,\mathbb{G}_m$, which has $\mathbb{Q}$-rank one.

The $\mathbb{Q}$-rational characters of A form $X(A)_\mathbb{Q} \cong \mathbb{Z}$, generated by $\chi_1 = N_{k/\mathbb{Q}} \circ t$. Put $\mathfrak{a} = \mathfrak{a}_B = \mathrm{Hom}(X(A)_\mathbb{Q},\mathbb{R})$, identified with $\mathbb{R}$ by $X \mapsto X(\chi_1)$. Then H of (2.5) and its global analog are $\langle H(x),\chi_1\rangle = \log|\chi_1(\tilde{a})|$, which is why H uses the norm. Let $d = [k:\mathbb{Q}]$. The simple root α of (B,A_B) is $a(s) \mapsto s^2$ ($s \in \mathbb{Q}^\times$), while $\chi_1(a(s)) = s^d$; so $\alpha = (2/d)\chi_1$ in $\mathfrak{a}^*$, and $\alpha(X) = 2X/d$ under $\mathfrak{a} \cong \mathbb{R}$. Let $\alpha^\vee = X_\alpha \in \mathfrak{a}$ be the coroot $s \mapsto a(s)$, so $X_\alpha = \langle \chi_1, \alpha^\vee\rangle = d$ and $\alpha(\alpha^\vee) = 2$, and let $\varpi = \frac{1}{2}\alpha^\vee$ be the fundamental coweight, so $\alpha(\varpi) = 1$. The modulus character is $\delta_B = e^{2H}$.

Let $\mathfrak{a}^+ = \{X \in \mathfrak{a} \mid \alpha(X) > 0\}$ and $(\mathfrak{a}^*)^+ = \{\Lambda \in \mathfrak{a}^* \mid \Lambda(\varpi) > 0\}$, with characteristic functions τ_B and $\hat{\tau}_B$, and set $\tau_G = \hat{\tau}_G = 1$. For standard $P_1 \subset P_2$ let τ_1^2, $\hat{\tau}_1^2$ and σ_1^2 be as in [Ar78]. With $\mathfrak{a} \cong \mathbb{R}$:

$$\tau_B = \tau_B^G = \hat{\tau}_B = \hat{\tau}_B^G = \text{characteristic function of } \{u \in \mathbb{R} \mid u > 0\}, \qquad \tau_1^2 = \begin{cases} \tau_1, & P_2 = G, \\ 1, & P_2 \neq G, \end{cases} \tag{2.6}$$

$$\sigma_B^G = \tau_B, \qquad \sigma_B^B = 0, \qquad \sigma_G^G = 1. \tag{2.7}$$

2.1.4 Algebras of functions

This subsection defines the test functions on the local cover. The definition uses the topology of Lemma 2.1(c), not the coordinates (g,ζ).

We follow Li [Li14a, Li14b]. A representation π of a cover is *genuine* if $\pi(1,\zeta) = \zeta$ for $\zeta \in \mu_2$, and a function f on the cover is *anti-genuine* if $f(g,\zeta) = \zeta^{-1} f(g,1)$. Test functions are anti-genuine. Then $\pi(f) = \int f(\tilde{g})\pi(\tilde{g})\,d\tilde{g}$ picks up the factor $\int_{\mu_2} \zeta^{-1}\zeta\,d\zeta = 1$, so it is in general nonzero on genuine representations, and it vanishes on representations that are trivial on μ_2.

For $n=2$, $\zeta^{-1}=\zeta$, so "anti-genuine" and "genuine" define the same functions. We keep Li's terminology so that the formulas remain correct for general n; see Remark 2026.3.

Let $C_c^\infty(\overline{G(\mathbb{Q}_p)})_-$ be the space of anti-genuine functions on $\overline{G(\mathbb{Q}_p)}$ that are smooth and compactly supported: locally constant if $p<\infty$, and C^∞ on the Lie group $\overline{G(\mathbb{R})}$ if $p=\infty$, in both cases for the topology of Lemma 2.1(c). For $p<\infty$ this means that f is anti-genuine, that its support has compact image in $G(\mathbb{Q}_p)$, and that f is invariant on both sides under the image of $K_{m,p}$ under $k\mapsto(k,1)$ for some m as in Lemma 2.3(c). The space is *not* described by the condition that $g\mapsto f(g,1)$ be smooth on $G(\mathbb{Q}_p)$. That function is Borel: on each translate g_0U it is the product of the smooth function $u\mapsto f((g_0,1)(u,1))$ and the Borel function $u\mapsto\beta_p(g_0,u)$. In general it is not continuous, because the section $g\mapsto(g,1)$ is not (Example 2.A); the unit element below is an example. Later sections use the coordinates only to integrate (in them f is bounded and Borel, with compact support) and to restrict f to the continuous sections of Lemma 2.2(i).

Let p be odd. The *spherical Hecke algebra* $\mathscr{H}(\overline{G(\mathbb{Q}_p)},K_p^*)$ consists of the $f\in C_c^\infty(\overline{G(\mathbb{Q}_p)})_-$ with $f(\tilde{k}x\tilde{k}')=f(x)$ for $\tilde{k},\tilde{k}'\in K_p^*$. Its unit element is

$$e_p(\tilde{k}\zeta)=\zeta^{-1}\ (\tilde{k}\in K_p^*,\ \zeta\in\mu_2),\qquad e_p=0\text{ off }\overline{K_p}. \tag{2.8}$$

It is locally constant, since K_p^* is an open subgroup (Lemma 2.2(i)). In coordinates, $e_p(k,\zeta)=\zeta^{-1}\prod_{v|p}\kappa_v(k_v)$ for $k\in K_p$. So $g\mapsto e_p(g,1)$ is $\prod_v\kappa_v$ on K_p, which is not locally constant: for $p=3$, $k=\mathbb{Q}$ and k_j as in Example 2.A, $k_j\to-1$ and $e_3(k_j,1)=(-1)^j$. The function $(k,\zeta)\mapsto\zeta^{-1}$ on $\overline{K_p}$ is a different function, $\prod_v\kappa_v\cdot e_p$; it is neither locally constant nor bi-K_p^*-invariant.

2.1.5 Measures

This subsection fixes the Haar measures on the local groups and on their covers.

Let dk be the Haar measure on K_p of total volume 1. Fix a nontrivial additive character ψ_v of k_v for each v, and let dn be the measure on $N(\mathbb{Q}_p)\cong\bigoplus_{v|p}k_v$ that is the product of the self-dual measures for the ψ_v. Let da be a Haar measure on $A(\mathbb{Q}_p)$. If T is a maximal torus of G over $\mathbb{Q}_p$, and $h\in G(\overline{\mathbb{Q}}_p)$ conjugates T to A, let dt be the pull-back of da as in [Ro90]; stably conjugate tori get compatible measures as in [Ro90]. Put

$$\delta_B(a(t))=|\alpha(a(t))|_p=|t|_p^2.$$

Normalize the Haar measure dg on $G(\mathbb{Q}_p)$ by

$$\int_{G(\mathbb{Q}_p)}\phi(g)\,dg=\int_N\int_A\int_K\phi(nak)\,\delta_B(a)^{-1}\,dk\,da\,dn=\int_K\int_A\int_N\phi(kan)\,\delta_B(a)\,dn\,da\,dk. \tag{2.9}$$

The two expressions agree because dg is invariant under $g\mapsto g^{-1}$ (as G is unimodular) and $(nak)^{-1}=k^{-1}a^{-1}n^{-1}$; see [Kn02, Ch. VIII] for the real case.

On a cover we use $d\tilde{g}=dg\,d\zeta$, where $d\zeta$ gives μ_2 total volume 1. Then $\overline{K_p}$ has volume 1, and e_p of (2.8) is idempotent: for $x=\tilde{k}\in K_p^*$,

$$(e_p*e_p)(x)=\int_{\overline{K_p}}e_p(y)\,e_p(y^{-1}x)\,dy=\int_{K_p^*}\int_{\mu_2}\zeta^{-1}\zeta\,d\zeta\,dy=1.$$

2.2 Global aspects

The adelic cover is not the cover defined by $\prod_v \beta_v$: that product does not converge, because β_v is not trivial on $K_v \times K_v$ (Lemma 2.2(b)). The standard remedy [Ku69, Ge76, KP84], used here, is to correct β_v by the coboundary of κ_v. This choice also fixes the splitting over $G(\mathbb{Q})$.

For each place v of k extend κ_v to $SL(2,k_v)$ by

$$\kappa_v(g) = \begin{cases} (c,d)_v, & v \text{ finite non-dyadic},\ g \in K_v,\ 0 < |c|_v < 1, \\ 1, & \text{otherwise}, \end{cases}$$

so $\kappa_v \equiv 1$ at dyadic and archimedean places, and put

$$\sigma_v(g_1,g_2) = \beta_v(g_1,g_2)\,\kappa_v(g_1)\kappa_v(g_2)\kappa_v(g_1g_2). \tag{2.10}$$

This cocycle is cohomologous to β_v; it defines the same cover $\overline{SL(2,k_v)}$, identified with the β_v-model by $(g,\zeta)_\sigma \mapsto (g,\kappa_v(g)\zeta)_\beta$. By Lemma 2.2(b), $\sigma_v \equiv 1$ on $K_v \times K_v$ at every finite non-dyadic v. Since $g \in G(\mathbb{A})$ has $g_v \in K_v$ for almost all v, the product

$$\sigma_{\mathbb{A}}(g_1,g_2) = \prod_v \sigma_v(g_{1,v},g_{2,v}) \tag{2.11}$$

has only finitely many factors different from 1. It is a Borel 2-cocycle on $G(\mathbb{A}) = SL(2,\mathbb{A}_k)$ (not a continuous one, since its factors are not continuous) and defines an extension of groups

$$1 \to \mu_2 \to \overline{G(\mathbb{A})} \xrightarrow{p} G(\mathbb{A}) \to 1.$$

Equivalently, $\overline{G(\mathbb{A})}$ is the restricted product of the $\overline{SL(2,k_v)}$ with respect to the subgroups K_v^*, divided by $\{(\zeta_v) \in \bigoplus_v \mu_2 \mid \prod_v \zeta_v = 1\}$. It is a *quotient* of the restricted product, not a subgroup. Factorization of genuine representations of $\overline{G(\mathbb{A})}$ (and of $\overline{A(\mathbb{A})}$ in Section 3) refers to this quotient. Group law, inverses and conjugation are given by (2.2) and (2.3) with $\sigma_{\mathbb{A}}$ in place of β. For $H \subset G(\mathbb{A})$ put $\overline{H} = p^{-1}(H)$.

Topology. Each $\overline{SL(2,k_v)}$ has the topology of Lemma 2.1(c), and K_v^* is a compact open subgroup of it for every finite non-dyadic v (Lemma 2.2(i)). So the restricted product is a locally compact group. The subgroup $Z_0 = \{(\zeta_v) \mid \prod_v \zeta_v = 1\}$ is central and discrete: it meets $\prod_{v\in S} W_v \times \prod_{v\notin S} K_v^*$ only in the identity when $W_v \cap \mu_2 = \{1\}$. We give $\overline{G(\mathbb{A})}$ the quotient topology. Then $\overline{G(\mathbb{A})}$ is a locally compact group, p is a two-fold covering map, and each $\overline{SL(2,k_v)}$ is a closed subgroup with its own topology. In the coordinates (g,ζ) of the cocycle $\sigma_{\mathbb{A}}$, the map $g \mapsto (g,1)$ is a homeomorphism of $\prod_{v|\infty} U_v \times K_f$ onto an open subset, for U_v as in Lemma 2.1(b) and K_f as in Lemma 2.6(c) below; it is not continuous on $G(\mathbb{A})$. As in the local case, the coordinates are Borel coordinates, and continuity and smoothness on $\overline{G(\mathbb{A})}$ refer to this topology. Since $G(\mathbb{Q})$ is discrete in $G(\mathbb{A})$ and p is a covering, $\overline{G(\mathbb{Q})}$ is discrete in $\overline{G(\mathbb{A})}$, and so is the image of $G(\mathbb{Q})$ under the splitting of Lemma 2.6(a).

For $\gamma \in SL(2,k)$ we have $\gamma \in K_v$ and $\mathrm{ord}_v(c) \le 0$ or $c = 0$ for almost all v, so

$$\kappa_{\mathbb{A}}(\gamma) = \prod_v \kappa_v(\gamma) = \prod_{\substack{v \text{ finite non-dyadic} \\ \gamma\in K_v,\ 0<|c|_v<1}} (c,d)_v \tag{2.12}$$

is a finite product.

Lemma 2.6. *(a) The map $\gamma \mapsto (\gamma, \kappa_{\mathbb{A}}(\gamma))$ is a homomorphism $G(\mathbb{Q}) \to \overline{G(\mathbb{A})}$, so $\overline{G(\mathbb{Q})} = G(\mathbb{Q}) \times \mu_2$.*

(b) $n \mapsto (n,1)$ splits $\overline{N(\mathbb{A})}$, and on $B(\mathbb{Q})$ the splitting of (a) is $b \mapsto (b,1)$.

(c) Let S be a finite set of places containing the archimedean and dyadic places, and let $K_f = \prod_{v \in S \smallsetminus \Sigma_k^\infty} K_{m_v,v} \times \prod_{v \notin S} K_v$, with $m_v \geq 1$, and $m_v \geq 2e_v + 1$ if v is dyadic. Then $k \mapsto (k,1)$ is a homomorphism $K_f \to \overline{G(\mathbb{A})}$; we write K_f^* for its image. (At $v \notin S$, translated to the β_v-model, its component is K_v^*.)

(d) The center of $\overline{G(\mathbb{A})}$ is $\overline{Z(G(\mathbb{A}))}$, where $Z(G(\mathbb{A})) = \prod_v \{\pm 1\}$. It is abelian, and a lift of $\varepsilon = (\varepsilon_v)$ has square $\prod_{v:\,\varepsilon_v = -1}(-1,-1)_v$, a finite product. It contains the subgroup $Z(G(\mathbb{Q})) \times \mu_2$ of order 4, a direct product, embedded by (a). It is not the direct product $Z(G(\mathbb{A})) \times \mu_2$ when k has a real place.

(e) $\sigma_{\mathbb{A}}(a(t_1), a(t_2)) = \prod_v (t_{1,v}, t_{2,v})_v$ for $t_1, t_2 \in \mathbb{A}_k^\times$, and the preimage of every abelian subgroup of $G(\mathbb{A})$, in particular $\overline{A(\mathbb{A})}$, is abelian.

Proof. (a) For $\gamma_1, \gamma_2 \in SL(2,k)$, $\prod_v \beta_v(\gamma_1, \gamma_2) = 1$ by the product formula (H4), because $\beta_v(\gamma_1, \gamma_2)$ is the local Hilbert symbol of two elements of $k^\times$. Hence $\sigma_{\mathbb{A}}(\gamma_1, \gamma_2) = \kappa_{\mathbb{A}}(\gamma_1)\kappa_{\mathbb{A}}(\gamma_2)\kappa_{\mathbb{A}}(\gamma_1\gamma_2)$, which is the condition for $\gamma \mapsto (\gamma, \kappa_{\mathbb{A}}(\gamma))$ to be a homomorphism.

(b) On N every β_v and κ_v is 1. On $B(\mathbb{Q})$ every κ_v is 1 because $c = 0$.

(c) It suffices that $\sigma_v \equiv 1$ on the v-component of $K_f \times K_f$ for each finite v. At $v \notin S$ this is Lemma 2.2(b). At finite $v \in S$, $\kappa_v \equiv 1$ on $K_{m_v,v}$ (non-dyadic: $c \in \varpi\mathscr{O}$ and d is a square; dyadic: by definition), so $\sigma_v = \beta_v \equiv 1$ there by Lemma 2.2(c).

(d) Lemma 2.2(g) applies to σ_v, since $\kappa_v(g_1)\kappa_v(g_2)\kappa_v(g_1g_2)$ is symmetric in commuting g_1, g_2; so preimages of abelian subgroups are abelian, and $Z(G(\mathbb{A}))$ is central. Its image under p is central, so equality holds. Since $\kappa_v(\pm 1) = 1$, the square of a lift of ε is $\prod_v \sigma_v(\varepsilon_v, \varepsilon_v) = \prod_{\varepsilon_v = -1}(-1,-1)_v$, which is finite by (H3). For $\varepsilon = -1 \in Z(G(\mathbb{Q}))$ it is $\prod_v (-1,-1)_v = 1$ by (H4). If v_0 is real, the lift of the element that is -1 at v_0 and 1 elsewhere has order 4.

(e) As in Lemma 2.2(f), since $\kappa_v \equiv 1$ on A. □

From now on we identify $G(\mathbb{Q})$ with its image under Lemma 2.6(a), and $N(\mathbb{A})$ with $N(\mathbb{A}) \times 1$. By (b), the two identifications agree on $N(\mathbb{Q})$. The splitting over $B(\mathbb{Q})$ by $b \mapsto (b,1)$ is what the constant terms, Eisenstein series and the truncation operator of Sections 4 and 5 use.

Example 2.C. Let $k = \mathbb{Q}$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z})$ with $c \neq 0$. Then $\gamma \in K_v$ for every v, and for odd $p \mid c$, $(c,d)_p = \left(\frac{d}{p}\right)^{\mathrm{ord}_p(c)}$ by (H3). So

$$\kappa_{\mathbb{A}}(\gamma) = \left(\frac{d}{c'}\right), \qquad c' = \text{the odd part of } |c|,$$

a Jacobi symbol: the factor behind the theta multiplier on $\Gamma_0(4)$ [Ku69, Ge76]. The script checks this on random elements of $SL(2,\mathbb{Z})$.

Let $K_0 = \prod_v K_v$ be the standard maximal compact subgroup of $G(\mathbb{A})$, with K_v as in Section 2.1; K_f and K_∞ denote its finite and archimedean parts.

Lemma 2.7 (Strong approximation; compare [Ku69, Prop. 5], [Ge76])**.** *Let K_f and K_f^* be as in Lemma 2.6(c), and put $\Gamma = G(\mathbb{Q}) \cap K_f$, a congruence subgroup of $SL(2, \mathscr{O}_k)$, viewed in $G(\mathbb{R}) = SL(2, k \otimes \mathbb{R})$. For $\gamma \in \Gamma$ let $\tilde{k}_\gamma \in K_f^*$ be the lift of γ_f. Then $\tilde{\gamma} = \gamma \tilde{k}_\gamma^{-1}$ lies in $\overline{G(\mathbb{R})}$, the map $\gamma \mapsto \tilde{\gamma}$ is an injective homomorphism, its image $\widetilde{\Gamma}$ is discrete of finite covolume, and*

$$G(\mathbb{Q}) \backslash \overline{G(\mathbb{A})} / K_f^* \;\cong\; \widetilde{\Gamma} \backslash \overline{G(\mathbb{R})}.$$

Proof. Since $SL(2)$ is simply connected and $G(\mathbb{R})$ is not compact, strong approximation gives $G(\mathbb{A}) = G(\mathbb{Q})\, G(\mathbb{R})\, K_f$. Lifting to the cover, with $\mu_2 \subset \overline{G(\mathbb{R})}$, gives $\overline{G(\mathbb{A})} = G(\mathbb{Q})\, \overline{G(\mathbb{R})}\, K_f^*$, so the double coset space has a single component. The stabilizer computation is the usual one: $\gamma \in G(\mathbb{Q})$ lies in $\overline{G(\mathbb{R})} K_f^*$ exactly when $\gamma_f \in K_f$, and then $\gamma = \tilde{\gamma} \tilde{k}_\gamma$. The map $\gamma \mapsto \tilde{\gamma}$ is a homomorphism because lifts of elements of $G(\mathbb{A})$ with disjoint supports commute: $\sigma_v(1,h) = \sigma_v(h,1) = 1$. □

2.2.1 *NAK* decompositions

The local decompositions of Section 2.1 assemble into an adelic one, which defines the global height.

Lemma 2.8. *(a) $\overline{G(\mathbb{A})} = N(\mathbb{A}) \cdot \overline{A(\mathbb{A})} \cdot \overline{K_0}$. In a decomposition $(n,1)\,\tilde{a}\tilde{k}$, $p(\tilde{a})$ is determined modulo $A(\mathbb{A}) \cap K_0$, and*

$$H(g,\zeta) = \log |t|_{\mathbb{A}_k} = \sum_v \log |t_v|_v, \qquad p(\tilde{a}) = a(t),$$

is well defined.

(b) H satisfies Lemma 2.5(a)–(c) globally, and $H(\gamma x) = H(x)$ for $\gamma \in B(\mathbb{Q})$ and $x \in \overline{G(\mathbb{A})}$.

Proof. (a) As for Lemma 2.4, place by place. (b) For $\gamma = n' a(s) \in B(\mathbb{Q})$, identified with $(\gamma, 1)$ by Lemma 2.6(b), and $x = (n,1)\tilde{a}\tilde{k}$, $\gamma x = (n'', 1)\,(a(s),1)\tilde{a}\tilde{k}$ with $n'' \in N(\mathbb{A})$, so $H(\gamma x) = \log |s|_{\mathbb{A}_k} + H(x) = H(x)$ by the product formula. □

2.2.2 Heights

The global norm is the product of the local norms. Its elementary properties are recorded here for later use.

For $g = (g_v) \in G(\mathbb{A})$ put $\|g\| = \prod_v \|g_v\|_v$ and $\|(g,\zeta)\| = \|g\|$.

Lemma 2.9. *For $x, y \in \overline{G(\mathbb{A})}$ and $\tilde{k}, \tilde{k}' \in \overline{K_0}$: (a) $\|x\| \geq 1$; (b) $\|\tilde{k} x \tilde{k}'\| = \|x\|$; (c) $\|xy\| \leq \|x\|\,\|y\|$; (d) $\|x^{-1}\| = \|x\|$.*

Proof. Each holds for $G(\mathbb{A})$ place by place, since $\det g_v = 1$ forces $\|g_v\|_v \geq 1$. For (d), g^{-1} has the same entries as g up to order and sign. The cover does not change $\|\cdot\|$. □

2.2.3 Algebras of functions

The global test functions are products of local ones, almost all of them equal to the unit element.

A function on $\overline{G(\mathbb{A})}$ is *anti-genuine* if $f(g,\zeta)=\zeta^{-1}f(g,1)$. Let $\mathscr{H}(\overline{G(\mathbb{A})})$ be the span of the functions $f=\bigotimes_v f_v$, $f_v\in C_c^\infty(\overline{SL(2,k_v)})_-$, with f_v equal to the unit element e_v of (2.8) (defined with K_v^*) for almost all v. Such a product is a function on the restricted product of the local covers which is invariant under $\{(\zeta_v)\mid\prod\zeta_v=1\}$, because each f_v is anti-genuine; so it descends to $\overline{G(\mathbb{A})}$ and is anti-genuine there. It is smooth on $\overline{G(\mathbb{A})}$: invariant under a compact open subgroup of the finite part, and C^∞ in the archimedean variables.

2.2.4 Measures

This subsection fixes the global measures. With them the quotient $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ has volume 1.

Let $dk=\prod_v dk_v$ on K_0. Let $dg=\prod_v dg_v$ be the Tamagawa measure on $G(\mathbb{A})$ [We82], with dg_v built from the self-dual measures of a fixed nontrivial character $\psi=\prod\psi_v$ of $\mathbb{A}_k/k$; then $\mathrm{vol}(G(\mathbb{Q})\backslash G(\mathbb{A}))=1$. For a maximal torus T of G over $\mathbb{Q}$ let dt be the Tamagawa measure on $T(\mathbb{A})$, written da for $T=A$, and let dn be the product measure on $N(\mathbb{A})$, for which $\mathrm{vol}(N(\mathbb{Q})\backslash N(\mathbb{A}))=1$. On the cover put $d\tilde{g}=dg\,d\zeta$ with $\mathrm{vol}(\mu_2)=1$, so $\mathrm{vol}(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})=1$.

2.3 The double cover of $GL(2)$

Section 7 compares $SL(2)$-conjugacy with $GL(2)$-conjugacy on the cover, so we need a double cover of $GL(2)$ which restricts to the one above. There are two natural choices for $n=2$ (Remark 2026.2); we fix one, and compute the conjugation action of $GL(2,F)$ on $\overline{SL(2,F)}$ for later use.

For $g_1,g_2\in GL(2,F)$, with $x(g)$ defined as before, put

$$\sigma(g_1,g_2)=\Big(\frac{x(g_1g_2)}{x(g_1)},\ \frac{x(g_1g_2)}{x(g_2)\det g_1}\Big)_v, \tag{2.13}$$

the $n=2$ case of Kubota's cocycle on $GL(2)$ [Ku69, Ge76]. Let $\overline{GL(2,F)}$ be the corresponding extension by μ_2.

Lemma 2.10. *(a) σ is a 2-cocycle on $GL(2,F)$ whose restriction to $SL(2,F)$ is β. Hence $p^{-1}(SL(2,F))\subset\overline{GL(2,F)}$ is the cover $\overline{SL(2,F)}$ of Section 2.1.*

(b) For v finite non-dyadic, $k\mapsto(k,\kappa'(k))$ is a homomorphism on $GL(2,\mathscr{O})$, where $\kappa'(k)=(c,d/\det k)_v$ if $0<|c|_v<1$ and $\kappa'(k)=1$ otherwise. On $SL(2,\mathscr{O})$, $\kappa'=\kappa$.

(c) Conjugation by any lift of $h\in GL(2,F)$ defines an automorphism $\mathrm{Ad}(h)$ of $\overline{SL(2,F)}$ lifting conjugation by h, and $h\mapsto\mathrm{Ad}(h)$ is an action of $GL(2,F)$. If h commutes with $\gamma\in SL(2,F)$, then

$$\mathrm{Ad}(h)(\gamma,\zeta)=\Big(\gamma,\ \Big(\frac{x(h\gamma)}{x(h)},\ \det h\Big)_v\zeta\Big). \tag{2.14}$$

In particular $\mathrm{Ad}(\mathrm{diag}(y,1))(a(t),\zeta)=(a(t),(y,t)_v\zeta)$, so the preimage of the diagonal torus of $GL(2,F)$ is not abelian unless $F=\mathbb{C}$. Likewise $E^\times\supset T_\theta$ (embedded as $\{\begin{pmatrix}a&b\theta\\b&a\end{pmatrix}\}$) acts on $\overline{T_\theta}$ through (2.14).

Proof. (a) On $SL(2)$, $\det g_1 = 1$ and (2.13) becomes (2.1). The cocycle identity is due to Kubota [Ku69]; we have also checked it numerically.

(b) We have checked this numerically for $F = \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7$; see Remark 2026.5.

(c) Since μ_2 is central, conjugation by (h,ζ) does not depend on ζ, and it preserves $p^{-1}(SL(2,F))$. If $h\gamma = \gamma h$, then $\mathrm{Ad}(h)(\gamma,\zeta) = (\gamma, \zeta\,\sigma(h,\gamma)\sigma(\gamma,h))$. With $x_3 = x(h\gamma)$, bimultiplicativity gives $\sigma(h,\gamma) = (x_3/x(h), x_3/x(\gamma))\,(x_3/x(h), \det h)$, and $\sigma(\gamma,h) = (x_3/x(\gamma), x_3/x(h))$ since $\det\gamma = 1$. By symmetry (H1) the first factors cancel. For $h = \mathrm{diag}(y,1)$ and $\gamma = a(t)$: $x(h\gamma) = t^{-1}$ and $x(h) = 1$, giving $(t^{-1},y) = (y,t)$. This is nontrivial for suitable y,t unless $F = \mathbb{C}$. □

The consequences are taken up in Section 7: the sign in (2.14) is $(\det h, \det(1+\gamma))_v$ (Lemma 7.2), conjugation by $GL(2,F)$ can exchange the two lifts of an element, and stable conjugacy on the cover is conjugation by $GL(2,F)$ twisted by this sign.

Remark 2026.2. The cocycle $\sigma'(g_1,g_2) = \sigma(g_1,g_2)\,(\det g_1, \det g_2)_v$ is another double cover of $GL(2,F)$ with the same restriction to $SL(2,F)$; these are the two Kazhdan–Patterson covers for $n=2$ [KP84]. We fix σ. Section 7 uses the cover of $GL(2,F)$ only through the automorphisms $\mathrm{Ad}(h)$ of $\overline{SL(2,F)}$, and these are the same for σ and σ': an automorphism of $\overline{SL(2,F)}$ is determined by the automorphism of $SL(2,F)$ that it induces (Lemma 7.1(a)). In particular the commutator formula (2.14) is the same for σ': the extra factor $(\det h, \det\gamma)(\det\gamma, \det h)$ is 1 by (H1). The adelic cover of $GL(2,\mathbb{A}_k)$ is built as in Section 2.2, with κ_v' in place of κ_v; Hilbert reciprocity for $\prod_v \sigma_v$ on $GL(2,k)$ is checked numerically for $k = \mathbb{Q}$.

Remark 2026.3. For general n (with $\mu_n \subset k$) the following change. The conjugation action of $\overline{A(F)}$ on itself is through $(t_1,t_2)_n^2$, so $\overline{A(F)}$ is abelian only for $n \le 2$; the argument of Lemma 2.2(g) uses the symmetry of the quadratic symbol and does not extend. Lemma 2.2(c) holds with $1 + \varpi^m\mathcal{O} \subset F^{\times n}$ (the same proof). Genuine (ζ) and anti-genuine (ζ^{-1}) differ, and with the 1995 convention $R(f)$ vanishes on L^2_{gen} unless $n \mid 2$.

2.4 Status of the results

This subsection lists the statements of Section 2 that rest on citation or on numerical checks.

Remark 2026.4. *What rests on citation.* Taken from the literature: Lemma 2.1(a) and the existence of the topological covering group in Lemma 2.1(c) [Ku67, Mo68]; Lemma 2.2(b) and the cocycle identity for (2.13) [Ku69, Ge76]; the statement on $\overline{SO(2)}$ in Lemma 2.2(d); Lemma 2.5(c) [Ar78]; strong approximation in Lemma 2.7; and the facts about Tamagawa measures in Section 2.2. Lemma 2.10(b) and Remark 2026.1 are supported by numerical checks only. Everything else in this section is proved in the text.

Remark 2026.5. The script `checks/sec2_n2_check.py` verifies, in exact rational arithmetic at $v = \infty, 2, 3, 5, 7$: the cocycle identity and the two forms of (2.1); Lemma 2.1(b) at $v = \infty$ and Example 2.A; Lemma 2.2(a)–(c), (e)–(i); Remark 2026.1; the product formula and the splitting of Lemma 2.6(a), (b), (d); Example 2.C; and Lemma 2.10(a)–(c) with Remark 2026.2. The checks are on random elements of $SL(2,\mathbb{Q})$, $GL(2,\mathbb{Q})$, $SL(2,\mathbb{Z}_{(p)})$ and the tori $T_\theta(\mathbb{Q})$; they are evidence, not proofs. Of the statements in this section, Lemma 2.2(b) and Lemma 2.10(a), (b) are supported here only by citation and these checks.

3 The rank 0 case

Before the trace formula for $\overline{G(\mathbb{A})}$ we treat the covers of the tori of G, where it is elementary. For $n=2$ these covers are abelian (Lemma 2.6(e)), but in general not split, so a genuine automorphic character cannot be written as an ordinary one times a sign (Example 3.A). The trace formula is then Poisson summation twisted by one fixed genuine automorphic character (Theorem 3.4, Lemma 3.5), and it gives a bijection between genuine automorphic characters and Hecke characters (Theorem 3.2). For the split torus, Weil's character χ_ψ (Lemma 3.3) makes the bijection canonical once ψ is fixed. We also define the distributions $I^A_{\mathscr{O}}$ and $I^A_{\tilde{\chi}}$ used in Section 6. We work with an arbitrary maximal torus because the same formula holds for anisotropic tori; for $SL(2)$ it is what Labesse and Langlands apply to the κ-orbital integrals, and Section 7.3 explains why the double cover does not need it there.

3.1 The cover of a torus

This subsection describes the cover of a maximal torus and its genuine characters. For the split torus, Weil's character gives a canonical choice among them.

Let T be a maximal torus of G defined over $\mathbb{Q}$. Then $T=\mathrm{Res}_{k/\mathbb{Q}}T_k$ for a maximal torus T_k of $SL(2)$ over k, which is either split, and then $GL(2,k)$-conjugate to A, or of the form T_θ of Section 2.1 with $E=k(\sqrt{\theta})$ a quadratic extension of k. Put

$$T(\mathbb{A})^1=\{t\in T(\mathbb{A})\mid |\chi(t)|=1 \text{ for all } \chi\in X(T)_{\mathbb{Q}}\}.$$

For $T=A$ this is $A(\mathbb{A})^1=\{a(t)\mid |t|_{\mathbb{A}_k}=1\}$, the kernel of H of Lemma 2.8; for $T_k=T_\theta$ the torus T is anisotropic over $\mathbb{Q}$ and $T(\mathbb{A})^1=T(\mathbb{A})$. In both cases $T(\mathbb{Q})\subset T(\mathbb{A})^1$ and $T(\mathbb{Q})\backslash T(\mathbb{A})^1$ is compact [PR94].

Let $\overline{T(\mathbb{A})}$ and $\overline{T(\mathbb{A})^1}$ be the preimages in $\overline{G(\mathbb{A})}$, and $\overline{T(k_v)}$ the preimage of $T_k(k_v)$ in $\overline{SL(2,k_v)}$. By Lemmas 2.2(g) and 2.6(e) these groups are abelian. On A the cocycle is

$$\sigma_{\mathbb{A}}(a(t_1),a(t_2))=\prod_v(t_{1,v},t_{2,v})_v,\qquad t_1,t_2\in\mathbb{A}_k^\times, \tag{3.1}$$

since $\kappa_v\equiv 1$ on A. Exactly as for G in Section 2.2, $\overline{T(\mathbb{A})}$ is the *quotient* of the restricted product of the $\overline{T(k_v)}$, taken with respect to the subgroups $T(\mathscr{O}_v)^*=\{(t,\kappa_v(t))\mid t\in T_k(k_v)\cap K_v\}$, by $\{(\zeta_v)\mid \prod_v\zeta_v=1\}$; it is not a subgroup of that restricted product. Each $\overline{T(k_v)}$ maps injectively into $\overline{T(\mathbb{A})}$. These are closed subgroups of the covers of Section 2, and they carry the induced topologies; for $\overline{T(\mathbb{A})}$ this is again the quotient topology of the restricted product. We regard $T(\mathbb{Q})$ as a subgroup of $\overline{T(\mathbb{A})^1}$ by Lemma 2.6(a), $\gamma\mapsto(\gamma,\kappa_{\mathbb{A}}(\gamma))$; for $T=A$ this is $a(t)\mapsto(a(t),1)$, $t\in k^\times$.

Conventions. As in Section 2.1.4, a representation π of one of these covers is *genuine* if $\pi(1,\zeta)=\zeta$, and a function f is *anti-genuine* if $f(x\zeta)=\zeta^{-1}f(x)$ for $\zeta\in\mu_2$. For $n=2$ the two conditions on functions coincide. Let $C_c^\infty(\overline{T(k_v)})_-$ and $C_c^\infty(\overline{T(\mathbb{A})^1})_-$ be the spaces of anti-genuine functions that are smooth and compactly supported on these locally compact groups: locally constant in the non-archimedean variables and C^∞ in the archimedean ones. As in Section 2.1.4, this is a condition on the topological group, not on the function $t\mapsto f(t,1)$. For the split torus the two agree, because $a\mapsto(a,1)$ is continuous on $A(k_v)$ and the cocycle (3.1) is locally constant

(Lemma 2.2(i)); for an anisotropic torus they do not, for example for $SO(2) \subset SL(2,\mathbb{R})$, whose preimage is connected (Lemma 2.2(d)). At places where $T(\mathscr{O}_v)^*$ is defined let $e_{T,v}$ be the anti-genuine function equal to 1 on $T(\mathscr{O}_v)^*$ and supported on $\overline{T(\mathscr{O}_v)}$. Let $\mathscr{H}(\overline{T(\mathbb{A})})$ be the span of the products $\bigotimes_v f_v$, $f_v \in C_c^\infty(\overline{T(k_v)})_-$, with $f_v = e_{T,v}$ for almost all v; these descend to $\overline{T(\mathbb{A})}$ as in Section 2.2. Measures on $T(\mathbb{A})$ are as in Section 2.2, and on covers μ_2 has total volume 1.

For a locally compact group H let $\Pi(H)$ be the set of (equivalence classes of) irreducible admissible genuine representations of H, and $\Pi_u(H)$ the unitary ones.

Lemma 3.1. *Let $F = k_v$.*

(a) Every element of $\Pi(\overline{T(F)})$, $\Pi(\overline{T(\mathbb{A})})$ or $\Pi(\overline{T(\mathbb{A})^1})$ is a genuine character.

(b) Genuine characters of $\overline{T(F)}$ exist, and any two differ by a character of $T(F)$: they form a torsor under $T(F)^\wedge$.

(c) Every genuine character $\bar\chi$ of $\overline{T(\mathbb{A})}$ factors uniquely as $\bar\chi = \prod_v \bar\chi_v$, where $\bar\chi_v$ is a genuine character of $\overline{T(k_v)}$, trivial on $T(\mathscr{O}_v)^*$ for almost all v. Conversely, every such family defines a genuine character of $\overline{T(\mathbb{A})}$. The same holds for unitary characters.

Proof. (a) The groups are abelian, so Schur's lemma applies; for unitary representations of locally compact abelian groups this is standard, and for admissible representations Schur's lemma holds because they have countable dimension. Since μ_2 is central, it acts by a character, and that character is the identity because the representation is genuine.

(b) μ_2 is a closed subgroup of the locally compact abelian group $\overline{T(F)}$, so the character $\zeta \mapsto \zeta$ of μ_2 extends [Ro83]. The quotient of two genuine characters is trivial on μ_2, hence is the pull-back of a character of $T(F)$.

(c) Put $\bar\chi_v = \bar\chi \circ \iota_v$, where ι_v is the inclusion of $\overline{T(k_v)}$. Since $\bar\chi$ is continuous, there is a neighborhood W of the identity in $\overline{T(\mathbb{A})}$ with $|\bar\chi(x) - 1| < 1$ for $x \in W$. By the definition of the topology, W contains the image of $\prod_{v \notin S} T(\mathscr{O}_v)^*$ for some finite set S of places. That image is a subgroup, so its image under $\bar\chi$ is a subgroup of $\mathbb{C}^\times$ contained in the disc $\{|z-1| < 1\}$, and the only such subgroup is $\{1\}$. Hence $\bar\chi_v$ is trivial on $T(\mathscr{O}_v)^*$ for $v \notin S$. (The kernel of $\bar\chi$ need not be open: at an archimedean place $\bar\chi_v$ can be nontrivial on every neighborhood of the identity.) Every $x \in \overline{T(\mathbb{A})}$ is the image of some (x_v) with $x_v \in T(\mathscr{O}_v)^*$ for $v \notin S'$, and then $\bar\chi(x) = \prod_{v \in S'} \bar\chi_v(x_v)$. Conversely, $\prod_v \bar\chi_v$ is a character of the restricted product, and it is trivial on $\{(\zeta_v) \mid \prod \zeta_v = 1\}$ because each $\bar\chi_v$ is genuine. So it descends to $\overline{T(\mathbb{A})}$. □

Let $X_T = T(\mathbb{Q})\backslash\overline{T(\mathbb{A})^1}$. It is a compact abelian group, an extension of the compact group $T(\mathbb{Q})\backslash T(\mathbb{A})^1$ by μ_2; μ_2 injects into X_T because $T(\mathbb{Q})$ meets μ_2 trivially in $\overline{G(\mathbb{A})}$. Let $\Pi_{\mathrm{aut}}(X_T)$ be the set of genuine characters of $\overline{T(\mathbb{A})^1}$ that are trivial on $T(\mathbb{Q})$, i.e. the genuine characters of X_T.

Theorem 3.2. *(a) $L^2(X_T)_{\mathrm{gen}} = \widehat{\bigoplus}_{\bar\chi \in \Pi_{\mathrm{aut}}(X_T)} \mathbb{C}\,\bar\chi$, each character occurring with multiplicity one.*

(b) $\Pi_{\mathrm{aut}}(X_T)$ is not empty. For any $\bar\chi_0 \in \Pi_{\mathrm{aut}}(X_T)$, the map $\chi \mapsto \bar\chi_0 \cdot (\chi \circ p)$ is a bijection from the characters of $T(\mathbb{Q})\backslash T(\mathbb{A})^1$ onto $\Pi_{\mathrm{aut}}(X_T)$. Replacing $\bar\chi_0$ by another element changes this bijection by a translation of $(T(\mathbb{Q})\backslash T(\mathbb{A})^1)^\wedge$.

(c) For $T = A$, the character χ_ψ of Lemma 3.3 lies in $\Pi_{\mathrm{aut}}(X_A)$; it depends only on the fixed additive character ψ of $\mathbb{A}_k/k$.

Proof. (a) Peter–Weyl for the compact abelian group X_T: $L^2(X_T)$ is the Hilbert sum of its characters, each once, and the genuine subspace is spanned by the genuine ones. (b) Apply Lemma 3.1(b) to the closed subgroup μ_2 of X_T; the second sentence follows as in that lemma. (c) is Lemma 3.3(c). □

The bijection in (b) depends on the choice of $\bar\chi_0$, and for a general torus there is no canonical one. The choice cannot be avoided:

Example 3.A. Let $k=\mathbb{Q}$, and let $\varepsilon\in A(\mathbb{A})^1$ be $a(-1)$ at the real place and 1 at every finite place. By Lemma 2.2(e) any lift $\tilde\varepsilon$ satisfies $\tilde\varepsilon^2=(1,(-1,-1)_\infty)=(1,-1)$. So every genuine character $\bar\chi$ of $\overline{A(\mathbb{A})^1}$ has $\bar\chi(\tilde\varepsilon)=\pm i$, and no genuine character is of the form $(\chi\circ p)\cdot\mathrm{sgn}$ for a splitting $A(\mathbb{A})^1\to\overline{A(\mathbb{A})^1}$; indeed no such splitting exists.

For the split torus there is a canonical choice, given by Weil indices [We64]. Let $\psi=\prod_v\psi_v$ be the nontrivial character of $\mathbb{A}_k/k$ fixed in Section 2.2. For $a\in k_v^\times$ let $\gamma_v(a)\in\mu_8$ be the Weil index of the character $x\mapsto\psi_v(ax^2)$ of k_v, normalized by

$$\int_{k_v}\psi_v(ax^2)\,dx=\gamma_v(a)\,|2a|_v^{-1/2} \tag{3.2}$$

(self-dual measure; for v finite the integral is the limit over the lattices $\varpi^{-M}\mathscr{O}_v$, $M\to\infty$).

Lemma 3.3 (Weil [We64]). *(a) $\gamma_v(a)$ depends only on the class of a in $k_v^\times/k_v^{\times2}$, and $\gamma_v(ab)\gamma_v(1)=\gamma_v(a)\gamma_v(b)(a,b)_v$.*

(b) If v is finite and non-dyadic and ψ_v is unramified, then $\gamma_v(u)=\gamma_v(1)=1$ for $u\in\mathscr{O}_v^\times$. For $a\in k^\times$, $\prod_v\gamma_v(a)=1$.

(c) Hence $\chi_{\psi,v}(a(t),\zeta)=\zeta\,\gamma_v(t)/\gamma_v(1)$ is a genuine character of $\overline{A(k_v)}$, trivial on $A(\mathscr{O}_v)^*=A(\mathscr{O}_v)\times\{1\}$ for almost all v, and $\chi_\psi=\prod_v\chi_{\psi,v}$ is a genuine character of $\overline{A(\mathbb{A})}$ trivial on $A(\mathbb{Q})$. Its restriction to $\overline{A(\mathbb{A})^1}$ lies in $\Pi_{\mathrm{aut}}(X_A)$.

(d) For $b\in k^\times$ and $\psi_b(x)=\psi(bx)$, $\chi_{\psi_b}=\chi_\psi\cdot\big(a(t)\mapsto\prod_v(b,t_v)_v\big)$.

Proof. (a) and (b) are due to Weil [We64, §§14, 28]. (c) By (a), $\chi_{\psi,v}(a(t_1),1)\chi_{\psi,v}(a(t_2),1)=(t_1,t_2)_v\,\chi_{\psi,v}(a(t_1t_2),1)=\chi_{\psi,v}((a(t_1),1)(a(t_2),1))$, using the group law (3.1). Unramifiedness is (b), and triviality on $(a(t),1)$, $t\in k^\times$, is the product formula in (b) applied to $\gamma(t)$ and to $\gamma(1)$. (d) $\psi_{b,v}(ax^2)=\psi_v(abx^2)$, so $\gamma_{\psi_b,v}(t)/\gamma_{\psi_b,v}(1)=\gamma_v(bt)/\gamma_v(b)=(\gamma_v(t)/\gamma_v(1))\,(b,t)_v$ by (a). □

Example 3.B. Let $k=\mathbb{Q}$, $\psi_p(x)=e^{2\pi i\{x\}_p}$ and $\psi_\infty(x)=e^{-2\pi ix}$, so ψ is trivial on $\mathbb{Q}$. Then $\gamma_\infty(a)=e^{-\pi i\,\mathrm{sgn}(a)/4}$, and for odd p, $\gamma_p(u)=1$ and $\gamma_p(pu)=\left(\frac{u}{p}\right)\varepsilon_p$ for $u\in\mathbb{Z}_p^\times$, where $\varepsilon_p=1$ if $p\equiv1\pmod4$ and $\varepsilon_p=i$ if $p\equiv3\pmod4$. So $\chi_{\psi,\infty}(a(-1),1)=i$, in accordance with Example 3.A, and $\chi_{\psi,p}(a(p),1)=\varepsilon_p$ for odd p. The dyadic values are tabulated by the check script, which computes γ_v from (3.2) as a finite Gauss sum and verifies Lemma 3.3(a)–(c) at $v=\infty,2,3,5,7$ and the product formula on random $a\in\mathbb{Q}^\times$.

3.2 The trace formula

This subsection proves the trace formula for the cover of a torus and shows that it is Poisson summation, twisted by a genuine automorphic character.

Let R be the right regular representation of $\overline{T(\mathbb{A})^1}$ on $L^2(X_T)_{\mathrm{gen}}$, and for $f \in C_c^\infty(\overline{T(\mathbb{A})^1})_-$ put $R(f) = \int_{\overline{T(\mathbb{A})^1}} f(y)R(y)\,dy$ and

$$\bar\chi(f) = \int_{\overline{T(\mathbb{A})^1}} f(y)\bar\chi(y)\,dy, \qquad \bar\chi \in \Pi(\overline{T(\mathbb{A})^1}).$$

The integrand $f\bar\chi$ is invariant under μ_2, so $\bar\chi(f)$ is an integral over $T(\mathbb{A})^1$; if $\bar\chi$ were trivial on μ_2 instead, the integral would vanish.

Theorem 3.4. *Let $f \in C_c^\infty(\overline{T(\mathbb{A})^1})_-$.*

(a) $R(f)$ is the integral operator on $L^2(X_T)_{\mathrm{gen}}$ with kernel

$$K_f(x,y) = \sum_{\delta \in T(\mathbb{Q})} f(x^{-1}\delta y),$$

and $R(f)\bar\chi = \bar\chi(f)\,\bar\chi$ for $\bar\chi \in \Pi_{\mathrm{aut}}(X_T)$.

(b) $R(f)$ is of trace class, and

$$\sum_{\bar\chi \in \Pi_{\mathrm{aut}}(X_T)} \bar\chi(f) = \mathrm{tr}\,R(f) = \int_{X_T} K_f(x,x)\,dx = \mathrm{vol}\big(T(\mathbb{Q})\backslash T(\mathbb{A})^1\big) \sum_{\delta \in T(\mathbb{Q})} f(\delta), \tag{3.3}$$

the sum on the left converging absolutely.

Proof. (a) For $\varphi \in L^2(X_T)_{\mathrm{gen}}$,

$$(R(f)\varphi)(x) = \int_{\overline{T(\mathbb{A})^1}} f(y)\varphi(xy)\,dy = \int_{\overline{T(\mathbb{A})^1}} f(x^{-1}y)\varphi(y)\,dy = \int_{X_T} \sum_{\delta \in T(\mathbb{Q})} f(x^{-1}\delta y)\varphi(y)\,dy,$$

because $\varphi(\delta y) = \varphi(y)$. For $\varphi = \bar\chi$ the first expression is $\bar\chi(x)\bar\chi(f)$.

(b) The group is abelian, so $f(x^{-1}\delta x) = f(\delta)$, and $\int_{X_T} K_f(x,x)\,dx = \mathrm{vol}(X_T)\sum_\delta f(\delta)$ with $\mathrm{vol}(X_T) = \mathrm{vol}(T(\mathbb{Q})\backslash T(\mathbb{A})^1)$ since μ_2 has volume 1. The function $F(x) = \sum_\delta f(\delta x)$ is smooth on X_T (the sum is locally finite, $T(\mathbb{Q})$ being discrete), so its Fourier series converges absolutely; its Fourier coefficient at $\bar\chi$ is $\int_{X_T} F\bar\chi = \bar\chi(f)$, which vanishes unless $\bar\chi$ is genuine. Evaluating the Fourier series at $x = 1$ gives the outer equality of (3.3), i.e. Poisson summation. Since $R(f)$ is diagonal in the orthonormal basis $\mathrm{vol}(X_T)^{-1/2}\bar\chi$ with summable eigenvalues $\bar\chi(f)$, it is of trace class with that trace. □

Every conjugacy class in $\overline{T(\mathbb{A})^1}$ is a point and every centralizer is the whole group, so the geometric side has no orbital integrals, only the values $f(\delta)$. Here $f(\delta)$ means f at the image $(\delta, \kappa_{\mathbb{A}}(\delta))$ of δ; for $T = A$ it is $f(a(\delta), 1)$.

The trace formula for $\overline{T(\mathbb{A})^1}$ is literally Poisson summation on $T(\mathbb{Q})\backslash T(\mathbb{A})^1$, after multiplying by a genuine automorphic character:

Lemma 3.5. *Fix $\bar\chi_0 \in \Pi_{\mathrm{aut}}(X_T)$ with local components $\bar\chi_{0,v}$, as in Lemma 3.1(c). For an anti-genuine $\bar f$ on $\overline{T(\mathbb{A})^1}$ (resp. $\overline{T(k_v)}$) define f on $T(\mathbb{A})^1$ (resp. $T_k(k_v)$) by $f(t) = \bar f(\tilde t)\bar\chi_0(\tilde t)$ (resp. $\bar\chi_{0,v}$), $\tilde t$ any lift of t.*

(a) $\bar f \mapsto f$ is well defined, bijective from $C_c^\infty(\overline{T(\cdot)})_-$ onto $C_c^\infty(T(\cdot))$, and carries convolution to convolution.

(b) At every v where $\bar\chi_{0,v}$ is trivial on $T(\mathcal{O}_v)^*$ (almost all v), it carries $e_{T,v}$ to the characteristic function of $T_k(k_v)\cap K_v$, which is the unit of the Hecke algebra of $T_k(k_v)$ when $T_k(k_v)\cap K_v$ has volume 1.

(c) $f(\delta) = \bar f(\delta)$ for $\delta \in T(\mathbb{Q})$, and $(\bar\chi_0\cdot(\chi\circ p))(\bar f) = \hat f(\chi) := \int_{T(\mathbb{A})^1} f(t)\chi(t)\,dt$ for every character χ of $T(\mathbb{A})^1$.

Consequently (3.3) for $\bar f$ is the Poisson summation formula for f,

$$\sum_{\chi}\hat f(\chi) = \mathrm{vol}\big(T(\mathbb{Q})\backslash T(\mathbb{A})^1\big)\sum_{\delta\in T(\mathbb{Q})} f(\delta),$$

with χ running over the characters of $T(\mathbb{Q})\backslash T(\mathbb{A})^1$.

Proof. (a) $\bar f\bar\chi_0$ is μ_2-invariant because $\bar f$ is anti-genuine and $\bar\chi_0$ genuine, and it is smooth because $\bar\chi_0$ is a continuous character; so it is the pull-back of a smooth function on T. The inverse map is $f\mapsto (f\circ p)\bar\chi_0^{-1}$. For convolution, $(\bar f_1 * \bar f_2)(x)\bar\chi_0(x) = \int \bar f_1(y)\bar\chi_0(y)\,\bar f_2(y^{-1}x)\bar\chi_0(y^{-1}x)\,dy$, and the integrand is μ_2-invariant in y, so the integral over the cover equals the integral over T. (b) is immediate. (c) $\bar\chi_0$ is trivial on $T(\mathbb{Q})$; the second identity is the definition, the integrand being μ_2-invariant. □

Together with Theorem 3.4, the lemma shows that the correspondence $\chi \leftrightarrow \bar\chi_0\cdot(\chi\circ p)$ of Theorem 3.2 matches the two trace formulas term by term.

3.3 Distributions for the invariant form

This subsection defines the rank-0 distributions that appear as correction terms in Section 6.

Let S be a finite set of places of $\mathbb{Q}$ and let $\overline{A(\mathbb{A}_S)}$ be the cover of $A(\mathbb{A}_S) = \prod_{p\in S}A(\mathbb{Q}_p)$ defined by $\prod_{p\in S}\beta_p$. Let $\mathscr{H}(\overline{A(\mathbb{A}_S)})$ be the algebra of functions $\bigotimes_{p\in S} f_p$, $f_p \in C_c^\infty(\overline{A(\mathbb{Q}_p)})_-$, and the analogous algebra $\mathscr{H}(\overline{A(\mathbb{A})})$ of Section 3.1. For a class $\mathscr{O} = \{\delta\}$ of $A(\mathbb{Q})$ (a single element, since the cover is abelian) and $\bar\chi \in \Pi_u(\overline{A(\mathbb{A}_S)})$, put

$$I^A_{\mathscr{O}}(f) = f(a(\delta),1), \qquad I^A_{\bar\chi}(f) = \bar\chi(f) = \int_{\overline{A(\mathbb{A}_S)}} f(y)\bar\chi(y)\,dy, \tag{3.4}$$

for $f\in\mathscr{H}(\overline{A(\mathbb{A}_S)})$, and likewise over $\mathbb{A}$. Here $(a(\delta),1)$ is the image of δ under the rational splitting (Lemma 2.6(b)). Both are invariant distributions: the cover is abelian, so conjugation acts trivially.

Remark 2026.6. For general n the cover of A is not abelian: by the computation in Lemma 2.2(f), $\overline{A(F)}$ is a Heisenberg-type group whose commutator pairing is $(t_1,t_2)_n^{-2}$. Its genuine irreducible representations are then parametrized by genuine characters of its center, the preimage of A^N with $N = n/\gcd(n,2)$. Matching them with characters of A^N requires a splitting over A^N, and the global argument requires a Hasse principle for N-th powers, which holds when $\mu_N \subset k$. These questions do not arise for $n = 2$, and this section does not treat $n > 2$.

3.4 Status of the results

This subsection lists the statements of Section 3 that rest on citation or on numerical checks.

Remark 2026.7. *What rests on citation.* Taken from the literature: Lemma 3.3(a)–(b) [We64]; the compactness of $T(\mathbb{Q})\backslash T(\mathbb{A})^1$ [PR94]; and the extension of characters from closed subgroups of locally compact abelian groups [Ro83]. Everything else in this section is proved in the text.

Remark 2026.8. The script `checks/sec3_n2_check.py` verifies: that the 1995 cocycle on A equals (3.1); that γ_v, computed from (3.2), is an eighth root of unity, depends only on the square class, is 1 on units at $p = 3,5,7$, and satisfies Lemma 3.3(a); that $\chi_{\psi,v}$ respects the group law (Lemma 3.3(c)) at $v = \infty,2,3,5,7$; the product formula of Lemma 3.3(b) for random $a \in \mathbb{Q}^\times$; the odd-p values of Example 3.B; and Example 3.A. The rest of the section is proved in the text.

4 The geometric side

This section computes the geometric side of the trace formula for $\overline{G(\mathbb{A})}$, class by class. The general theory is due to Arthur [Ar78, Ar80, Ar86] for connected reductive groups and to Li [Li14a] for covering groups; we specialize it to the double cover of $G = \mathrm{Res}_{k/\mathbb{Q}} SL(2)$. For this group one inequality between heights (Lemma 4.3) makes the reduction theory explicit, with truncation parameter $T_0 = 0$. With it, every step of Arthur's coarse expansion can be checked directly, and each class contributes a polynomial in T of degree at most one that we write down (Theorem 4.5, Propositions 4.6, 4.7 and 4.8). The new parts are the corrected weight $2T + v_A$ for hyperbolic classes, the unipotent contribution of the central classes, given as the finite part of a zeta integral, and the local distribution $J_A(z,f)$ at a central element (Lemma 4.10). Section 4.5 states the fine expansion [Ar86, Li14a] in the form that Sections 5–7 use.

Roadmap. The argument has four steps.

1. *Classes* (Section 4.1). $G(\mathbb{Q})$ is sorted into classes $\mathcal{O}$ by semisimple part: two central classes, the hyperbolic classes and the elliptic classes.

2. *Truncation* (Section 4.2). The kernel $K_G(x,x) = \sum_\gamma f(x^{-1}\gamma x)$ of $R(f)$ is not integrable over $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$. The divergence comes from the cusp, the region where $H(\delta x) > T$ for some $\delta \in G(\mathbb{Q})$. There one subtracts the kernel K_B of the Borel subgroup; the result is the truncated kernel $k^T(x,f)$. The height inequality (Lemma 4.3) says that for $T \geq 0$ at most one coset $B(\mathbb{Q})\delta$ is involved at each x, so no further reduction theory is needed.

3. *Integration, class by class* (Section 4.3). For an elliptic class nothing is truncated and $J_{\mathcal{O}}$ is an orbital integral. For a hyperbolic class the divergent integral over $A(\mathbb{Q})\backslash A(\mathbb{A})$ is cut down to an interval of length $2T + v_A(y)$, and $J^T_{\mathcal{O}}$ is a weighted orbital integral. For a central class the unipotent elements give a zeta integral, of which $J^T_{\mathcal{O}}$ keeps the finite part at $s = 1$ and a term linear in T.

4. *Localization* (Sections 4.4 and 4.5). For $f = f_S \otimes e^S$ the terms are rewritten as local distributions on $\overline{G(\mathbb{Q}_S)}$ with global coefficients.

The objects that occur are the following.

K_P, $K_{P,\mathcal{O}}$	kernel of $R_P(f)$ and its part on the class $\mathcal{O}$, (4.1)
$J_{P,\mathcal{O}}$	variant of $K_{P,\mathcal{O}}$ with the same integral, (4.2)
$\hat{\tau}_B(H(\delta x) - T)$	cutoff: 1 in the cusp above height T, 0 elsewhere
k^T, $k^T_{\mathcal{O}}$, $j^T_{\mathcal{O}}$	truncated kernels, (4.4)
$J^T_{\mathcal{O}}(f)$, $J_{\mathcal{O}}(f)$	their integrals, polynomials in T, and the value at $T = 0$
v_A, $v_{A,v}$	the weight $-H(x) - H(wx)$ and its local form, (4.3)
$Z(f,s)$, $\mathrm{FP}_{s=1}$	zeta integral of the unipotent terms and its finite part, (4.9)
$J_G(\gamma, f_S)$, $J_A(\gamma, f_S)$	local orbital and weighted orbital integrals, (4.11)–(4.12)
$J_A(z, f_S)$	the weighted orbital integral at a central element (Lemma 4.10)
$a^M(S,\gamma)$	global coefficients (Theorem 4.11); the $a^G(S, u_\xi)$ are computed in Proposition 7.10

Throughout, $G(\mathbb{Q})$ is identified with its image in $\overline{G(\mathbb{A})}$ under Lemma 2.6(a), and $N(\mathbb{A})$ with $N(\mathbb{A}) \times 1$; on $B(\mathbb{Q})$ the two identifications agree, and $b \mapsto (b, 1)$ (Lemma 2.6(b)). Test functions are anti-genuine (Section 2.1.4). We write $w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in G(\mathbb{Q})$, and $d = [k : \mathbb{Q}]$.

4.1 Equivalence classes

The geometric side is a sum over classes of rational elements. This subsection sorts $G(\mathbb{Q})$ into these classes.

Following [Ar78, §2], call $\gamma, \gamma' \in G(\mathbb{Q})$ *equivalent* if their semisimple parts are $G(\mathbb{Q})$-conjugate, and let $\mathscr{C}$ be the set of equivalence classes. A class is *unramified* [Ar78, p. 949] if it consists of semisimple elements.

Lemma 4.1. *The classes $\mathcal{O} \in \mathscr{C}$ are of three kinds.*

(a) Central*: for $z \in \{\pm 1\}$, $\mathcal{O}_z = \{zu \mid u \in G(\mathbb{Q})$ unipotent$\}$. It is not unramified: it contains z and $zn(\nu)$, $\nu \in k^\times$.*

(b) Hyperbolic*: the $G(\mathbb{Q})$-conjugacy class of $\mu = a(t)$, $t \in k^\times \smallsetminus \{\pm 1\}$. It contains $a(t^{-1}) = w\mu w^{-1}$ and meets $A(\mathbb{Q})$ in $\{a(t), a(t^{-1})\}$; it is determined by $\{t, t^{-1}\}$. The centralizer of μ is A.*

(c) Elliptic*: the $G(\mathbb{Q})$-conjugacy class of a regular $\mu \in G(\mathbb{Q})$ whose characteristic polynomial is irreducible over k. Its centralizer is a maximal torus $T_\mu = \mathrm{Res}_{k/\mathbb{Q}} T_k$, anisotropic over $\mathbb{Q}$, and $\mathcal{O} \cap A(\mathbb{Q}) = \emptyset$.*

Classes of kinds (b) and (c) are unramified. In all three cases the centralizer of (the image of) μ in $\overline{G(\mathbb{A})}$ is the full preimage $\overline{G_\mu(\mathbb{A})}$, which is abelian in cases (b) and (c).

Proof. An element of $SL(2,k)$ with semisimple part ± 1 is $\pm$ a unipotent element. A regular semisimple element with eigenvalues in k is $GL(2,k)$-conjugate to $a(t)$, and then $SL(2,k)$-conjugate to it, because $\mathrm{diag}(y,1)$ centralizes $a(t)$. The centralizers are standard. For the cover: if $h \in G_\mu(\mathbb{A})$, lifts of h and μ commute by Lemma 2.6(d)–(e) (the argument of Lemma 2.2(g), applied to the abelian group generated by h and μ); conversely the centralizer in $\overline{G(\mathbb{A})}$ maps into $G_\mu(\mathbb{A})$. □

For general n the centralizer of a lift of a split γ can be smaller than the preimage of A, and then the genuine orbital integral vanishes. For $n = 2$ this does not happen, by the last sentence of the lemma.

4.2 Kernels, heights and truncation

This subsection defines the kernels of $R(f)$ and of its analog for the Borel subgroup, and the truncated kernel whose integral is the geometric side. The key tool is the height inequality of Lemma 4.3.

Let $f \in \mathscr{H}(\overline{G(\mathbb{A})})$. We use Arthur's convention: R is the right regular representation on $L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{\mathrm{gen}}$, the space of φ with $\varphi(\gamma x\zeta) = \zeta\varphi(x)$, and $(R(f)\varphi)(x) = \int_{\overline{G(\mathbb{A})}} f(y)\varphi(xy)\,dy$. The integrand is invariant under $y \mapsto y\zeta$ because f is anti-genuine. For a standard parabolic $P = MN_P \in \{B, G\}$ and $\mathscr{O} \in \mathscr{C}$ put

$$K_P(x,y) = \sum_{\gamma\in M(\mathbb{Q})} \int_{N_P(\mathbb{A})} f(x^{-1}\gamma n y)\,dn, \qquad K_{P,\mathscr{O}}(x,y) = \sum_{\gamma\in M(\mathbb{Q})\cap\mathscr{O}} \int_{N_P(\mathbb{A})} f(x^{-1}\gamma n y)\,dn, \tag{4.1}$$

$$J_{P,\mathscr{O}}(x,y) = \sum_{\mu\in M(\mathbb{Q})\cap\mathscr{O}} \sum_{\nu\in N_{P,\mu_s}(\mathbb{Q})\backslash N_P(\mathbb{Q})} \int_{N_{P,\mu_s}(\mathbb{A})} f(x^{-1}\nu^{-1}\mu n\nu y)\,dn, \tag{4.2}$$

where $\mu = \mu_s\mu_u$ is the Jordan decomposition and N_{P,μ_s} the centralizer of μ_s in N_P. For $P = G$ these are $K_G = \sum_{\gamma\in G(\mathbb{Q})} f(x^{-1}\gamma y)$ and $K_{G,\mathscr{O}} = J_{G,\mathscr{O}} = \sum_{\gamma\in\mathscr{O}} f(x^{-1}\gamma y)$.

Lemma 4.2. *The sums (4.1) and (4.2) are finite, uniformly for x, y in compact sets. $K_P(x,y) = \sum_{\mathscr{O}} K_{P,\mathscr{O}}(x,y)$, and K_P is the kernel of $R_P(f)$, where R_P is the right regular representation on $L^2(N_P(\mathbb{A})M(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{\mathrm{gen}}$.*

Proof. $G(\mathbb{Q})$ is discrete in $G(\mathbb{A})$ and f has compact support modulo μ_2, so only finitely many γ contribute for x, y in a compact set; the integrals over $N_P(\mathbb{A})$ are over compact sets for the same reason. For upper triangular $b_1, b_2 \in B(\mathbb{A})$ with lower right entries d_1, d_2, (2.1) and $\kappa_v|_B \equiv 1$ give $\sigma_{\mathbb{A}}(b_1,b_2) = \prod_v (d_2,d_1)_v$, which is 1 when $d_1, d_2 \in k^\times$. So $b \mapsto (b,1)$ is a homomorphism on $N(\mathbb{A})A(\mathbb{Q})$, and $N_P(\mathbb{A})M(\mathbb{Q})$ is a subgroup of $\overline{G(\mathbb{A})}$. For φ in that space,

$$(R_P(f)\varphi)(x) = \int_{\overline{G(\mathbb{A})}} f(x^{-1}y)\varphi(y)\,dy = \int_{N_P(\mathbb{A})M(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \sum_{\gamma\in M(\mathbb{Q})} \int_{N_P(\mathbb{A})} f(x^{-1}\gamma n y)\,dn\ \varphi(y)\,dy,$$

using $\varphi(\gamma n y) = \varphi(y)$. The decomposition over $\mathscr{O}$ is a regrouping of the finite sum. □

Heights. For a row vector $u=(c,d)\in k_v^2$ put $\|u\|_v=\max(|c|_v,|d|_v)$ if v is finite and $\|u\|_v=(|c|^2+|d|^2)^{[k_v:\mathbb{R}]/2}$ if v is archimedean, and for $u\in\mathbb{A}_k^2$ put $\|u\|=\prod_v\|u_v\|_v$. Then $\|\lambda u\|_v=|\lambda|_v\|u\|_v$, and $\|e_2k\|_v=1$ for $k\in K_v$, where $e_2=(0,1)$. Recall $H=H_B$ from Lemma 2.8.

Lemma 4.3. *(a) For $x\in\overline{G(\mathbb{A})}$ with image $g\in G(\mathbb{A})$, $H(x)=-\log\|e_2g\|$. The same holds at a single place: $H_v(g)=-\log\|e_2g\|_v$ for $g\in SL(2,k_v)$.*

(b) If $\gamma\in G(\mathbb{Q})\smallsetminus B(\mathbb{Q})$, then

$$H(x)+H(\gamma x)\le 0\qquad(x\in\overline{G(\mathbb{A})}).$$

Consequently, if $T\ge 0$, then for each x there is at most one $\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})$ with $H(\delta x)>T$.

(c) $H(ax)=H(a)+H(x)$ and $H(wax)=-H(a)+H(wx)$ for $a\in\overline{A(\mathbb{A})}$. Hence

$$v_A(x)=-H(x)-H(wx)\tag{4.3}$$

is left $\overline{A(\mathbb{A})}$-invariant, right $\overline{K_0}$-invariant, and $v_A(x)\ge 0$.

Proof. (a) If $g=n\,a(t)\,k$, then $e_2g=t^{-1}e_2k$, so $\|e_2g\|=|t|^{-1}=e^{-H(x)}$. The local statement is the same computation.

(b) Let $e_2\gamma=(c,d)$, with $c\neq 0$ since $\gamma\notin B(\mathbb{Q})$. The matrix with rows e_2g and $e_2\gamma g$ has determinant $\det\begin{pmatrix}0&1\\c&d\end{pmatrix}\det g=-c$. At each place, $|\det(u_1;u_2)|_v\le\|u_1\|_v\|u_2\|_v$: this is the ultrametric inequality at finite places and Hadamard's inequality (squared at complex places) at archimedean ones. Taking the product over v, $1=|c|_{\mathbb{A}_k}\le\|e_2g\|\,\|e_2\gamma g\|=e^{-H(x)-H(\gamma x)}$ by (a) and the product formula. If $\delta_1\neq\delta_2$ in $B(\mathbb{Q})\backslash G(\mathbb{Q})$ both had $H(\delta_i x)>T\ge 0$, then (b), applied to δ_1x and $\gamma=\delta_2\delta_1^{-1}\notin B(\mathbb{Q})$, would give $0<2T<H(\delta_1x)+H(\delta_2x)\le 0$.

(c) The first identity is Lemma 2.5(a). For the second, $waw^{-1}=a^{-1}$, so $H(wax)=H(a^{-1}\cdot wx)=-H(a)+H(wx)$. Right K_0-invariance is Lemma 2.5(b), and $v_A\ge 0$ is (b) with $\gamma=w$. □

For Lemma 4.3(b) compare [Ar78, Lemma 5.1]. Reduction theory [Go62] gives a Siegel set $\mathfrak{S}=\{(n,1)\,\tilde a\,\tilde k\mid n\in\omega,\ H(\tilde a)>-c,\ p(\tilde a)\in\Omega A_B(\mathbb{R})^0,\ \tilde k\in\overline{K_0}\}$, with $\omega\subset N(\mathbb{A})$ and $\Omega\subset A(\mathbb{A})^1$ compact and $c\ge 0$, such that $\overline{G(\mathbb{A})}=G(\mathbb{Q})\mathfrak{S}$. In the coordinates $x=nak$ the measure is $\delta_B(a)^{-1}dn\,da\,dk$ by (2.9), and $\mathfrak{S}$ has finite volume.

Truncation. Let $\hat\tau_B=\tau_B$ be the characteristic function of $(0,\infty)$ (Section 2.1.3), $\hat\tau_G=1$, and $\dim A_G=0$, $\dim A_B=1$. For $T\in\mathbb{R}\cong\mathfrak{a}$ put

$$k^T(x,f)=K_G(x,x)-\sum_{\delta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}K_B(\delta x,\delta x)\,\hat\tau_B(H(\delta x)-T),\tag{4.4}$$

and define $k^T_{\mathscr{O}}(x,f)$ and $j^T_{\mathscr{O}}(x,f)$ in the same way with $K_{P,\mathscr{O}}$, respectively $J_{P,\mathscr{O}}$, in place of K_P. By Lemma 4.3(b), for $T\ge 0$ the sum over δ has at most one nonzero term. Let $F^G(x,T)$ be the characteristic function of $\{x\mid H(\delta x)\le T\text{ for all }\delta\in G(\mathbb{Q})\}$; its support is compact modulo $G(\mathbb{Q})$ by reduction theory [Go62].

Lemma 4.4. *Let $T \geq 0$. Then $F^G(x,T) + \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \tau_B(H(\delta x) - T) = 1$, and*

$$k^T(x,f) = F^G(x,T)\,K_G(x,x) + \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \tau_B(H(\delta x) - T)\big(K_G(\delta x, \delta x) - K_B(\delta x, \delta x)\big). \tag{4.5}$$

Proof. The first identity says that either $H(\delta x) \leq T$ for all δ, or $H(\delta x) > T$ for exactly one coset, which is Lemma 4.3(b). Multiply it by $K_G(x,x) = K_G(\delta x, \delta x)$ and subtract the δ-sum of (4.4). □

In general [Ar78, §6], [Ar05, §8], the partition holds for T sufficiently regular; here $T \geq 0$ suffices.

Example 4.A. Let $k = \mathbb{Q}$, take $x \in G(\mathbb{R})$, viewed in $\overline{G(\mathbb{A})}$ with trivial finite components, and write $x = n(u)a(y^{1/2})k_\infty$ with $y > 0$ and $k_\infty \in SO(2)$. Let $z = u + iy$ be the corresponding point of the upper half plane. By Lemma 4.3(a), $H(x) = -\log \|(0, y^{-1/2})\| = \frac{1}{2}\log y$. Every coset in $B(\mathbb{Q})\backslash G(\mathbb{Q})$ has a representative $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z})$. The finite components of γx then lie in K_p, so only the real place contributes, and $e_2 \gamma x = (cy^{1/2}, (cu+d)y^{-1/2})k_\infty$ gives

$$H(\gamma x) = -\log|cz+d| + \tfrac{1}{2}\log y = \tfrac{1}{2}\log \mathrm{Im}(\gamma z).$$

So the height of x is half the logarithm of the imaginary part of z. For $c \neq 0$ the height inequality of Lemma 4.3(b) becomes $\mathrm{Im}(z)\,\mathrm{Im}(\gamma z) \leq 1$. It holds because $|cz+d| \geq |c|\,y \geq y$. The truncation (4.4) subtracts the constant term on the horoballs $\{\mathrm{Im}(\delta z) > e^{2T}\}$, and for $T \geq 0$ these horoballs are disjoint. On the standard fundamental domain $\{|u| \leq \frac{1}{2},\ |z| \geq 1\}$ the largest of the $\mathrm{Im}(\delta z)$ is y itself. So there $F^G(x,T) = 1$ exactly when $y \leq e^{2T}$, and $F^G(\cdot,T)$ cuts off the cusp at height e^{2T}.

4.3 The coarse geometric expansion

This subsection integrates the truncated kernel class by class. Each class contributes a polynomial in T of degree at most one.

Let dx be a Haar measure on $G(\mathbb{A})$ with $dx = \delta_B(a)^{-1}\,dn\,da\,dk$ for $x = nak$ as in (2.9), where dk gives K_0 volume 1, dn gives $N(\mathbb{Q})\backslash N(\mathbb{A})$ volume 1, and da is a Haar measure on $A(\mathbb{A}) \cong \mathbb{A}_k^\times$, written $d^\times t$ for $a = a(t)$. On covers μ_2 has volume 1. Write $A(\mathbb{A}) = A(\mathbb{A})^1 \times A_B(\mathbb{R})^0$, where $A_B(\mathbb{R})^0 = \{a(u) \mid u \in \mathbb{R}_{>0}\}$ (u embedded diagonally in $k \otimes \mathbb{R}$), let dX be Lebesgue measure on $\mathbb{R} \cong \mathfrak{a}$ transported to $A_B(\mathbb{R})^0$ by H, and let da^1 be the Haar measure on $A(\mathbb{A})^1$ with $da = da^1\,dX$. Put

$$c_k = \mathrm{vol}\big(A(\mathbb{Q})\backslash A(\mathbb{A})^1\big) = \mathrm{vol}\big(k^\times \backslash \mathbb{A}_k^1\big).$$

Theorem 4.5. *Let $f \in \mathscr{H}(\overline{G(\mathbb{A})})$ and $T \geq 0$.*

(a) Only finitely many $\mathscr{O} \in \mathscr{C}$ have $k^T_{\mathscr{O}}(\cdot,f) \not\equiv 0$, and $k^T(x,f) = \sum_{\mathscr{O}} k^T_{\mathscr{O}}(x,f)$.

(b) Each $k^T_{\mathscr{O}}(\cdot,f)$ and each $j^T_{\mathscr{O}}(\cdot,f)$ is absolutely integrable over $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$, and

$$J^T_{\mathscr{O}}(f) := \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} k^T_{\mathscr{O}}(x,f)\,dx = \int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} j^T_{\mathscr{O}}(x,f)\,dx.$$

(c) Hence $k^T(\cdot,f)$ is absolutely integrable, and $J^T(f) := \int k^T(x,f)\,dx = \sum_{\mathscr{O}} J^T_{\mathscr{O}}(f)$.

(d) Each $J^T_{\mathscr{O}}(f)$ is a polynomial in T of degree at most 1, given by Propositions 4.6, 4.7 and 4.8.

We put $J_{\mathscr{O}}(f) = J^0_{\mathscr{O}}(f)$ and $J(f) = J^0(f) = \sum_{\mathscr{O}} J_{\mathscr{O}}(f)$. In general one defines $J_{\mathscr{O}}$ as the value at $T = 0$ of the polynomial $J^T_{\mathscr{O}}$, which is known only for T sufficiently regular [Ar80, Ar05]. Here the integral itself converges at $T = 0$, so the two definitions agree. Theorem 4.5 is the case of $\overline{G(\mathbb{A})}$ of Arthur's coarse expansion [Ar78, §§7–8], [Ar80], extended to covers by Li [Li14a]; for this group we prove it directly.

Proposition 4.6 (Hyperbolic classes)**.** *Let $\mathscr{O}$ be the class of $\mu = a(t)$, $t \in k^\times \smallsetminus \{\pm 1\}$, and $T \geq 0$. Then*

$$j^T_{\mathscr{O}}(x,f) = \sum_{\eta \in A(\mathbb{Q})\backslash G(\mathbb{Q})} f(x^{-1}\eta^{-1}\mu\eta x)\big(1 - \hat\tau_B(H(\eta x) - T) - \hat\tau_B(H(w\eta x) - T)\big), \tag{4.6}$$

and

$$J^T_{\mathscr{O}}(f) = c_k \int_{\overline{A(\mathbb{A})}\backslash\overline{G(\mathbb{A})}} f(y^{-1}\mu y)\,\big(2T + v_A(y)\big)\,dy, \tag{4.7}$$

with $v_A(y) = -H(y) - H(wy) \geq 0$ as in (4.3). The integral converges absolutely.

Proof. Here $\mathscr{O} \cap A(\mathbb{Q}) = \{\mu, \mu^{-1}\}$ and $N_{\mu_s} = 1$, so

$$J_{B,\mathscr{O}}(y,y) = \sum_{\mu' \in \{\mu,\mu^{-1}\}} \sum_{\nu \in N(\mathbb{Q})} f(y^{-1}\nu^{-1}\mu'\nu y).$$

As δ runs over $B(\mathbb{Q})\backslash G(\mathbb{Q})$ and ν over $N(\mathbb{Q})$, $\eta = \nu\delta$ runs over $A(\mathbb{Q})\backslash G(\mathbb{Q})$, and $H(\nu\delta x) = H(\delta x)$. Writing $\mu^{-1} = w^{-1}\mu w$ and replacing η by $w^{-1}\eta$ in the μ^{-1}-term (note that $H(w^{-1}y) = H(-wy) = H(wy)$) gives (4.6), since $K_{G,\mathscr{O}}(x,x) = \sum_\eta f(x^{-1}\eta^{-1}\mu\eta x)$.

By Lemma 4.3(b), for $T \geq 0$ the two $\hat\tau_B$ in (4.6) are never both 1, so the bracket is 0 or 1. Unfolding, and writing $x = \tilde a y$ with $\tilde a \in \overline{A(\mathbb{A})}$,

$$\int |j^T_{\mathscr{O}}| \leq \int_{\overline{A(\mathbb{A})}\backslash\overline{G(\mathbb{A})}} |f(y^{-1}\mu y)| \int_{A(\mathbb{Q})\backslash\overline{A(\mathbb{A})}} \big(1 - \hat\tau_B(H(\tilde a y) - T) - \hat\tau_B(H(w\tilde a y) - T)\big)\,d\tilde a\,dy,$$

with equality if $f \geq 0$, and we used $f(y^{-1}\tilde a^{-1}\mu\tilde a y) = f(y^{-1}\mu y)$, because $\overline{A(\mathbb{A})}$ is abelian (Lemma 2.6(e)). Put $X = H(\tilde a)$, $h_1 = H(y)$, $h_2 = H(wy)$; by Lemma 4.3(c) the bracket is $1 - \mathbf{1}[X + h_1 > T] - \mathbf{1}[-X + h_2 > T]$, whose integral over $X \in \mathbb{R}$ is $2T - h_1 - h_2 = 2T + v_A(y)$. The integral over $A(\mathbb{Q})\backslash A(\mathbb{A})^1$ gives c_k. This proves (4.7) for $\int j^T_{\mathscr{O}}$ and its absolute convergence: in the coordinates $y = n(z)k$ on $\overline{A(\mathbb{A})}\backslash\overline{G(\mathbb{A})}$ (measure $dz\,dk$), $y^{-1}\mu y = k^{-1}\mu\, n((1-t^{-2})z)k$, so z runs over a compact set and $v_A(n(z)k)$ is bounded there (Lemma 4.9).

It remains to show $\int k^T_{\mathscr{O}} = \int j^T_{\mathscr{O}}$. By definition

$$k^T_{\mathscr{O}}(x) - j^T_{\mathscr{O}}(x) = \sum_\delta \hat\tau_B(H(\delta x) - T)\,\big(J_{B,\mathscr{O}} - K_{B,\mathscr{O}}\big)(\delta x, \delta x).$$

Let $x = (n(u_0),1)\,\tilde a\tilde k$ with $p(\tilde a) = a(s)$. For $\mu' = a(t')$ one has $n(-\nu)\mu' n(\nu) = \mu' n((1-t'^{-2})\nu)$, a bijection of $N(\mathbb{Q})$, and $\tilde a^{-1}\mu'(n(z),1)\tilde a = \mu'(n(s^{-2}z),1)$ (Lemma 2.2(h)). So with $g_{\mu'}(z) = f(\tilde k^{-1}\mu'(n(z),1)\tilde k)$ and $c = (1-t'^{-2})u_0$,

$$(J_{B,\mathscr{O}} - K_{B,\mathscr{O}})(x,x) = \sum_{\mu'}\Big(\sum_{\nu \in k} g_{\mu'}(s^{-2}(\nu + c)) - \int_{\mathbb{A}_k} g_{\mu'}(s^{-2}(z+c))\,dz\Big).$$

By Poisson summation over $k \subset \mathbb{A}_k$ [Ta67], this is $|s|^2 \sum_{\mu'} \sum_{\nu \in k^\times} \hat{g}_{\mu'}(s^2\nu)\psi(\nu c)$, which is $O(|s|^{2-M})$ for every M, uniformly for $\tilde{k} \in \overline{K_0}$ and s in $\Omega \times \mathbb{R}_{\geq 1}$, $\Omega \subset \mathbb{A}_k^1$ compact. Since $\delta_B(a)^{-1} = |s|^{-2}$, the function $\hat{\tau}_B(H(x)-T)(J_{B,\mathscr{O}} - K_{B,\mathscr{O}})(x,x)$ is absolutely integrable over $B(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$, and unfolding is justified. Integrating first over $u_0 \in k\backslash\mathbb{A}_k$ gives 0, because $\int_{k\backslash\mathbb{A}_k} \sum_{\nu\in k} g(s^{-2}(\nu+c))\,du_0 = |1-t'^{-2}|_{\mathbb{A}_k}^{-1} \int_{\mathbb{A}_k} g(s^{-2}z)\,dz$ and $|1-t'^{-2}|_{\mathbb{A}_k} = 1$ by the product formula. □

In the form (4.7), $J_{\mathscr{O}}(f)$ is c_k times Arthur's weighted orbital integral, v_A being the volume of the convex hull of $\{-H_B(y), -H_{\bar{B}}(y)\}$ with $H_{\bar{B}}(y) = -H(wy)$. For the measure on $\mathfrak{a}$ see Remark 2026.9.

Proposition 4.7 (Elliptic classes)**.** *Let $\mathscr{O}$ be elliptic, with $\mu \in \mathscr{O}$ and $T_\mu = G_\mu$. Then $k_{\mathscr{O}}^T = j_{\mathscr{O}}^T = K_{G,\mathscr{O}}(x,x)$ for every T, and*

$$J_{\mathscr{O}}(f) = J_{\mathscr{O}}^T(f) = \operatorname{vol}\big(T_\mu(\mathbb{Q})\backslash T_\mu(\mathbb{A})\big) \int_{\overline{T_\mu(\mathbb{A})}\backslash\overline{G(\mathbb{A})}} f(y^{-1}\mu y)\,dy,$$

the integral converging absolutely.

Proof. $\mathscr{O} \cap A(\mathbb{Q}) = \emptyset$, so the B-terms vanish. Unfold $\sum_{\eta \in T_\mu(\mathbb{Q})\backslash G(\mathbb{Q})}$ and use that $\overline{T_\mu(\mathbb{A})}$ is abelian, so $f(y^{-1}\tilde{t}^{-1}\mu\tilde{t}y) = f(y^{-1}\mu y)$. The quotient $T_\mu(\mathbb{Q})\backslash T_\mu(\mathbb{A})$ is compact (T_μ is anisotropic, so $T_\mu(\mathbb{A})^1 = T_\mu(\mathbb{A})$), and the orbit of μ is closed, so the orbital integral of the compactly supported $|f|$ converges. □

Central classes. For $z \in \{\pm 1\}$ let $f_z(x) = f(zx)$, with z identified with its image $(z,1)$ in $\overline{G(\mathbb{A})}$ (Lemma 2.6(a), $\kappa_{\mathbb{A}}(\pm 1) = 1$). It is central, so $f_z \in \mathscr{H}(\overline{G(\mathbb{A})})$, and at almost all v its local component is again e_v, because $(-1,1) \in K_v^*$. For $f \in \mathscr{H}(\overline{G(\mathbb{A})})$ define a Schwartz–Bruhat function on $\mathbb{A}_k$ by

$$\Phi_f(y) = \int_{\overline{K_0}} f\big(\tilde{k}^{-1}(n(y),1)\tilde{k}\big)\,d\tilde{k}, \tag{4.8}$$

let $\hat{\Phi}_f$ be its Fourier transform with respect to ψ, and for $\operatorname{Re} s > 1$ put

$$Z(f,s) = \int_{k^\times\backslash\mathbb{A}_k^\times} |t|^{-2s} \sum_{\nu\in k^\times} \Phi_f(t^{-2}\nu)\,d^\times t = \frac{1}{2} \sum_{\xi \in k^\times/k^{\times 2}} \int_{\mathbb{A}_k^\times} \Phi_f(\xi t^2)\,|t|^{2s}\,d^\times t. \tag{4.9}$$

For $f = \bigotimes_v f_v$ the last integral is the product of the local integrals

$$Z_v(\Phi_{f,v}, \xi, s) = \int_{k_v^\times} \Phi_{f,v}(\xi t^2)|t|_v^{2s} d^\times t,$$

which are, at $s = 1$, twice the local unipotent orbital integrals of f_v at $n(\xi)$ (Lemma 4.10(c)).

Proposition 4.8 (Central classes)**.** *Let $z \in \{\pm 1\}$. Then $k_{\mathscr{O}_z}^T = j_{\mathscr{O}_z}^T$, and $J_{\mathscr{O}_z}^T(f) = J_{\mathrm{unip}}^T(f_z)$, where for every $T \in \mathbb{R}$*

$$J_{\mathrm{unip}}^T(f) = \operatorname{vol}\big(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}\big) f(1) + c_k \hat{\Phi}_f(0)\,T + \operatorname*{FP}_{s=1} Z(f,s). \tag{4.10}$$

Here $Z(f,s)$, absolutely convergent for $\operatorname{Re} s > 1$, extends to a meromorphic function on $\mathbb{C}$ whose only possible poles are simple ones at $s=1$ and $s=0$; the residue at $s=1$ is $c_k\hat{\Phi}_f(0)/2$, and $\mathrm{FP}_{s=1}$ denotes the constant term of the Laurent expansion. In particular

$$J_{\mathscr{O}_z}(f) = J_{\mathrm{unip}}(f_z), \qquad J_{\mathrm{unip}}(f) = \mathrm{vol}\big(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}\big)f(1) + \operatorname*{FP}_{s=1} Z(f,s).$$

Proof. Since z is central, $K_{P,\mathscr{O}_z}(x,y)$ for f equals $K_{P,\mathscr{O}_1}(x,y)$ for f_z, and likewise for J_P; and $J_{P,\mathscr{O}_1} = K_{P,\mathscr{O}_1}$ because $\mu_s = 1$ and $N_{P,1} = N_P$. So it suffices to treat $z=1$. The nontrivial unipotent elements of $G(\mathbb{Q})$ are the $\delta^{-1}n(\nu)\delta$, $\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})$, $\nu \in k^\times$, each written once, so

$$k^T_{\mathscr{O}_1}(x) = f(1) + \sum_{\delta}\Big(F(\delta x) - \hat{\tau}_B(H(\delta x)-T)\,C(\delta x)\Big),$$

where $F(x) = \sum_{\nu\in k^\times} f(x^{-1}n(\nu)x)$ and $C(x) = \int_{N(\mathbb{A})} f(x^{-1}nx)\,dn$. For $x = (n_0,1)\tilde{a}\tilde{k}$ with $p(\tilde{a}) = a(t)$ we have $x^{-1}(n(\nu),1)x = \tilde{k}^{-1}(n(t^{-2}\nu),1)\tilde{k}$, so F and C depend only on t and $\tilde{k}$. With $g_{\tilde{k}}(y) = f(\tilde{k}^{-1}(n(y),1)\tilde{k})$, Poisson summation gives

$$\delta_B(a)^{-1}\big(F-C\big)(x) = \sum_{\nu\in k^\times} \hat{g}_{\tilde{k}}(t^2\nu) - |t|^{-2}g_{\tilde{k}}(0),$$

which is integrable over $|t| > e^T$, uniformly in $\tilde{k}$; and $F(x) = 0$ once $|t|$ is small, because $g_{\tilde{k}}$ has compact support uniformly in $\tilde{k}$ and k is discrete in $\mathbb{A}_k$. Hence the three pieces $f(1)$, $\sum_\delta(1-\hat{\tau}_B)F$ and $\sum_\delta \hat{\tau}_B(F-C)$ of $k^T_{\mathscr{O}_1}$ are absolutely integrable, and unfolding each over $B(\mathbb{Q})\backslash\overline{G(\mathbb{A})} = N(\mathbb{Q})\backslash N(\mathbb{A})\times A(\mathbb{Q})\backslash A(\mathbb{A})\times\overline{K_0}$ gives, with $G(t) = |t|^{-2}\sum_{\nu\in k^\times}\Phi_f(t^{-2}\nu)$ and $\delta_B(a)^{-1}C = \hat{\Phi}_f(0)$ after the $\tilde{k}$-integration,

$$J^T_{\mathrm{unip}}(f) = \mathrm{vol}\big(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}\big)f(1) + \int_{|t|\le e^T} G(t)\,d^\times t + \int_{|t|>e^T}\big(G(t)-\hat{\Phi}_f(0)\big)\,d^\times t.$$

Now $Z(f,s) = \int G(t)|t|^{2-2s}d^\times t$. Split it at $|t| = e^T$ and write $G = (G-\hat{\Phi}_f(0)) + \hat{\Phi}_f(0)$ on $|t| > e^T$. Since $\int_{|t|>e^T}|t|^{2-2s}d^\times t = c_k\,e^{(2-2s)T}/(2s-2)$,

$$Z(f,s) = \int_{|t|\le e^T} G|t|^{2-2s} + \int_{|t|>e^T}\big(G-\hat{\Phi}_f(0)\big)|t|^{2-2s} + c_k\hat{\Phi}_f(0)\frac{e^{(2-2s)T}}{2s-2}.$$

The first integral is entire in s (compact range of $|t|$). By the Poisson formula above, $G-\hat{\Phi}_f(0) = \sum_{\nu\neq 0}\hat{\Phi}_f(t^2\nu) - |t|^{-2}\Phi_f(0)$, so the second integral is an entire function minus $c_k\Phi_f(0)e^{-2sT}/(2s)$. This gives the continuation and the poles. Expanding $e^{(2-2s)T}/(2s-2) = 1/(2(s-1)) - T + O(s-1)$ gives (4.10). The second form of (4.9) follows by writing $k^\times$ as the union of the cosets $\xi k^{\times 2}$ and unfolding $\sum_{c\in k^\times/\{\pm1\}}$ against $k^\times\backslash\mathbb{A}_k^\times$, using $|ct| = |t|$. □

The linear term of (4.10) agrees with the fine expansion: the term $\frac{1}{2}a^A(S,1)J^T_A(1,f)$ there has weight $v_A + 2T$, and $a^A(S,1) = c_k$, J_G of the class induced from 1 tends to $\hat{\Phi}_f(0)$ (Lemma 4.10).

Example 4.B. Let $k = \mathbb{Q}$ and $\Phi = \Phi_\infty \otimes \mathbf{1}_{\hat{\mathbb{Z}}}$ with $\Phi_\infty(y) = e^{-\pi y^2}$, so $\hat{\Phi}(0) = 1$, and take $d^\times t$ with $\mathrm{vol}(\hat{\mathbb{Z}}^\times) = 1$, so $c_\mathbb{Q} = 1$. Representing $\mathbb{Q}^\times\backslash\mathbb{A}^\times$ by $\mathbb{R}_{>0}\times\hat{\mathbb{Z}}^\times$, $Z(s) = \int_0^\infty u^{-2s}\sum_{\nu\neq0}e^{-\pi\nu^2/u^4}\,du/u =$

$\zeta(s)M(s)$ with $M(s)=\int_0^\infty e^{-\pi w^2}w^{s-1}dw=\frac{1}{2}\pi^{-s/2}\Gamma(s/2)$. The residue at $s=1$ is $M(1)=\frac{1}{2}=c_{\mathbb{Q}}\hat{\Phi}(0)/2$, and

$$\operatorname*{FP}_{s=1}Z(s)=\gamma_E M(1)+M'(1)=\frac{\gamma_E}{4}-\frac{\log 4\pi}{4}\approx -0.48845.$$

In the second form of (4.9), ξ runs over $\pm m$ with m squarefree, each contributing $m^{-s}M(s)\zeta(2s)$, and $\sum_m m^{-s}=\zeta(s)/\zeta(2s)$. The check script computes the truncated integrals of the proof of Proposition 4.8 at $T=-0.3,0,0.5,1.2$ and finds that $J^T_{\mathrm{unip}}-\mathrm{vol}\cdot f(1)-T$ equals this constant to 14 digits.

Proof of Theorem 4.5. (a) $k^T(x)=\sum_{\mathscr{O}}k^T_{\mathscr{O}}(x)$ by Lemma 4.2. If $k^T_{\mathscr{O}}\not\equiv 0$, then some element of $\mathscr{O}$, or some μn with $\mu\in\mathscr{O}\cap A(\mathbb{Q})$, is $G(\mathbb{A})$-conjugate into the support of f; its trace lies in k and in a compact subset of $\mathbb{A}_k$, so in a finite set. This leaves the two central classes, finitely many hyperbolic classes, and the elliptic classes with finitely many characteristic polynomials. The $SL(2,k)$-classes with a given regular elliptic characteristic polynomial, splitting field E, form a torsor under $k^\times/N_{E/k}E^\times$, which injects into $\bigoplus_v k_v^\times/NE_v^\times$ by the Hasse norm theorem [CF67]. Let v be a finite place outside a finite set, where $f_v=e_v$, E_v/k_v is unramified and μ is a unit with $|D(\mu)|_v=1$. If v splits in E, the local class is unique. If v is inert, an element $\gamma\in K_v$ with the characteristic polynomial of μ has irreducible reduction, so it fixes no edge of the Bruhat–Tits tree and exactly one vertex, that of K_v. If $\gamma'=m\gamma m^{-1}\in K_v$ with $m\in GL(2,k_v)$, then m fixes that vertex, so $m\in k_v^\times GL(2,\mathscr{O}_v)$ and $\det m\in k_v^{\times 2}\mathscr{O}_v^\times=NE_v^\times$; correcting m by an element of the centralizer $E_v^\times$ makes it lie in $SL(2,k_v)$. So one local class meets K_v. Hence only finitely many local classes, and so finitely many global classes, meet the support of f.

(b) and (d) are Propositions 4.6, 4.7 and 4.8. (c) follows from (a) and (b). □

4.4 Local distributions

The global terms are now rewritten as local distributions on a finite product of local covers. This subsection defines these distributions.

Test functions. Let S be a finite set of places of $\mathbb{Q}$ containing ∞ and 2, and let S_k be the set of places of k above S. Let $\overline{G(\mathbb{Q}_S)}$ be the cover of $G(\mathbb{Q}_S)=\prod_{v\in S_k}SL(2,k_v)$ defined by $\prod_{v\in S_k}\sigma_v$, the quotient of $\prod_{v\in S_k}\overline{SL(2,k_v)}$ as in Section 2.1. Let $\mathscr{H}(\overline{G(\mathbb{Q}_S)})$ be the span of the products $\bigotimes_{v\in S_k}f_v$, $f_v\in C_c^\infty(\overline{SL(2,k_v)})_-$, with f_v bi-$\overline{K_v}$-finite at archimedean v; here $C_c^\infty(\cdot)_-$ is the space of Section 2.1.4, defined on the topological cover. Let $\mathscr{H}_N(\overline{G(\mathbb{Q}_S)})_\Gamma$, for $N>0$ and a finite set Γ of irreducible representations of $\overline{K_S}$, be the subspace of functions supported on $\{\|g\|\le e^N\}$ whose left and right $\overline{K_S}$-translates span representations with constituents in Γ. With the seminorms $\sup|Df|$, D a differential operator on $\overline{G(\mathbb{R})}$, it is a Fréchet space, and $\mathscr{H}(\overline{G(\mathbb{Q}_S)})$ is the union over N and Γ. For $f_S\in\mathscr{H}(\overline{G(\mathbb{Q}_S)})$ put $f=f_S\otimes e^S$, $e^S=\bigotimes_{v\notin S_k}e_v$. For $\gamma\in G(\mathbb{Q})$ let $\gamma_S=(\gamma,\kappa_{\mathbb{A}}(\gamma))\in\overline{G(\mathbb{Q}_S)}$; since $e_v((k,\zeta)_\sigma)=\zeta$ for $k\in K_v$, $f(\gamma)=f_S(\gamma_S)$ when $\gamma\in K_v$ for all $v\notin S_k$.

Arthur's spaces of almost compactly supported and moderate functions [Ar88b] coincide with $\mathscr{H}$ for G, because $A_G=1$. For A they do not: $\mathfrak{a}_A=\mathfrak{a}\neq 0$, and a function that is compactly supported on each fiber of H need not be compactly supported. This matters in Section 6.3.

Orbital integrals. At each $v \in S_k$ we use the local measures of Section 2.1.5, and on $G(\mathbb{Q}_S)$ their product. For $\gamma = a(t) \in A(\mathbb{Q}_S)$ with $t_v^2 \neq 1$ for all v, and γ also denoting a lift in $\overline{A(\mathbb{Q}_S)}$, let $D(\gamma) = \prod_v (t_v - t_v^{-1})^2$, and

$$J_G(\gamma, f_S) = |D(\gamma)|^{1/2} \int_{A(\mathbb{Q}_S)\backslash G(\mathbb{Q}_S)} f_S(x^{-1}\gamma x)\, dx, \tag{4.11}$$

$$J_A(\gamma, f_S) = |D(\gamma)|^{1/2} \int_{A(\mathbb{Q}_S)\backslash G(\mathbb{Q}_S)} f_S(x^{-1}\gamma x)\, v_A(x)\, dx, \qquad v_A(x) = \sum_{v \in S_k} v_{A,v}(x_v), \tag{4.12}$$

where $v_{A,v}(x) = -H_v(x) - H_v(wx)$ is the local weight. For $\gamma = \mu_S$ regular elliptic, $J_G(\gamma, f_S)$ is defined by (4.11) with the centralizer $T_\mu(\mathbb{Q}_S)$ in place of $A(\mathbb{Q}_S)$. For a unipotent $u \in G(\mathbb{Q}_S)$ and $z \in \{\pm 1\}$, $J_G(zu, f_S) = \int_{G_u(\mathbb{Q}_S)\backslash G(\mathbb{Q}_S)} f_S(x^{-1}zux)\, dx$. These integrals converge absolutely, and the quotient measures are those of (2.9): on $A(\mathbb{Q}_S)\backslash G(\mathbb{Q}_S)$, $dx = dn\, dk$ for $x = nk$.

Lemma 4.9. *Let v be a place of k. Then $v_{A,v} \geq 0$, it is left $A(k_v)$-invariant and right K_v-invariant, and*

$$v_{A,v}(n(y)k) = \log \|(1,y)\|_v = \begin{cases} \log \max(1, |y|_v), & v \text{ finite}, \\ \tfrac{1}{2}\log(1+y^2), & v \text{ real}, \\ \log(1+|y|^2), & v \text{ complex}. \end{cases}$$

For $x \in G(\mathbb{A})$, $v_A(x) = \sum_v v_{A,v}(x_v)$, a finite sum.

Proof. As for Lemma 4.3, at one place: $|\det(e_2 x; e_2 wx)|_v = |-1|_v = 1$ gives $v_{A,v} \geq 0$, and $H_v(n(y)) = 0$, $e_2 wn(y) = (1,y)$. The global identity is $H = \sum_v H_v$. □

Example 4.C. Let $k = \mathbb{Q}$, let p be an odd prime in S with ψ_p unramified, and let $f_p = e_p$. Take $\gamma = (a(t), 1)$ with $t \in \mathbb{Z}_p^\times$ and $t^2 \neq 1$, and put $p^{-m} = |1 - t^{-2}|_p$. For $x = n(y)k$ with $k \in K_p$ the element $x^{-1}\gamma x$ is conjugate under K_p^* to

$$(n(y),1)^{-1}(a(t),1)(n(y),1) = \big(a(t)\, n((1-t^{-2})y), 1\big).$$

This element lies in K_p^* when $|y|_p \leq p^m$, and outside $\overline{K_p}$ otherwise. Since $|D(\gamma)|_p^{1/2} = |t - t^{-1}|_p = p^{-m}$ and $\mathbb{Z}_p$ has volume 1,

$$J_G(\gamma, e_p) = p^{-m} \cdot p^m = 1.$$

By Lemma 4.9, $v_{A,p}(n(y)k) = \log \max(1, |y|_p)$, so

$$J_A(\gamma, e_p) = p^{-m} \int_{|y|_p \leq p^m} \log \max(1, |y|_p)\, dy = \left(m - \frac{1 - p^{-m}}{p-1}\right) \log p.$$

If $t^2 \not\equiv 1 \pmod{p}$, then $m = 0$ and $J_A(\gamma, e_p) = 0$. This is the vanishing of the weight on the support of the integrand that is used after Theorem 4.11. For $p = 3$ and $t = 2$, $1 - t^{-2} = \frac{3}{4}$, so $m = 1$ and $J_A(\gamma, e_3) = \frac{2}{3}\log 3$.

Lemma 4.10. *Let $f_S = \bigotimes_{v \in S_k} f_v$, $z \in \{\pm 1\}$, and let $\Phi_v(y) = \int_{K_v} f_v(\tilde{k}^{-1} z(n(y),1)\tilde{k})\, d\tilde{k}$, a Schwartz–Bruhat function on k_v, with $\hat{\Phi}_v(0) = \int_{k_v} \Phi_v$. For $t \in k_S^\times$ near 1 let $a_t = (a(t), 1)z$ and*

$$r(t) = \sum_{v \in S_k} \log \big|t_v^2 - t_v^{-2}\big|_v.$$

(a) $J_G(a_t, f_S) \to \prod_v \hat{\Phi}_v(0)$ *as* $t \to 1$.

(b) The limit

$$J_A(z, f_S) := \lim_{t\to 1}\big(J_A(a_t, f_S) + r(t)\, J_G(a_t, f_S)\big) = \sum_{v\in S_k}\Big(\int_{k_v} \Phi_v(y)\log|2y|_v\, dy\Big)\prod_{w\in S_k\smallsetminus\{v\}} \hat{\Phi}_w(0)$$

exists. Without the term $r(t)J_G$, $J_A(a_t, f_S) = -\log|1-t^{-2}|_S \prod_v \hat{\Phi}_v(0) + O(1)$, *which diverges unless* $\prod_v \hat{\Phi}_v(0) = 0$.

(c) For $\xi \in k_S^\times$, $J_G(z\,n(\xi), f_S) = \prod_{v\in S_k} \frac{1}{2}\int_{k_v^\times} \Phi_v(\xi t^2)|t|_v^2\, d^\times t$.

Proof. Everything factors over $v \in S_k$, J_A by the Leibniz rule $J_A = \sum_v J_{A,v}\prod_{w\neq v} J_{G,w}$, and $r(t) = \sum_v r_v(t_v)$; so it suffices to treat one place. For $x = n(y)k$, $x^{-1}a_t x = \tilde{k}^{-1}a_t(n(\varepsilon y),1)\tilde{k}$ with $\varepsilon = 1-t^{-2}$, and $|D(a(t))|^{1/2} = |t|\,|\varepsilon|$. Put $y' = \varepsilon y$ and $h_t(y') = \int f_v(\tilde{k}^{-1}a_t(n(y'),1)\tilde{k})\, d\tilde{k}$. By Lemma 4.9 and homogeneity of $\|\cdot\|_v$,

$$J_{G,v} = |t|\int h_t(y')\, dy', \qquad J_{A,v} = |t|\int h_t(y')\big(\log\|(\varepsilon, y')\|_v - \log|\varepsilon|_v\big)\, dy'.$$

This gives (a), since $h_t \to \Phi_v$ with uniformly compact support, and the asymptotics in (b). Since $t^2 - t^{-2} = t^2\varepsilon(1+t^{-2})$,

$$J_{A,v} + r_v J_{G,v} = |t|\int h_t(y')\big(\log\|(\varepsilon, y')\|_v + \log|t^2(1+t^{-2})|_v\big)\, dy' \to \int \Phi_v(y)\big(\log|y|_v + \log|2|_v\big)\, dy$$

by dominated convergence ($\log|y|_v$ is locally integrable). For (c), $G_{n(\xi)} = \{\pm 1\}N$, so the integral is over $\{\pm 1\}\backslash A(k_v)\times K_v$ with measure $\frac{1}{2}\delta_B(a)^{-1}da\,dk$, and $a(t)^{-1}n(\xi)a(t) = n(t^{-2}\xi)$; substitute $t \mapsto t^{-1}$. □

Lemma 4.10(b) is Arthur's definition [Ar88a, (8.1)] of $J_M(\gamma, f)$ at a non-regular point, as the limit of $\sum_{L\in\mathscr{L}(M)} r_M^L(\gamma, a)J_L(a\gamma, f)$, specialized to $M = A$, $\gamma = z$: $r_A^A = 1$, and $r_A^G(1,a)$ is the function $r(t)$ above (Remark 2026.9). The term $L = G$ is present because the class induced from $1 \in A$ to G is the regular unipotent class; without it the limit diverges, as (b) shows.

Example 4.D. Let $k = \mathbb{Q}$, let p be an odd prime in S with ψ_p unramified, and let $f_p = e_p$. For $z = \pm 1$ the element $z\,(n(y),1)$ lies in K_p^* exactly when $y \in \mathbb{Z}_p$, and conjugation by $\overline{K_p}$ preserves K_p^*. Hence $\Phi_p = \mathbf{1}_{\mathbb{Z}_p}$ and $\hat{\Phi}_p(0) = 1$. In the formula of Lemma 4.10(b) the term of $v = p$ is

$$\int_{\mathbb{Z}_p} \log|2y|_p\, dy = -\log p \sum_{j\geq 1} j\,(p^{-j} - p^{-j-1}) = -\frac{\log p}{p-1},$$

multiplied by $\prod_{w\neq p}\hat{\Phi}_w(0)$. For $p = 3$ it is $-\frac{1}{2}\log 3$. So at a central element the unit element contributes a nonzero term, while at a regular element $a(t)$ with $t^2 \not\equiv 1 \pmod p$ it contributes nothing (Example 4.C).

4.5 The fine expansion

This subsection states the fine geometric expansion. It writes each term of the coarse expansion through local distributions with global coefficients.

For S as above let $\mathscr{U}_S$ be the set of (G,S)-equivalence classes of unipotent elements of $G(\mathbb{Q})$, that is, the classes $\{1\}$ and $u_\xi = n(\xi)$, $\xi \in k_S^\times/k_S^{\times 2}$ (each meets $G(\mathbb{Q})$, by weak approximation). The integral $\int f_S(x^{-1}zn(\xi)x)\,dx$ of Lemma 4.10(c) is taken with the fixed measure on $N(\mathbb{Q}_S)$, and it changes by $|c|_S^{-2}$ when ξ is replaced by ξc^2. For the class u_ξ we therefore put

$$J_G(zu_\xi, f_S) = |\xi|_S \int_{\{\pm1\}N(\mathbb{Q}_S)\backslash G(\mathbb{Q}_S)} f_S\big(x^{-1}zn(\xi)x\big)\,dx,$$

which depends only on the class of ξ modulo squares; this is the orbital integral for the measure on the centralizer of $n(\xi)$ transported from that of $n(1)$ by conjugation. For $v \notin S_k$ we use Haar measures giving K_v, $A(\mathscr{O}_v)$ and $T_\mu(\mathscr{O}_v)$ volume 1, and global measures that are products of these with the measures at S_k; the volumes below are taken with respect to them.

Theorem 4.11 (Arthur [Ar86, Thm. 8.1]; Li [Li14a] for covers)**.** *Let $f_S \in \mathscr{H}(\overline{G(\mathbb{Q}_S)})$ and $f = f_S \otimes e^S$. If S is large enough (depending on the support of f_S), then:*

(a) for $z \in \{\pm1\}$,

$$J_{\mathscr{O}_z}(f) = a^G(S,1)f_S(z) + \sum_{\xi \in k_S^\times/k_S^{\times 2}} a^G(S,u_\xi)J_G(zu_\xi,f_S) + \tfrac{1}{2}a^A(S,1)J_A(z,f_S),$$

with $a^G(S,1) = \mathrm{vol}(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})$, $a^A(S,1) = c_k$, and $J_A(z,f_S)$ as in Lemma 4.10(b); the coefficients $a^G(S,u_\xi)$ are those of [Ar86, Li14a], and they are computed in Proposition 7.10;

(b) for $\mathscr{O}$ elliptic with $\mu \in \mathscr{O}$, $J_{\mathscr{O}}(f) = a^G(S,\mu)J_G(\mu_S,f_S)$ with $a^G(S,\mu) = \mathrm{vol}(T_\mu(\mathbb{Q})\backslash T_\mu(\mathbb{A}))$;

(c) for $\mathscr{O}$ hyperbolic with $\mu = a(t) \in \mathscr{O}$, $J_{\mathscr{O}}(f) = a^A(S,\mu)J_A(\mu_S,f_S)$ with $a^A(S,\mu) = c_k$.

In Arthur's notation the right-hand sides are $\sum_{M\in\{A,G\}} |W_0^M||W_0^G|^{-1}\sum_\gamma a^M(S,\gamma)J_M(\gamma,f_S)$, with γ running over the (M,S)-classes in $M(\mathbb{Q})\cap\mathscr{O}$; in (c) the factor $\frac{1}{2}$ cancels against the two elements $\mu^{\pm1}$, since $J_A(\mu^{-1},f_S) = J_A(\mu,f_S)$. (Substitute $x = wy$, and use $v_A(wy) = -H(wy) - H(-y) = v_A(y)$.) Parts (b) and (c) follow from Propositions 4.7 and 4.6: for S large, $|D(\mu)|_S = |D(\mu)|_{\mathbb{A}_k} = 1$, and at $v \notin S_k$ the orbital integrals of e_v equal 1 while $v_{A,v} = 0$ on their support. For the same reason $f(\gamma) = f_S(\gamma_S)$, which is why the lift μ_S carries the global factor $\kappa_{\mathbb{A}}(\mu)$. In (a) the coefficient of $f_S(z)$ and the linear term agree with Proposition 4.8. The coefficients $a^G(S,u_\xi)$ exist by the theorem and are determined by (a). They are computed in Section 7.6: Proposition 7.10 derives (a) from Proposition 4.8, for the sets S such that the ring of S_k-integers of k is principal, and finds $a^G(S,u_\xi) = \frac{1}{2}\big(c_S + \sum_{\kappa\neq1}\kappa_S(\xi)L^S(1,\kappa)\big)$, the sum over the quadratic Hecke characters unramified outside S_k.

The coefficients are defined with centralizers in the cover, which for $n = 2$ are the full preimages (Lemma 4.1), and the sums run over (G,S)-classes, not over the rational class $\mathscr{O}$.

Remark 2026.9. *Normalizations on* $\mathfrak{a}$. Identify $\mathfrak{a}$ with $\mathbb{R}$ by $X \mapsto X(\chi_1)$, $\chi_1 = N_{k/\mathbb{Q}} \circ t$, as in Section 2.1.3. The coroot is $\alpha^\vee : s \mapsto a(s)$, $s \in \mathbb{Q}^\times$, so $\alpha^\vee = \langle \chi_1, \alpha^\vee \rangle = d$, and the simple root α of (B, A_B), $a(s) \mapsto s^2$, is $\alpha = (2/d)\chi_1$, i.e. $\alpha(X) = 2X/d$. The modulus is $\delta_B = e^{2H}$ for every k. We use Lebesgue measure dX on $\mathfrak{a}$, for which $\mathfrak{a}/\mathbb{Z}\alpha^\vee$ has volume d; Arthur normalizes it to volume 1. With Arthur's normalization, $\theta_B(\lambda) = \lambda(\alpha^\vee)$, the weight v_A and the function r_A^G are divided by d, and $\mathrm{vol}(A(\mathbb{Q})\backslash A(\mathbb{A})^1)$ is multiplied by d; the products in (4.7) and Theorem 4.11 do not change. With $\theta_P(\lambda)$ in our normalization, Arthur's $r_A^G(1,a) = \lim_{\lambda \to 0} \sum_{P \in \mathscr{P}(A)} r_P(\lambda, 1, a)\theta_P(\lambda)^{-1}$ [Ar88a, §3] with $r_P(\lambda, 1, a) = |a^\beta - a^{-\beta}|^{\lambda(\beta^\vee)/2}$ evaluates to $r(t)$ for $a = a(t)$, $t \in \mathbb{Q}_S^\times$. We have checked this only for the formula of [Ar88a] as quoted in `book7d.tex`, not against the general definition of the exponents $\rho(\beta, u)$ there. The independent check is that the limit in Lemma 4.10(b) exists. The trace formula of this paper does not depend on the comparison: Proposition 7.10(c) derives the central terms of Theorem 4.11 from Proposition 4.8, with $J_A(z, f)$ as defined in Lemma 4.10(b), when the ring of S_k-integers is principal.

4.6 Status of the results

This subsection lists the statements of Section 4 that rest on citation or on numerical checks.

Remark 2026.10. *What rests on citation.* Theorem 4.5 and Propositions 4.6–4.8 are proved here. Theorem 4.11(a) is stated here as taken from [Ar86, Li14a]. It is derived from Proposition 4.8 in Proposition 7.10, which also computes the coefficients $a^G(S, u_\xi)$, when the ring of S_k-integers is principal. (Grouping the second form of (4.9) by the image of ξ in $k_S^\times / k_S^{\times 2}$ does not work directly: each fiber is infinite and contributes a Dirichlet series that diverges at $s = 1$. One expands in quadratic Hecke characters instead.) The unramified computations used for (b)–(c) of Theorem 4.11 are standard and not written out. For an elliptic μ they use the uniqueness of the local class meeting K_v from the proof of Theorem 4.5(a).

Remark 2026.11. The script `checks/sec4_n2_check.py` verifies, for $k = \mathbb{Q}$: that the Iwasawa height computed from an explicit decomposition equals $-\log \|e_2 g\|_v$ at $v = \infty, 2, 3, 5, 7$; the inequality $H(x) + H(\gamma x) \le 0$ on random adelic x and $\gamma \in SL(2,\mathbb{Q}) \smallsetminus B(\mathbb{Q})$, and $H(\gamma x) = H(x)$ for $\gamma \in B(\mathbb{Q})$; Lemma 4.3(c) and Lemma 4.9; the weight integral in the proof of Proposition 4.6; the conjugation identities used there (symbolically); Lemma 4.10(b) at $v = \mathbb{R}$ (quadrature) and $v = \mathbb{Q}_3$ (exactly), including the divergence without $r(t)$; the symmetry $J_A(\mu^{-1}) = J_A(\mu)$ at $\mathbb{R}$; and Example 4.B. The script `checks/examples_n2_check.py` checks Examples 4.A, 4.C and 4.D.

5 The spectral side

This section computes the spectral side of the trace formula for $\overline{G(\mathbb{A})}$, class by class, and assembles the trace formula. The general theory is due to Langlands [La76] and Arthur [Ar80, Ar82] for connected reductive groups, and to Mœglin–Waldspurger [MW95] and Li [Li13] for covering groups; as in Section 4, we state it for the double cover of $G = \mathrm{Res}_{k/\mathbb{Q}} SL(2)$ and make the terms explicit. Three features of the double cover stand out. The residual spectrum is not empty: it consists of one even Weil representation for each class in $k^\times / k^{\times 2}$ (Theorem 5.8). These classes are exactly the cuspidal data (A, μ) with $\mu^w = \mu$ (Lemma 5.3, Theorem 5.14(c)). And the operator

M_B^T is $2T - M^{-1}M'$ (Theorem 5.14), whose T-term we check against the geometric side (Proposition 5.17). The normalizing factor of the intertwining operator, $L^S(2\lambda,\chi^2)/L^S(2\lambda+1,\chi^2)$, is computed in Lemma 5.6.

Roadmap. The objects attached to a genuine automorphic character μ of the cover of A are related as follows.

$$\begin{array}{ccc} \varphi \in \mathscr{H}_B(\mu) & \xrightarrow{\text{sum over } B(\mathbb{Q})\backslash G(\mathbb{Q})} & E(x,\varphi,\lambda) \\ & & \Big\downarrow \text{constant term} \\ \theta_\varphi = \operatorname{Res}_{\lambda=1/2} E(x,\varphi,\lambda) \in V_b & \xleftarrow{\text{pole at } \lambda=\frac{1}{2}\ (\mu^w=\mu)} & \varphi_\lambda + (M(w,\lambda)\varphi)_{-\lambda} \end{array}$$

For $\lambda = it$ the Eisenstein series are not square integrable; their wave packets span the continuous part $L^2_{\chi,\mathrm{cont}}$ of the spectrum. Section 5.1 sorts the spectrum by cuspidal data χ. Section 5.2 constructs the Eisenstein series E, whose constant term defines the intertwining operator $M(w,\lambda)$. The operator factors as a quotient of partial L-functions times finitely many local operators (Lemma 5.6); its only pole in $\operatorname{Re}\lambda > 0$ is at $\lambda = \frac{1}{2}$ and occurs exactly for the ramified classes, and the residues of E there span the residual spectrum, which consists of even Weil representations (Theorem 5.8). Section 5.3 truncates the kernels as on the geometric side and computes the integral $J^T_\chi(f)$ of each truncated kernel from the inner products of truncated Eisenstein series (the Maass–Selberg relations): the diagonal terms give $2T - M^{-1}M'$, and for a ramified class the cross terms give $\frac{1}{4}\operatorname{tr}(M(w,0)I(\mu_0,f))$. Section 5.4 equates the two sides.

Conventions. $L^2_{\mathrm{gen}} = L^2(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{\mathrm{gen}}$ is the space of φ with $\varphi(\gamma x\zeta) = \zeta\varphi(x)$, with the measure of Section 2.2 (total volume 1), and R is the right regular representation; test functions $f \in \mathscr{H}(\overline{G(\mathbb{A})})$ are anti-genuine and $R(f) = \int f(y)R(y)\,dy$, as in Section 4.2. As in Section 2.1.3 and Remark 2026.9, $\mathfrak{a} \cong \mathbb{R}$ by $X \mapsto X(\chi_1)$, with Lebesgue measure dX, and we identify $\mathfrak{a}^*_{\mathbb{C}}$ with $\mathbb{C}$ so that $\lambda(X) = \lambda X$. Then $e^{\lambda(H(x))} = e^{\lambda H(x)}$,

$$\rho = \rho_B = 1, \qquad \alpha = 2/d, \qquad \alpha^\vee = d, \qquad d = [k:\mathbb{Q}],$$

since $\delta_B = e^{2H}$, and $\lambda \in i\mathbb{R}$ is written $\lambda = it$ with the measure $dt/2\pi$, dual to dX. For $k = \mathbb{Q}$ and $x = n(u)a(t)k$ the point $x\cdot i$ of the upper half plane has imaginary part $y = t^2$, so $e^{(\lambda+1)H(x)} = y^s$ with $s = (\lambda+1)/2$. We write $c_k = \operatorname{vol}(A(\mathbb{Q})\backslash A(\mathbb{A})^1)$ as in Section 4.3, and $X_A = A(\mathbb{Q})\backslash\overline{A(\mathbb{A})^1}$ and $\Pi_{\mathrm{aut}}(X_A)$ as in Section 3.1; $\Pi_u(H)$ is the set of irreducible unitary genuine representations of H, as there.

5.1 The spectral decomposition and cuspidal data

This subsection decomposes the space of genuine square-integrable functions and sorts it by cuspidal data.

Lemma 5.1. *$L^2_{\mathrm{gen}} = L^2_{\mathrm{disc}} \oplus L^2_{\mathrm{cont}}$, where L^2_{disc} is the closed span of the irreducible closed invariant subspaces. Each $\pi \in \Pi_u(\overline{G(\mathbb{A})})$ occurs in L^2_{disc} with finite multiplicity $m(\pi)$, and $L^2_{\mathrm{disc}} = L^2_{\mathrm{cusp}} \oplus$*

L^2_{res}, *where* L^2_{cusp} *is the space of cusp forms,*

$$\int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \varphi(nx)\,dn = 0 \quad \textit{for almost all } x,$$

and L^2_{res} *is its orthogonal complement in* L^2_{disc}.

Proof. By Lemma 2.7, $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}/K_f^* \cong \widetilde{\Gamma}\backslash\overline{G(\mathbb{R})}$ with $\widetilde{\Gamma}$ a lattice, so this is Langlands' theory [La76] for $\widetilde{\Gamma}\backslash\overline{G(\mathbb{R})}$; in adelic form, see [MW95], whose setting includes finite central covers. The finiteness of multiplicities in L^2_{disc} is Harish-Chandra's finiteness theorem for automorphic forms. □

Lemma 5.2. *Every* $\pi \in \Pi_u(\overline{G(\mathbb{A})})$ *is a restricted tensor product* $\pi \cong \bigotimes'_v \pi_v$, *with* $\pi_v \in \Pi_u(\overline{SL(2,k_v)})$ *genuine and* π_v *having a nonzero* K_v^**-fixed vector for almost all* v*; the restricted product is taken with respect to these vectors. Conversely every such family defines an element of* $\Pi_u(\overline{G(\mathbb{A})})$.

Proof. A representation of $\overline{G(\mathbb{A})}$ is a representation of the restricted product of the $\overline{SL(2,k_v)}$ (with respect to the K_v^*) that is trivial on $Z_0 = \{(\zeta_v) \mid \prod_v \zeta_v = 1\}$ (Section 2.2). Flath's theorem [Fl79] applies to the restricted product and gives $\pi \cong \bigotimes' \pi_v$. Each μ_2 at v acts by a sign ε_v; triviality on Z_0 forces all ε_v to be equal, and genuineness forces them to be the identity. Conversely, if every π_v is genuine, $\bigotimes' \pi_v$ is trivial on Z_0 and descends. □

Cuspidal data. The *cuspidal data* are the pairs (G,π), with $\pi \in \Pi_u(\overline{G(\mathbb{A})})$ occurring in L^2_{cusp}, and the pairs (A,μ), with $\mu \in \Pi_{\mathrm{aut}}(X_A)$ a genuine character of X_A (Theorem 3.2). Among the standard parabolic subgroups, B is associate only to itself, with $W(\mathfrak{a}) = \Omega(\mathfrak{a},\mathfrak{a}) = \{1,w\}$. (The opposite $\bar{B} = wBw^{-1}$ also has Levi A and is associate to B, but it is not standard.) For μ as above put $\mu^w(\tilde{a}) = \mu(w^{-1}\tilde{a}w)$, where w stands for its image $(w,\kappa_{\mathbb{A}}(w)) = (w,1)$ in $\overline{G(\mathbb{A})}$. Two data are *equivalent* if they are equal or of the form (A,μ), (A,μ^w). Let $\mathscr{X}$ be the set of classes, $\mathscr{X}_G$ the classes of the (G,π) and $\mathscr{X}_A$ those of the (A,μ); for $\chi \in \mathscr{X}_A$ we write $\mu \in \chi$ for the members of the set $\{\mu,\mu^w\}$, which has one or two elements. A class $\chi \in \mathscr{X}_A$ is *unramified* if $\mu^w \neq \mu$, and *ramified* if $\mu^w = \mu$ [Ar80, §4].

Lemma 5.3. *(a) For* $t \in \mathbb{A}_k^\times$ *and* $\zeta \in \mu_2$, $(w,1)\,(a(t),\zeta)\,(w,1)^{-1} = (a(t^{-1}),\zeta)$ *in* $\overline{G(\mathbb{A})}$, *and the same holds in each* $\overline{SL(2,k_v)}$.

(b) $(a(t),1)^{-1} = (a(t^{-1}),\prod_v(t_v,-1)_v)$*; so* w*-conjugation is not inversion on* $\overline{A(\mathbb{A})}$, *and* $\mu^w(a(t),1) = \mu(a(t),1)^{-1}\prod_v(t_v,-1)_v$.

(c) $\chi_\psi^w = \chi_\psi$ *for Weil's character* χ_ψ *(Lemma 3.3). Hence, writing* $\mu = \chi_\psi\cdot(\chi\circ p)$ *with* χ *a character of* $A(\mathbb{Q})\backslash A(\mathbb{A})^1$ *(Theorem 3.2),* $\mu^w = \chi_\psi\cdot(\chi^{-1}\circ p)$.

(d) $\mu^w = \mu$ *if and only if* $\chi^2 = 1$, *and then* $\mu = \chi_{\psi_b}$ *on* $\overline{A(\mathbb{A})^1}$ *for a unique* $b \in k^\times/k^{\times 2}$. *So the ramified classes in* $\mathscr{X}_A$ *are the classes* χ_b *of* (A,χ_{ψ_b}), $b \in k^\times/k^{\times 2}$, *each with one member, and every other class in* $\mathscr{X}_A$ *has two members.*

Proof. (a) At one place, with $x(\cdot)$ as in (2.1): $x(w)=1$, $x(a(t))=t^{-1}$, $wa(t)=\begin{pmatrix}0 & -t^{-1}\\ t & 0\end{pmatrix}$ has $x=t$, $wa(t)w^{-1}=a(t^{-1})$ has $x=t$, and $x(w^{-1})=-1$. So by the conjugation formula (2.3), $\beta(w,a(t))\beta(wa(t),w^{-1})\beta(w,w^{-1})=(t,t^2)(1,-t)(1,-1)=1$. At non-dyadic finite v the correction κ_v is 1 on w, w^{-1}, $a(t)$, $a(t^{-1})$ and $wa(t)$ (the lower left entry is 0 or of absolute value ≥ 1), so $\sigma_v=\beta_v$ on these pairs. Taking the product over v gives the global statement.

(b) $(a(t),1)(a(t^{-1}),\varepsilon)=(1,\sigma_{\mathbb{A}}(a(t),a(t^{-1}))\varepsilon)$ and $(t_v,t_v^{-1})_v=(t_v,t_v)_v=(t_v,-1)_v$ by (H1) and (H2). The formula for μ^w follows from (a).

(c) $t^{-1}=t\cdot t^{-2}$ is in the square class of t, so $\gamma_v(t^{-1})=\gamma_v(t)$ by Lemma 3.3(a), and $\chi_\psi(a(t^{-1}),\zeta)=\chi_\psi(a(t),\zeta)$. By (a), $\mu^w(a(t),\zeta)=\mu(a(t^{-1}),\zeta)=\chi_\psi(a(t),\zeta)\,\chi(a(t))^{-1}$.

(d) By (c), $\mu^w=\mu$ if and only if $\chi^2=1$. A quadratic character of $k^\times\backslash\mathbb{A}_k^1$ extends uniquely to a quadratic character of $k^\times\backslash\mathbb{A}_k^\times$ trivial on $\mathbb{R}_{>0}$ (embedded diagonally at the archimedean places). By class field theory the quadratic characters of $k^\times\backslash\mathbb{A}_k^\times$ are the $t\mapsto\prod_v(b,t_v)_v$, $b\in k^\times$, and two values of b give the same character exactly when they differ by an element of $k^\times\cap\prod_v k_v^{\times 2}=k^{\times 2}$; each such character is trivial on $\mathbb{R}_{>0}$, since $(b,u)_v=1$ for $u>0$ at a real v. By Lemma 3.3(d), $\chi_\psi\cdot(b,\cdot)=\chi_{\psi_b}$. □

For $SL(2)$ itself the ramified classes are (A,μ) with $\mu^2=1$, because there w acts on A by inversion [LL79]. For the cover, inversion is twisted by $(t,-1)$, and Weil's character absorbs the twist.

Example 5.A. Let $k=\mathbb{Q}$. Then $k^\times/k^{\times 2}$ is represented by the squarefree integers b, positive and negative, and the ramified classes are the χ_{ψ_b}. If $f=f_S\otimes e^S$ as in Section 4.4, only the classes with $\chi_{\psi_b,p}$ trivial on $A(\mathbb{Z}_p)$ for every $p\notin S$ can contribute; since $(b,u)_p=1$ for all units u exactly when $\mathrm{ord}_p(b)$ is even, these are the b whose prime factors lie in S. For $S=\{\infty,2\}$ they are $b\in\{\pm1,\pm2\}$.

Proposition 5.4. *$L^2_{\mathrm{gen}}=\widehat{\bigoplus}_{\chi\in\mathscr{X}}L^2_\chi$, an orthogonal decomposition into invariant subspaces, where:*

(a) for χ the class of (G,π), L^2_χ is the π-isotypic subspace of L^2_{cusp}, of multiplicity $m(\pi)$;

(b) for $\chi\in\mathscr{X}_A$ with $\mu\in\chi$, $L^2_\chi=L^2_{\chi,\mathrm{res}}\oplus L^2_{\chi,\mathrm{cont}}$, where $L^2_{\chi,\mathrm{cont}}$ is the closed span of the wave packets $\int_{\mathbb{R}}E(x,\Phi(t),it)\,dt$ of the Eisenstein series of Section 5.2, $\Phi\in C_c^\infty(\mathbb{R},\mathscr{H}_B(\mu))$ with values in a finite sum of $\overline{K_0}$-types, and $L^2_{\chi,\mathrm{res}}=L^2_{\mathrm{res}}\cap L^2_\chi$ is spanned by residues of these Eisenstein series at poles with $\mathrm{Re}\,\lambda>0$. In particular $L^2_{\mathrm{cusp}}=\widehat{\bigoplus}_{\chi\in\mathscr{X}_G}L^2_\chi$.

This is Langlands' decomposition [La76, Ch. 7] along cuspidal data [MW95, Ch. II, V]. By Lemma 5.3(d) and Lemma 5.6(c) below, $L^2_{\chi,\mathrm{res}}=0$ unless χ is ramified.

5.2 Eisenstein series and intertwining operators

This subsection constructs the Eisenstein series and the intertwining operator, computes the operator at the unramified places, and identifies the residual spectrum.

For $\mu\in\Pi_{\mathrm{aut}}(X_A)$ let $\mathscr{H}_B(\mu)$ be the space of genuine functions φ on $N(\mathbb{A})A(\mathbb{Q})A_B(\mathbb{R})^0\backslash\overline{G(\mathbb{A})}$ that are right $\overline{K_0}$-finite and satisfy

$$\varphi(\tilde{a}x)=\mu(\tilde{a})\varphi(x)\quad(\tilde{a}\in\overline{A(\mathbb{A})^1}),\qquad\langle\varphi,\varphi'\rangle=\int_{X_A}\int_{\overline{K_0}}\varphi(\tilde{a}\tilde{k})\overline{\varphi'(\tilde{a}\tilde{k})}\,d\tilde{k}\,d\tilde{a}=c_k\int_{\overline{K_0}}\varphi\overline{\varphi'}.\quad(5.1)$$

Here $A(\mathbb{Q})$ and $N(\mathbb{A})$ are embedded by $b \mapsto (b,1)$ (Lemma 2.6(b)), μ is extended to $\overline{A(\mathbb{A})}$ trivially on $A_B(\mathbb{R})^0$, and restriction to $\overline{K_0}$ is injective by Lemma 2.8. For $\lambda \in \mathbb{C}$ put $\varphi_\lambda(x) = \varphi(x)e^{(\lambda+1)H(x)}$, and let $I(\mu_\lambda)$ be the representation on $\mathscr{H}_B(\mu)$ given by $(I(\mu_\lambda,y)\varphi)_\lambda(x) = \varphi_\lambda(xy)$, and $I(\mu_\lambda,f) = \int_{\overline{G(\mathbb{A})}} f(y)I(\mu_\lambda,y)\,dy$. It is the induced representation $\mathrm{Ind}_{\overline{B(\mathbb{A})}}^{\overline{G(\mathbb{A})}}(\mu\, e^{\lambda H})$ (normalized induction), unitary for $\lambda \in i\mathbb{R}$. Note that φ is $A_B(\mathbb{R})^0$-invariant and that λ enters only through φ_λ.

For $\varphi \in \mathscr{H}_B(\mu)$ and $\mathrm{Re}\,\lambda > 1$ put

$$E(x,\varphi,\lambda) = \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \varphi_\lambda(\delta x), \tag{5.2}$$

$$\big(M(w,\lambda)\varphi\big)_{-\lambda}(x) = \int_{N(\mathbb{A})} \varphi_\lambda(w^{-1}nx)\,dn. \tag{5.3}$$

The sum is well defined because φ_λ is left $B(\mathbb{Q})$-invariant (Lemma 2.8(b), and μ is trivial on $A(\mathbb{Q})$). By Lemma 5.3(a), $M(w,\lambda)$ maps $\mathscr{H}_B(\mu)$ to $\mathscr{H}_B(\mu^w)$, and it intertwines $I(\mu_\lambda)$ with $I(\mu^w_{-\lambda})$. The constant term of a function ϕ on $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ is $\phi_B(x) = \int_{N(\mathbb{Q})\backslash N(\mathbb{A})} \phi(nx)\,dn$.

Theorem 5.5 (Langlands [La76]; Mœglin–Waldspurger [MW95] for covers)**.** *Let $\mu \in \Pi_{\mathrm{aut}}(X_A)$ and $\varphi \in \mathscr{H}_B(\mu)$.*

(a) The series (5.2) and the integral (5.3) converge absolutely for $\mathrm{Re}\,\lambda > 1$, and $E(x,\varphi,\lambda)$ and $M(w,\lambda)\varphi$ continue to meromorphic functions of $\lambda \in \mathbb{C}$.

(b) $E_B(x,\varphi,\lambda) = \varphi_\lambda(x) + (M(w,\lambda)\varphi)_{-\lambda}(x)$.

(c) $E(x,M(w,\lambda)\varphi,-\lambda) = E(x,\varphi,\lambda)$ and $M(w,-\lambda)M(w,\lambda) = 1$.

(d) On $i\mathbb{R}$, $E(x,\varphi,\lambda)$ and $M(w,\lambda)$ are holomorphic and $M(w,\lambda)$ is unitary. In $\mathrm{Re}\,\lambda > 0$ they have at most finitely many poles, all simple and in $(0,1]$, and the poles of E are those of $M(w,\lambda)$.

(e) $E(x,I(\mu_\lambda,f)\varphi,\lambda) = \int_{\overline{G(\mathbb{A})}} f(y)E(xy,\varphi,\lambda)\,dy$, and $M(w,\lambda)I(\mu_\lambda,f) = I(\mu^w_{-\lambda},f)M(w,\lambda)$.

The measure on $i\mathfrak{a}^*$ is $dt/2\pi$, and the number $n(A_B)$ of chambers in $\mathfrak{a}$ is 2; it produces the factor $\frac{1}{2}$ in Proposition 5.9. The convergence of (5.2) for $\mathrm{Re}\,\lambda > \rho = 1$ is Godement's criterion; the integral (5.3) in fact converges for $\mathrm{Re}\,\lambda > \frac{1}{2}$, by Lemma 5.6.

Lemma 5.6. *Let $\mu = \chi_\psi \cdot (\chi \circ p) \in \Pi_{\mathrm{aut}}(X_A)$. Let S be a finite set of places of k containing the archimedean and dyadic places and those where ψ_v has conductor $\neq \mathscr{O}_v$ or χ_v is ramified. For $v \notin S$ let $\varphi_v^0 \in I(\mu_{v,\lambda})$ be the K_v^*-fixed vector with $\varphi_v^0 = 1$ on K_v^*, and let $M_v(\lambda)$ be the local intertwining operator, $(M_v(\lambda)\varphi_v)(x) = \int_{k_v} \varphi_{v,\lambda}(w^{-1}n(y)x)\,dy$ with the self-dual measure.*

(a) For $\varphi = \varphi_S \otimes \bigotimes_{v\notin S}\varphi_v^0$, $M(w,\lambda)\varphi = (M_S(\lambda)\varphi_S) \otimes \bigotimes_{v\notin S} M_v(\lambda)\varphi_v^0$, with $M_S = \bigotimes_{v\in S} M_v$.

(b) For $v \notin S$, with $q = q_v$ and $\chi_v(\varpi)$ the value at a uniformizer,

$$M_v(\lambda)\varphi_v^0 = \frac{1-\chi_v(\varpi)^2 q^{-1-2\lambda}}{1-\chi_v(\varpi)^2 q^{-2\lambda}}\,\varphi_v^{0,w} = \frac{L_v(2\lambda,\chi_v^2)}{L_v(2\lambda+1,\chi_v^2)}\,\varphi_v^{0,w},$$

where $\varphi_v^{0,w}$ is the K_v^*-fixed vector of $I(\mu_{v,-\lambda}^w)$. Hence $M(w,\lambda) = m^S(\mu,\lambda)\,M_S(\lambda)\otimes 1$ on such φ, with

$$m^S(\mu,\lambda) = \frac{L^S(2\lambda,\chi^2)}{L^S(2\lambda+1,\chi^2)}. \tag{5.4}$$

(c) In $\operatorname{Re}\lambda > 0$, $M(w,\lambda)$ is holomorphic unless $\chi^2 = 1$ (χ extended to $k^\times\backslash\mathbb{A}_k^\times$ trivially on $\mathbb{R}_{>0}$). If $\chi^2 = 1$, its only pole there is a simple pole at $\lambda = \frac12$, with residue $\operatorname{Res}_{s=1}\zeta_k^S(s)\,\big(2\zeta_k^S(2)\big)^{-1} M_S(\frac12)\otimes 1$, which is not 0.

Proof. (a) The lift $(w,1)$ of $w \in G(\mathbb{Q})$ is the product of the local lifts $(w,1)_v$ ($\kappa_v(w) = 1$ for all v), $N(\mathbb{A})$ is the restricted product of the $N(k_v)$ with $dn = \prod dn_v$, and a factorizable φ is a product of local functions (Lemma 5.2).

(b) Since $\chi_{\psi,v}$ is trivial on $A(\mathscr{O}_v)^*$ (Lemma 3.3(b)), $\varphi_v^0((n,1)(a(t),\zeta)\tilde{k}) = \zeta\,\mu_v(a(t),1)|t|^{\lambda+1}$ for $\tilde{k} \in K_v^*$ is well defined. If $|y| \le 1$, then $w^{-1}n(y) \in K_v$, its lift lies in K_v^* because $\sigma_v \equiv 1$ on $K_v \times K_v$, and the integrand is 1. If $|y| = q^m$ with $m \ge 1$, write $y = \varpi^{-m}u$, $u \in \mathscr{O}^\times$; then

$$w^{-1}n(y) = n(-y^{-1})\,a(-y^{-1})\begin{pmatrix} 1 & 0 \\ y^{-1} & 1 \end{pmatrix},$$

the last factor in K_v. Computing the cocycle of this decomposition, the integrand is $\chi_v(\varpi)^m|y|^{-\lambda-1}$ times 1 if m is even, and times $c\,(u,\varpi)_v$ with a constant c of absolute value 1 if m is odd. (The odd case uses $\gamma_v(\varpi u) = (u,\varpi)_v\gamma_v(\varpi)$, Lemma 3.3(a)–(b); both cases are checked in `checks/sec5_n2_check.py` at $p = 3,5,7$.) The shell $\{|y| = q^m\}$ has volume $q^m(1-q^{-1})$, and $u \mapsto (u,\varpi)_v$ is a nontrivial character of $\mathscr{O}^\times$, so the odd shells contribute 0 and

$$M_v(\lambda)\varphi_v^0(1) = 1 + (1-q^{-1})\sum_{j\ge1}\big(\chi_v(\varpi)^2q^{-2\lambda}\big)^j = \frac{1-\chi_v(\varpi)^2q^{-1-2\lambda}}{1-\chi_v(\varpi)^2q^{-2\lambda}}.$$

$M_v(\lambda)\varphi_v^0$ is K_v^*-fixed, so it is this multiple of $\varphi_v^{0,w}$.

(c) The local integrals at $v \in S$ converge absolutely for $\operatorname{Re}\lambda > 0$, since the integrand is $O(|y|^{-\operatorname{Re}\lambda-1})$, so $M_S(\lambda)$ is holomorphic there. If $\chi^2 \ne 1$, $L^S(s,\chi^2)$ is entire (the removed Euler factors have no poles in $\operatorname{Re}s > 0$) and $L^S(s,\chi^2) \ne 0$ for $\operatorname{Re}s \ge 1$. If $\chi^2 = 1$, $m^S = \zeta_k^S(2\lambda)/\zeta_k^S(2\lambda+1)$, whose only pole in $\operatorname{Re}\lambda > 0$ is at $2\lambda = 1$. The operator $M_S(\frac12)$ is not 0: if φ_S is supported in $\overline{B(\mathbb{Q}_S)}\,w^{-1}N(U)$ for a small compact neighborhood U of 0, then $(M_S(\frac12)\varphi_S)_{-1/2}(1) = \int_U \varphi_{S,1/2}(w^{-1}n(y))\,dy$, and on the big cell φ_S may be any smooth compactly supported function of y. □

For $SL(2)$ itself the same computation gives $L_v(\lambda,\chi_v)/L_v(\lambda+1,\chi_v)$, with the pole at $\lambda = 1 = \rho$ for $\chi = 1$ (the script checks this too). On the double cover the variable is doubled and the character squared; this is the Gindikin–Karpelevich formula for the double cover [KP84]; for general covering groups see [Mc12, Ga14]. The factor (5.4) is what produces the logarithmic derivative m'/m in Section 6.

The Weil representation. The residual spectrum consists of Weil representations, and to say which ones we fix a normalization. Let v be a place of k and ψ' a nontrivial character of k_v. The *Weil representation* $\omega_{\psi'}$ attached to ψ' and the quadratic form x^2 [We64, Ge76] is a genuine representation of $\widetilde{SL(2,k_v)}$ on the Schwartz–Bruhat space $\mathscr{S}(k_v)$ in which

$$\begin{gathered}\omega_{\psi'}(n(u),1)\phi(x)=\psi'(ux^2)\phi(x),\qquad \omega_{\psi'}(w,1)\phi(x)=c\,(\mathscr{F}\phi)(-x),\\ \mathscr{F}\phi(x)=\int_{k_v}\phi(y)\psi'(2xy)\,dy,\end{gathered}\tag{5.5}$$

with a constant c of absolute value 1 and the measure that is self-dual for $\psi'(2xy)$. (In Kubota's model $c=\gamma'(1)^{-1}$, with γ' as in Lemma 5.7; see Lemma 7.4.) It is the sum of its restrictions $r^+_{\psi'}$ and $r^-_{\psi'}$ to the even and the odd functions, which are irreducible. In some references $n(u)$ acts by $\psi'(\frac{1}{2}ux^2)$; that representation is $\omega_{\psi'_{1/2}}$ in the present notation.

Lemma 5.7. *Let $\omega=\omega_{\psi'}$ be as in (5.5), and let $\gamma'(a)$ be the Weil index of $x\mapsto\psi'(ax^2)$, as in (3.2). In the model of Section 2.1, for $t\in k_v^\times$ and $\zeta\in\mu_2$,*

$$\omega(a(t),\zeta)\phi(x)=\zeta\,\frac{\gamma'(1)}{\gamma'(t)}\,|t|_v^{1/2}\phi(tx)=\zeta\,\chi_{\psi'_{-1}}(a(t),1)\,|t|_v^{1/2}\phi(tx),$$

where $\psi'_{-1}(x)=\psi'(-x)$. Consequently $\phi\mapsto\big(x\mapsto(\omega(x)\phi)(0)\big)$ maps $r^+_{\psi'}$ into the induced representation $I(\chi_{\psi'_{-1},v},-\frac{1}{2})$: the function $F(x)=(\omega(x)\phi)(0)$ satisfies $F((n,1)\tilde{a}x)=\chi_{\psi'_{-1}}(\tilde{a})e^{H_v(\tilde{a})/2}F(x)$.

Proof. Let $\bar{n}(u)=wn(-u)w^{-1}$, the lower triangular unipotent matrix with entry u. By Lemma 2.2(h), $(\bar{n}(u),1)=(w,1)(n(-u),1)(w,1)^{-1}$, so by (5.5) $\omega(\bar{n}(u),1)=\mathscr{F}m(-u)\mathscr{F}^{-1}$, where $m(u)$ is multiplication by $\psi'(ux^2)$; the constant c and the reflection $x\mapsto-x$ of (5.5) cancel, because the reflection commutes with $m(u)$. Completing the square and using (3.2) for ψ' (with $\gamma'(1/t)=\gamma'(t)$; the measure self-dual for $\psi'(2xy)$ is $|2|_v^{1/2}$ times the one in (3.2)),

$$\big(\mathscr{F}m(1/t)\mathscr{F}^{-1}\phi\big)(x)=\gamma'(t)\,|t|_v^{1/2}\int_{k_v}\phi(z)\,\psi'\big(-t(x-z)^2\big)\,dz.$$

Let $w(t)=n(t)\bar{n}(-t^{-1})n(t)=\begin{pmatrix}0&t\\-t^{-1}&0\end{pmatrix}$. Multiplying by $m(t)$ on both sides, the terms $\psi'(tx^2)$ and $\psi'(tz^2)$ cancel against the square, and

$$\omega(n(t),1)\,\omega(\bar{n}(-t^{-1}),1)\,\omega(n(t),1)\phi(x)=\gamma'(t)\,|t|_v^{1/2}\,(\mathscr{F}\phi)(tx).$$

Now compute in the cover with (2.1). The values of $x(\cdot)$ on $n(t)$, $\bar{n}(-t^{-1})$, $n(t)\bar{n}(-t^{-1})$ and $w(t)$ are 1, $-t^{-1}$, $-t^{-1}$, $-t^{-1}$, so both cocycles in the triple product are 1 and $(n(t),1)(\bar{n}(-t^{-1}),1)(n(t),1)=(w(t),1)$. Next $(w(1),1)^{-1}=(w(1)^{-1},1)$, and $\beta(w(t),w(1)^{-1})=\big(\frac{t^{-1}}{-t^{-1}},t^{-1}\big)_v=(-1,t)_v$, because $w(t)w(1)^{-1}=a(t)$ has $x=t^{-1}$. Hence $(w(t),1)(w(1),1)^{-1}=(a(t),(-1,t)_v)$ and

$$\omega\big(a(t),(-1,t)_v\big)\phi(x)=\frac{\gamma'(t)}{\gamma'(1)}\,|t|_v^{1/2}\phi(tx).$$

By Lemma 3.3(a) with $a=b=t$, $(\gamma'(t)/\gamma'(1))^2=(t,t)_v=(t,-1)_v$. So $(-1,t)_v\gamma'(t)/\gamma'(1)=(\gamma'(t)/\gamma'(1))^3=\gamma'(1)/\gamma'(t)$, which is the first formula. The second is Lemma 3.3(d) with $b=-1$: $\chi_{\psi'_{-1}}=\chi_{\psi'}\cdot(-1,\cdot)_v=\chi_{\psi'}^{-1}$. For the last statement, $(\omega((n(u),1)x)\phi)(0)=\psi'(0)(\omega(x)\phi)(0)$, and $e^{(-\frac12+1)H_v(\tilde a)}=|t|_v^{1/2}$. □

For example, let v be real and $\psi'(x)=e^{2\pi iax}$. The operator $d\omega\begin{pmatrix}0&1\\-1&0\end{pmatrix}=i\big(2\pi ax^2-\frac{1}{8\pi a}\frac{d^2}{dx^2}\big)$ generates the action of the subgroup $\tilde k_\theta$ of Lemma 6.6(b); for $a>0$ its eigenvalues on even functions are $i(\frac12+2j)$, $j\ge0$, with ground state $e^{-2\pi ax^2}$, and for $a<0$ they are $-i(\frac12+2j)$. So $r^+_{\psi'}$ has lowest weight $\frac12$ when $a>0$ and highest weight $-\frac12$ when $a<0$. This agrees with the lemma: $\chi_{\psi'_{-1}}(a(-1),1)=i$ for $a>0$, which is the condition $m_0=\frac12$ of Lemma 6.6(b).

Globally, for a nontrivial character ψ' of $\mathbb{A}_k/k$ let $\omega_{\psi'}=\bigotimes_v\omega_{\psi'_v}$ on $\mathscr{S}(\mathbb{A}_k)$, a genuine representation of $\overline{G(\mathbb{A})}$, and for $\phi\in\mathscr{S}(\mathbb{A}_k)$ let

$$\theta_\phi^{\psi'}(x)=\sum_{\xi\in k}\big(\omega_{\psi'}(x)\phi\big)(\xi). \tag{5.6}$$

By Weil [We64], $\theta_\phi^{\psi'}$ is left invariant under $G(\mathbb{Q})$. By (5.5) its constant term is $x\mapsto(\omega_{\psi'}(x)\phi)(0)$, because $\int_{k\backslash\mathbb{A}_k}\psi'(u\xi^2)\,du=0$ for $\xi\neq0$.

Theorem 5.8. *(a) $L^2_{\chi,\mathrm{res}}=0$ for every $\chi\in\mathscr{X}_A$ that is unramified.*

(b) For $b\in k^\times$, $V_b:=L^2_{\chi_b,\mathrm{res}}$ is the closed span of the residues $\theta_\varphi(x)=\mathrm{Res}_{\lambda=1/2}E(x,\varphi,\lambda)$, $\varphi\in\mathscr{H}_B(\chi_{\psi_b})$; it is not 0, and $\theta_{\varphi,B}(x)=(\mathrm{Res}_{\lambda=1/2}M(w,\lambda)\varphi)_{-1/2}(x)$, of order $e^{H(x)/2}$.

(c) $L^2_{\mathrm{res}}=\widehat{\bigoplus}_{b\in k^\times/k^{\times2}}V_b$, and each V_b is irreducible. With the normalization (5.5),

$$V_b\cong r^+_{\psi_{-b}}=\bigotimes{}'_v\,r^+_{\psi_{-b,v}},$$

the even Weil representation attached to $\psi_{-b}(x)=\psi(-bx)$, and the elements of V_b are the theta functions $\theta_\phi^{\psi_{-b}}$ of (5.6), ϕ an even Schwartz–Bruhat function on $\mathbb{A}_k$.

Proof. (a) and the first part of (b): by Proposition 5.4(b) and Theorem 5.5(d), $L^2_{\chi,\mathrm{res}}$ is spanned by residues of E at poles of $M(w,\lambda)$ with $\mathrm{Re}\,\lambda>0$, and by Lemma 5.6(c) there are none unless $\chi^2=1$, i.e. $\mu=\chi_{\psi_b}$ (Lemma 5.3(d)), and then only $\lambda=\frac12$. The constant term of the residue is the residue of Theorem 5.5(b), in which φ_λ is holomorphic; it is nonzero by Lemma 5.6(c), and it is $O(e^{(-\frac12+1)H(x)})$. Since $\int_T^\infty e^Xe^{-2X}\,dX<\infty$ for the measure $\delta_B^{-1}=e^{-2X}$, θ_φ is square integrable (its difference with the constant term is rapidly decreasing on a Siegel set).

(c) By Gelbart [Ge76], Gelbart and Piatetski-Shapiro [GPS80] and Waldspurger [Wa80], L^2_{res} is the orthogonal sum, over the classes $b'\in k^\times/k^{\times2}$, of the spaces $\Theta_{b'}$ of theta functions $\theta_\phi^{\psi_{b'}}$, ϕ even, and $\Theta_{b'}$ is irreducible and isomorphic to $r^+_{\psi_{b'}}$; the identification of the residues with theta functions is the Siegel–Weil formula in this case. It remains to match b' with b. The constant term of an element of V_b transforms on the left under $\overline{A(\mathbb{A})}$ by $\chi_{\psi_b}e^{H/2}$, by (b) and $\chi^w_{\psi_b}=\chi_{\psi_b}$. The constant term of $\theta_\phi^{\psi_{b'}}$ is $x\mapsto(\omega_{\psi_{b'}}(x)\phi)(0)$, which by Lemma 5.7, applied at each place (on A and N the

cocycles σ_v and β_v agree), transforms by $\chi_{\psi_{-b'}} e^{H/2}$. So $\Theta_{b'} \subset L^2_{\chi_{-b'}}$, and therefore $\Theta_{b'} = V_{-b'}$, by the uniqueness in Lemma 5.3(d). □

Example 5.B. Let $k = \mathbb{Q}$, $b = 1$, and ψ as in Example 3.B, so $\psi_{-1,\infty}(x) = e^{2\pi i x}$. The residual representation $V_1 \cong r^+_{\psi_{-1}}$ contains the theta function of $\phi = e^{-2\pi\xi^2} \otimes \mathbf{1}_{\hat{\mathbb{Z}}}$: at $x = n(u)a(y^{1/2})$ in the real component it is $y^{1/4}\theta(u+iy)$ with the classical $\theta(z) = \sum_{n\in\mathbb{Z}} e^{2\pi i n^2 z}$, of weight $\frac{1}{2}$ on $\Gamma_0(4)$, in agreement with the lowest weight $\frac{1}{2}$ found above. The space V_{-1} contains its complex conjugate. The multiplier of θ involves the Jacobi symbol of Example 2.C. In the classical variable $s = (\lambda+1)/2$ the pole is at $s = \frac{3}{4}$, and the constant term of θ is a multiple of $y^{1/4}$. For $SL(2)$ the only residual representation is the trivial one, the residue of E at $\lambda = 1$ ($s = 1$), whose constant term is a constant. With $S = \{\infty, 2\}$ and $\mu = \chi_\psi$, $m^S(\mu,\lambda) = (1-2^{-2\lambda})\zeta(2\lambda)/\big((1-2^{-1-2\lambda})\zeta(2\lambda+1)\big)$, with residue $2/\pi^2$ at $\lambda = \frac{1}{2}$.

The residual term is the analog for the cover of the term [LL79, (5.5)] for $SL(2)$. The odd Weil representations, the products $\bigotimes'_v r^{\varepsilon_v}_{\psi_{b,v}}$ with some $\varepsilon_v = -$, are not residual: those that are automorphic occur in L^2_{cusp} [GPS80, Wa80], and they are counted among the (G,π).

5.3 Kernels, truncation and the terms J_χ

This subsection truncates the spectral kernels as on the geometric side and computes the integral J^T_χ of each truncated kernel.

For $\chi \in \mathscr{X}$ let $K_{G,\chi}(x,y)$ be the kernel of $R(f)$ on L^2_χ, and $K_{B,\chi}(x,y)$ the kernel of $R_B(f)$ (Lemma 4.2) on the χ-part of $L^2(N(\mathbb{A})A(\mathbb{Q})\backslash\overline{G(\mathbb{A})})_{\text{gen}}$, which is 0 for $\chi \in \mathscr{X}_G$ and the closed span of the $\mathscr{H}_B(\mu)e^{(\lambda+1)H}$, $\mu \in \chi$, $\lambda \in i\mathbb{R}$, for $\chi \in \mathscr{X}_A$. For $\mu \in \Pi_{\text{aut}}(X_A)$ let $\mathscr{B}(\mu)$ be an orthonormal basis of $\mathscr{H}_B(\mu)$, for (5.1), consisting of $\overline{K_0}$-finite vectors.

Proposition 5.9. *(a) If χ is the class of (G,π), then $K_{B,\chi} = 0$ and*

$$K_{G,\chi}(x,y) = \sum_\phi (R(f)\phi)(x)\overline{\phi(y)},$$

ϕ running over an orthonormal basis of L^2_χ of $\overline{K_0}$-finite vectors.

(b) If $\chi \in \mathscr{X}_A$, then

$$K_{B,\chi}(x,y) = \sum_{\mu\in\chi} \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{\varphi\in\mathscr{B}(\mu)} \big(I(\mu_{it},f)\varphi\big)_{it}(x)\,\overline{\varphi_{it}(y)}\,dt, \tag{5.7}$$

$$K_{G,\chi}(x,y) = K_{\chi,\text{res}}(x,y) + \frac{1}{2}\sum_{\mu\in\chi} \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{\varphi\in\mathscr{B}(\mu)} E\big(x, I(\mu_{it},f)\varphi, it\big)\,\overline{E(y,\varphi,it)}\,dt, \tag{5.8}$$

where $K_{\chi,\text{res}}(x,y) = \sum_\theta (R(f)\theta)(x)\overline{\theta(y)}$ over an orthonormal basis of $L^2_{\chi,\text{res}}$; it is 0 unless $\chi = \chi_b$. If $\mu^w \neq \mu$, the two terms of the sum over $\mu \in \chi$ in (5.8) are equal.

(c) $\sum_{\chi\in\mathscr{X}} K_{P,\chi}(x,y) = K_P(x,y) = \sum_{\mathscr{O}\in\mathscr{C}} K_{P,\mathscr{O}}(x,y)$ for $P \in \{B,G\}$, the sum over χ converging absolutely and uniformly on compact sets.

Proof. (a) is the spectral theorem on the discrete space L^2_χ, and L^2_χ has no constant terms. For (5.7), write $x = n\tilde{a}^1 e^X k$ with $\tilde{a}^1 \in \overline{A(\mathbb{A})^1}$; for a function F on $N(\mathbb{A})A(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$, $e^{-X}F$ is a function on $X_A \times \mathbb{R} \times \overline{K_0}$ with $\int |F|^2 dx = \int |e^{-X}F|^2 \, d\tilde{a}^1 \, dX \, dk$, because $dx = \delta_B^{-1} dn\, da\, dk$. Fourier analysis on X_A (Theorem 3.2(a); only genuine characters occur) and on $\mathbb{R}$ (with dX and $dt/2\pi$) writes F as $\sum_\mu \frac{1}{2\pi} \int \varphi_{\mu,t,it}\, dt$ with $\varphi_{\mu,t} \in \mathscr{H}_B(\mu)$ and $\|F\|^2 = \sum_\mu \frac{1}{2\pi} \int \|\varphi_{\mu,t}\|^2 dt$, the norm being (5.1). So $R_B(f)$ is the direct integral of the $I(\mu_{it}, f)$, which gives (5.7). Formula (5.8) is Langlands' spectral decomposition of $L^2_{\chi,\mathrm{cont}}$ by wave packets of Eisenstein series [La76, Ch. 7], [MW95, Ch. V], written as in [Ar80, §4]; the factor $\frac{1}{2} = n(A_B)^{-1}$ is the reciprocal of the number of chambers $\pm\mathfrak{a}^+$ in $\mathfrak{a}$ [Ar78, p. 927]. (Proposition 5.17 checks this factor against the geometric side.) The equality of the two terms is Theorem 5.5(c),(e) and the unitarity of $M(w,it)$. (c) Both sides are kernels of the same operator; the absolute convergence is [Ar80, §3], [MW95], [Li13]. □

Truncation. For a continuous function ϕ on $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ and $T \geq 0$ put

$$\Lambda^T \phi(x) = \phi(x) - \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} \hat{\tau}_B\big(H(\delta x) - T\big)\, \phi_B(\delta x). \tag{5.9}$$

By Lemma 4.3(b) the sum has at most one nonzero term.

Lemma 5.10. *Let $T \geq 0$.*

(a) $(\Lambda^T \phi)_B(x) = 0$ *if* $H(x) > T$.

(b) $\Lambda^T \Lambda^T = \Lambda^T$, *and* $\langle \Lambda^T \phi, \phi' \rangle = \langle \phi, \Lambda^T \phi' \rangle$ *whenever both sides converge absolutely.*

(c) If ϕ is an automorphic form, for example $E(\cdot, \varphi, \lambda)$ at a regular point λ, then $\Lambda^T \phi$ is rapidly decreasing on $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$. It is not compactly supported in general.

Proof. (a) Integrating (5.9) over $n \in N(\mathbb{Q})\backslash N(\mathbb{A})$, the term $\delta = 1$ gives $\hat{\tau}_B(H(x) - T)\phi_B(x) = \phi_B(x)$. For $\delta \neq 1$ and every n, $H(\delta nx) \leq -H(nx) = -H(x) < -T \leq T$ by Lemma 4.3(b), so these terms vanish. (b) Idempotence follows from (a), since $\Lambda^T \phi$ then has no constant term where $\hat{\tau}_B(H(x) - T) \neq 0$. For self-adjointness, unfold: $\langle \phi - \Lambda^T \phi, \phi' \rangle = \int_{B(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \hat{\tau}_B(H(x) - T)\phi_B(x)\overline{\phi'(x)}\, dx = \int_{N(\mathbb{A})B(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} \hat{\tau}_B(H(x) - T)\phi_B(x)\overline{\phi'_B(x)}\, dx$, because $\hat{\tau}_B(H(x) - T)\phi_B(x)$ is left $N(\mathbb{A})$-invariant; this is symmetric. (c) is in [Ar80, §1]; the cover splits over $N(\mathbb{A})$ (Lemma 2.6(b)), so the proof applies verbatim. □

Define, for $\chi \in \mathscr{X}$ and $T \geq 0$,

$$k^T_\chi(x, f) = K_{G,\chi}(x,x) - \sum_{\delta \in B(\mathbb{Q})\backslash G(\mathbb{Q})} K_{B,\chi}(\delta x, \delta x)\, \hat{\tau}_B\big(H(\delta x) - T\big). \tag{5.10}$$

Lemma 5.11. *For $T \geq 0$, $k^T_\chi(x,f) = F^G(x,T)K_{G,\chi}(x,x) + \sum_\delta \tau_B(H(\delta x) - T)\big(K_{G,\chi} - K_{B,\chi}\big)(\delta x, \delta x)$, and $\sum_\chi k^T_\chi(x,f) = k^T(x,f)$.*

Proof. As for Lemma 4.4, whose first identity is used; the sum over χ is Proposition 5.9(c) and (4.4). □

Theorem 5.12 (Arthur [Ar80, §§2–4]; Li [Li13] for covers)**.** *Let $f \in \mathscr{H}(\overline{G(\mathbb{A})})$.*

(a) *For T sufficiently large, $\sum_{\chi\in\mathscr{X}}\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}}|k_\chi^T(x,f)|\,dx<\infty$.*

(b) *$J_\chi^T(f):=\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}}k_\chi^T(x,f)\,dx$ is, for such T, a polynomial in T, and $\sum_\chi J_\chi^T(f)=J^T(f)$, the geometric side of Theorem 4.5.*

(c) *$J_\chi^T(f)-\int_{G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}}\Lambda_2^T K_{G,\chi}(x,x)\,dx\to 0$ as $T\to\infty$, where Λ_2^T is Λ^T applied to the second variable.*

We put $J_\chi(f)=J_\chi^0(f)$, the value of the polynomial at $T=0$.

By Theorem 5.14 below each J_χ^T has degree at most 1, so $J^T(f)=\sum_\chi J_\chi^T(f)$, with Theorem 4.5, is an identity of polynomials of degree at most 1 and holds for every T, in particular at $T=0$.

Lemma 5.13 (Maass–Selberg relations)**.** *Let $\mu,\mu'\in\Pi_{\mathrm{aut}}(X_A)$, $\varphi\in\mathscr{H}_B(\mu)$, $\varphi'\in\mathscr{H}_B(\mu')$, $T\ge 0$, and let λ,λ' be regular points with $\lambda+\bar\lambda'\neq 0$ and $\lambda-\bar\lambda'\neq 0$. Then, with $M=M(w,\lambda)$ and $\check M=M(w,\lambda')$,*

$$\big\langle\Lambda^T E(\varphi,\lambda),\Lambda^T E(\varphi',\lambda')\big\rangle=\frac{\langle\varphi,\varphi'\rangle e^{(\lambda+\bar\lambda')T}-\langle M\varphi,\check M\varphi'\rangle e^{-(\lambda+\bar\lambda')T}}{\lambda+\bar\lambda'}+\frac{\langle M\varphi,\varphi'\rangle e^{(-\lambda+\bar\lambda')T}}{-\lambda+\bar\lambda'}+\frac{\langle\varphi,\check M\varphi'\rangle e^{(\lambda-\bar\lambda')T}}{\lambda-\bar\lambda'}.\tag{5.11}$$

The inner products are those of (5.1) and vanish unless the characters agree; in particular the last two terms vanish unless $\mu'=\mu^w$.

Sketch. For $\operatorname{Re}\lambda'$ large, Lemma 5.10(b) gives $\langle\Lambda^T E,\Lambda^T E'\rangle=\langle\Lambda^T E,E'\rangle$. Unfold E' and integrate over $N(\mathbb{Q})\backslash N(\mathbb{A})$ first. By the proof of Lemma 5.10, $(\Lambda^T E)_B=E_B$ on $\{H\le T\}$ minus the terms $\delta\neq 1$; folding these back into a sum over $G(\mathbb{Q})$ and unfolding the other way, they contribute $-\int_{H>T}E_B\,\overline{(\check M\varphi')_{-\lambda'}}$, because the constant term of $E'-\varphi'_{\lambda'}$ is $(\check M\varphi')_{-\lambda'}$. So

$$\langle\Lambda^T E,E'\rangle=\int_{H(x)\le T}E_B\,\overline{\varphi'_{\lambda'}}-\int_{H(x)>T}E_B\,\overline{(\check M\varphi')_{-\lambda'}},$$

integrals over $N(\mathbb{A})A(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$. Insert Theorem 5.5(b); in the coordinates $x=\tilde a^1 e^X k$ each of the four products is $e^{\nu X}$ times a product of values on $\overline{K_0}$, with $\nu\in\{\pm\lambda\pm\bar\lambda'\}$, and $\int_{-\infty}^T e^{\nu X}dX=e^{\nu T}/\nu$, $\int_T^\infty e^{\nu X}dX=-e^{\nu T}/\nu$ in the appropriate half planes. The integral over $X_A\times\overline{K_0}$ gives the inner products (5.1). The general case follows by analytic continuation. In general the relations are due to Langlands; see [Ar80, §4], [MW95], [Li13]. □

Theorem 5.14. *Let $f\in\mathscr{H}(\overline{G(\mathbb{A})})$, and for $\mu\in\Pi_{\mathrm{aut}}(X_A)$ and $\lambda\in i\mathbb{R}$ put*

$$M_B^T(\mu,\lambda)=2T-M(w,\lambda)^{-1}\frac{d}{d\lambda}M(w,\lambda),\tag{5.12}$$

an operator on $\mathscr{H}_B(\mu)$, self-adjoint for $\lambda\in i\mathbb{R}$.

(a) *If χ is the class of (G,π), then $J_\chi^T(f)=m(\pi)\operatorname{tr}\pi(f)$.*

(b) If $\chi \in \mathscr{X}_A$ is unramified and $\mu \in \chi$, then

$$J^T_\chi(f) = \frac{1}{2\pi}\int_{\mathbb{R}} \operatorname{tr}\big(M^T_B(\mu,it)\,I(\mu_{it},f)\big)\,dt.$$

(c) If $\chi = \chi_b$ is ramified and $\mu = \chi_{\psi_b}$, then

$$J^T_\chi(f) = \operatorname{tr}\big(R(f)|_{V_b}\big) + \frac{1}{2}\cdot\frac{1}{2\pi}\int_{\mathbb{R}} \operatorname{tr}\big(M^T_B(\mu,it)\,I(\mu_{it},f)\big)\,dt + \frac{1}{4}\operatorname{tr}\big(M(w,0)\,I(\mu_0,f)\big).$$

Proof. By Theorem 5.12(c) it suffices to compute $\int \Lambda^T_2 K_{G,\chi}(x,x)\,dx$ up to $o(1)$ as $T\to\infty$. By Lemma 5.10(b) and Proposition 5.9 this is $\sum_\phi \langle R(f)\phi, \Lambda^T\phi\rangle$ in case (a) and

$$\sum_\theta \langle R(f)\theta, \Lambda^T\theta\rangle + \frac{1}{2}\sum_{\mu\in\chi}\frac{1}{2\pi}\int_{\mathbb{R}}\sum_{\varphi\in\mathscr{B}(\mu)} \big\langle \Lambda^T E(I(\mu_{it},f)\varphi,it), \Lambda^T E(\varphi,it)\big\rangle\,dt$$

in cases (b)–(c), the inner product on $i\mathbb{R}$ being the value at $\lambda' = \lambda$ of (5.11), which is regular there; the interchange of the integrals is justified as in [Ar80, §4]. (a) Cusp forms satisfy $\Lambda^T\phi = \phi$. For the residual part, $\Lambda^T\theta \to \theta$ in L^2, so the first sum tends to $\operatorname{tr}(R(f)|_{V_b})$.

Now let $\lambda = it$, $\lambda' = \lambda + \varepsilon$ with $\varepsilon \in i\mathbb{R}$, so $\lambda + \bar\lambda' = -\varepsilon$. Expanding $M(w,\lambda') = M + \varepsilon\frac{dM}{d\lambda} + O(\varepsilon^2)$ and using $M^* = M^{-1}$ on $i\mathbb{R}$, the first fraction of (5.11) tends to $\langle \varphi, M^T_B(\mu,\lambda)\varphi'\rangle$ as $\varepsilon\to 0$. With $\varphi = I(\mu_\lambda,f)\varphi_j$, $\varphi' = \varphi_j$ and the sum over j this gives $\operatorname{tr}(M^T_B(\mu,\lambda)I(\mu_\lambda,f))$, using that $M^{-1}\frac{dM}{d\lambda}$ is self-adjoint on $i\mathbb{R}$ (differentiate $M(it)^*M(it)=1$). For $\mu^w \neq \mu$ the two $\mu\in\chi$ give equal contributions, and the factor $\frac12$ cancels; this proves (b), the expression being already linear in T.

For $\mu^w = \mu$ the last two terms of (5.11) also contribute, with $-\lambda + \bar\lambda' \to -2it$ and $\lambda - \bar\lambda' \to 2it$. Put $F(t) = \operatorname{tr}(M(w,it)I(\mu_{it},f))$; by Theorem 5.5(c),(e), $\operatorname{tr}(M(w,-it)I(\mu_{it},f)) = F(-t)$. The contribution is

$$\frac{1}{2}\cdot\frac{1}{2\pi}\operatorname{PV}\int_{\mathbb{R}}\frac{F(-t)e^{2itT} - F(t)e^{-2itT}}{2it}\,dt = \frac{1}{4\pi}\Big(\int_{\mathbb{R}} F(t)\frac{\sin 2tT}{t}\,dt + i\operatorname{PV}\int_{\mathbb{R}} F(t)\frac{\cos 2tT}{t}\,dt\Big),$$

which tends to $\frac{1}{4\pi}\cdot\pi F(0) = \frac14 F(0)$ as $T\to\infty$, by Dirichlet's integral and the Riemann–Lebesgue lemma applied to $(F(t)-F(-t))/t$. Since J^T_χ is a polynomial, it equals the polynomial part of these limits, which is (c). □

Two remarks on the constants. First, M^T_B is the case of $SL(2)$ of Arthur's formula

$$M^T_B = \operatorname{vol}(\mathfrak{a}/\mathbb{Z}\alpha^\vee)\lim_{\zeta\to 0}\sum_{s\in\{1,w\}}\frac{e^{(s\zeta)(T)}M(s,\lambda)^{-1}M(s,\lambda+\zeta)}{\theta(s\zeta)},$$

in which $\theta(\zeta) = \zeta(\alpha^\vee) = d\zeta$ and $\operatorname{vol}(\mathfrak{a}/\mathbb{Z}\alpha^\vee) = d$ for our dX (Remark 2026.9); the factor d cancels and the limit is (5.12) for every k. Proposition 5.17 confirms the coefficient of T against the geometric side. Second, in Arthur's fine expansion [Ar82, Thm. 8.2], [Li13] the last term of (c) is $|W^A_0||W^G_0|^{-1}|\det(w-1)_{\mathfrak{a}}|^{-1}\operatorname{tr}(M(w,0)I(\mu_0,f))$, with $|W^A_0||W^G_0|^{-1} = \frac12$ and $|\det(w-1)_{\mathfrak{a}}| = 2$. In the trace formula for $GL(2)$ of [GJ79, §6] the same term is written $-\frac14\operatorname{tr}(M(0)\pi_0(f))$, next to $+\frac{1}{4\pi}\int\operatorname{tr}(M(-iy)M'(iy)\pi_{iy}(f))\,dy$: that formula has the trace of $R(f)$ on the discrete spectrum alone on one side, and moving both terms to the spectral side gives the signs of Theorem 5.16(e). The sign of the cross terms in (5.11), from which it comes, is also checked numerically on the Eisenstein series of $SL(2,\mathbb{Z})$ (Remark 2026.17), and $M(w,0)$ is computed in Proposition 6.13.

Corollary 5.15. *Let S and $\mu = \chi_\psi \cdot (\chi \circ p)$ be as in Lemma 5.6, $f = f_S \otimes e^S$, and $\lambda \in i\mathbb{R}$. Then $I(\mu_\lambda, f) = 0$ unless μ_v is trivial on $A(\mathscr{O}_v)^*$ for all $v \notin S$, and in that case*

$$\operatorname{tr}\Big(M(w,\lambda)^{-1}\tfrac{dM}{d\lambda}(w,\lambda)I(\mu_\lambda,f)\Big) = \frac{(m^S)'}{m^S}(\mu,\lambda)\operatorname{tr} I_S(\mu_{S,\lambda},f_S) + \operatorname{tr}\Big(M_S(\lambda)^{-1}\tfrac{dM_S}{d\lambda}(\lambda)I_S(\mu_{S,\lambda},f_S)\Big),$$

where $I_S = \bigotimes_{v\in S} I_v$ and $\frac{(m^S)'}{m^S}(\mu,\lambda) = 2\frac{(L^S)'}{L^S}(2\lambda,\chi^2) - 2\frac{(L^S)'}{L^S}(2\lambda+1,\chi^2)$.

Proof. $I(\mu_\lambda, e_v)$ is the projection onto the K_v^*-fixed vectors, which are 0 unless μ_v is trivial on $A(\mathscr{O}_v)^*$. Then apply Lemma 5.6(a)–(b) and the Leibniz rule. □

Example 5.C. For $k = \mathbb{Q}$, $S = \{\infty, 2\}$ and $\mu = \chi_\psi$, $\frac{(m^S)'}{m^S}(\lambda) = 2\frac{\zeta'}{\zeta}(2\lambda) - 2\frac{\zeta'}{\zeta}(2\lambda+1) + \frac{2\log 2\cdot 2^{-2\lambda}}{1-2^{-2\lambda}} - \frac{2\log 2\cdot 2^{-1-2\lambda}}{1-2^{-1-2\lambda}}$. For $SL(2)$ the corresponding factor is $\frac{\zeta'}{\zeta}(\lambda) - \frac{\zeta'}{\zeta}(\lambda+1)$ plus the 2-adic terms.

5.4 The trace formula

This subsection equates the geometric side of Section 4 with the spectral side computed above.

Theorem 5.16. *Let S be a finite set of places of $\mathbb{Q}$ as in Section 4.4, large enough for Theorem 4.11 and containing the places below which ψ is ramified; let $f_S \in \mathscr{H}(\overline{G(\mathbb{Q}_S)})$ and $f = f_S \otimes e^S$. Then $\sum_{\mathscr{O}} J_{\mathscr{O}}(f) = \sum_\chi J_\chi(f)$, where both sides converge absolutely and:*

(a) for the central classes $\mathscr{O}_z$, $z \in \{\pm 1\}$,

$$J_{\mathscr{O}_z}(f) = a^G(S,1)f_S(z) + \sum_{\xi \in k_S^\times / k_S^{\times 2}} a^G(S,u_\xi) J_G(zu_\xi, f_S) + \tfrac{1}{2} c_k J_A(z, f_S),$$

with the coefficients $a^G(S,u_\xi)$ of Theorem 4.11, which are computed in Proposition 7.10;

(b) for $\mathscr{O}$ elliptic, $J_{\mathscr{O}}(f) = \operatorname{vol}(T_\mu(\mathbb{Q})\backslash T_\mu(\mathbb{A})) J_G(\mu_S, f_S)$;

(c) for $\mathscr{O}$ hyperbolic, the class of $a(t)$, $J_{\mathscr{O}}(f) = c_k J_A(a(t)_S, f_S)$;

(d) for χ the class of (G,π), $J_\chi(f) = m(\pi)\operatorname{tr}\pi(f)$;

(e) summed over $\chi \in \mathscr{X}_A$,

$$\sum_{\chi\in\mathscr{X}_A} J_\chi(f) = \sum_b \operatorname{tr}\big(R(f)|_{V_b}\big) - \frac{1}{2}\sum_\mu \frac{1}{2\pi}\int_{\mathbb{R}} \operatorname{tr}\Big(M(w,it)^{-1}\tfrac{dM}{d\lambda}(w,it)\, I(\mu_{it},f)\Big)\, dt + \frac{1}{4}\sum_b \operatorname{tr}\big(M(w,0)\, I(\chi_{\psi_b,0},f)\big),$$

where b runs over $k^\times/k^{\times 2}$ and μ over $\Pi_{\mathrm{aut}}(X_A)$; only finitely many b, and only μ unramified outside S, contribute (Example 5.A, Corollary 5.15).

Proof. At $T=0$, Theorem 4.5, Theorem 4.11 and Theorem 5.12(b) give $J(f)=\sum_{\mathscr{O}} J_{\mathscr{O}}(f)=\sum_{\chi} J_{\chi}(f)$. (a)–(c) are Theorem 4.11. (d)–(e) are Theorem 5.14 at $T=0$: in (b) of that theorem, write the integral as half the sum over the two members of χ; in (c) the class has one member. The absolute convergence of the spectral side is [Ar80], [Li13]. □

Proposition 5.17. *The coefficients of T in $\sum_{\chi} J_{\chi}^{T}(f)$ and in $\sum_{\mathscr{O}} J_{\mathscr{O}}^{T}(f)$ agree. Both equal $c_k \sum_{\gamma\in A(\mathbb{Q})} F_f(\gamma)$, where $F_f(x)=\int_{\overline{K_0}}\int_{N(\mathbb{A})} f(\tilde{k}^{-1}x n\tilde{k})\,dn\,d\tilde{k}$.*

Proof. Spectral side. By Theorem 5.14, only the term $2T$ of M_B^T contributes, and summing over χ the coefficient is $\sum_{\mu}\frac{1}{2\pi}\int_{\mathbb{R}} \operatorname{tr} I(\mu_{it},f)\,dt$ (each unramified class contributes twice $\frac{1}{2}$ through its two members). Computing $I(\mu_\lambda,f)$ on $\overline{K_0}$ in the coordinates $y=n\tilde{a}k'$ of (2.9) gives the kernel $\int_{\overline{A(\mathbb{A})}}\int_{N(\mathbb{A})} f(\tilde{k}^{-1}\tilde{a}n\tilde{k}')\mu(\tilde{a})e^{(\lambda+1)H(\tilde{a})}\,dn\,d\tilde{a}$, so $\operatorname{tr} I(\mu_\lambda,f)=\int_{\overline{A(\mathbb{A})}} e^{H(\tilde{a})}F_f(\tilde{a})\mu(\tilde{a})e^{\lambda H(\tilde{a})}\,d\tilde{a}$. Fourier inversion on $\mathbb{R}$ (dX, $dt/2\pi$) and then Theorem 3.4 on X_A, applied to the anti-genuine function F_f on $\overline{A(\mathbb{A})^1}$, give $\sum_{\mu}\frac{1}{2\pi}\int \operatorname{tr} I(\mu_{it},f)\,dt = c_k\sum_{\gamma\in A(\mathbb{Q})} F_f(\gamma)$.

Geometric side. By Propositions 4.6–4.8, the coefficient is $c_k\big(\hat{\Phi}_f(0)+\hat{\Phi}_{f_{-1}}(0)\big)+2c_k\sum_{\mathscr{O}\text{ hyp}}\int_{\overline{A(\mathbb{A})}\backslash\overline{G(\mathbb{A})}} f(y^{-1}\mu y)\,dy$. Here $\hat{\Phi}_{f_z}(0)=F_f(z)$ for $z=\pm1$. For $\mu=a(t)$, $t\in k^\times\smallsetminus\{\pm1\}$, the coordinates $y=n(z)\tilde{k}$ and $y^{-1}\mu y=\tilde{k}^{-1}\mu\, n((1-t^{-2})z)\tilde{k}$ give $\int f(y^{-1}\mu y)\,dy=|1-t^{-2}|_{\mathbb{A}_k}^{-1}F_f(\mu)=F_f(\mu)$ by the product formula. The integral is the same for $a(t^{-1})=w\,a(t)\,w^{-1}$ (Lemma 5.3(a)), so $2F_f(a(t))=F_f(a(t))+F_f(a(t^{-1}))$, and the hyperbolic class of $a(t)$ accounts for the two elements $a(t^{\pm1})$ of $A(\mathbb{Q})$. □

Proposition 5.17 is a consistency check between Sections 4 and 5: with $4T$ in place of $2T$ in M_B^T, or with $n(A_B)=1$ in place of 2, the spectral coefficient would be twice the geometric one.

Remark 2026.12. *Normalizations.* Everything here uses $\lambda(X)=\lambda X$ on $\mathfrak{a}\cong\mathbb{R}$, Lebesgue dX and $dt/2\pi$ on $i\mathbb{R}$, and the inner product (5.1), which includes c_k. Arthur normalizes the measure on $\mathfrak{a}$ so that $\mathfrak{a}/\mathbb{Z}\alpha^\vee$ has volume 1, i.e. dX/d, and takes the dual measure on $i\mathfrak{a}^*$; the operator M_B^T then carries the factor $\operatorname{vol}(\mathfrak{a}/\mathbb{Z}\alpha^\vee)\theta(\zeta)^{-1}$, and the products in Theorem 5.14 do not change (Remark 2026.9). In the classical variable $s=(\lambda+1)/2$ of the Conventions above, for $k=\mathbb{Q}$, $\frac{d}{d\lambda}=\frac{1}{2}\frac{d}{ds}$, and $2T=\log y_T$, where $y_T=e^{2T}$ is the height at which Λ^T truncates.

5.5 Status of the results

This subsection lists the statements of Section 5 that rest on citation or on numerical checks.

Remark 2026.13. *What rests on citation.* Proved here: Lemma 5.2 (given Flath's theorem); Lemmas 5.3, 5.10(a)–(b) and 5.11; Lemma 5.6, with (b) also checked numerically; Theorem 5.8(a)–(b), the computations in the proofs of Lemma 5.13 and Theorem 5.14 from (5.11), and Proposition 5.17. Taken from the literature: Lemma 5.1, Proposition 5.4, Theorem 5.5 [La76, MW95]; the wave-packet formula (5.8) and the absolute convergence in Proposition 5.9; Lemma 5.10(c); Theorem 5.12 [Ar80, Li13], including the interchange of integrals used for Theorem 5.14; the analytic continuation in Lemma 5.13; and Theorem 5.8(c) [Ge76, GPS80, Wa80], except for the matching $V_b\cong r^+_{\psi_{-b}}$ of the parameters, which is proved here from Lemma 5.7. The matching depends on the normalization (5.5): if $n(u)$ acts by $\psi'(\frac{1}{2}ux^2)$, it reads $V_b\cong r^+_{\psi_{-2b}}$. We use from the cited papers that the residual spectrum is the sum of the spaces of even theta functions (5.6) and that these are irreducible. The range "T sufficiently large" in Theorem 5.12 is Arthur's; for this group we expect $T\geq 0$ to suffice, as on the geometric side, but have not checked it.

Remark 2026.14. The script `checks/sec5_n2_check.py` verifies, for $k=\mathbb{Q}$: Lemma 5.3(a)–(c) at $v=\infty,2,3,5,7$ (exactly, with σ_v and the Weil index); the unramified intertwining integral of Lemma 5.6(b) at $p=3,5,7$, by an explicit decomposition of the lift of $w^{-1}n(y)$, shell by shell, including the shape of the integrand on odd and even shells, and the same integral for $SL(2)$; Arthur's limit for M_B^T for $d=1,2,3$ and the 1995 normalization (symbolically); the diagonal limit of (5.11) and the self-adjointness of $M^{-1}M'$ for a model unitary family; the limit $\frac{1}{4}F(0)$ of the cross terms; the residue $2/\pi^2$ of Example 5.B; at $v=\mathbb{R}$, the two local identities behind Proposition 5.17; and, for Lemma 5.7, the product $(w(t),1)(w(1),1)^{-1}=(a(t),(-1,t)_v)$ of unipotent lifts and the identity $(\gamma_v(t)/\gamma_v(1))^2=(t,-1)_v$ at $v=\infty,2,3,5,7$, and the weights $\pm\frac{1}{2}$ of the ground state at $v=\mathbb{R}$.

6 The invariant form

The trace formula of Theorem 5.16 is an identity $\sum_{\mathscr{O}} J_{\mathscr{O}}(f)=\sum_\chi J_\chi(f)$ between distributions, but not all of the terms are invariant: $J(f^y)\neq J(f)$ in general, where $f^y(x)=f(yxy^{-1})$. The elliptic terms $J_{\mathscr{O}}$ and the cuspidal terms J_χ are invariant; the hyperbolic and central terms, and the terms J_χ with $\chi\in\mathscr{X}_A$, are not. This section computes the failure of invariance, defines the invariant distributions $I_A(\gamma,f)$ that replace the weighted orbital integrals $J_A(\gamma,f)$, following Arthur [Ar81, Ar88b, Ar88c], and puts the trace formula in invariant form (Theorem 6.11 and Corollary 6.14).

For covering groups the general theory is due to Li. The normalization of the local intertwining operators, which the construction needs, is in [Li12a]. The invariant trace formula is in [Li14b], under the hypothesis that the invariant Paley–Wiener theorem holds at the real places for the Levi subgroups; Li verifies the hypothesis for the two-fold covers of $Sp(2n)$, hence for the group here. For $G=\mathrm{Res}_{k/\mathbb{Q}}SL(2)$ much less is needed. The only proper Levi subgroup is A, and its cover is abelian (Lemma 2.6(e)), so the one Paley–Wiener theorem that enters is Fourier inversion on the abelian group $\overline{A(\mathbb{Q}_S)}$. Given the normalized intertwining operators, we can therefore prove the variance formulas, the invariance of $I_A(\gamma,f)$ and the invariant trace formula directly; we do so, and cite Arthur and Li for the general statements.

Why a correction term, and why this one. The weighted orbital integral $J_A(\gamma,f)$ integrates $f(x^{-1}\gamma x)$ against the weight $v_A(x)=-H(x)-H(wx)$. The height H is defined through the Iwasawa decomposition, so it depends on the choice of the maximal compact subgroup: $H(xy)\neq H(x)+H(y)$ in general, but $H(xy)=H(x)+H(\tilde{k}y)$ when $x=n\tilde{a}\tilde{k}$. Replacing f by f^y replaces $v_A(x)$ by $v_A(xy)$, and the difference $-H(\tilde{k}y)-H(\tilde{k}'y)$ does not vanish. This is the geometric source of the non-invariance (Lemma 6.3). Since the difference depends only on the K-components, the change in $J_A(\gamma,f)$ is not a new kind of object: it is the value at γ and at $w\gamma w^{-1}$ of one function $f_{B,y}$ on the cover of A.

On the spectral side the same function appears. The logarithmic derivative $\mathscr{M}^{-1}\mathscr{M}'$ of an intertwining operator between $I(\pi_\lambda)$ and $I(\pi^w_{-\lambda})$ is not an intertwining operator, because the two representations depend on λ through the same height: differentiating $e^{(\lambda+1)H(\tilde{k}y)}$ brings down $H(\tilde{k}y)$. Conjugating $\mathscr{M}^{-1}\mathscr{M}'$ by $I(\pi_\lambda,y)$ therefore changes it by multiplication by $H(\tilde{k}y)$, once directly and once through $\mathscr{M}$ (Lemma 6.4), and the trace against $I(\pi_\lambda,f)$ changes by the Fourier transform of $f_{B,y}$ at π_λ and at $\pi^w_{-\lambda}$.

So the function $\varphi_A(f,\pi) = -\operatorname{tr}(R^{-1}R' I(\pi,f))$ on the unitary dual of the cover of A, built from the normalized operators R, has a variation which is the Fourier transform of the variation of $\gamma \mapsto J_A(\gamma,f)$. Its inverse Fourier transform $\Phi_A(\gamma,f)$ then varies exactly as $J_A(\gamma,f)$ does, and

$$I_A(\gamma,f) = J_A(\gamma,f) - \Phi_A(\gamma,f)$$

is invariant (Theorem 6.9). The two functions live on opposite sides of Fourier inversion: $\varphi_A(f,\cdot)$ on characters π of the cover of $A(\mathbb{Q}_S)$, $\Phi_A(\cdot,f)$ on its elements. Globally the corrections on the two sides are matched by Poisson summation on the cover of A, which is the rank-0 trace formula of Section 3, so the identity is preserved. The normalized operators R_v are used, rather than M, because they are holomorphic and unitary on $i\mathbb{R}$ and trivial on the unramified vectors, so that φ_A is given by finitely many local terms. The scalar that remains, m in $M = m \cdot R$, contributes $\frac{m'}{m} \operatorname{tr} I(\pi,f)$, which is invariant; this is the term that survives on the spectral side.

Results. The main points of the section are these.

- The weight in the variance formulas is $u'_B(k,y) = -H(ky)$ (Lemma 6.1).
- The normalization of the intertwining operators is a theorem of Li [Li12a], and no trace Paley–Wiener theorem is needed for this group.
- The invariant spectral terms attached to $\chi \in \mathscr{X}_A$ do not vanish. The logarithmic derivative m'/m of the global normalizing factor survives, and m is identified up to finitely many local factors: $m'/m = 2\frac{L'}{L}(2\lambda,\chi^2) - 2\frac{L'}{L}(2\lambda+1,\chi^2)$ plus the terms of the places in $S_{\mathrm{bad}}(\mu)$ (Corollary 6.7).
- The residual terms $\operatorname{tr}(R(f)|_{V_b})$ and the ramified terms of Theorem 5.14(c) are already invariant. The ramified term is evaluated: $M(w,0) = -1$ on $\mathscr{H}_B(\chi_{\psi_b})$, so it equals $-\frac{1}{4}\operatorname{tr} I(\chi_{\psi_b,0},f)$ (Proposition 6.13).
- The function $\varphi_A(f)$ that carries the correction is not the Fourier transform of a compactly supported function on $\overline{A(\mathbb{Q}_S)}$ (Example 6.C).

Conventions. S is a finite set of places of $\mathbb{Q}$ containing ∞, 2 and the places below which ψ is ramified, S_k is the set of places of k above S, and $\overline{G(\mathbb{Q}_S)}$, $\overline{\mathscr{H}(G(\mathbb{Q}_S))}$, $f = f_S \otimes e^S$ and γ_S are as in Section 4.4. Let $K_S = \prod_{v\in S_k} K_v$, with preimage $\overline{K_S}$ in $\overline{G(\mathbb{Q}_S)}$, and let $\overline{A(\mathbb{Q}_S)}$ be the preimage of $A(\mathbb{Q}_S) = \prod_{v\in S_k} A(k_v)$, an abelian group (it is the group $\overline{A(\mathbb{A}_S)}$ of Section 3.3). Put $H_S(x) = \sum_{v\in S_k} H_v(x_v)$ for $x \in \overline{G(\mathbb{Q}_S)}$; on the image of $\overline{G(\mathbb{Q}_S)}$ in $\overline{G(\mathbb{A})}$ it is the height H of Lemma 2.8. Let $\Pi_u(\overline{A(\mathbb{Q}_S)})$ be the group of genuine unitary characters of $\overline{A(\mathbb{Q}_S)}$. It is a principal homogeneous space under the unitary dual of $A(\mathbb{Q}_S)$ (Lemma 3.1), and $d\pi$ is the measure on it that corresponds to the Plancherel measure dual to the Haar measure da on $A(\mathbb{Q}_S)$. For an anti-genuine integrable function h on $\overline{A(\mathbb{Q}_S)}$ put

$$\hat{h}(\pi) = \int_{\overline{A(\mathbb{Q}_S)}} h(\tilde{a})\pi(\tilde{a})\,d\tilde{a}, \qquad \text{so that} \qquad h(\tilde{a}) = \int_{\Pi_u(\overline{A(\mathbb{Q}_S)})} \hat{h}(\pi)\,\pi(\tilde{a})^{-1}\,d\pi$$

when $\hat{h}$ is integrable and h continuous. For $\pi \in \Pi_u(\overline{A(\mathbb{Q}_S)})$ and $\lambda \in \mathbb{C}$ put $\pi_\lambda = \pi e^{\lambda H_S}$, and $\pi^w(\tilde{a}) = \pi(w^{-1}\tilde{a}w)$, where w stands for $\prod_{v\in S_k}(w,1)_v$; by Lemma 5.3(a), $(\pi_\lambda)^w = (\pi^w)_{-\lambda}$. Let $V(\pi)$ be the space of smooth functions φ on $\overline{K_S}$ with $\varphi(\tilde{m}\tilde{k}) = \pi(\tilde{m})\varphi(\tilde{k})$ for $\tilde{m} \in \overline{A(\mathbb{Q}_S)} \cap \overline{K_S}$, and let $I(\pi_\lambda)$ be the induced representation of $\overline{G(\mathbb{Q}_S)}$ on $V(\pi)$: $(I(\pi_\lambda, y)\varphi)(\tilde{k}) = \varphi_\lambda(\tilde{k}y)$, where $\varphi_\lambda(n\tilde{a}\tilde{k}) = \pi(\tilde{a})e^{(\lambda+1)H_S(\tilde{a})}\varphi(\tilde{k})$. These are the local analogs of $\mathscr{H}_B(\mu)$ and $I(\mu_\lambda)$ in Section 5.2: for $\mu \in \Pi_{\mathrm{aut}}(X_A)$ with μ_v trivial on $A(\mathscr{O}_v)^*$ for all $v \notin S_k$, the restriction μ_S of μ to $\overline{A(\mathbb{Q}_S)}$ lies in $\Pi_u(\overline{A(\mathbb{Q}_S)})$, and $\mathrm{tr}\, I(\mu_\lambda, f) = \mathrm{tr}\, I(\mu_{S,\lambda}, f_S)$.

Invariance refers to conjugation by all $y \in \overline{G(\mathbb{Q}_S)}$. At an archimedean place f^y need not be K-finite when f is, so in Sections 6.1 and 6.3 test functions are taken in $C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$, the space of smooth, compactly supported, anti-genuine functions, which is stable under $f \mapsto f^y$ and to which the distributions $J_G(\gamma,\cdot)$, $J_A(\gamma,\cdot)$ and $\mathrm{tr}\, I(\pi,\cdot)$ extend by the same formulas.

6.1 Non-invariance

This subsection computes $J(f^y) - J(f)$ for each of the distributions in the trace formula. In every case the difference is a rank-0 distribution on the cover of A, evaluated at the function $f_{B,y}$ defined below. The geometric and the spectral differences are the two sides of the trace formula for $\overline{A(\mathbb{A})^1}$ (Theorem 3.4), which is why the correction terms of Section 6.4 cancel in the sum.

As in [Ar81, §§2–3], let

$$\Gamma'_B(X,Y) = \tau_B(X) - \hat{\tau}_B(X-Y), \qquad X, Y \in \mathfrak{a} \cong \mathbb{R}, \tag{6.1}$$

the case $Q = B$ of $\Gamma'_Q(X,Y) = \sum_{R\supset Q}(-1)^{\dim(A_R/A_G)}\tau_Q^R(X)\hat{\tau}_R(X-Y)$, in the notation of (2.6); for fixed Y it is compactly supported in X. For $\tilde{k} \in \overline{K_S}$, $y \in \overline{G(\mathbb{Q}_S)}$ and $f \in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$ put

$$u'_B(\tilde{k}, y) = \int_{\mathfrak{a}} \Gamma'_B\big(X, -H_S(\tilde{k}y)\big)\, dX, \tag{6.2}$$

$$f_{B,y}(\tilde{a}) = \delta_B(\tilde{a})^{1/2} \int_{\overline{K_S}} \int_{N(\mathbb{Q}_S)} f(\tilde{k}^{-1}\tilde{a}n\tilde{k})\, u'_B(\tilde{k}, y)\, dn\, d\tilde{k}, \qquad \tilde{a} \in \overline{A(\mathbb{Q}_S)}, \tag{6.3}$$

and likewise for $y \in \overline{G(\mathbb{A})}$, $f \in \mathscr{H}(\overline{G(\mathbb{A})})$ and $\tilde{a} \in \overline{A(\mathbb{A})}$, with $\overline{K_0}$, $N(\mathbb{A})$ and H. Here dX is Lebesgue measure, as in Remark 2026.9.

Lemma 6.1. *(a)* $u'_B(\tilde{k}, y) = -H_S(\tilde{k}y)$.

(b) $f_{B,y}$ *is smooth, compactly supported and anti-genuine on* $\overline{A(\mathbb{Q}_S)}$, *and* $f_{B,1} = 0$.

(c) For $\pi \in \Pi_u(\overline{A(\mathbb{Q}_S)})$ *and* $\lambda \in \mathbb{C}$ *let* U_y *be the operator of multiplication by* $\tilde{k} \mapsto u'_B(\tilde{k}, y)$ *on* $V(\pi)$. *Then* $\mathrm{tr}\big(U_y I(\pi_\lambda, f)\big) = \widehat{f_{B,y}}(\pi_\lambda)$.

Proof. (a) $\int_{\mathbb{R}} \big(\tau_B(X) - \tau_B(X-Y)\big)\, dX = Y$ for every real Y: the integrand is the characteristic function of $(0,Y]$ if $Y > 0$ and minus that of $(Y,0]$ if $Y < 0$. (b) $u'_B(\cdot, y)$ is smooth and left invariant under $\overline{A(\mathbb{Q}_S)} \cap \overline{K_S}$, and $u'_B(\tilde{k}, 1) = -H_S(\tilde{k}) = 0$. (c) As in the proof of Proposition 5.17, $I(\pi_\lambda, f)$ has the kernel $\int_{\overline{A(\mathbb{Q}_S)}} \int_{N(\mathbb{Q}_S)} f(\tilde{k}^{-1}\tilde{a}n\tilde{k}')\pi(\tilde{a})e^{(\lambda+1)H_S(\tilde{a})}\, dn\, d\tilde{a}$ on $\overline{K_S}$; multiply by $u'_B(\tilde{k}, y)$ and integrate over the diagonal, using $\delta_B^{1/2} = e^{H_S}$. □

The sign matters: $u'_B(\cdot,y)$ is not of constant sign (Example 6.A), and with $|H|$ in place of $-H$ the formulas of Lemmas 6.2 and 6.3 fail.

Example 6.A. Let v be a real place, $k_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ and $y = a(s)$ with $s > 0$. Since $H_v(g) = -\log\|e_2 g\|$ (Lemma 4.3),

$$u'_B(k_\theta, a(s)) = \tfrac{1}{2}\log\bigl(s^2\sin^2\theta + s^{-2}\cos^2\theta\bigr),$$

which is $-\log s$ at $\theta = 0$ and $+\log s$ at $\theta = \frac{\pi}{2}$.

For a class $\mathscr{O} \in \mathscr{C}$, a class $\chi \in \mathscr{X}$ and an anti-genuine function h on $\overline{A(\mathbb{A})}$ whose restriction to $\overline{A(\mathbb{A})^1}$ is smooth and compactly supported, put

$$J^A_{\mathscr{O}}(h) = c_k \sum_{\gamma \in \mathscr{O} \cap A(\mathbb{Q})} h(\gamma), \qquad J^A_\chi(h) = \sum_{\mu \in \chi} \int_{\overline{A(\mathbb{A})^1}} h(\tilde{a})\mu(\tilde{a})\, d\tilde{a}, \tag{6.4}$$

with $J^A_\chi = 0$ for $\chi \in \mathscr{X}_G$. These are the distributions (3.4), summed over $\mathscr{O} \cap A(\mathbb{Q})$ and over χ, and Theorem 3.4 says that $\sum_{\mathscr{O}} J^A_{\mathscr{O}}(h) = \sum_\chi J^A_\chi(h)$. By Fourier inversion on $\mathfrak{a} \cong \mathbb{R}$, $\int_{\overline{A(\mathbb{A})^1}} h\mu = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{h}(\mu_{it})\, dt$ when $\hat{h}(\mu_{it}) = \int_{\overline{A(\mathbb{A})}} h\,\mu\, e^{itH}$ is integrable in t.

Lemma 6.2. *Let $f \in \mathscr{H}(\overline{G(\mathbb{A})})$ and $y \in \overline{G(\mathbb{A})}$ with $f^y \in \mathscr{H}(\overline{G(\mathbb{A})})$. Then, for all $\mathscr{O} \in \mathscr{C}$ and $\chi \in \mathscr{X}$,*

$$J_{\mathscr{O}}(f^y) = J_{\mathscr{O}}(f) + J^A_{\mathscr{O}}(f_{B,y}), \qquad J_\chi(f^y) = J_\chi(f) + J^A_\chi(f_{B,y}).$$

In particular $J_{\mathscr{O}}$ is invariant when $\mathscr{O}$ is elliptic, and J_χ is invariant when $\chi \in \mathscr{X}_G$.

Proof. This is the formula of [Ar81, §3] for this group; the proof is short. Since $f^y(x^{-1}\gamma n x) = f\bigl((xy^{-1})^{-1}\gamma n (xy^{-1})\bigr)$, the kernels satisfy $K^{f^y}_{P,\mathscr{O}}(x,x) = K^f_{P,\mathscr{O}}(xy^{-1}, xy^{-1})$. Substituting $x \mapsto xy$ in $J^T_{\mathscr{O}}(f^y) = \int k^T_{\mathscr{O}}(x, f^y)\, dx$ (Theorem 4.5) replaces $\hat{\tau}_B(H(\delta x) - T)$ in (4.4) by $\hat{\tau}_B(H(\delta xy) - T)$. Write $\delta x = n\tilde{a}\tilde{k}$; then $H(\delta xy) = H(\delta x) + H(\tilde{k}y)$, and by (6.1),

$$\hat{\tau}_B\bigl(H(\delta x) + H(\tilde{k}y) - T\bigr) = \hat{\tau}_B\bigl(H(\delta x) - T\bigr) - \Gamma'_B\bigl(H(\delta x) - T, -H(\tilde{k}y)\bigr).$$

So $J^T_{\mathscr{O}}(f^y) - J^T_{\mathscr{O}}(f) = \int_{B(\mathbb{Q})\backslash\overline{G(\mathbb{A})}} K_{B,\mathscr{O}}(x,x)\,\Gamma'_B(H(x) - T, -H(\tilde{k}y))\, dx$. In the coordinates $x = n\tilde{a}\tilde{k}$, with $dx = \delta_B(\tilde{a})^{-1} dn\, d\tilde{a}\, d\tilde{k}$,

$$K_{B,\mathscr{O}}(x,x) = \delta_B(\tilde{a}) \sum_{\gamma \in \mathscr{O} \cap A(\mathbb{Q})} \int_{N(\mathbb{A})} f(\tilde{k}^{-1}\gamma n \tilde{k})\, dn,$$

and $N(\mathbb{Q})\backslash N(\mathbb{A})$ has volume 1. The integral over $A(\mathbb{Q})\backslash\overline{A(\mathbb{A})}$ of $\Gamma'_B(H(\tilde{a}) - T, -H(\tilde{k}y))$ is $c_k\, u'_B(\tilde{k}, y)$, for every T. Since $\delta_B(\gamma) = 1$ for $\gamma \in A(\mathbb{Q})$, the difference is $c_k \sum_\gamma f_{B,y}(\gamma)$. All integrals converge absolutely: the sum over γ is finite and $\Gamma'_B(\cdot, Y)$ is compactly supported.

For χ the argument is the same, with (5.10) and $K_{B,\chi}$. By (5.7), and because an orthonormal basis for (5.1) satisfies $c_k \int_{\overline{K_0}} |\varphi|^2 = 1$, the difference is

$$\sum_{\mu \in \chi} \frac{1}{2\pi} \int_{\mathbb{R}} \operatorname{tr}\bigl(U_y I(\mu_{it}, f)\bigr)\, dt = \sum_{\mu \in \chi} \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f_{B,y}}(\mu_{it})\, dt = J^A_\chi(f_{B,y})$$

by Lemma 6.1(c); the interchange of integrals is justified as in [Ar81, §3], by Theorem 5.12. □

Written as a sum over all parabolic subgroups $Q \supset A$, as in [Ar88b, Ar88c], the formula is $J(f^y) = \sum_Q |W_0^{M_Q}||W_0|^{-1} J^{M_Q}(f_{Q,y})$. Here Q runs over G, B and $\bar{B}$; the last two contribute $\frac{1}{2}J^A(f_{B,y})$ each, because $J^A(f_{\bar{B},y}) = J^A(f_{B,y})$.

The local distributions vary in the same way. Recall J_G, J_A and the weight v_A from (4.11)–(4.12), and $J_A(z,f)$ from Lemma 4.10.

Lemma 6.3. *Let $f \in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$ and $x, y \in \overline{G(\mathbb{Q}_S)}$.*

(a) If $x = n\tilde{a}\tilde{k}$ and $wx = n'\tilde{a}'\tilde{k}'$ are Iwasawa decompositions, then $v_A(xy) = v_A(x) - H_S(\tilde{k}y) - H_S(\tilde{k}'y)$.

(b) If $\gamma \in \overline{A(\mathbb{Q}_S)}$ is regular (that is, γ lies over $a(t)$ with $t_v^2 \neq 1$ for all $v \in S_k$), then $J_G(\gamma, f^y) = J_G(\gamma, f)$ and

$$J_A(\gamma, f^y) = J_A(\gamma, f) + f_{B,y}(\gamma) + f_{B,y}(w\gamma w^{-1}).$$

(c) For $z \in \{\pm 1\}$, $J_A(z, f^y) = J_A(z, f) + 2f_{B,y}(z)$.

Proof. (a) $H_S(xy) = H_S(x) + H_S(\tilde{k}y)$ and $H_S(wxy) = H_S(wx) + H_S(\tilde{k}'y)$; subtract from $v_A(x) = -H_S(x) - H_S(wx)$.

(b) Substituting $x \mapsto xy$ in (4.12) gives $J_A(\gamma, f^y) = |D(\gamma)|^{1/2} \int f(x^{-1}\gamma x) v_A(xy)\, dx$, and likewise for J_G without the weight. By (a) the difference from $J_A(\gamma, f)$ is the sum of two terms. In the first, with $x = n\tilde{k}$ and $n^{-1}\gamma n = \gamma n'$, where $dn = |1 - t^{-2}|^{-1} dn'$ and $|D(\gamma)|^{1/2} = \delta_B(\gamma)^{1/2}|1 - t^{-2}|$, we get $f_{B,y}(\gamma)$. In the second, substitute $x \mapsto w^{-1}x$; this replaces γ by $w\gamma w^{-1}$ and $\tilde{k}'$ by the $\overline{K_S}$-component of x, and $D(w\gamma w^{-1}) = D(\gamma)$.

(c) Apply (b) to a_t in Lemma 4.10(b): $J_G(a_t, \cdot)$ and $r(t)$ do not change, and $f_{B,y}(a_t) + f_{B,y}(wa_t w^{-1}) \to 2f_{B,y}(z)$ as $t \to 1$, because $f_{B,y}$ is continuous and z is central. □

Parts (b) and (c) are the case $M = A$ of Arthur's formula $J_M(\gamma, f^y) = \sum_{Q \in \mathscr{F}(M)} J_M^{M_Q}(\gamma, f_{Q,y})$ [Ar81, Ar88b], with $\mathscr{F}(A) = \{G, B, \bar{B}\}$ and $f_{\bar{B},y}(\gamma) = f_{B,y}(w\gamma w^{-1})$.

On the spectral side everything follows from one identity. It holds for the global operator $M(w,\lambda)$, for the local operators and for their normalized versions alike, so we state it for any intertwining family.

Lemma 6.4. *Let $\pi \in \Pi_u(\overline{A(\mathbb{Q}_S)})$, and let $\mathscr{M}(\lambda)\colon V(\pi) \to V(\pi^w)$ be a meromorphic family of invertible operators with $\mathscr{M}(\lambda) I(\pi_\lambda, y) = I(\pi_{-\lambda}^w, y)\mathscr{M}(\lambda)$ for all $y \in \overline{G(\mathbb{Q}_S)}$. For $y \in \overline{G(\mathbb{Q}_S)}$ let H_y be multiplication by $\tilde{k} \mapsto H_S(\tilde{k}y)$, on $V(\pi)$ and on $V(\pi^w)$. Then, with $\mathscr{M}' = d\mathscr{M}/d\lambda$,*

$$I(\pi_\lambda, y)\mathscr{M}(\lambda)^{-1}\mathscr{M}'(\lambda) I(\pi_\lambda, y)^{-1} = \mathscr{M}(\lambda)^{-1}\mathscr{M}'(\lambda) + H_y + \mathscr{M}(\lambda)^{-1} H_y \mathscr{M}(\lambda), \tag{6.5}$$

and for $f \in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$,

$$-\operatorname{tr}\big(\mathscr{M}^{-1}\mathscr{M}' I(\pi_\lambda, f^y)\big) = -\operatorname{tr}\big(\mathscr{M}^{-1}\mathscr{M}' I(\pi_\lambda, f)\big) + \widehat{f_{B,y}}(\pi_\lambda) + \widehat{f_{B,y}}(\pi_{-\lambda}^w). \tag{6.6}$$

The same holds over $\mathbb{A}$, for $\mathscr{H}_B(\mu)$, $I(\mu_\lambda)$ and $\overline{G(\mathbb{A})}$.

Proof. In the compact picture, $(I(\pi_\lambda,y)\varphi)(\tilde{k}) = \pi(\tilde{a})e^{(\lambda+1)H_S(\tilde{k}y)}\varphi(\tilde{k}_1)$ for $\tilde{k}y = n\tilde{a}\tilde{k}_1$, so $\frac{d}{d\lambda}I(\pi_\lambda,y) = H_yI(\pi_\lambda,y)$ and $\frac{d}{d\lambda}I(\pi^w_{-\lambda},y) = -H_yI(\pi^w_{-\lambda},y)$. Differentiating the intertwining relation,

$$\mathscr{M}'I(\pi_\lambda,y) + \mathscr{M}H_yI(\pi_\lambda,y) = -H_yI(\pi^w_{-\lambda},y)\mathscr{M} + I(\pi^w_{-\lambda},y)\mathscr{M}'.$$

Multiply by $\mathscr{M}^{-1}$ on the left and by $I(\pi_\lambda,y)^{-1}$ on the right, and use $\mathscr{M}^{-1}I(\pi^w_{-\lambda},y) = I(\pi_\lambda,y)\mathscr{M}^{-1}$; this is (6.5). Since $I(\pi_\lambda,f^y) = I(\pi_\lambda,y)^{-1}I(\pi_\lambda,f)I(\pi_\lambda,y)$, the left side of (6.6) is $-\operatorname{tr}\big(I(\pi_\lambda,y)\mathscr{M}^{-1}\mathscr{M}'I(\pi_\lambda,y)^{-1}I(\pi_\lambda,f)\big)$. Insert (6.5); the last term contributes $-\operatorname{tr}(H_y\mathscr{M}I(\pi_\lambda,f)\mathscr{M}^{-1}) = -\operatorname{tr}(H_yI(\pi^w_{-\lambda},f))$, and $-H_y = U_y$, so Lemma 6.1(c) applies to both terms. □

With $\mathscr{M} = M(w,\lambda)$, formula (6.6), Theorem 5.14 and the invariance of $\operatorname{tr}(R(f)|_{V_b})$ and of $\operatorname{tr}(M(w,0)I(\mu_0,f))$ give a second proof of the formula for $J_\chi(f^y)$ in Lemma 6.2. (The operator $M(w,0)$ commutes with $I(\mu_0,y)$ when $\mu^w = \mu$, so $f \mapsto \operatorname{tr}(M(w,0)I(\mu_0,f))$ is invariant.)

6.2 Normalized intertwining operators

The correction terms are built from the logarithmic derivative of the intertwining operator. The global operator $M(w,\lambda)$ is not a product of local operators that are holomorphic on $i\mathbb{R}$: the local operators have poles there, and the Euler product of their unramified eigenvalues does not converge. One divides each local operator by a scalar normalizing factor; the product of these factors is the global function $m(\mu,\lambda)$, and what remains is a finite product of normalized local operators, which are unitary and holomorphic on $i\mathbb{R}$.

Let v be a place of k, π_v a genuine unitary character of $\overline{A(k_v)}$, and

$$\big(M_v(\pi_v,\lambda)\varphi\big)_{-\lambda}(x) = \int_{k_v}\varphi_\lambda\big((w,1)^{-1}(n(u),1)x\big)\,du, \qquad M_v(\pi_v,\lambda)\colon V(\pi_v)\to V(\pi_v^w),$$

as in Lemma 5.6; the integral converges for $\operatorname{Re}\lambda > 0$ and continues meromorphically.

Theorem 6.5 (Arthur [Ar89, Thm. 2.1]; Li [Li12a, §3] for covers)**.** *There are meromorphic functions $r_v(\pi_v,\lambda)$ of $\lambda\in\mathbb{C}$, with $r_v(\pi_{v,\nu},\lambda) = r_v(\pi_v,\lambda+\nu)$ for $\nu\in i\mathbb{R}$, such that the operators $R_v(\pi_v,\lambda) = r_v(\pi_v,\lambda)^{-1}M_v(\pi_v,\lambda)$ have the following properties.*

(N1) $R_v(\pi_v,\lambda)I(\pi_{v,\lambda},x) = I(\pi^w_{v,-\lambda},x)R_v(\pi_v,\lambda)$ *for* $x\in\overline{SL(2,k_v)}$.

(N2) $R_v(\pi_v^w,-\lambda)R_v(\pi_v,\lambda) = 1$, *and for* $\lambda\in i\mathbb{R}$ *the operator* $R_v(\pi_v,\lambda)$ *is holomorphic and unitary.*

(N3) On each $\overline{K_v}$-isotypic subspace, $R_v(\pi_v,\lambda)$ is a rational function of $q_v^{-\lambda}$ if v is finite, and a rational function of λ if v is archimedean. In the archimedean case the norms of $R_v^{-1}\frac{d}{d\lambda}R_v$ on the $\overline{K_v}$-types, for $\lambda\in i\mathbb{R}$, are bounded by a polynomial in the highest weight of the type and in $|\lambda|$ (for real v this is explicit in (6.9)).

(N4) If v is finite and non-dyadic, ψ_v is unramified and π_v is trivial on $A(\mathscr{O}_v)^*$, then $R_v(\pi_v,\lambda)\varphi_v^0 = \varphi_v^{0,w}$ for all λ.

These are Arthur's conditions (R_1)–(R_8) of [Ar89, §2] for the pair $(B,\bar{B})$, written with w: $M_v(\pi_v,\lambda)$ is the composition of the operator $J_{\bar{B}|B}(\pi_{v,\lambda})$ of [Ar89] with left translation by w, which does not depend on λ. At the real places everything is explicit for the double cover (Lemma 6.6(b),(c)); for the universal cover of $SL(2,\mathbb{R})$ see Sally [Sa67]. The factors r_v are not unique. Three statements about them should be kept apart.

(i) *Existence.* At every place there are factors r_v with (N1)–(N4). This is Theorem 6.5, and it is all that the construction of the invariant distributions uses.

(ii) *An explicit formula at a given place.* We have one at the finite places covered by (N4), at the real places and at the complex places (Lemma 6.6 and the remark after Example 6.B). We do not have one at the places of $S_{\mathrm{bad}}(\mu)$ defined below (Remark 2026.15).

(iii) *Compatibility with the global operator.* For any choice of local factors, the product $m(\mu,\lambda)$ in (6.10) is defined, $M(w,\lambda) = m\cdot R_S$, and m has the properties of Corollary 6.7(b). So no compatibility condition has to be imposed on the local choices; but m, m'/m and the distributions $I_A(\gamma,f)$ depend on them, and m is a known function only up to the unknown factors at $S_{\mathrm{bad}}(\mu)$.

At most places there is a natural choice:

Lemma 6.6. *Write $\pi_v = \chi_{\psi,v}\cdot(\chi_v\circ p)$.*

(a) Let v, ψ_v and π_v be as in (N4). Then

$$r_v(\pi_v,\lambda) = \frac{L_v(2\lambda,\chi_v^2)}{L_v(2\lambda+1,\chi_v^2)}$$

satisfies (N1)–(N4).

(b) Let v be real and $\chi_v = \mathrm{sgn}^{\varepsilon}$, $\varepsilon\in\{0,1\}$. Let $\theta\mapsto\tilde{k}_\theta$ be the one-parameter subgroup of $\overline{K_v}$ above k_θ (Example 6.A). The $\overline{K_v}$-types of $I(\pi_v)$ are the characters $\tilde{k}_\theta\mapsto e^{im\theta}$, $m\in m_0+2\mathbb{Z}$, each with multiplicity one, where $m_0\in\{\frac12,-\frac12\}$ is determined by $e^{i\pi m_0}=\pi_v(a(-1),1)$. On the line of weight m, with du Lebesgue measure,

$$M_v(\pi_v,\lambda) = e^{i\pi m/2}\,\frac{\pi\,2^{1-\lambda}\,\Gamma(\lambda)}{\Gamma\!\left(\frac{\lambda+1+m}{2}\right)\Gamma\!\left(\frac{\lambda+1-m}{2}\right)}. \tag{6.7}$$

With $r_v(\pi_v,\lambda) = \sqrt{2}\,e^{i\pi m_0/2}\,L_v(2\lambda)/L_v(2\lambda+1)$, where $L_v(s)=\pi^{-s/2}\Gamma(s/2)$, the operator $R_v(\pi_v,\lambda)$ is the identity on the weight m_0, and on the weight m_0+2j it is

$$\prod_{i=0}^{j-1}\frac{a_i-\lambda}{a_i+\lambda}\quad(j>0,\ a_i=1+m_0+2i),\qquad \prod_{i=0}^{|j|-1}\frac{b_i-\lambda}{b_i+\lambda}\quad(j<0,\ b_i=1-m_0+2i). \tag{6.8}$$

These operators satisfy (N1)–(N3).

(c) *Let v be real. Every genuine unitary character of $\overline{A(k_v)}$ is of the form $\pi_{v,i\tau} = \pi_v e^{i\tau H_v}$ with π_v as in (b) and a unique $\tau \in \mathbb{R}$; that is, $\chi_v = \operatorname{sgn}^\varepsilon |\cdot|^{i\tau}$. Then $V(\pi_{v,i\tau}) = V(\pi_v)$, $I(\pi_{v,i\tau},\lambda) = I(\pi_v,\lambda+i\tau)$ and $M_v(\pi_{v,i\tau},\lambda) = M_v(\pi_v,\lambda+i\tau)$. With $r_v(\pi_{v,i\tau},\lambda) = r_v(\pi_v,\lambda+i\tau)$, the normalized operator $R_v(\pi_{v,i\tau},\lambda)$ is given by (6.8) with $\lambda+i\tau$ in place of λ, and satisfies (N1)–(N3). In terms of χ_v,*

$$r_v(\pi_{v,i\tau},\lambda) = \sqrt{2}\, e^{i\pi m_0/2}\, \frac{L_v(2\lambda,\chi_v^2)}{L_v(2\lambda+1,\chi_v^2)}, \qquad L_v(s,|\cdot|^{2i\tau}) = L_v(s+2i\tau).$$

Proof. (a) (N1) holds for M_v, and (N4) is Lemma 5.6(b). The induced representation $I(\pi_{v,\lambda})$ is irreducible for λ outside a discrete set, so $M_v(\pi_v^w,-\lambda)M_v(\pi_v,\lambda)$, which commutes with it, is a scalar; on φ_v^0 it is $r_v(\pi_v^w,-\lambda)r_v(\pi_v,\lambda)$ by Lemma 5.6(b). So $R_v(\pi_v^w,-\lambda)R_v(\pi_v,\lambda) = 1$. For $\lambda \in i\mathbb{R}$ the adjoint of $M_v(\pi_v,\lambda)$ is $M_v(\pi_v^w,-\lambda)$ [Ar89, §1]. (This uses that $(w,1)_v^2$ acts trivially on $I(\pi_{v,\lambda})$. Here it does: $(-1,-1)_v = 1$ at a non-dyadic finite place, so $(w,1)_v^2 = (-1,1)$, a central element lying in $A(\mathscr{O}_v)^*$, and it acts by the scalar $\pi_v(-1,1) = 1$. At a real place $(-1,1)$ has order 4 and acts by $\pm i$ on a genuine representation, by Lemma 2.2(e); part (b) does not use the adjoint formula.) Moreover $\overline{r_v(\pi_v,\lambda)} = r_v(\pi_v^w,-\lambda)$ for $\lambda \in i\mathbb{R}$; hence R_v is unitary on $i\mathbb{R}$ wherever it is defined, so it is bounded there and has no poles. Rationality is [Li12a].

(b) In the model of Section 2.1, $\tilde{k}_\theta = (k_\theta,1)$ for $0 < \theta < \pi$: for $\theta_1,\theta_2,\theta_1+\theta_2 \in (0,\pi)$ the three values of $x(\cdot)$ in (2.1) are $-\sin$ of the angles, all negative, so $\beta(k_{\theta_1},k_{\theta_2}) = 1$. Then $\tilde{k}_\pi = (k_{\pi/2},1)^2 = (-1,1) = (a(-1),1)$, which gives the $\overline{K_v}$-types, since $\overline{A(k_v)} \cap \overline{K_v}$ is generated by $\tilde{k}_\pi$. Also $(w,1)^{-1} = \tilde{k}_{\pi/2}$, and $\tilde{k}_{\pi/2}(n(u),1) = (n',1)(a(s),1)\tilde{k}_{\theta(u)}$ with $s = (1+u^2)^{-1/2} > 0$ and $\theta(u) = \frac{\pi}{2} + \arctan u$, by continuity in u. So on a vector of weight m the integral is $e^{i\pi m/2}\int_{\mathbb{R}}(1+u^2)^{-(\lambda+1)/2}e^{im\arctan u}\,du$, and $\int_{-\pi/2}^{\pi/2}\cos^{\lambda-1}\vartheta\, e^{im\vartheta}\,d\vartheta$ is the classical integral in (6.7). By the duplication formula, $\Gamma(\frac{\lambda}{2}+\frac{3}{4})\Gamma(\frac{\lambda}{2}+\frac{1}{4}) = 2^{\frac{1}{2}-\lambda}\sqrt{\pi}\,\Gamma(\lambda+\frac{1}{2})$, so (6.7) at $m = \pm\frac{1}{2}$ is $\sqrt{2}\,e^{i\pi m/2}\sqrt{\pi}\,\Gamma(\lambda)/\Gamma(\lambda+\frac{1}{2})$. The quotient of (6.7) at $m+2$ and at m is $-(\lambda-1-m)/(\lambda+1+m)$, which gives (6.8). Each factor in (6.8) has absolute value 1 on $i\mathbb{R}$ and no pole there, because a_i and b_i are nonzero real numbers; (N2) and (N3) follow, and (N1) holds for M_v.

(c) The unitary characters of $\mathbb{R}^\times$ are the $\operatorname{sgn}^\varepsilon|\cdot|^{i\tau}$, and $H_v(a(t),\zeta) = \log|t|$. The space $V(\pi)$ depends only on the restriction of π to $\overline{A(k_v)} \cap \overline{K_v}$, on which $e^{i\tau H_v} = 1$, and φ_λ formed with $\pi_{v,i\tau}$ is $\varphi_{\lambda+i\tau}$ formed with π_v; this gives the first three statements. The factors in (6.8) with $\lambda+i\tau$ in place of λ still have absolute value 1 and no poles for $\lambda \in i\mathbb{R}$. The last formula is $L_v(s,|\cdot|^{2i\tau}) = \pi^{-(s+2i\tau)/2}\Gamma((s+2i\tau)/2)$. □

Example 6.B. In Lemma 6.6(b) take $m_0 = \frac{1}{2}$, which is the case $\pi_v = \chi_{\psi,v}$ for $\psi_v(x) = e^{-2\pi ix}$ (Example 3.B). The weights are $\frac{1}{2},\frac{5}{2},\frac{9}{2},\dots$ and $-\frac{3}{2},-\frac{7}{2},\dots$, and

$$R_v(\lambda)\big|_{5/2} = \frac{\frac{3}{2}-\lambda}{\frac{3}{2}+\lambda}, \qquad R_v(\lambda)\big|_{-3/2} = \frac{\frac{1}{2}-\lambda}{\frac{1}{2}+\lambda}, \qquad R_v^{-1}R_v'(\lambda)\big|_{5/2} = -\frac{3}{\frac{9}{4}-\lambda^2}.$$

$R_v(\lambda)$ has a zero at $\lambda = \frac{1}{2}$ on every weight below $\frac{1}{2}$ and is 1 on the weight $\frac{1}{2}$: this is the reducibility of $I(\chi_{\psi,v,1/2})$, whose quotient with lowest weight $\frac{1}{2}$ is the even Weil representation of Theorem 5.8. In general, by (6.8), on the weight m_0+2j and for $\lambda = it$,

$$-R_v^{-1}R_v'(it) = \rho_j(t) := \sum_i \frac{2c_i}{c_i^2+t^2} \ge 0, \tag{6.9}$$

the sum over $c_i = a_i$, $0 \le i < j$, if $j > 0$, and over $c_i = b_i$, $0 \le i < |j|$, if $j < 0$; and $\rho_0 = 0$. For $\pi_{v,i\tau}$ as in Lemma 6.6(c) the value is $\rho_j(t+\tau)$. For $SL(2,\mathbb{R})$ the weights are integers and the same computation gives the familiar formula with $L_v(\lambda)/L_v(\lambda+1)$.

At a complex place the cover splits and R_v is the normalized operator of $SL(2,\mathbb{C})$ [Ar89]; the factor $L_v(\lambda,\chi_v)/L_v(\lambda+1,\chi_v)$ there equals $2L_v(2\lambda,\chi_v^2)/L_v(2\lambda+1,\chi_v^2)$. So the factor of Lemma 6.6 is available at every place outside the finite set

$$S_{\mathrm{bad}}(\mu) = \{v \text{ dyadic}\} \cup \{v \mid \psi_v \text{ ramified}\} \cup \{v \text{ finite} \mid \mu_v \text{ nontrivial on } A(\mathscr{O}_v)^*\}.$$

The global normalizing factor. Let $\mu = \chi_\psi \cdot (\chi \circ p) \in \Pi_{\mathrm{aut}}(X_A)$, and fix factors $r_v(\mu_v,\lambda)$ as in Theorem 6.5 for $v \in S_{\mathrm{bad}}(\mu)$ and as in Lemma 6.6 (and the remark on complex places) for all other v. At a real place the component μ_v is $\chi_{\psi,v} \cdot (\chi_v \circ p)$ with $\chi_v = \mathrm{sgn}^\varepsilon |\cdot|^{i\tau_v}$, and Lemma 6.6(c) applies; for $k = \mathbb{Q}$ one has $\tau_\infty = 0$, because μ is trivial on $A_B(\mathbb{R})^0$, but for other fields the τ_v need not vanish. For a finite set $S_1 \supset S_{\mathrm{bad}}(\mu)$ of places of k containing the archimedean ones, put

$$m(\mu,\lambda) = m^{S_1}(\mu,\lambda) \prod_{v \in S_1} r_v(\mu_v,\lambda), \qquad m^{S_1}(\mu,\lambda) = \frac{L^{S_1}(2\lambda,\chi^2)}{L^{S_1}(2\lambda+1,\chi^2)} \tag{6.10}$$

as in (5.4). By Lemma 6.6(a) this does not depend on S_1.

Corollary 6.7. *Let $f = f_S \otimes e^S$ and $\mu \in \Pi_{\mathrm{aut}}(X_A)$ with μ_v trivial on $A(\mathscr{O}_v)^*$ for $v \notin S_k$, and let $R_S(\mu_S,\lambda) = \bigotimes_{v\in S_k} R_v(\mu_v,\lambda)$.*

(a) On $\bigotimes_{v\in S_k} V(\mu_v) \otimes \bigotimes_{v\notin S_k} \varphi_v^0$, $M(w,\lambda) = m(\mu,\lambda) R_S(\mu_S,\lambda) \otimes 1$, and for $\lambda \in i\mathbb{R}$

$$\mathrm{tr}\big(M(w,\lambda)^{-1} M'(w,\lambda) I(\mu_\lambda, f)\big) = \frac{m'}{m}(\mu,\lambda)\,\mathrm{tr}\, I(\mu_{S,\lambda}, f_S) + \mathrm{tr}\big(R_S^{-1} R_S'(\mu_S,\lambda) I(\mu_{S,\lambda}, f_S)\big).$$

(b) $m(\mu^w,-\lambda)m(\mu,\lambda) = 1$; for $\lambda \in i\mathbb{R}$, $m(\mu,\lambda)$ is holomorphic of absolute value 1, $\frac{m'}{m}(\mu,\lambda)$ is real, and $\frac{m'}{m}(\mu^w,-\lambda) = \frac{m'}{m}(\mu,\lambda)$.

(c) With $\Lambda^{\mathrm{bad}}(s,\chi^2) = \prod_{v\notin S_{\mathrm{bad}}(\mu)} L_v(s,\chi_v^2)$, the product over all places outside $S_{\mathrm{bad}}(\mu)$, archimedean ones included,

$$\frac{m'}{m}(\mu,\lambda) = 2\frac{(\Lambda^{\mathrm{bad}})'}{\Lambda^{\mathrm{bad}}}(2\lambda,\chi^2) - 2\frac{(\Lambda^{\mathrm{bad}})'}{\Lambda^{\mathrm{bad}}}(2\lambda+1,\chi^2) + \sum_{v\in S_{\mathrm{bad}}(\mu)} \frac{r_v'}{r_v}(\mu_v,\lambda).$$

Proof. (a) Lemma 5.6(a)–(b), with $S_1 = S_k$, and $M_v = r_v R_v$; then the Leibniz rule, as for Corollary 5.15. (b) $M(w,-\lambda)M(w,\lambda) = 1$ (Theorem 5.5(c)) and (N2) give the first statement, and the unitarity of $M(w,\lambda)$ and of R_S on $i\mathbb{R}$ the second; differentiate. (c) is the logarithmic derivative of (6.10). □

Remark 2026.15. *The factors at the bad places.* For $v \in S_{\mathrm{bad}}(\mu)$ (dyadic places, and places where ψ_v or μ_v is ramified) we use the factors r_v of [Li12a] without making them explicit. The expected choice is again $L_v(2\lambda,\chi_v^2)/L_v(2\lambda+1,\chi_v^2)$ times an exponential factor, which would give

$m(\mu,\lambda) = c\,q^{-\lambda}\Lambda(2\lambda,\chi^2)/\Lambda(2\lambda+1,\chi^2)$ with the completed Hecke L-function and constants c, $q>0$, as the Shimura correspondence suggests. The distributions $I_A(\gamma,f)$ and the function m'/m below depend on the choice of the r_v; the invariant trace formula holds for each choice, and this section does not use more. The question is taken up again in Section 7.5: Proposition 7.8′ there gives a canonical choice at every finite place, and by Szpruch's computation of the local coefficients [Sz09, Sz16] it is a constant times $L_v(2\lambda,\chi_v^2)/L_v(2\lambda+1,\chi_v^2)$, with no exponential factor, also at the places of $S_{\mathrm{bad}}(\mu)$. With that choice $m(\mu,\lambda)$ is a constant times $\Lambda(2\lambda,\chi^2)/\Lambda(2\lambda+1,\chi^2)$.

6.3 The map φ_A and the distributions $I_A(\gamma,f)$

We now define the invariant distributions. Fix normalizing factors as in Section 6.2, and for $\pi\in\Pi_u(\overline{A(\mathbb{Q}_S)})$ let $R_S(\pi,\lambda)=\bigotimes_{v\in S_k}R_v(\pi_v,\lambda)$. For $f\in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$ put

$$\varphi_A(f,\pi) = -\operatorname{tr}\Big(R_S(\pi,0)^{-1}\frac{dR_S}{d\lambda}(\pi,0)\,I(\pi,f)\Big),\qquad \pi\in\Pi_u(\overline{A(\mathbb{Q}_S)}). \tag{6.11}$$

By Theorem 6.5, $\varphi_A(f,\pi_\lambda) = -\operatorname{tr}(R_S^{-1}R_S'(\pi,\lambda)I(\pi_\lambda,f))$ for $\lambda\in i\mathbb{R}$, and $R_S^{-1}R_S' = \sum_{v\in S_k}R_v^{-1}R_v'$. This is Arthur's map $\phi_M(f,\pi)=\operatorname{tr}(\mathscr{R}_M(\pi,P)I_P(\pi,f))$ [Ar81, §7], [Ar88b] for $M=A$, $P=B$: the (G,A)-family is $\mathscr{R}_B(\Lambda)=1$, $\mathscr{R}_{\bar B}(\Lambda)=R_S(\pi,0)^{-1}R_S(\pi,\Lambda)$, and, as for (5.12),

$$\mathscr{R}_A(\pi,B)=\lim_{\Lambda\to0}\Big(\frac{1}{\theta_B(\Lambda)}+\frac{R_S(\pi,0)^{-1}R_S(\pi,\Lambda)}{\theta_{\bar B}(\Lambda)}\Big) = -R_S(\pi,0)^{-1}R_S'(\pi,0),$$

with $\theta_B(\Lambda)=\Lambda=-\theta_{\bar B}(\Lambda)$ in the normalization of Remark 2026.9.

Proposition 6.8. *Let $f\in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$ and $y\in\overline{G(\mathbb{Q}_S)}$.*

(a) $\varphi_A(f,\cdot)$ is continuous and integrable on $\Pi_u(\overline{A(\mathbb{Q}_S)})$. More precisely, it vanishes unless the conductors of the π_v, v finite, are bounded in terms of f, and it decreases rapidly in the continuous parameters of π.

(b) $\varphi_A(f,\pi^w)=\varphi_A(f,\pi)$.

(c) $\varphi_A(f^y,\pi)=\varphi_A(f,\pi)+\widehat{f_{B,y}}(\pi)+\widehat{f_{B,y}}(\pi^w)$.

(d) If $v\notin S_k$ is as in (N4) and $S'=S\cup\{v\}$, then $\varphi_A(f\otimes e_v,\pi\otimes\pi_v)=\varphi_A(f,\pi)$ if π_v is trivial on $A(\mathscr{O}_v)^*$, and $=0$ otherwise.

Proof. (a) f is bi-invariant under an open subgroup at the finite places, which bounds the conductors. $\pi\mapsto I(\pi,f)$ is the operator-valued Fourier transform of a compactly supported smooth function, so its trace norm on each $\overline{K_S}$-type decreases rapidly in π and in the $\overline{K_S}$-type; by (N2) and (N3), $R_S^{-1}R_S'$ is continuous in π and of polynomial growth. (b) By (N2), $R_S(\pi^w,\nu)=R_S(\pi,-\nu)^{-1}$, so $R_S(\pi^w,\nu)^{-1}\frac{d}{d\nu}R_S(\pi^w,\nu)$ at $\nu=0$ is $R_S'R_S^{-1}$ at $(\pi,0)$, and $\operatorname{tr}(R_S'R_S^{-1}I(\pi^w,f)) = \operatorname{tr}(R_S^{-1}R_S'R_S^{-1}I(\pi^w,f)R_S) = \operatorname{tr}(R_S^{-1}R_S'I(\pi,f))$ by (N1). (c) is (6.6) with $\mathscr{M}=R_S(\pi,\lambda)$ at $\lambda=0$. (d) $I(\pi_v,e_v)$ is the projection onto $\mathbb{C}\varphi_v^0$ (or 0), and $R_v^{-1}R_v'\varphi_v^0=0$ by (N4). □

For $\tilde{a} \in \overline{A(\mathbb{Q}_S)}$ put

$$\Phi_A(\tilde{a}, f) = \int_{\Pi_u(\overline{A(\mathbb{Q}_S)})} \varphi_A(f, \pi)\, \pi(\tilde{a})^{-1}\, d\pi, \tag{6.12}$$

the function on $\overline{A(\mathbb{Q}_S)}$ whose Fourier transform is $\varphi_A(f, \cdot)$, and for $\gamma \in \overline{A(\mathbb{Q}_S)}$ regular and $z \in \{\pm 1\}$ define

$$I_A(\gamma, f) = J_A(\gamma, f) - \Phi_A(\gamma, f), \qquad I_A(z, f) = J_A(z, f) - \Phi_A(z, f). \tag{6.13}$$

Theorem 6.9. *Let $f \in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$.*

(a) $\Phi_A(\cdot, f)$ is a continuous, bounded, anti-genuine function on $\overline{A(\mathbb{Q}_S)}$, smooth in the archimedean coordinates and locally constant in the others, with $\Phi_A(w\tilde{a}w^{-1}, f) = \Phi_A(\tilde{a}, f)$. If $f = \bigotimes_v f_v$, its restriction to each set $\{\tilde{a} \in \overline{A(\mathbb{Q}_S)} \mid H_S(\tilde{a}) = X\}$ is compactly supported. It is not compactly supported on $\overline{A(\mathbb{Q}_S)}$ in general.

(b) The distributions $I_A(\gamma, \cdot)$ and $I_A(z, \cdot)$ are invariant: $I_A(\gamma, f^y) = I_A(\gamma, f)$ for all $y \in \overline{G(\mathbb{Q}_S)}$.

(c) $I_A(w\gamma w^{-1}, f) = I_A(\gamma, f)$.

(d) If $v \notin S_k$ is as in (N4), $S' = S \cup \{v\}$, and $\gamma_v \in A(\mathscr{O}_v)^*$ lies over $a(t)$ with $|t^2 - 1|_v = 1$, then $I_A(\gamma\gamma_v, f \otimes e_v) = I_A(\gamma, f)$.

Proof. (a) Continuity and boundedness follow from Proposition 6.8(a), smoothness from the rapid decrease, local constancy from the bound on the conductors, and the symmetry from Proposition 6.8(b) and the invariance of $d\pi$ under $\pi \mapsto \pi^w$. For $f = \bigotimes f_v$, by the Leibniz rule $\varphi_A(f, \pi) = \sum_{v \in S_k} \varphi_{A,v}(f_v, \pi_v) \prod_{u \neq v} \operatorname{tr} I(\pi_u, f_u)$, and $\operatorname{tr} I(\pi_u, f_u) = \widehat{F_u}(\pi_u)$ with $F_u(\tilde{a}) = \delta_B(\tilde{a})^{1/2} \int\int f_u(\tilde{k}^{-1}\tilde{a}n\tilde{k})\, dn\, d\tilde{k}$ compactly supported on $\overline{A(k_u)}$. So $\Phi_A = \sum_v \Phi_{A,v} \otimes \bigotimes_{u \neq v} F_u$ with $\Phi_{A,v}$ continuous. On $\{H_S = X\}$ the v-th summand is supported where $\tilde{a}_u \in \operatorname{supp} F_u$ for $u \neq v$, and then $H_v(\tilde{a}_v) = X - \sum_{u \neq v} H_u(\tilde{a}_u)$ is bounded; this set is compact. The last statement is Example 6.C.

(b) By Proposition 6.8(c) and Fourier inversion for $f_{B,y} \in C_c^\infty(\overline{A(\mathbb{Q}_S)})_-$ (Lemma 6.1(b)),

$$\Phi_A(\tilde{a}, f^y) = \Phi_A(\tilde{a}, f) + f_{B,y}(\tilde{a}) + f_{B,y}(w\tilde{a}w^{-1}),$$

since the inverse transform of $\pi \mapsto \hat{h}(\pi^w)$ is $\tilde{a} \mapsto h(w\tilde{a}w^{-1})$. Subtract this from Lemma 6.3(b), and at $\tilde{a} = z$ from Lemma 6.3(c).

(c) $J_A(w\gamma w^{-1}, f) = J_A(\gamma, f)$ by the substitution $x \mapsto wx$ in (4.12), as after Theorem 4.11; and (a). (d) For such γ_v the orbital integral of e_v is 1 and its weighted orbital integral is 0, because $v_{A,v} = 0$ on the support (the proof of Theorem 4.11); and Proposition 6.8(d) gives $\Phi_A(\gamma\gamma_v, f \otimes e_v) = \Phi_A(\gamma, f)$. □

For connected reductive groups, $I_M(\gamma, f)$ is defined in [Ar88b, §2] by $I_M(\gamma, f) = J_M(\gamma, f) - \sum_{L \in \mathscr{L}(M), L \neq G} \hat{I}_M^L(\gamma, \phi_L(f))$, and for covers in [Li14b, §4.5]. Here $\mathscr{L}(A) = \{A, G\}$, the only term is $L = A$, and $\hat{I}_A^A(\gamma, \phi_A(f))$ is the value at γ of the function with Fourier transform $\phi_A(f)$, which is $\Phi_A(\gamma, f)$. For a general group the corresponding step needs the trace Paley–Wiener theorem for L; for $L = A$ it is Fourier inversion on an abelian group. Two further remarks.

The space containing $\varphi_A(f)$. Arthur's ϕ_M maps $\mathscr{H}(G(F_S))$ to a space $\mathscr{I}_{\mathrm{ac}}(M(F_S))$ of functions ϕ on $\Pi_{\mathrm{temp}}(M(F_S))$ for which

$$\phi(\pi,X)=\int_{i\mathfrak{a}_M^*}\phi(\pi_\Lambda)\,e^{-\Lambda(X)}\,d\Lambda,\qquad X\in\mathfrak{a}_M,$$

is, for each X, a Paley–Wiener function of π [Ar89, §11]. In our case $\phi(\pi,X)$ is the Fourier transform of the restriction of $\Phi_A(\cdot,f)$ to $\{H_S=X\}$, and Theorem 6.9(a) is the statement that $\varphi_A(f)\in\mathscr{I}_{\mathrm{ac}}$. This space is strictly larger than the space $\mathscr{I}(\overline{A(\mathbb{A}_S)})$ of Fourier transforms of compactly supported functions, for the reason given in Section 4.4 for the spaces of test functions: $\mathfrak{a}_A\neq 0$ (Example 6.C).

Weighted characters. The weighted character of [Ar88b] is $J_A(\pi,f)=\mathrm{tr}(\mathscr{R}_A(\pi,B)I(\pi,f))=\varphi_A(f,\pi)$, so the invariant distribution $I_A(\pi,X,f)=J_A(\pi,X,f)-\hat{I}_A^A(\pi,X,\varphi_A(f))$ vanishes for every $\pi\in\Pi_u(\overline{A(\mathbb{Q}_S)})$. This does not make the spectral side of the invariant formula vanish: the terms that remain are characters, with the coefficients of Theorem 6.11.

Example 6.C. Let v be real, π_v and m_0 as in Lemma 6.6(b), and let P_j be the projection of $V(\pi_v)$ onto the weight m_0+2j. For $f_v\in C_c^\infty(\overline{SL(2,\mathbb{R})})_-$ let F_j be the function on $\overline{A(\mathbb{R})}$ with $\widehat{F_j}(\pi)=\mathrm{tr}(P_jI(\pi,f_v))$; then $\sum_j F_j=F_{f_v}$, the Abel transform in the proof of Theorem 6.9(a). By (6.9), $\varphi_{A,v}(f_v,\pi_{v,it})=\sum_j\rho_j(t)\widehat{F}_j(\pi_{v,it})$, and since $\frac{1}{2\pi}\int_{\mathbb{R}}\frac{2c}{c^2+t^2}e^{-itX}\,dt=e^{-c|X|}$ for $c>0$,

$$\Phi_{A,v}(\tilde{a},f_v)=\sum_j\int_{\mathbb{R}}\kappa_j(X)\,F_j\big(\tilde{a}\,a(e^X)\big)\,dX,\qquad \kappa_j(X)=\sum_i e^{-c_i|X|},$$

with c_i as in (6.9). So $\Phi_{A,v}(\cdot,f_v)$ is the Abel transform smoothed, on each $\overline{K_v}$-type, by a kernel $\kappa_j\geq 0$ that decays exponentially but is not compactly supported; it vanishes on the weight m_0. For example, if f_v transforms on both sides under $\overline{K_v}$ by the character of weight m_0+2j with $j\neq 0$, and $F_j=F_{f_v}\geq 0$ is not zero, then $\Phi_{A,v}(\tilde{a},f_v)>0$ at every $\tilde{a}$ in the cosets of $\{a(s)\mid s>0\}$ that meet the support of F_{f_v}, although F_{f_v} is compactly supported.

6.4 The invariant trace formula

This subsection assembles the invariant trace formula from the distributions of Section 6.3.

We return to $f=f_S\otimes e^S$ with $f_S\in\mathscr{H}(\overline{G(\mathbb{Q}_S)})$, and to the situation of Theorem 5.16. For $\gamma\in A(\mathbb{Q})$ put $\Phi_A(\gamma,f)=\Phi_A(\gamma_S,f_S)$ if $\gamma\in K_v$ for all $v\notin S_k$, and $\Phi_A(\gamma,f)=0$ otherwise; for $\mu\in\Pi_{\mathrm{aut}}(X_A)$ put $\varphi_A(f,\mu_\lambda)=\varphi_A(f_S,\mu_{S,\lambda})$ if μ_v is trivial on $A(\mathscr{O}_v)^*$ for all $v\notin S_k$, and 0 otherwise.

Lemma 6.10. *$\Phi_A(\gamma,f)=0$ for all but finitely many $\gamma\in A(\mathbb{Q})$, and*

$$\sum_{\mu\in\Pi_{\mathrm{aut}}(X_A)}\frac{1}{2\pi}\int_{\mathbb{R}}\varphi_A(f,\mu_{it})\,dt=c_k\sum_{\gamma\in A(\mathbb{Q})}\Phi_A(\gamma,f),$$

the left side converging absolutely.

Proof. We may take $f_S=\bigotimes_v f_v$. Let $h=\Phi_A(\cdot,f_S)\otimes e_A^S$ on $\overline{A(\mathbb{A})}$, where $e_A^S=\bigotimes_{v\notin S_k}e_{A,v}$ is the product of the units of Lemma 3.5(b). By Theorem 6.9(a) the restriction of h to $\overline{A(\mathbb{A})^1}$

is compactly supported; it is smooth at the archimedean places and locally constant at the finite ones, because $\varphi_A(f_S,\cdot)$ decreases rapidly and is supported on finitely many components. So $\gamma\mapsto h(\gamma)=\Phi_A(\gamma,f)$ has finite support on the discrete subgroup $A(\mathbb{Q})$, and Theorem 3.4 applies to $h|_{\overline{A(\mathbb{A})^1}}$: $\sum_\mu \int_{\overline{A(\mathbb{A})^1}} h\mu = c_k\sum_\gamma h(\gamma)$. Finally $\int_{\overline{A(\mathbb{A})^1}} h\mu = \frac{1}{2\pi}\int_{\mathbb{R}}\hat{h}(\mu_{it})\,dt$ and $\hat{h}(\mu_{it})=\varphi_A(f,\mu_{it})$, by Fourier inversion on $\mathbb{R}$. The left side converges absolutely by Proposition 6.8(a): the μ that contribute form finitely many cosets of a lattice in the unitary dual of $A(\mathbb{Q}_S)$ modulo $i\mathbb{R}$. □

For $\mathscr{O}\in\mathscr{C}$ and $\chi\in\mathscr{X}$ define

$$I_{\mathscr{O}}(f)=J_{\mathscr{O}}(f)-\frac{c_k}{2}\sum_{\gamma\in\mathscr{O}\cap A(\mathbb{Q})}\Phi_A(\gamma,f),\qquad I_\chi(f)=J_\chi(f)-\frac{1}{2}\sum_{\mu\in\chi}\frac{1}{2\pi}\int_{\mathbb{R}}\varphi_A(f,\mu_{it})\,dt, \tag{6.14}$$

the second sum being empty for $\chi\in\mathscr{X}_G$. That is, $I=J-J'$ with $J'=|W_G|^{-1}(I^A)^\wedge(\varphi_A(f))$, where $(I^A_{\mathscr{O}})^\wedge$ and $(I^A_\chi)^\wedge$ are the distributions (6.4) transported to the Fourier transform side.

Theorem 6.11 (The invariant trace formula)**.** *Let S and $f=f_S\otimes e^S$ be as in Theorem 5.16. Then*

$$\sum_{\mathscr{O}\in\mathscr{C}} I_{\mathscr{O}}(f)=\sum_{\chi\in\mathscr{X}} I_\chi(f),$$

both sides converging absolutely, and:

(a) for the central classes $\mathscr{O}_z$, $z\in\{\pm1\}$,

$$I_{\mathscr{O}_z}(f)=a^G(S,1)f_S(z)+\sum_{\xi\in k_S^\times/k_S^{\times2}} a^G(S,u_\xi)\,J_G(zu_\xi,f_S)+\tfrac{1}{2}c_k\,I_A(z,f_S);$$

(b) for $\mathscr{O}$ elliptic, $I_{\mathscr{O}}(f)=J_{\mathscr{O}}(f)=\operatorname{vol}(T_\mu(\mathbb{Q})\backslash T_\mu(\mathbb{A}))J_G(\mu_S,f_S)$;

(c) for $\mathscr{O}$ hyperbolic, the class of $\gamma=a(t)$, $I_{\mathscr{O}}(f)=c_k I_A(\gamma_S,f_S)$ if $t\in\mathscr{O}_v^\times$ for all $v\notin S_k$, and $I_{\mathscr{O}}(f)=0$ otherwise;

(d) for χ the class of (G,π), $I_\chi(f)=m(\pi)\operatorname{tr}\pi(f)$;

(e) for $\chi\in\mathscr{X}_A$ unramified and $\mu\in\chi$,

$$I_\chi(f)=-\frac{1}{2\pi}\int_{\mathbb{R}}\frac{m'}{m}(\mu,it)\operatorname{tr}I(\mu_{it},f)\,dt;$$

(f) for $\chi=\chi_b$ ramified and $\mu=\chi_{\psi_b}$,

$$I_\chi(f)=\operatorname{tr}\big(R(f)|_{V_b}\big)-\frac{1}{2}\cdot\frac{1}{2\pi}\int_{\mathbb{R}}\frac{m'}{m}(\mu,it)\operatorname{tr}I(\mu_{it},f)\,dt+\frac{1}{4}\operatorname{tr}\big(M(w,0)I(\mu_0,f)\big).$$

Each of the distributions (a)–(f) is the restriction to $\mathscr{H}(\overline{G(\mathbb{Q}_S)})$ of an invariant distribution in $f_S\in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$. Here $a^G(S,u_\xi)$ are the coefficients of Theorem 4.11, and m and I_A are formed with the normalizing factors fixed in Section 6.2.

Proof. By Theorem 5.16 and Lemma 6.10, $\sum_{\mathscr{O}} I_{\mathscr{O}}(f) - \sum_{\chi} I_{\chi}(f) = -\frac{1}{2}\big(c_k \sum_{\gamma} \Phi_A(\gamma,f) - \sum_{\mu} \frac{1}{2\pi}\int \varphi_A(f,\mu_{it})\,dt\big) = 0$. (a) is Theorem 5.16(a) and (6.13), since $\mathscr{O}_z \cap A(\mathbb{Q}) = \{z\}$. (b) $\mathscr{O} \cap A(\mathbb{Q})$ is empty. (c) $\mathscr{O} \cap A(\mathbb{Q}) = \{\gamma, \gamma^{-1}\}$ and $\Phi_A(\gamma^{-1}, f) = \Phi_A(\gamma, f)$ by Theorem 6.9(a), because $\gamma^{-1} = w\gamma w^{-1}$; with Theorem 5.16(c) this gives $c_k(J_A - \Phi_A)(\gamma_S, f_S)$. If t is not a unit at some $v \notin S_k$, both $J_{\mathscr{O}}(f)$ and $\Phi_A(\gamma, f)$ vanish. (d) is Theorem 5.16(d). (e), (f) By Theorem 5.14 at $T = 0$ and Corollary 6.7(a),

$$-\operatorname{tr}\big(M^{-1}M'(w,it)I(\mu_{it},f)\big) = -\frac{m'}{m}(\mu,it)\operatorname{tr} I(\mu_{it},f) + \varphi_A(f,\mu_{it})$$

when μ is unramified outside S, and both sides vanish otherwise. In the unramified case the two members of χ contribute equally to (6.14), by Proposition 6.8(b) and $t \mapsto -t$, so φ_A cancels; in the ramified case J_χ has the factor $\frac{1}{2}$ and χ has one member. Invariance: (b) and (d) are an orbital integral and a character; the terms of (a) and (c) are invariant by Theorem 6.9(b), and $f_S(z)$ and $J_G(zu_\xi, \cdot)$ are orbital integrals. In (e) and (f), $\operatorname{tr} I(\mu_{it}, \cdot)$ and $\operatorname{tr}(R(\cdot)|_{V_b})$ are characters, and $\operatorname{tr}(M(w,0)I(\mu_0,\cdot))$ is invariant because $M(w,0)$ commutes with $I(\mu_0)$. The absolute convergence is that of Theorem 5.16 and Lemma 6.10. □

Theorem 6.11 is the invariant trace formula of [Li14b] for this group. Its terms are explicit with one exception. (The unipotent coefficients $a^G(S, u_\xi)$ in (a) are computed in Proposition 7.10.) The normalizing factors r_v at the places of $S_{\mathrm{bad}}(\mu)$, which always include the dyadic places, are those of Theorem 6.5, defined but not evaluated in this section (for their evaluation by means of [Sz09, Sz16] see Proposition 7.8′); the function m'/m in (e) and (f) and the distributions I_A in (a) and (c) depend on them (Remark 2026.15). The elliptic terms (b), the discrete terms (d), the residual term and the ramified term in (f) do not involve them.

It remains to evaluate $M(w,0)$ in (f). For $a \in k^\times$ let η_a be the quadratic character $t \mapsto \prod_v (-a, t_v)_v$ of $k^\times \backslash \mathbb{A}_k^\times$.

Lemma 6.12. *Let $\mu = \chi_\psi \cdot (\chi \circ p) \in \Pi_{\mathrm{aut}}(X_A)$, $\varphi = \bigotimes_v \varphi_v \in \mathscr{H}_B(\mu)$ and $a \in k^\times$. For $\operatorname{Re}\lambda > 1$ put*

$$W_a(x,\varphi,\lambda) = \int_{k\backslash\mathbb{A}_k} E\big(n(u)x,\varphi,\lambda\big)\,\psi(-au)\,du.$$

(a) $E(x,\varphi,\lambda) = E_B(x,\varphi,\lambda) + \sum_{a\in k^\times} W_a(x,\varphi,\lambda)$, and $W_a(x,\varphi,\lambda) = \prod_v W_{a,v}(x_v,\varphi_v,\lambda)$ with $W_{a,v}(x_v,\varphi_v,\lambda) = \int_{k_v} \varphi_{v,\lambda}\big((w,1)^{-1}(n(u),1)x_v\big)\psi_v(-au)\,du$.

(b) Each $W_{a,v}(x_v,\varphi_v,\lambda)$ is an entire function of λ.

(c) If v is as in (N4), $|a|_v = 1$, $\varphi_v = \varphi_v^0$ and $x_v \in K_v^*$, then

$$W_{a,v}(x_v,\varphi_v^0,\lambda) = 1 + \eta_{a,v}(\varpi)\chi_v(\varpi)q^{-\lambda-\frac{1}{2}} = \frac{L_v(\lambda+\frac{1}{2},\chi_v\eta_{a,v})}{L_v(2\lambda+1,\chi_v^2)}.$$

(d) Hence, for a finite set S_1 of places containing those not covered by (c),

$$W_a(x,\varphi,\lambda) = \frac{L^{S_1}(\lambda+\frac{1}{2},\chi\eta_a)}{L^{S_1}(2\lambda+1,\chi^2)}\prod_{v\in S_1} W_{a,v}(x_v,\varphi_v,\lambda),$$

which continues W_a meromorphically to $\mathbb{C}$.

Proof. (a) The first formula is the Fourier expansion on $k\backslash\mathbb{A}_k$. By the Bruhat decomposition $G(\mathbb{Q}) = B(\mathbb{Q}) \sqcup B(\mathbb{Q})w^{-1}N(\mathbb{Q})$, for $a \neq 0$ the series (5.2) unfolds to $\int_{\mathbb{A}_k} \varphi_\lambda(w^{-1}n(u)x)\psi(-au)\,du$, which is the product. (b) For v finite and $|u|$ large, $w^{-1}n(u) = n(-u^{-1})a(-u^{-1})\bar{n}(u^{-1})$ with $\bar{n}(u^{-1})$ in a subgroup that fixes $I(x_v)\varphi_v$, so the integrand is $|u|^{-\lambda-1}$ times a function of u that is invariant under multiplication by a subgroup $1+\varpi^c\mathscr{O}$; its integral against $\psi_v(-au)$ over a shell $|u| = q^m$ vanishes for m large, and $W_{a,v}$ is a polynomial in $q^{\pm\lambda}$. For v archimedean the integrand is smooth in u with an asymptotic expansion in powers of u^{-1} times $|u|^{-\lambda-1}$, and repeated integration by parts against $\psi_v(-au)$ continues the integral to all λ. (c) Proceed as for Lemma 5.6(b). On $|u| \leq 1$ the integrand is 1. On the shell $|u| = q^m$, $m \geq 1$, write $u = \varpi^{-m}u_0$; the decomposition $(w,1)^{-1}(n(u),1) = (n(-u^{-1}),1)(a(-u^{-1}),1)(\bar{n}(u^{-1}),1)$ holds in the cover, because every value of Kubota's cocycle that occurs is of the form $(1,\cdot)_v$ or $(u,-u)_v$, hence 1, and $\kappa_v = 1$ on all factors. So the integrand is $\mu_v(a(-u^{-1}),1)|u|^{-\lambda-1} = \chi_v(\varpi)^m|u|^{-\lambda-1}\gamma_v(-\varpi^m u_0)$, and by Lemma 3.3(a)–(b) $\gamma_v(-\varpi^m u_0)$ is 1 for m even and $(-u_0,\varpi)_v\gamma_v(\varpi)$ for m odd. For m even the integral of $\psi_v(-au)$ over the shell is 0 if $m \geq 2$. For m odd the integral of $(u_0,\varpi)_v\psi_v(-au)$ is 0 if $m \geq 3$, and for $m = 1$ it is the Gauss sum $\sum_{u_0 \in (\mathscr{O}/\varpi)^\times}(u_0,\varpi)_v\psi_v(-a\varpi^{-1}u_0) = (-a,\varpi)_v q^{1/2}\gamma_v(\varpi)$, by (3.2). The shell $m = 1$ therefore contributes $(-1,\varpi)_v(-a,\varpi)_v\gamma_v(\varpi)^2\chi_v(\varpi)q^{-\lambda-\frac{1}{2}}$, and $\gamma_v(\varpi)^2 = (-1,\varpi)_v$. (d) follows. □

Part (c) is the unramified Whittaker function of a genuine principal series at the identity; compare [GPS80]. For $SL(2)$ the same computation gives $1-\chi_v(\varpi)q^{-\lambda-1} = L_v(\lambda+1,\chi_v)^{-1}$. The script checks both at $p = 3,5,7,11$.

Proposition 6.13. *Let $b \in k^\times$ and $\mu = \chi_{\psi_b}$. Then $E(x,\varphi,0) = 0$ for all $\varphi \in \mathscr{H}_B(\mu)$, and*

$$M(w,0) = -1 \quad \text{on } \mathscr{H}_B(\chi_{\psi_b}).$$

Consequently the ramified term of Theorem 5.14(c) is $\frac{1}{4}\operatorname{tr}(M(w,0)I(\mu_0,f)) = -\frac{1}{4}\operatorname{tr} I(\chi_{\psi_b,0},f)$.

Proof. Write $\mu = \chi_\psi\cdot(\chi\circ p)$ with $\chi = \prod_v(b,\cdot)_v$ (Lemma 3.3(d)), so $\chi^2 = 1$. It suffices to treat factorizable φ. By Theorem 5.5(d), $E(x,\varphi,\lambda)$ and $M(w,\lambda)$ are holomorphic at $\lambda = 0$.

Step 1: the Whittaker coefficients vanish at $\lambda = 0$. In Lemma 6.12(d) the denominator is $L^{S_1}(2\lambda+1,\chi^2) = \zeta_k^{S_1}(2\lambda+1)$, because $\chi^2 = 1$. The Dedekind zeta function has a pole at $s = 1$, and removing finitely many Euler factors does not remove it; so $\zeta_k^{S_1}(2\lambda+1)^{-1}$ has a zero at $\lambda = 0$. The numerator $L^{S_1}(\lambda+\frac{1}{2},\chi\eta_a)$ is the L-function of a quadratic Hecke character at the center of the critical strip, which is finite (for $\chi\eta_a = 1$ it is $\zeta_k^{S_1}(\frac{1}{2})$), and the local factors at S_1 are entire. So $W_a(x,\varphi,0) = 0$ for every $a \in k^\times$ and every x. The zero is specific to the cover: the unramified Whittaker function has the denominator $L_v(2\lambda+1,\chi_v^2)$, in which the character is squared, so every ramified class sees the pole of ζ_k.

Step 2: the Eisenstein series is its constant term. By Lemma 6.12(a) and Theorem 5.5(b),

$$F(x) := E(x,\varphi,0) = E_B(x,\varphi,0) = \big(\varphi + M(w,0)\varphi\big)(x)\,e^{H(x)}.$$

Step 3: $|F|$ is constant. F is left invariant under $G(\mathbb{Q})$. Being equal to its constant term $\int_{N(\mathbb{Q})\backslash N(\mathbb{A})} F(nx)\,dn$, it is left invariant under $N(\mathbb{A})$. Since $w \in G(\mathbb{Q})$, it is then left invariant under $wN(\mathbb{A})w^{-1}$ as well: $F(wnw^{-1}x) = F(nw^{-1}x) = F(w^{-1}x) = F(x)$. Here $wN(\mathbb{A})w^{-1}$ is the image of the lower triangular unipotent group $\bar{N}(\mathbb{A})$ under its canonical section (Lemma 2.2(h)). Now

$N(\mathbb{A})$ and $\bar{N}(\mathbb{A})$ generate $G(\mathbb{A})$: over a field or a local ring every element of $SL(2)$ is a product of a bounded number of elementary matrices, the bound being independent of the ring, so the factorization can be carried out at all places simultaneously, with integral factors at almost all places. Hence the subgroup of $\overline{G(\mathbb{A})}$ generated by the lifts of $N(\mathbb{A})$ and $\bar{N}(\mathbb{A})$ maps onto $G(\mathbb{A})$, and F is left invariant under it. Since F is genuine, $|F|$ is invariant under μ_2 too. So $|F|$ is left invariant under all of $\overline{G(\mathbb{A})}$, that is, constant.

Step 4: the constant is 0. For $\tilde{a} \in \overline{A(\mathbb{A})}$ the formula of Step 2 gives $|F(\tilde{a}x)| = e^{H(\tilde{a})}|F(x)|$, because φ and $M(w,0)\varphi$ both lie in $\mathscr{H}_B(\mu)$ when $\mu^w = \mu$. Choosing $H(\tilde{a}) \neq 0$ we get $F = 0$, and restricting to $\overline{K_0}$ gives $\varphi + M(w,0)\varphi = 0$. □

The statement is about representations, not only about L-functions: $M(w,0)$ is a self-intertwining operator of the unitary representation $I(\mu_0)$ with $M(w,0)^2 = 1$, and the proposition says that it is the scalar -1 for every ramified class. The proof finds the sign from a global fact: an automorphic function that is equal to its constant term and homogeneous of degree one under A must vanish.

For $SL(2)$ the same argument applies to the trivial character, where $L^{S_1}(\lambda+1,1)^{-1}$ vanishes at $\lambda = 0$: this is the classical fact that the scattering matrix of $SL(2,\mathbb{Z})$ is -1 at $s = \frac{1}{2}$ and $E(z,\frac{1}{2}) = 0$. For a quadratic character $\eta \neq 1$ of $SL(2)$ it does not apply, and $M(w,0)$ is then a product of nonscalar local operators [LL79]; that is the source of the multiplicity formula in [LL79]. On the double cover the denominator is always $\zeta_k^{S_1}(2\lambda+1)$, and every ramified class behaves like the trivial character of $SL(2)$. This agrees with the irreducibility of the unitary principal series of $\overline{SL(2,k_v)}$ [Mo88]. For $k = \mathbb{Q}$ and $b \in \{\pm 1, \pm 2\}$ the script confirms $M(w,0) = -1$ by a separate computation, from the residues at $\lambda = 0$ of the real and the 2-adic local operators and the zero of order $|S|$ of $m^S(\mu,\lambda)$, $S = \{\infty, 2\}$.

Corollary 6.14. *In Theorem 6.11,*

$$\sum_{\chi \in \mathscr{X}} I_\chi(f) = \sum_\pi m_{\mathrm{disc}}(\pi) \operatorname{tr}\pi(f) - \frac{1}{4} \sum_{b \in k^\times/k^{\times 2}} \operatorname{tr} I(\chi_{\psi_b,0}, f) \\ - \frac{1}{2} \sum_{\mu \in \Pi_{\mathrm{aut}}(X_A)} \frac{1}{2\pi} \int_{\mathbb{R}} \frac{m'}{m}(\mu, it) \operatorname{tr} I(\mu_{it}, f)\, dt,$$

where π runs over the irreducible constituents of $L^2_{\mathrm{disc}} = L^2_{\mathrm{cusp}} \oplus \bigoplus_b V_b$ with their multiplicities $m_{\mathrm{disc}}(\pi)$. Every term is a character of $\overline{G(\mathbb{Q}_S)}$ evaluated at f_S.

Proof. Theorem 6.11(d)–(f), Proposition 6.13, and Corollary 6.7(b) to write (e) as half the sum over the two members of χ. □

In Arthur's notation [Ar88c, §4] this is $\sum_t \int_{\Pi(G,t)} a^G(\pi) I_G(\pi, f)\, d\pi$: the coefficient $a^G(\pi)$ is $m_{\mathrm{disc}}(\pi)$ on the discrete spectrum, $-\frac{1}{4}$ at $I(\chi_{\psi_b,0})$, and $-\frac{1}{4\pi}\frac{m'}{m}(\mu,it)\,dt$ on the unitary principal series; the first two make up the discrete part I_{disc}. Since $V_b \cong r^+_{\psi_{-b}}$ (Theorem 5.8(c)), $\operatorname{tr}(R(f)|_{V_b}) = \prod_v \operatorname{tr} r^+_{\psi_{-b},v}(f_v)$, the character of the even Weil representation.

Example 6.D. Let $k=\mathbb{Q}$ and $S=\{\infty,2\}$, so $f=f_\infty\otimes f_2\otimes e^S$. By Example 5.A the ramified classes that contribute are χ_{ψ_b}, $b\in\{\pm1,\pm2\}$, and the discrete correction is $-\frac14\sum_{b=\pm1,\pm2}\operatorname{tr}I(\chi_{\psi_b,0},f)$. For $\mu=\chi_{\psi_b}$ we have $S_{\mathrm{bad}}(\mu)=\{2\}$ and, with $\xi(s)=\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$,

$$m(\mu,\lambda)=c\,\frac{\xi(2\lambda)}{\xi(2\lambda+1)}\cdot\frac{1-2^{-2\lambda}}{1-2^{-1-2\lambda}}\cdot r_2(\mu_2,\lambda),\qquad \frac{\xi(2\lambda)}{\xi(2\lambda+1)}\Big|_{\lambda=0}=-1,$$

where r_2 is the 2-adic factor of Theorem 6.5 (Remark 2026.15) and $c=\sqrt{2}\,e^{\pm i\pi/4}$ comes from Lemma 6.6(b), the sign being that of b. The hyperbolic classes that contribute are those of $a(t)$, $t=\pm2^j$, $j\geq1$, and their terms are $c_{\mathbb{Q}}\,I_A(a(t)_S,f_S)$. The archimedean part of the correction Φ_A is given by Example 6.C.

6.5 Status of the results

This subsection lists the statements of Section 6 that rest on citation or on numerical checks.

Remark 2026.16. *What rests on citation.* Proved here, given Sections 3–5 and Theorem 6.5: Lemmas 6.1–6.4; Lemma 6.6(b), and (a) except for rationality; Corollary 6.7; Proposition 6.8; Theorem 6.9; Lemma 6.10; Theorem 6.11; Lemma 6.12; Proposition 6.13; and Corollary 6.14. Taken from the literature: Theorem 6.5 [Ar89, Li12a], of which only the bad places are not made explicit (Remark 2026.15); the generic irreducibility of $I(\pi_{\nu,\lambda})$ and the adjoint formula used in Lemma 6.6(a); the estimates behind Proposition 6.8(a) for functions that are not $\overline{K_S}$-finite; and, in the proof of Lemma 6.2, the interchange of integrals on the spectral side. We have not compared the definition (6.13) with the definitions of [Li14b] beyond the remark after Theorem 6.9. As in Theorem 5.16, the set S must be large for the given f_S; the coefficients $a^G(S,u_\xi)$ are computed in Proposition 7.10.

Remark 2026.17. The script `checks/sec6_n2_check.py` verifies, for $k=\mathbb{Q}$: the sign of the cross terms in the Maass–Selberg relation (5.11), hence of the ramified term in Theorem 5.14(c), on the Eisenstein series of $SL(2,\mathbb{Z})$ (by numerical integration over the modular surface); Lemma 6.1(a) and Example 6.A; Lemma 6.3(a)–(b) at $v=\mathbb{R}$, with explicit Iwasawa decompositions and by quadrature for a Gaussian f; the identity (6.5) for the principal series of $SL(2,\mathbb{R})$ in the compact picture; formulas (6.7)–(6.9) and the Fourier transform in Example 6.C; Lemma 6.12(c) at $p=3,5,7,11$, shell by shell in the cover, and its analog for $SL(2)$; Proposition 6.13 for $b=\pm1,\pm2$, from local residues; and the value -1 in Example 6.D.

7 The trace formula in stable(r) form

The invariant trace formula of Theorem 6.11 is an identity between distributions that are invariant under conjugation by the cover itself. For $SL(2)$, Labesse and Langlands [LL79] went one step further: they regrouped the terms according to conjugacy under $GL(2)$, which for $SL(2)$ is stable conjugacy, and identified the part of the formula that is not stably invariant with contributions of the elliptic tori. This section carries out the first half of that program for the double cover: it determines how $GL(2)$ acts on the cover, what stable conjugacy is, and how each term of the trace formula behaves. We call the result the *stable(r) form*: the terms are organized by stable classes,

and the elliptic regular terms are written as stable orbital integrals, but nothing is transferred to another group.

Three properties of a distribution on the cover occur below and should be kept apart. *Equivariance*: it is multiplied by a character of $\det h$ when f is replaced by $f^h = f \circ \mathrm{Ad}(h)$, $h \in GL(2)$; this is about ordinary conjugation, and it is what Section 7.5 proves for the weighted terms. *Stability*: it is invariant under the twisted action $f \mapsto f^{[h]}$ of Section 7.2; this is proved for the regular elliptic terms, the characters of the principal series and the theta functions. *Endoscopic transfer*: it is identified with a stable distribution on $SO(3)$ applied to a transfer of f; this is Li's theorem, and nothing of the kind is proved here.

For the double cover of $Sp(2n)$ the full theory exists and is due to Li: stable conjugacy and the transfer of orbital integrals [Li11], following Adams [Ad98], Renard [Re99] and, for the group here over a p-adic field, Schultz [Sc98] (see also [Ho10]); the stabilization of the elliptic terms [Li15]; and the stabilization of the whole invariant trace formula [Li21]. The elliptic endoscopic groups of the double cover of $Sp(2n)$ are the groups $SO(2n'+1) \times SO(2n''+1)$, $n'+n''=n$. For $n=1$ these are two copies of $SO(3) = PGL(2)$, attached to the pairs $(1,0)$ and $(0,1)$; the tori of $SL(2)$, which are the endoscopic groups of $SL(2)$ in [LL79], do not occur. We do not reprove any of this. What can be done by hand for the group of this paper, and is done below, is the following.

Results.

- $GL(2,F)$ acts on $\overline{SL(2,F)}$ by automorphisms $\mathrm{Ad}(h)$, with an explicit formula (Lemma 7.1). An element h that commutes with a semisimple γ multiplies the lifts of γ by the sign $(\det h, \det(1+\gamma))_v$ (Lemma 7.2). So conjugation by $GL(2)$ can carry a lift of γ to the other lift, and it is not stable conjugacy.

- Stable conjugacy on the cover is $GL(2)$-conjugacy *twisted* by that sign: $h \star x = (\det h, \det(1+x))_v \, \mathrm{Ad}(h)x$. Its classes are the level sets, over the stable classes of $SL(2,F)$, of an explicit genuine function Δ_ψ with values in μ_4 (Lemma 7.3). This function is $|\det(1+x)|^{1/2}$ times the character of the difference $r_\psi^+ - r_\psi^-$ of the two halves of the Weil representation (Proposition 7.5), so the definition agrees with that of Adams and Li.

- For a regular elliptic $\gamma \in G(\mathbb{Q})$ with splitting field E, the terms of the trace formula attached to the $GL(2,k)$-conjugates of γ add up to $\frac{1}{2}$ of a stable orbital integral plus $\frac{1}{2}$ of a κ-orbital integral, as for $SL(2)$. For the cover the κ-orbital integral at γ is again a stable orbital integral: that of the translate f_- of f by the central element -1, at $-\gamma$ (Lemma 7.6). Hence the elliptic regular part of the trace formula is $S_{\mathrm{ell}}(f) + S_{\mathrm{ell}}(f_-)$ with one stable distribution S_{ell} (Theorem 7.7). This is the two-term pattern, with coefficients $\frac{1}{2}$ and $\frac{1}{2}$, of the elliptic part of Li's stable trace formula for $n=1$ with its two endoscopic data; the terms are not matched with the transferred distributions on $SO(3)$.

- The remaining geometric terms transform under $GL(2,k_S)$ by explicit characters (Proposition 7.9), and the unipotent coefficients $a^G(S,u_\xi)$, left open in Section 4, are computed: $a^G(S,u_\xi) = \frac{1}{2}\big(c_S + \sum_{\kappa \neq 1} \kappa_S(\xi) L^S(1,\kappa)\big)$, the sum over the quadratic Hecke characters unramified outside S (Proposition 7.10).

- On the spectral side, $GL(2,k)$ permutes the pieces of the spectrum, and $V_b \mapsto V_{ab}$. The theta functions attached to ψ_{-b} span a subspace of the discrete spectrum whose trace is

$\frac{1}{2}\Theta(f)+\frac{1}{2}\Theta(f_-)$ with Θ a stable character (Proposition 7.12): the simplest instance of the same pattern on the spectral side.

The hyperbolic terms, the unipotent terms and the cuspidal spectrum are *not* stabilized here; Section 7.8 says what is missing, and Remark 2026.19 what rests on citation.

Conventions. $F=k_v$ is a local field of characteristic 0, as in Section 2.1; we write $G_1=GL(2)$, Z for its center, and $\delta_a=\mathrm{diag}(a,1)$. For $h\in G_1(F)$, $\mathrm{Ad}(h)$ is the automorphism of $\overline{SL(2,F)}$ of Lemma 2.10(c), and for a function f on the cover

$$f^h(x)=f(\mathrm{Ad}(h)x),$$

which for $h\in SL(2,F)$ is the function f^y of Section 6, y a lift of h. For a central element z of a cover, $f_z(x)=f(zx)$, as in Section 4.3. The element $-1\in G(\mathbb{Q})$ is identified with its image $s(-1)=(-1,1)$ in $\overline{G(\mathbb{A})}$, and

$$f_-(x)=f_{-1}(x)=f\big(s(-1)x\big),\qquad f\in\mathscr{H}(\overline{G(\mathbb{A})}).$$

Since $s(-1)^2=1$, $(f_-)_-=f$. For a quadratic extension $E=F(\sqrt{\theta})$ let $\kappa_E=(\cdot,\theta)_v$ be the quadratic character of $F^\times$ with kernel $NE^\times=N_{E/F}(E^\times)$; for $E=F\times F$ put $NE^\times=F^\times$ and $\kappa_E=1$.

7.1 Conjugation by $GL(2)$

Conjugation by $h\in G_1(F)$ is an automorphism of $SL(2,F)$, and it lifts to the cover in exactly one way (for the metaplectic cover of $Sp(V)$ and the similitude group in general, see [Th13, §3]). This subsection makes the lift explicit and computes it on the subgroups that matter: N, A, the center, the maximal compact subgroup, the tori, and globally on $G(\mathbb{Q})$.

Lemma 7.1. *Let $h\in G_1(F)$.*

(a) $\mathrm{Ad}(h)$ *is the only automorphism of the group $\overline{SL(2,F)}$ that induces $g\mapsto hgh^{-1}$ on $SL(2,F)$. It is a homeomorphism for the topology of Lemma 2.1(c), and it fixes μ_2 pointwise.*

(b) $\mathrm{Ad}(h)$ *depends only on the image of h in $PGL(2,F)$. For $h\in SL(2,F)$ it is conjugation by either lift of h. So* $\mathrm{Ad}(h)$ *is inner when* $\det h\in F^{\times 2}$*, and the group $F^\times/F^{\times 2}$ acts through the* $\mathrm{Ad}(\delta_a)$ *modulo inner automorphisms.*

(c) For $a\in F^\times$ and $g=\begin{pmatrix}\alpha & b\\ c & d\end{pmatrix}\in SL(2,F)$,

$$\mathrm{Ad}(\delta_a)(g,\zeta)=\big(\delta_a g\delta_a^{-1},\ v_a(g)\,\zeta\big),\qquad v_a(g)=\begin{cases}1, & c\neq 0,\\ (a,d)_v, & c=0.\end{cases}\tag{7.1}$$

(d) In particular

$$\mathrm{Ad}(\delta_a)(n(u),1)=(n(au),1),\qquad \mathrm{Ad}(\delta_a)(a(t),\zeta)=(a(t),(a,t)_v\zeta),$$
$$\mathrm{Ad}(\delta_a)(-1,\zeta)=(-1,(a,-1)_v\zeta),\qquad \mathrm{Ad}(\delta_a)(w,1)=(a(a),1)(w,1).$$

(e) *Let v be finite and non-dyadic. If $h \in Z(F)GL(2,\mathscr{O})$, then* $\mathrm{Ad}(h)K^* = K^*$ *and* $e_v^h = e_v$ *for the unit element e_v of* (2.8).

(f) *(Global.) For $h = (h_v) \in G_1(\mathbb{A}_k)$ the automorphisms* $\mathrm{Ad}(h_v)$ *define an automorphism* $\mathrm{Ad}(h)$ *of $\overline{G(\mathbb{A})}$, and $f^h \in \mathscr{H}(\overline{G(\mathbb{A})})$ for $f \in \mathscr{H}(\overline{G(\mathbb{A})})$. If $h \in G_1(k)$, then* $\mathrm{Ad}(h)$ *preserves the image of $G(\mathbb{Q})$:* $\mathrm{Ad}(h)s(\gamma) = s(h\gamma h^{-1})$ *for $\gamma \in G(\mathbb{Q})$, where $s(\gamma) = (\gamma, \kappa_{\mathbb{A}}(\gamma))$ is the splitting of Lemma 2.6(a).*

Proof. (a) Two automorphisms that induce the same automorphism of $SL(2,F)$ differ by a homomorphism $SL(2,F) \to \mu_2$, which is trivial because $SL(2,F)$ is perfect. $\mathrm{Ad}(h)$ fixes μ_2 because μ_2 is central in the cover of $GL(2,F)$. For continuity it suffices to look near the identity. Let U be as in Lemma 2.1(b), and let $V \subset U$ be a neighborhood of 1 with $hVh^{-1} \subset U$. For $g \in V$ write $\mathrm{Ad}(h)(g,1) = (hgh^{-1}, \varepsilon(g))$. Since $\beta \equiv 1$ on $U \times U$, $\varepsilon(g_1g_2) = \varepsilon(g_1)\varepsilon(g_2)$ whenever $g_1, g_2, g_1g_2 \in V$; and every g in a smaller neighborhood is a square of an element of V, as in the proof of Lemma 2.1(c). So $\varepsilon = 1$ near 1, and $\mathrm{Ad}(h)$ maps the basic neighborhoods $\{(g,1) \mid g \in V'\}$ to basic neighborhoods.

(b) A scalar matrix induces the identity on $SL(2,F)$, so by (a) it acts trivially on the cover; and conjugation by a lift of $h \in SL(2,F)$ is an automorphism over $g \mapsto hgh^{-1}$. If $\det h = c^2$, then $c^{-1}h \in SL(2,F)$.

(c) By definition $\mathrm{Ad}(\delta_a)(g,\zeta) = (\delta_a,1)(g,\zeta)(\delta_a,1)^{-1}$ in $\overline{GL(2,F)}$, which is

$$\left(\delta_a g \delta_a^{-1},\ \zeta\, \sigma(\delta_a, g)\, \sigma(\delta_a g, \delta_a^{-1})\, \sigma(\delta_a, \delta_a^{-1})\right)$$

with the cocycle (2.13). Here $x(\delta_a) = x(\delta_a^{-1}) = 1$ and $x(\delta_a g) = x(g)$, so $\sigma(\delta_a, g) = (x(g), a)_v$ and $\sigma(\delta_a, \delta_a^{-1}) = 1$. If $c \neq 0$, then $x(\delta_a g \delta_a^{-1}) = c/a$ and $\sigma(\delta_a g, \delta_a^{-1}) = (a^{-1}, c/a^2)_v = (a,c)_v$, so the product is 1. If $c = 0$, then $x(\delta_a g \delta_a^{-1}) = d = x(g)$ and $\sigma(\delta_a g, \delta_a^{-1}) = (1, d/a)_v = 1$, so the product is $(d,a)_v$. (The script also checks directly that (7.1) is a homomorphism for β, which by (a) proves (c) without reference to the cover of $GL(2)$.)

(d) The first three are (7.1) with $c = 0$ and $d = 1$, t^{-1}, -1. For the last, $\delta_a w \delta_a^{-1} = a(a)w$ has lower left entry $a^{-1} \neq 0$, so $\mathrm{Ad}(\delta_a)(w,1) = (a(a)w, 1)$, and $\beta(a(a), w) = (1, a^{-1})_v = 1$.

(e) By (b) we may take $h \in GL(2,\mathscr{O})$, so $hKh^{-1} = K$. Then $k \mapsto \mathrm{Ad}(h)^{-1}(hkh^{-1}, \kappa(hkh^{-1}))$ is a homomorphic section of $\overline{K} \to K$. It differs from $k \mapsto (k, \kappa(k))$ by a homomorphism $K = SL(2,\mathscr{O}) \to \mu_2$, and there is none except the trivial one: K is generated by elementary matrices, which are squares.

(f) By (e), $\mathrm{Ad}(h_v)$ preserves K_v^* for almost all v, so the product acts on the restricted product of the local covers; it fixes every μ_2, hence the subgroup Z_0 of Section 2.2, and descends to $\overline{G(\mathbb{A})}$. For $f = \bigotimes f_v$ we get $f^h = \bigotimes f_v^{h_v}$ with $f_v^{h_v} = e_v$ for almost all v. If $h \in G_1(k)$, then $\gamma \mapsto \mathrm{Ad}(h)^{-1} s(h\gamma h^{-1})$ is a homomorphic section of $\overline{G(\mathbb{Q})} \to G(\mathbb{Q})$; it equals s because $SL(2,k)$ is perfect. □

Part (d) already shows that $\mathrm{Ad}(h)$ is not inner in general: it acts on the abelian group $\overline{A(F)}$ by the character $a(t) \mapsto (a,t)_v$, and it exchanges the two lifts of -1 when $(\det h, -1)_v = -1$. The next lemma gives the general statement. For a semisimple $\gamma \in SL(2,F)$, $\gamma \neq -1$, put

$$D^+(\gamma) = \det(1+\gamma) = 2 + \operatorname{tr}\gamma \ \in F^\times, \qquad \eta_\gamma(a) = \left(a, D^+(\gamma)\right)_v \quad (a \in F^\times). \tag{7.2}$$

The definition of D^+ and η_γ makes sense for every γ with $\operatorname{tr}\gamma \neq -2$, and depends only on the trace.

Lemma 7.2. *Let $\gamma \in SL(2,F)$ be semisimple, $\gamma \neq -1$, and let $h \in G_1(F)$ commute with γ. Then*

$$\mathrm{Ad}(h)(\gamma,\zeta) = \big(\gamma,\ \eta_\gamma(\det h)\,\zeta\big).$$

For $\gamma = -1$ and every $h \in G_1(F)$, $\mathrm{Ad}(h)(-1,\zeta) = (-1,(\det h,-1)_v\zeta)$.

Proof. The second statement is Lemma 7.1(b),(d). For $\gamma = 1$ there is nothing to prove ($D^+ = 4$). Let γ be regular. Its centralizer in $M_2(F)$ is the commutative algebra $E = F[\gamma]$, and $h \in E^\times$. Put $z = 1+\gamma \in E$. Then $\det z = D^+(\gamma) \neq 0$, and $z\gamma^{-1} = 1+\gamma^{-1}$ is the conjugate $\bar z$ of z in E, so $z\bar z = \det z$ and

$$\gamma\cdot\big(D^+(\gamma)\cdot 1\big) = z^2 \qquad \text{in } E^\times \subset G_1(F).$$

For commuting $x,y \in G_1(F)$ let $[x,y] \in \mu_2$ be the commutator of any two lifts to $\overline{GL(2,F)}$. On the commutative group $E^\times$ it is bimultiplicative, so $[h,z^2] = [h,z]^2 = 1$ and therefore $[h,\gamma] = [h,D^+(\gamma)\cdot 1]$. For a scalar λ and any h, (2.13) gives $\sigma(h,\lambda) = (\lambda, x(h)/\det h)_v$ and $\sigma(\lambda,h) = (x(h),\lambda)_v$, so $[h,\lambda] = (\lambda,\det h)_v$. Hence $\mathrm{Ad}(h)(\gamma,\zeta) = (\gamma,[h,\gamma]\zeta)$ with $[h,\gamma] = (D^+(\gamma),\det h)_v$. □

By (2.14) the sign is also $(x(h\gamma)/x(h),\det h)_v$; the lemma says that this equals $(\det h, 2+\mathrm{tr}\,\gamma)_v$, which the script confirms on random elements of the tori $T_\theta(\mathbb{Q})$. If γ is regular with centralizer $E^\times$ in $G_1(F)$, then $\det h$ runs through $NE^\times$, and $D^+(\gamma) = N_{E/F}(1+\gamma)$ is itself a norm. So the lifts of γ are exchanged by the centralizer of γ in $G_1(F)$ exactly when η_γ is not trivial on $NE^\times$.

Example 7.A. (a) *Split torus.* For $\gamma = a(t)$, $t \neq \pm 1$: $D^+ = (1+t)^2/t$, so $\eta_\gamma = (\cdot,t)_v$ and $NE^\times = F^\times$. The two lifts of $a(t)$ are conjugate under $G_1(F)$ if and only if $t \notin F^{\times 2}$.

(b) $F = \mathbb{R}$, *γ elliptic.* $NE^\times = \mathbb{R}_{>0}$ and $D^+ = 2+2\cos\theta > 0$, so $\eta_\gamma = 1$. In $Mp(2,\mathbb{R})$ the automorphism $\mathrm{Ad}(\delta_{-1})$ sends the element $\tilde k_\theta$ of Lemma 6.6(b) to $\tilde k_{-\theta}$; on the center $\{\tilde k_{j\pi}\} \cong \mathbb{Z}/4$ it is inversion, in agreement with $(-1,-1)_\mathbb{R} = -1$.

(c) *F non-dyadic, E/F unramified, $\gamma \in K$.* The norms are the elements of even valuation, and η_γ is trivial on them if $D^+(\gamma)$ is a unit. But $\eta_\gamma(\varpi) = (\varpi, D^+(\gamma))_v$ is $+1$ or -1 according as the unit $D^+(\gamma)$ is a square or not; both occur.

(d) $F = \mathbb{Q}_3$, $E = \mathbb{Q}_3(\sqrt{3})$. Let $h = \begin{pmatrix} 0 & 3 \\ 1 & 0 \end{pmatrix}$, which is $\sqrt{3} \in E^\times$, of determinant -3. Then $\mathrm{Ad}(h)(-1,\zeta) = (-1,(-3,-1)_3\zeta) = (-1,-\zeta)$. By continuity the elements of $\overline{T_3}$ close to a lift of -1 are conjugate under $G_1(\mathbb{Q}_3)$, indeed under the centralizer $E^\times$, to their negatives; here $D^+(\gamma) \in -3\,\mathbb{Q}_3^{\times 2}$ and $(-3,-3)_3 = -1$.

(e) *A rational example at 2 and ∞.* Let

$$\gamma = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Both lie in $\mathbb{Q}[w]$, so they commute; $\det h = 2$ and $D^+(\gamma) = 16/5$. Over $\mathbb{Q}_2$, $(2,16/5)_2 = (2,5)_2 = -1$: ordinary conjugation by h exchanges the two lifts of γ, and $h\star x = x$ for both. Over $\mathbb{R}$ the sign is $(2,16/5)_\infty = +1$, and over $\mathbb{Q}_5$ it is $(2,5)_5 = -1$, in agreement with the product formula.

7.2 Stable conjugacy

Two elements of $SL(2,F)$ are *stably conjugate* if they are conjugate in $SL(2,\bar{F})$; for semisimple elements this is the same as conjugacy under $G_1(F)$ [LL79]. On the cover, conjugacy under $G_1(F)$ is too coarse, by Example 7.A: it can identify the two lifts of an element, on which every genuine function takes opposite values. The remedy is to twist the action by the sign of Lemma 7.2.

Let $\widetilde{SL(2,F)}^\flat$ be the set of $x \in \widetilde{SL(2,F)}$ whose image $p(x)$ has trace different from -2; it is open, dense, invariant under every $\mathrm{Ad}(h)$, and contains the regular semisimple elements. For $x \in \widetilde{SL(2,F)}^\flat$ write $\eta_x = \eta_{p(x)}$, and for $h \in G_1(F)$ put

$$h \star x = \eta_x(\det h) \cdot \mathrm{Ad}(h)x, \qquad \eta_x(\det h) = \big(\det h,\ \det(1+p(x))\big)_v \in \mu_2. \tag{7.3}$$

Since η_x depends only on the trace of $p(x)$, which $\mathrm{Ad}(h)$ preserves, $(h_1h_2)\star x = h_1 \star (h_2 \star x)$: this is an action of $G_1(F)$ on $\widetilde{SL(2,F)}^\flat$. Scalars act trivially, and $h \in SL(2,F)$ acts by conjugation. Two elements of $\widetilde{SL(2,F)}^\flat$ are called *stably conjugate* if they lie in the same orbit of this action. For a function f on $\widetilde{SL(2,F)}$ put

$$f^{[h]}(x) = f(h\star x) = \eta_x(\det h)\, f^h(x) \qquad (x \in \widetilde{SL(2,F)}^\flat),$$

the last equality for anti-genuine f. The complement of $\widetilde{SL(2,F)}^\flat$ has measure 0, so $f^{[h]}$ is defined almost everywhere; it is smooth on $\widetilde{SL(2,F)}^\flat$ but in general not across $\mathrm{tr} = -2$. A locally integrable class function Θ on $\widetilde{SL(2,F)}$, and the invariant distribution $f \mapsto \int f\Theta$ it defines, are called *stable* if $\Theta(h\star x) = \Theta(x)$ for all $h \in G_1(F)$ and almost all x. Since $x \mapsto h\star x$ preserves the Haar measure, this means $\int f^{[h]}\Theta = \int f\Theta$.

By Lemma 7.2, a regular semisimple x over γ is fixed by $h\star$ for every h in the centralizer $E^\times$ of γ. So the stable class of x maps bijectively onto the $G_1(F)$-conjugacy class of γ; it never contains the other lift of γ; and it is the union of the conjugacy classes of the $\delta_a \star x$, $a \in F^\times/NE^\times$: two classes if E is a field, and one if γ is hyperbolic.

That this is the right notion is shown by the following function, which separates the two lifts and is constant on stable classes. Let $\gamma_v(a)$ be the Weil index of (3.2), for the fixed character $\psi = \psi_v$.

Lemma 7.3. *Let $\widetilde{SL(2,F)}_{\mathrm{reg}}$ be the set of regular semisimple elements of $\widetilde{SL(2,F)}$.*

(a) There is exactly one function $\Delta_\psi \colon \widetilde{SL(2,F)}_{\mathrm{reg}} \to \mu_4$ that is invariant under conjugation by $\widetilde{SL(2,F)}$ and satisfies

$$\Delta_\psi(g,\zeta) = \zeta\,\big(c, D^+(g)\big)_v \,\frac{\gamma_v(D^+(g))}{\gamma_v(1)} \qquad \text{for } g = \begin{pmatrix} \alpha & b \\ c & d \end{pmatrix},\ c \neq 0. \tag{7.4}$$

It is genuine: $\Delta_\psi(x\zeta) = \zeta\,\Delta_\psi(x)$.

(b) $\Delta_\psi(\mathrm{Ad}(h)x) = \eta_x(\det h)\,\Delta_\psi(x)$ for $h \in G_1(F)$; that is, $\Delta_\psi(h\star x) = \Delta_\psi(x)$. So two elements of $\widetilde{SL(2,F)}_{\mathrm{reg}}$ are stably conjugate if and only if their images in $SL(2,F)$ are conjugate under $G_1(F)$ and Δ_ψ takes the same value on them.

(c) *On the split torus,* $\Delta_\psi(a(t),\zeta) = \zeta\,\gamma_v(1)/\gamma_v(t) = \chi_{\psi_{-1},v}(a(t),\zeta)$*, the complex conjugate of Weil's character* $\chi_{\psi,v}$ *(Lemma 3.3).*

(d) *For* $a \in F^\times$, $\Delta_{\psi_a} = \eta_x(a)\,\Delta_\psi$*, where* $\psi_a(u) = \psi(au)$.

Proof. Write Δ for the right side of (7.4), a function on the set X of (g,ζ) with $c \neq 0$ and $\operatorname{tr} g \neq -2$; and $D = D^+(g)$, which is invariant under conjugation. We show that $\Delta(yxy^{-1}) = \Delta(x)$ when x and yxy^{-1} lie in X.

Conjugation by $(n(u),1)$. The lower left entry is unchanged, and by (2.3) the sign is $\beta(n(u),g)\beta(n(u)g,n(-u))\beta(n(u),n(-u)) = (c,1)_v(1,c)_v = 1$.

Conjugation by $(a(s),1)$. The lower left entry becomes c/s^2, and the sign is

$$\beta(a(s),g)\,\beta(a(s)g,a(s)^{-1})\,\beta(a(s),a(s)^{-1}) = (c,s)_v\cdot(s,cs)_v\cdot(s,-1)_v = 1.$$

So Δ is unchanged.

Conjugation by $(w,1)$*, when* $b \neq 0$. Here $wgw^{-1} = \begin{pmatrix} d & -c \\ -b & \alpha \end{pmatrix}$. Suppose $\alpha \neq 0$. Then $x(wg) = \alpha$, and the sign is $\beta(w,g)\beta(wg,w^{-1})\beta(w,w^{-1}) = (\alpha,\alpha/c)_v(-b/\alpha,b)_v = (\alpha,-c)_v(\alpha,b)_v = (\alpha,-bc)_v$. So invariance amounts to $(-b,D)_v(\alpha,-bc)_v = (c,D)_v$, that is, to $(-bc,\ \alpha D)_v = 1$. Put $X_0 = -bc = 1-\alpha d$ and $Y_0 = \alpha+1$. Then $\alpha D = \alpha^2+\alpha d+2\alpha = Y_0^2 - X_0$, and $(X_0, Y_0^2 - X_0)_v = 1$ because $Y_0^2 - X_0$ is the norm of $Y_0 + \sqrt{X_0}$. If $\alpha = 0$, then $bc = -1$, $x(wg) = b$, the sign is $(b,b/c)_v(-1,b)_v = (b,-b^2)_v(-1,b)_v = 1$, and the identity to be checked is $(-b,D)_v = (c,D)_v$, i.e. $(-bc,D)_v = 1$, which holds because $-bc = 1$.

Now let $y \in SL(2,F)$ be arbitrary with $x, yxy^{-1} \in X$. If $p(y) \in B$, then y is a product of elements of the first two kinds and of μ_2. Otherwise $p(y) = b_1 wn$ with $b_1 \in B(F)$ and $n \in N(F)$ (Bruhat). The element $x' = (n,1)x(n,1)^{-1}$ is in X, and its upper right entry is not 0: otherwise $(w,1)x'(w,1)^{-1}$, hence yxy^{-1}, would be upper triangular. So the three steps apply in turn. Finally, every regular semisimple element is conjugate to an element of X (if $c = 0 \neq b$, conjugate by w; if $c = b = 0$, first by $n(1)$), which gives existence and uniqueness of Δ_ψ. The values lie in μ_4, because $(\gamma_v(D)/\gamma_v(1))^2 = (D,-1)_v$ by Lemma 3.3(a), and genuineness is clear.

(b) For $h \in SL(2,F)$ this is (a). For $h = \delta_a$ and $x = (g,\zeta) \in X$, (7.1) gives $\mathrm{Ad}(\delta_a)x = (\delta_a g \delta_a^{-1}, \zeta)$ with lower left entry c/a, so $\Delta(\mathrm{Ad}(\delta_a)x) = (a,D)_v\Delta(x)$. Since $G_1(F)$ is generated by $Z(F)$, $SL(2,F)$ and the δ_a, the formula follows. For the last statement, if $p(x') = hp(x)h^{-1}$, then $x' = \pm h\star x$, and $\Delta_\psi(x') = \pm\Delta_\psi(x)$ accordingly.

(c) $(a(t),1)$ is conjugate, by $(n(1),1)$ and then $(w,1)$, to $(g_2,(t^2-1,t)_v)$ with $g_2 = \begin{pmatrix} t^{-1} & 0 \\ t-t^{-1} & t \end{pmatrix}$: the first conjugation has sign 1, and the second has sign $\beta(w,g_1)\beta(wg_1,w^{-1}) = (1-t^{-2},t)_v = (t^2-1,t)_v$, where $g_1 = n(1)a(t)n(-1)$. Here $D = (t+1)^2/t$, so $(c,D)_v = ((t^2-1)/t,t)_v = (t^2-1,t)_v(t,-1)_v$ and $\Delta_\psi(a(t),1) = (t,-1)_v\gamma_v(t)/\gamma_v(1)$. By Lemma 3.3(a), $(\gamma_v(t)/\gamma_v(1))^2 = (t,-1)_v$, so this is $\gamma_v(1)/\gamma_v(t)$; and $\chi_{\psi_{-1}} = \chi_\psi\cdot(-1,\cdot)_v$ by Lemma 3.3(d).

(d) $\gamma_{\psi_a,v}(D)/\gamma_{\psi_a,v}(1) = \gamma_v(aD)/\gamma_v(a) = (a,D)_v\,\gamma_v(D)/\gamma_v(1)$, by Lemma 3.3(a). □

So stable conjugacy can be described without $GL(2)$: *x and x' are stably conjugate when $p(x)$, $p(x')$ are stably conjugate in $SL(2,F)$ and $\Delta_\psi(x) = \Delta_\psi(x')$*; by (d) the condition does not depend on ψ. This is the definition of Adams [Ad98] and Li [Li11, §5], who use, in place of Δ_ψ, the character of the difference of the two halves of the Weil representation. The two functions agree up to a positive factor:

Lemma 7.4. *Let $\omega = \omega_{\psi'}$ be the Weil representation of (5.5), and γ' the Weil index of ψ'. Then $\omega(w,1)\phi(x) = \gamma'(1)^{-1}\int \phi(y)\psi'(-2xy)\,dy$, and for $g = \begin{pmatrix} \alpha & b \\ c & d \end{pmatrix}$ with $c \neq 0$,*

$$\omega(g,1)\phi(x) = \gamma'(c)^{-1}\,|c|_v^{-1/2}\int_F \psi'\Big(\frac{\alpha x^2 - 2xy + dy^2}{c}\Big)\phi(y)\,dy, \tag{7.5}$$

with the measure that is self-dual for $\psi'(2xy)$. Moreover $\omega_{\psi'}\circ \mathrm{Ad}(\delta_a) = \omega_{\psi'_a}$ for $a \in F^\times$, as representations on $\mathscr{S}(F)$; in particular $r^\pm_{\psi'}\circ \mathrm{Ad}(\delta_a) = r^\pm_{\psi'_a}$.

Proof. In the proof of Lemma 5.7 it is shown that $\omega(w(t),1)\phi(x) = \gamma'(t)|t|^{1/2}(\mathscr{F}\phi)(tx)$ for $w(t) = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}$; for $t=-1$ this is the first formula, since $w(-1) = w$ and $\gamma'(-1) = \gamma'(1)^{-1}$. For $c \neq 0$, $g = n(\alpha/c)\,a(c^{-1})\,wn(d/c)$, and the same holds for the lifts $(\cdot,1)$, because the three values of Kubota's cocycle are $(c,1)_v$, $(1,c)_v$ and $(1,c)_v$. Apply (5.5), Lemma 5.7 with $t = c^{-1}$, and $\gamma'(-1)\gamma'(1) = 1$. For the last statement it suffices to compare the two representations on $(n(u),1)$ and $(w,1)$, which generate $\overline{SL(2,F)}$ together with μ_2. By Lemma 7.1(d), $\omega_{\psi'}(\mathrm{Ad}(\delta_a)(n(u),1))$ is multiplication by $\psi'(aux^2)$, and $\omega_{\psi'}(\mathrm{Ad}(\delta_a)(w,1))\phi(x) = \frac{\gamma'(1)}{\gamma'(a)}|a|^{1/2}\cdot\gamma'(1)^{-1}\int\phi(y)\psi'(-2axy)\,dy$. This is $\omega_{\psi'_a}(w,1)\phi(x)$: the Weil index of ψ'_a at 1 is $\gamma'(a)$, and the measure self-dual for $\psi'_a(2xy)$ is $|a|^{1/2}$ times the one for $\psi'(2xy)$. □

As a check on the constant in (7.5): composing the kernels of $(g_1,1)$ and $(g_2,1)$ by a Gaussian integral gives the kernel of g_1g_2 times $\gamma'(c_3/c_1c_2)\gamma'(c_3)/\gamma'(c_1)\gamma'(c_2)$, where c_3 is the lower left entry of g_1g_2; by Lemma 3.3(a) this is $(c_3/c_1, c_3/c_2)_v = \beta(g_1,g_2)$. So (7.5) is a genuine representation of the cover defined by Kubota's cocycle, with no further constant.

Proposition 7.5. *Let $\Theta^\pm_\psi$ be the characters of $r^\pm_\psi$. On $\overline{SL(2,F)}_{\mathrm{reg}}$ the distribution $\Theta^+_\psi - \Theta^-_\psi$ is the function*

$$\Theta^+_\psi(x) - \Theta^-_\psi(x) = \frac{\Delta_\psi(x)}{|D^+(x)|_v^{1/2}}; \tag{7.6}$$

that is, $\mathrm{tr}\, r^+_\psi(f) - \mathrm{tr}\, r^-_\psi(f) = \int f(x)\Delta_\psi(x)|D^+(x)|_v^{-1/2}dx$ for $f \in C_c^\infty(\overline{SL(2,F)})$ with support in $\overline{SL(2,F)}_{\mathrm{reg}}$. Hence $\Theta^+_\psi - \Theta^-_\psi$ is invariant under $\star$ on $\overline{SL(2,F)}_{\mathrm{reg}}$, it does not vanish there, and stable conjugacy as defined above coincides with stable conjugacy in the sense of [Ad98, Li11]. The right side of (7.6) is locally integrable on $\overline{SL(2,F)}$, and by the regularity theorem for the characters of $r^\pm_\psi$ ([Ma99] for the Weil representation; [Li12a, §4] for irreducible genuine representations of p-adic covers; Harish-Chandra's theorem at the real places) the formula holds for all $f \in C_c^\infty(\overline{SL(2,F)})$: $\Theta^+_\psi - \Theta^-_\psi$ is a stable character.

Proof. Let $P\phi(x) = \phi(-x)$, so that $\mathrm{tr}\, r^+_\psi(f) - \mathrm{tr}\, r^-_\psi(f) = \mathrm{tr}(\omega_\psi(f)P)$. Both sides of the identity to be proved are unchanged when f is replaced by a conjugate f^y. Let X be the set of the proof of Lemma 7.3; it is open, and every element of $\overline{SL(2,F)}_{\mathrm{reg}}$ is conjugate to an element of it. So each point of the support of f has an open neighborhood of the form yUy^{-1} with $U \subset X$ open. Finitely many of these cover the support, and a partition of unity subordinate to them writes f as a finite sum of functions $f_i \in C_c^\infty(\overline{SL(2,F)})$ with $f_i^{y_i}$ supported in X. We may therefore assume that f has

compact support in X. Then c and $D = D^+(g)$ are bounded away from 0 and ∞ on the support of f.

For $x = (g, \zeta) \in X$ let $k_x(\xi, \eta)$ be ζ times the kernel of (7.5), so that $|k_x| = |c|_v^{-1/2}$ and

$$k_x(\xi, -\xi) = \zeta\, \gamma_v(c)^{-1} |c|_v^{-1/2}\, \psi\big(D\xi^2/c\big).$$

By Fubini's theorem $\omega_\psi(f)$ has the kernel $K_f(\xi, \eta) = \int f(x) k_x(\xi, \eta)\, dx$, which is continuous and bounded.

Let v be finite. Then f is bi-invariant under the image K' of some K_m, and $\omega_\psi(f) = \omega_\psi(e')\omega_\psi(f)\omega_\psi(e')$ for the idempotent e' of K'. The space V' of the vectors of $L^2(F)$ fixed by K' is finite-dimensional and consists of Schwartz–Bruhat functions, by a two-lattice argument. The elements $(n(u), 1)$, $u \in \varpi^m \mathscr{O}$, act by multiplication by $\psi(u\xi^2)$, so a fixed vector is supported in the bounded set $L_1 = \{\xi \mid \psi(u\xi^2) = 1 \text{ for all } u \in \varpi^m \mathscr{O}\}$. The conjugates by $(w, 1)$ of the $(n(u), 1)$ with $u \in 2\varpi^m \mathscr{O}$ lie in K' (they lie over K_m, and the sign is a character of $u \in \varpi^m \mathscr{O}$, hence trivial on $2\varpi^m \mathscr{O}$), and they act in the same way on the Fourier transform, by (5.5); so the Fourier transform of a fixed vector is supported in a bounded set as well, and the vector is constant on the cosets of a lattice L_2. Functions supported in a bounded set and constant modulo a lattice form a finite-dimensional space. So $\omega_\psi(f)$ has finite rank: $\omega_\psi(f)\phi = \sum_i \langle \phi, w_i \rangle u_i$ with finitely many $u_i, w_i \in V'$ (the functionals are of this form because $\omega_\psi(e')$ is the orthogonal projection onto V'). Its kernel is $\sum_i u_i(\xi)\overline{w_i(\eta)}$, a Schwartz–Bruhat function, which equals the continuous function K_f, and $\operatorname{tr}(\omega_\psi(f)P) = \sum_i \langle Pu_i, w_i \rangle = \int_F K_f(\xi, -\xi)\, d\xi$. Hence

$$\operatorname{tr}(\omega_\psi(f)P) = \lim_{M \to \infty} \int f(x) \Big(\int_{\varpi^{-M}\mathscr{O}} k_x(\xi, -\xi)\, d\xi \Big) dx.$$

By Lemma 7.5′below, applied to $a = D/c$, the inner integral does not depend on M for $M \geq M_0$, with M_0 uniform on the support of f, and by (3.2) (the measure being $|2|_v^{1/2}$ times the one there) its value is ζ times

$$\gamma_v(c)^{-1} |c|_v^{-1/2} \int_F \psi\Big(\frac{D\xi^2}{c}\Big) d\xi = \frac{\gamma_v(cD)}{\gamma_v(c)} |D|_v^{-1/2} = (c, D)_v \frac{\gamma_v(D)}{\gamma_v(1)} |D|_v^{-1/2};$$

the last step is Lemma 3.3(a). This is (7.6).

Let v be archimedean. We show that K_f is a Schwartz function on $F \times F$. On X use the coordinates (α, c, d). The section $g \mapsto (g, 1)$ is continuous on each connected component of $\{c \neq 0\}$: $g = n(\alpha/c)\, a(-c^{-1})\, w\, n(d/c)$, the sections of Lemma 2.2(i) are continuous on the factors, and the cocycle of the factors is a Hilbert symbol in c, which is locally constant. The Haar measure is a smooth multiple of $d\alpha\, dc\, dd$, and $\gamma_v(c)$ is locally constant. So

$$K_f(\xi, \eta) = \int_{F^\times} \widehat{\Phi}_c\Big(\frac{\xi^2}{c}, \frac{\eta^2}{c}\Big)\, \psi\Big(-\frac{2\xi\eta}{c}\Big)\, dc,$$

where $\Phi(\alpha, c, d)$ is smooth with compact support in $\{c \neq 0\}$ and $\widehat{\Phi}_c$ is its Fourier transform in (α, d). The functions $\widehat{\Phi}_c$ and their derivatives in c are Schwartz functions, uniformly for c in the compact set in question, and $|\xi|^2 + |\eta|^2$ is bounded by a constant times $|\xi^2/c| + |\eta^2/c|$ there. So K_f decays faster than any power of $1 + |\xi| + |\eta|$, and so do its derivatives, which

are integrals of the same kind with polynomial factors. An integral operator on $L^2(F)$ with a Schwartz kernel is of trace class, and its trace is the integral of the kernel over the diagonal. So $\operatorname{tr}(\omega_\psi(f)P) = \int K_f(\xi,-\xi)\,d\xi = \lim_{\varepsilon\to 0}\int f(x)\big(\int k_x(\xi,-\xi)e^{-\varepsilon|\xi|^2}d\xi\big)dx$. The inner integrals are Gaussian; they are bounded by a constant times $|D|_v^{-1/2}$, uniformly in ε, and converge to the value above. Dominated convergence gives (7.6).

The remaining statements follow from Lemma 7.3(b) and $\Delta_\psi \neq 0$. Local integrability: $|D^+|_v^{-1/2}$ is the inverse square root of the function $2+\operatorname{tr}$, which vanishes to first order on the smooth part of its zero set and is a nondegenerate quadratic form in three variables at -1. □

Lemma 7.5′. *Let F be non-archimedean, let ψ be trivial on $\varpi^{n_0}\mathscr{O}$ and not on $\varpi^{n_0-1}\mathscr{O}$, and let $e = \operatorname{ord}(2)$. For $a \in F^\times$ and $j \in \mathbb{Z}$ with*

$$2j \;\geq\; \operatorname{ord}(a) - n_0 + 2 + 2e,$$

the integral of $\psi(a\xi^2)$ over the shell $\{\operatorname{ord}(\xi) = -j\}$ vanishes. Consequently $\int_{\varpi^{-M}\mathscr{O}} \psi(a\xi^2)\,d\xi$ does not depend on M once $2(M+1) \geq \operatorname{ord}(a) - n_0 + 2 + 2e$, and the bound is uniform for a in a compact subset of $F^\times$.

Proof. Put $m = \operatorname{ord}(a) - 2j$, the order of $a\xi^2$ on the shell, and $r = n_0 - 1 - m - e$. The hypothesis is $m \leq n_0 - 2 - 2e$; it gives $r \geq 1 + e \geq 1$ and $m + 2r \geq n_0$. For $u \in \varpi^r\mathscr{O}$ the substitution $\xi \mapsto \xi(1+u)$ preserves the shell and the measure, and

$$\psi\big(a\xi^2(1+u)^2\big) = \psi(a\xi^2)\,\psi(2a\xi^2 u)\,\psi(a\xi^2u^2) = \psi(a\xi^2)\,\psi(2a\xi^2u),$$

because $\operatorname{ord}(a\xi^2u^2) \geq m + 2r \geq n_0$. Averaging over $u \in \varpi^r\mathscr{O}$ multiplies the integrand by the average of the character $u \mapsto \psi(2a\xi^2u)$ of $\varpi^r\mathscr{O}$, which is not trivial because $2a\xi^2\varpi^r\mathscr{O} = \varpi^{m+e+r}\mathscr{O} = \varpi^{n_0-1}\mathscr{O}$. So the integral vanishes. □

For $F = \mathbb{Q}_2$ and $\psi(x) = e^{2\pi i\{x\}}$ the lemma says that the integrals over $2^{-M}\mathbb{Z}_2$ are constant from $M = 1$ on for $a = 1$ and from $M = 2$ on for $a = 2$; the values are $1+i$ and $\sqrt{2}(1+i)$.

The same computation without P gives the character of ω_ψ itself on the set where $D^- = \operatorname{tr} g - 2 \neq 0$:

$$\Theta_{\omega_\psi}(g,\zeta) = \zeta\,(c,D^-)_v\,\frac{\gamma_v(D^-)}{\gamma_v(1)}\,|D^-|_v^{-1/2}, \qquad c \neq 0. \tag{7.7}$$

Two comparisons. First, $\det(g-1) = -D^-$ and $\gamma_v(1)\gamma_v(-D^-) = (-1,D^-)_v\gamma_v(D^-)/\gamma_v(1)$, so (7.7) is $(-c,D^-)_v \cdot \gamma_v(1)\gamma_v(\det(g-1))|\det(g-1)|_v^{-1/2}$. This is the formula of Thomas [Th08, Theorem 1C], for his additive character $x \mapsto \psi(2x)$: in [Th08] the Weil index $\gamma(a)$ is that of $x \mapsto \psi(\frac{1}{2}ax^2)$ and $n(u)$ acts by $\psi(\frac{1}{2}ux^2)$. That theorem is stated up to a sign, so the comparison confirms the absolute value and the Weil index and says nothing about the sign $(-c,D^-)_v$, which rests on the computation above. (Theorem 2A of [Th08] gives the character without the ambiguity, on a model of the metaplectic group built from the Maslov index; we have not carried our section $g \mapsto (g,1)$ over to that model. Its proof, in §12.2 there, shows that $\omega(f)$ is of trace class for f smooth of compact support and computes the trace as the integral of the kernel over the diagonal, as is done above.) Second, $\omega_\psi(-1,1) = \gamma_v(1)^2P$ (Lemma 5.7), so $\Theta_\psi^+(x) - \Theta_\psi^-(x) = \gamma_v(1)^{-2}\,\Theta_{\omega_\psi}\big((-1,1)x\big)$; and indeed (7.7) at $(-1,1)(g,1) = (-g,(c,-1)_v)$ gives back (7.6), by Lemma 3.3(a) and $\gamma_v(1)^4 =$

$(-1,-1)_v$. The character of the difference in [Th13, Theorem D] stands in the same relation to the character of the Weil representation at $-g$; that theorem is written in other coordinates on the metaplectic group and with a Weil index of a quadratic form, and we have not verified an equality between its factors and ours, nor matched the normalization of the Weil index in [Th13] with ours. Lemma 7.3, on which the rest of the section depends, is algebraic and uses none of this (Remark 2026.19).

Example 7.B. Let $F=\mathbb{R}$. For $0<\theta<\pi$ the element $\tilde{k}_\theta=(k_\theta,1)$ has $c=-\sin\theta<0$ and $D^+=4\cos^2\frac{\theta}{2}>0$, so $\Delta_\psi(\tilde{k}_\theta)=1$ and (7.6) gives $\Theta^+-\Theta^-=1/(2\cos\frac{\theta}{2})$. Directly: if r_ψ^+ has lowest weight $\frac{1}{2}$, its weights are $\frac{1}{2}+2j$ and those of r_ψ^- are $\frac{3}{2}+2j$, $j\geq 0$, and

$$\sum_{j\geq 0} e^{i(\frac{1}{2}+2j)\theta}-\sum_{j\geq 0} e^{i(\frac{3}{2}+2j)\theta}=\frac{e^{i\theta/2}}{1+e^{i\theta}}=\frac{1}{2\cos\frac{\theta}{2}}$$

(Abel summation). The function is even in θ, in agreement with Example 7.A(b): $\tilde{k}_\theta$ and $\tilde{k}_{-\theta}=\mathrm{Ad}(\delta_{-1})\tilde{k}_\theta$ are stably conjugate, and they are not conjugate. On the hyperbolic elements over $a(t)$, $t<0$, the stable class is the conjugacy class, and $\mathrm{Ad}(\delta_{-1})$ exchanges the two lifts.

Example 7.C. Let v be finite and non-dyadic, ψ unramified, and $x=(g,\kappa(g))\in K^*$ with c and $D^+(g)$ units. Then $\Delta_\psi(x)=1$ and $\Delta_{\psi_\varpi}(x)=(\varpi,D^+(g))_v$, the quadratic residue symbol of $\det(1+g)$. The second value can be seen in the Weil representation of $SL(2,\mathbb{F}_q)$ on the functions on $\mathbb{F}_q$, which is the space of vectors of ω_{ψ_ϖ} fixed by K_1 (the functions on $\varpi^{-1}\mathscr{O}/\mathscr{O}$): there the trace of $\omega(g)P$ is $\left(\frac{\det(1+g)}{q}\right)$ for every g with $\det(1+g)\neq 0$. The script verifies this for $q=3,5,7,11$ by computing the matrices. (The character of ω_{ψ_ϖ} also involves the vectors of higher level, which we have not computed separately.)

The dual group. The unramified genuine principal series of $\overline{SL(2,F)}$ are parametrized by $\chi_v(\varpi)\in\mathbb{C}^\times$ modulo inversion, through $\pi_v=\chi_{\psi,v}\cdot(\chi_v\circ p)$, and the intertwining operator has the normalizing factor $L_v(2\lambda,\chi_v^2)/L_v(2\lambda+1,\chi_v^2)$ (Lemma 5.6). This is the normalizing factor of the principal series of $PGL(2,F)$ induced from $\mathrm{diag}(t_1,t_2)\mapsto\chi_v(t_1/t_2)$, whose Satake parameter is $\mathrm{diag}(\chi_v(\varpi),\chi_v(\varpi)^{-1})\in SL(2,\mathbb{C})$. So the dual group of the double cover of $SL(2)$ is $SL(2,\mathbb{C})$, the dual group of $SO(3)=PGL(2)$, and not $PGL(2,\mathbb{C})$, the dual group of $SL(2)$ [Sa88, Li11, GGW18, We18]. The elliptic endoscopic data correspond to the two central elements ± 1 of $SL(2,\mathbb{C})$. This is why the endoscopic groups are $SO(3)$ twice, why the translation by -1 appears below, and why no tori occur.

7.3 Stable and κ-orbital integrals

We now sort the orbital integrals in a $G_1(F)$-conjugacy class by the characters of $F^\times/NE^\times$. For $SL(2)$ this gives the stable orbital integral and one κ-orbital integral [LL79]. On the cover there are again two combinations, and the second is a stable orbital integral at another point.

Fix Haar measures on the centralizers so that they are compatible under every $\mathrm{Ad}(h)$, $h\in G_1(F)$ (Section 2.1.5). For $x\in\overline{SL(2,F)}_{\mathrm{reg}}$ over γ, with centralizer $T_x=T_\gamma$ in $SL(2,F)$, and $f\in C_c^\infty(\overline{SL(2,F)})_-$, let

$$J_G(x,f)=|D(\gamma)|_v^{1/2}\int_{T_x\backslash SL(2,F)} f(y^{-1}xy)\,dy,\qquad D(\gamma)=(\mathrm{tr}\,\gamma)^2-4,$$

as in (4.11). Then $J_G(x,f^h) = J_G(\mathrm{Ad}(h)x,f)$ and $J_G(x,f^{[h]}) = J_G(h\star x,f)$ for $h \in G_1(F)$. Let $E = F[\gamma]$, and define

$$J_G^{\mathrm{st}}(x,f) = \sum_{a\in F^\times/NE^\times} J_G(\delta_a\star x,f), \qquad J_G^{\kappa}(x,f) = \sum_{a\in F^\times/NE^\times} \kappa_E(a)\, J_G(\delta_a\star x,f). \tag{7.8}$$

The terms depend only on a modulo $NE^\times$, because $\delta_{aN(e)}\star x$ is conjugate to $\delta_a\star(e\star x) = \delta_a\star x$ for $e\in E^\times$. If γ is hyperbolic, both are $J_G(x,f)$.

Lemma 7.6. *Let $x \in \overline{SL(2,F)}_{\mathrm{reg}}$ lie over γ, $E = F[\gamma]$, and $f \in C_c^\infty(\overline{SL(2,F)})_-$.*

(a) $J_G^{\mathrm{st}}(h\star x,f) = J_G^{\mathrm{st}}(x,f)$ and $J_G^{\kappa}(h\star x,f) = \kappa_E(\det h)J_G^{\kappa}(x,f)$ for $h\in G_1(F)$. Equivalently, $J_G^{\mathrm{st}}(x,f^{[h]}) = J_G^{\mathrm{st}}(x,f)$: the distribution $J_G^{\mathrm{st}}(x,\cdot)$ is stable.

(b) For a character κ' of $F^\times/F^{\times 2}$,

$$\sum_{a\in F^\times/F^{\times 2}} \kappa'(a)\, J_G(x,f^{\delta_a}) = [NE^\times : F^{\times 2}]\cdot \begin{cases} J_G^{\mathrm{st}}(x,f), & \kappa' = \eta_\gamma, \\ J_G^{\kappa}(x,f), & \kappa' = \eta_\gamma\kappa_E, \\ 0, & \text{otherwise.} \end{cases}$$

In particular the average of $J_G(x,f^h)$ over $h\in G_1(F)/Z(F)SL(2,F)$ vanishes unless $D^+(\gamma)\in F^{\times 2}\cup\theta F^{\times 2}$; for $\gamma = a(t)$, unless t is a square.

(c) Let E be a field, and let $\tilde z$ be a lift of -1. Then

$$J_G^{\kappa}(x,f_{\tilde z}) = J_G^{\mathrm{st}}(\tilde z x,f), \qquad \text{that is,} \qquad J_G^{\kappa}(x,f) = J_G^{\mathrm{st}}(\tilde z x, f_{\tilde z^{-1}}).$$

(d) Let v be finite and non-dyadic, $f = e_v$, $x\in K^*$ and $|D(\gamma)|_v = 1$, and let $T_\gamma\cap K$ have volume 1. Then $J_G^{\mathrm{st}}(x,e_v) = J_G^{\kappa}(x,e_v) = J_G(x,e_v) = 1$.

Proof. (a) $\delta_a\star(h\star x) = (\delta_a h)\star x$. Write $\delta_a h = \delta_{a\det h}h_0$ with $h_0\in SL(2,F)$; then $(\delta_a h)\star x$ is conjugate to $\delta_{a\det h}\star x$, by $\delta_{a\det h}h_0\delta_{a\det h}^{-1}$. So the terms of (7.8) are permuted by $a\mapsto a\det h$. The reformulation is $J_G(y,f^{[h]}) = J_G(h\star y,f)$.

(b) $J_G(x,f^{\delta_a}) = J_G(\mathrm{Ad}(\delta_a)x,f) = \eta_\gamma(a)J_G(\delta_a\star x,f)$, since f is anti-genuine. So the left side is $\sum_a \kappa'(a)\eta_\gamma(a)J_G(\delta_a\star x,f)$, and the summand $J_G(\delta_a\star x,f)$ depends only on a modulo $NE^\times$. The sum of $\kappa'\eta_\gamma$ over a coset of $NE^\times/F^{\times 2}$ is 0 unless $\kappa'\eta_\gamma$ is trivial on $NE^\times$, that is, unless $\kappa'\eta_\gamma\in\{1,\kappa_E\}$. For $\kappa' = 1$ the condition is $\eta_\gamma\in\{1,\kappa_E\}$.

(c) Write $\gamma = \begin{pmatrix} x_0 & y_0\theta \\ y_0 & x_0\end{pmatrix}$ in a suitable basis, $x_0^2 - \theta y_0^2 = 1$. Then $D^+(\gamma)D^+(-\gamma) = 4-(2x_0)^2 = -4\theta y_0^2$, so

$$\eta_\gamma\eta_{-\gamma} = (\cdot,-\theta)_v = \kappa_E\cdot(\cdot,-1)_v. \tag{7.9}$$

By Lemma 7.2, $\mathrm{Ad}(\delta_a)\tilde z = (a,-1)_v\tilde z$. Hence

$$\delta_a\star(\tilde z x) = \eta_{-\gamma}(a)\,(a,-1)_v\tilde z\,\mathrm{Ad}(\delta_a)x = \eta_{-\gamma}(a)(a,-1)_v\eta_\gamma(a)\cdot\tilde z(\delta_a\star x) = \kappa_E(a)\,\tilde z(\delta_a\star x).$$

Since $J_G(\tilde z y,f) = J_G(y,f_{\tilde z})$, $J_G(\varepsilon y,f) = \varepsilon J_G(y,f)$ for $\varepsilon\in\mu_2$, and $D(-\gamma) = D(\gamma)$, summing over a gives $J_G^{\mathrm{st}}(\tilde z x,f) = J_G^{\kappa}(x,f_{\tilde z})$.

(d) $J_G(x,e_v) = 1$ is the unramified computation used for Theorem 4.11. If E is a field, it is unramified, and $\delta_\varpi\star x$ lies over $\delta_\varpi\gamma\delta_\varpi^{-1}$, which has no $SL(2,F)$-conjugate in K (proof of Theorem 4.5(a)); so $J_G(\delta_\varpi\star x,e_v) = 0$. □

By part (b), for the double cover the orbital integral of a genuine function at a lift of $a(t)$ does not vanish when t is not a square, but its average over the $G_1(F)$-orbit does. Part (c) is the form that the relation between the endoscopic data $(1,0)$ and $(0,1)$ takes here: Li's transfer factors for the two data differ by the translation by the central element -1 [Li11, §5], and correspondingly the κ-orbital integral of f at x is the stable orbital integral of a translate of f at $\tilde{z}x$. For $SL(2)$ there is no such identity: there J^κ is a new function on the torus T_γ, and [LL79] apply Poisson summation on T_γ to it.

By Lemma 7.3(b), $x \mapsto \Delta_\psi(x)^{-1} J_G^{\mathrm{st}}(x,f)$ is invariant under μ_2 and under $\star$, so it is a function of the stable conjugacy class of $p(x)$ in $SL(2,F)$, that is, of the trace. Li's transfer theorem [Li11] says that it is the stable orbital integral of a function f' on $SO(3,F)$ at the element with eigenvalues $\varepsilon, 1, \varepsilon^{-1}$, where $\varepsilon^{\pm 1}$ are the eigenvalues of $p(x)$, up to a normalization of the transfer factor. We do not use this.

Example 7.D. Let v be finite and non-dyadic, E/F the unramified quadratic extension, and $\gamma \in K$ regular elliptic with $F[\gamma] \cong E$ and $|D(\gamma)|_v = 1$. Let $x \in K^*$ be the lift of γ, and let $T_\gamma \cap K$ have volume 1. Here $F^\times / NE^\times = \{1, \varpi\}$, so each sum in (7.8) has two terms:

$$J_G^{\mathrm{st}}(x, e_v) = J_G(x, e_v) + J_G(\delta_\varpi \star x, e_v), \qquad J_G^\kappa(x, e_v) = J_G(x, e_v) - J_G(\delta_\varpi \star x, e_v).$$

The second term vanishes. Indeed $\delta_\varpi \star x$ lies over $\delta_\varpi \gamma \delta_\varpi^{-1}$, which has no $SL(2,F)$-conjugate in K (proof of Lemma 7.6(d)), and e_v is supported on $\overline{K}$. With $J_G(x, e_v) = 1$ this gives $J_G^{\mathrm{st}}(x, e_v) = J_G^\kappa(x, e_v) = 1$, as Lemma 7.6(d) states. Part (c) of the lemma can be seen here as well. Take $\tilde{z} = (-1, 1)$, which lies in K^*. Then $(e_v)_{\tilde{z}^{-1}} = e_v$, because e_v is left K^*-invariant. The element $\tilde{z}x \in K^*$ lies over $-\gamma$, which again satisfies the hypotheses of part (d). So both sides of $J_G^\kappa(x, e_v) = J_G^{\mathrm{st}}(\tilde{z}x, e_v)$ equal 1.

7.4 The elliptic regular terms

This subsection applies the local results of Sections 7.1–7.3 to the regular elliptic terms of the trace formula.

Let $\gamma \in G(\mathbb{Q}) = SL(2,k)$ be regular elliptic, $E = k[\gamma]$ its splitting field, T_γ its centralizer and $\kappa_E = \prod_v \kappa_{E_v}$ the quadratic character of $k^\times \backslash \mathbb{A}_k^\times$ attached to E. Everything in Sections 7.1–7.3 has an adelic version. Let

$$\eta_\gamma(a) = \prod_v \big(a_v, D^+(\gamma)\big)_v, \qquad a \in \mathbb{A}_k^\times,$$

a quadratic character of $\mathbb{A}_k^\times$ which is trivial on $k^\times$ by the product formula (H4), because $D^+(\gamma) = 2 + \operatorname{tr}\gamma \in k^\times$. For $h \in G_1(\mathbb{A}_k)$ and x in the orbit of $\pm s(\gamma)$ under $\mathrm{Ad}(G_1(\mathbb{A}_k))$ put $h \star x = \eta_\gamma(\det h)\,\mathrm{Ad}(h)x$; its component at v is $h_v \star x_v$ in the sense of (7.3). So for $a \in k^\times$,

$$s(\delta_a \gamma \delta_a^{-1}) = \mathrm{Ad}(\delta_a) s(\gamma) = \delta_a \star s(\gamma) \tag{7.10}$$

by Lemma 7.1(f): *rational elements that are conjugate under $G_1(k)$ are stably conjugate in $\overline{G(\mathbb{A})}$, at every place.* This is what makes a stabilization possible, and it is the reason for the twist in (7.3): the untwisted local conjugates $\mathrm{Ad}(\delta_a)\gamma_v$ are not coherent in v. (Compare [Li15, Lemme 4.1.7].)

For $f \in \mathscr{H}(\overline{G(\mathbb{A})})$ and $x = a\star s(\gamma)$, $a \in \mathbb{A}_k^\times$ (we write $a\star$ for $\delta_a\star$), let $J_G(x,f) = \int f(y^{-1}xy)\,dy$ over $T_x(\mathbb{A})\backslash G(\mathbb{A})$, with the Tamagawa measures on $G(\mathbb{A})$ and on $T_x(\mathbb{A}) = \delta_a T_\gamma(\mathbb{A})\delta_a^{-1}$; no factor $|D|^{1/2}$ is needed, since $|D(\gamma)|_{\mathbb{A}_k} = 1$. Put

$$J_G^{\mathrm{st}}(\gamma,f) = \sum_{a\in \mathbb{A}_k^\times/N\mathbb{A}_E^\times} J_G\big(a\star s(\gamma), f\big), \qquad J_G^{\kappa}(\gamma,f) = \sum_{a\in \mathbb{A}_k^\times/N\mathbb{A}_E^\times} \kappa_E(a)\, J_G\big(a\star s(\gamma), f\big). \tag{7.11}$$

Here $\mathbb{A}_k^\times/N\mathbb{A}_E^\times = \bigoplus_v k_v^\times/NE_v^\times$. By Lemma 7.6(d) only finitely many a contribute, and for $f = \bigotimes f_v$ and local measures whose products are the Tamagawa measures,

$$J_G^{\mathrm{st}}(\gamma,f) = \prod_v J_G^{\mathrm{st}}(\gamma_v, f_v), \qquad J_G^{\kappa}(\gamma,f) = \prod_v J_G^{\kappa}(\gamma_v, f_v),$$

where (γ_v) are local components of $s(\gamma)$; the products do not depend on their choice, and almost all factors are 1.

Theorem 7.7. *For $f \in \mathscr{H}(\overline{G(\mathbb{A})})$ let*

$$S_{\mathrm{ell}}(f) = \frac{1}{2}\sum_{\gamma} \mathrm{vol}\big(T_\gamma(\mathbb{Q})\backslash T_\gamma(\mathbb{A})\big)\, J_G^{\mathrm{st}}(\gamma,f),$$

the sum over a set of representatives γ of the $G_1(k)$-conjugacy classes of regular elliptic elements of $SL(2,k)$. The sum is finite, and

$$\sum_{\mathscr{O}\ \text{elliptic}} J_{\mathscr{O}}(f) = S_{\mathrm{ell}}(f) + S_{\mathrm{ell}}(f_-).$$

More precisely, for each γ the classes $\mathscr{O}$ contained in the $G_1(k)$-conjugacy class of γ contribute

$$\sum_{\mathscr{O}\subset G_1(k)\cdot\gamma} J_{\mathscr{O}}(f) = \frac{1}{2}\,\mathrm{vol}\big(T_\gamma(\mathbb{Q})\backslash T_\gamma(\mathbb{A})\big)\Big(J_G^{\mathrm{st}}(\gamma,f) + J_G^{\kappa}(\gamma,f)\Big), \qquad J_G^{\kappa}(\gamma,f) = J_G^{\mathrm{st}}(-\gamma, f_-). \tag{7.12}$$

The distribution S_{ell} is stable: for $f = \bigotimes_v f_v$ it is a finite sum of products of the stable local distributions $J_G^{\mathrm{st}}(\gamma_v,\cdot)$ of Lemma 7.6(a).

Proof. The $SL(2,k)$-conjugacy classes in the $G_1(k)$-class of γ are the classes $\mathscr{O}_a$ of $\delta_a\gamma\delta_a^{-1}$, $a \in k^\times/NE^\times$. By Proposition 4.7 and (7.10), $J_{\mathscr{O}_a}(f) = \mathrm{vol}\cdot J_G(a\star s(\gamma), f)$, with the same volume for all a, the Tamagawa measure being compatible with $\mathrm{Ad}(\delta_a)$. Let $\Phi(a) = J_G(a\star s(\gamma), f)$ for $a \in \mathbb{A}_k^\times$. It depends only on the image of a in $\mathbb{A}_k^\times/N\mathbb{A}_E^\times$, and it vanishes outside a finite set. By the Hasse norm theorem, $k^\times/NE^\times$ injects into $\mathbb{A}_k^\times/N\mathbb{A}_E^\times$, and by class field theory its image is the kernel of κ_E, of index 2 [CF67]. Therefore

$$\sum_{a\in k^\times/NE^\times} \Phi(a) = \frac{1}{2}\sum_{a\in \mathbb{A}_k^\times/N\mathbb{A}_E^\times} \big(1+\kappa_E(a)\big)\Phi(a) = \frac{1}{2}\Big(J_G^{\mathrm{st}}(\gamma,f) + J_G^{\kappa}(\gamma,f)\Big).$$

For the second formula of (7.12) we may take $f = \bigotimes f_v$. Choose local components γ_v of $s(\gamma)$ and $\tilde z_v$ of $s(-1)$. Then $(\tilde z_v\gamma_v)$ are local components of $s(-1)s(\gamma) = s(-\gamma)$, and $\bigotimes_v (f_v)_{\tilde z_v^{-1}} = f_{s(-1)^{-1}} = f_-$. By Lemma 7.6(c) at the places where E_v is a field (at the others both sides are J_G),

$$J_G^{\kappa}(\gamma,f) = \prod_v J_G^{\mathrm{st}}\big(\tilde z_v\gamma_v, (f_v)_{\tilde z_v^{-1}}\big) = J_G^{\mathrm{st}}(-\gamma, f_-).$$

Finally $\gamma \mapsto -\gamma$ is a bijection of the set of $G_1(k)$-classes of regular elliptic elements, with $T_{-\gamma} = T_\gamma$. Summing (7.12) over γ gives the first formula; the sums are finite by Theorem 4.5(a). Stability is Lemma 7.6(a) at each place. □

For $SL(2)$ the first formula of (7.12) is the pre-stabilization of [LL79, §5]. The second has no analog there, and it is the content of the theorem: the unstable half of the elliptic terms of f is the stable half of the elliptic terms of f_-. In the notation of [Li15, Théorème 5.2.6], where the elliptic part of the trace formula is $\sum_{(n',n'')} \iota(\tilde{G},H)\, ST^H_{\mathrm{ell}}(f^H)$ with $\iota = \frac{1}{2}$ for both pairs when $n = 1$, there are likewise two summands with coefficient $\frac{1}{2}$, those of $(1,0)$ and $(0,1)$. This is a comparison of the decomposition and of the coefficients only. Li's terms are stable distributions on $SO(3)$ applied to transfers of f, with transfer factors; ours are distributions on the cover. Matching them requires the transfer and its normalization, which are not carried out here (Remark 2026.19).

Example 7.E. Let $k = \mathbb{Q}$ and $\gamma = w$, of order 4, with $E = \mathbb{Q}(i)$, $D^+(w) = 2$ and $\eta_w = \prod_v (\cdot, 2)_v$. The $G_1(\mathbb{Q})$-conjugacy class of w is the union of the $SL(2,\mathbb{Q})$-classes of the $\delta_a w \delta_a^{-1}$, $a \in \mathbb{Q}^\times / N E^\times$; there are infinitely many, and for a given f finitely many contribute. The class $a = -1$ is that of $w^{-1} = -w$. So $-w$ lies in the $G_1(\mathbb{Q})$-class of w, and (7.12) reads

$$\sum_{\mathscr{O} \subset G_1(\mathbb{Q})\cdot w} J_{\mathscr{O}}(f) = \tfrac{1}{2}\,\mathrm{vol}\big(T_w(\mathbb{Q})\backslash T_w(\mathbb{A})\big)\big(J^{\mathrm{st}}_G(w,f) + J^{\mathrm{st}}_G(w,f_-)\big).$$

At the real place the stable class of a lift of w consists of two conjugacy classes, exchanged by $\mathrm{Ad}(\delta_{-1})$ (Example 7.B); at a prime $p \equiv 3 \pmod 4$ it consists of two classes, of which one meets K_p^*; at $p \equiv 1 \pmod 4$ it is one class.

7.5 The hyperbolic terms

The stable class of a hyperbolic element is its conjugacy class, so the orbital integrals $J_G(x,f)$, $x \in \overline{A(\mathbb{Q}_S)}$ regular, are stable distributions. The invariant distributions $I_A(x,f)$ of Section 6.3 are not orbital integrals, and their stabilization is a substantial part of [LL79] and of [Li21]. Here we only determine how they transform under the untwisted action of $G_1(k_S)$.

Let S, $\overline{G(\mathbb{Q}_S)}$, $\overline{A(\mathbb{Q}_S)}$ be as in Section 6, and $G_1(k_S) = \prod_{v\in S_k} GL(2,k_v)$, acting on $\overline{G(\mathbb{Q}_S)}$ by Ad. Extend the height to G_1: for $g \in GL(2,k_v)$ put

$$H_{1,v}(g) = -\log \|e_2 g\|_v + \tfrac{1}{2}\log|\det g|_v,$$

so that $H_{1,v} = H_v$ on $SL(2,k_v)$ (Lemma 4.3(a)), $H_{1,v}(zng) = H_{1,v}(g)$ for z scalar and $n \in N$, and $H_{1,v}(\mathrm{diag}(t_1,t_2)g) = \frac{1}{2}\log|t_1/t_2|_v + H_{1,v}(g)$; let $H_1 = \sum_{v\in S_k} H_{1,v}$. For $\delta \in G_1(k_S)$ define $u'_B(\tilde{k},\delta) = -H_1(k\delta)$, $k = p(\tilde{k})$, and $f_{B,\delta}$ by (6.3) with this weight; as in Lemma 6.1(b), $f_{B,\delta} \in C_c^\infty(\overline{A(\mathbb{Q}_S)})_-$. For $a \in k_S^\times$ let κ_a be the character $\tilde{a} \mapsto \prod_{v\in S_k}(a_v,t_v)_v$ of $\overline{A(\mathbb{Q}_S)}$, $p(\tilde{a}) = a(t)$; by Lemma 7.1(d), $\pi \circ \mathrm{Ad}(\delta_a) = \pi\kappa_a$ for $\pi \in \Pi_u(\overline{A(\mathbb{Q}_S)})$. We need one more property of the normalizing factors of Theorem 6.5:

(N5) for every $v \in S_k$, $a \in k_v^\times$ and π_v, the quotient $r_v(\pi_v\kappa_a,\lambda)/r_v(\pi_v,\lambda)$ does not depend on λ.

Since $\pi_v\kappa_a = \chi_{\psi,v}\cdot\chi_v(a,\cdot)_v$, the condition holds whenever $r_v(\pi_v,\lambda)$ depends on $\pi_v = \chi_{\psi,v}\chi_v$ only through χ_v^2, up to a constant factor. The explicit factors of Lemma 6.6 are of this form; so (N5)

holds where π_v and $\pi_v\kappa_a$ are both covered by that lemma, which is always the case at the real and complex places. At the other places (N5) is a condition on the choice of the factors, in addition to (N1)–(N4). It can always be met: Proposition 7.8′ below constructs factors with (N1)–(N5) at every finite place, and identifies them with $L_v(2\lambda,\chi_v^2)/L_v(2\lambda+1,\chi_v^2)$ up to a constant by Szpruch's computation of the local coefficients. The statements below say where (N5) is used, and Lemma 7.8(c) says what happens for factors without it.

Example 7.F. Let $k=\mathbb{Q}$, let p be an odd prime in S, and let $\delta=\delta_a$ with a equal to p at p and to 1 at the other places of S. For $k=\begin{pmatrix}\alpha & b\\ c & d\end{pmatrix}\in K_p$ we have $e_2k\delta=(pc,d)$ and $\det(k\delta)=p$. If $d\in\mathbb{Z}_p^\times$, then $\|e_2k\delta\|_p=1$. If $p\mid d$, then c is a unit and $\|e_2k\delta\|_p=p^{-1}$. At the other places of S the component of $k\delta$ lies in K_v and contributes nothing. Hence

$$u_B'(\tilde{k},\delta)=-H_1(k\delta)=\begin{cases}\frac{1}{2}\log p, & d\in\mathbb{Z}_p^\times,\\ -\frac{1}{2}\log p, & d\in p\mathbb{Z}_p.\end{cases}$$

So the weight in $f_{B,\delta}$ is a step function on K_p. The bottom row of k reduces to a uniformly distributed nonzero vector of $\mathbb{F}_p^2$, so the second case has measure $1/(p+1)$. As a check of Lemma 7.8(a) below, take $x=1$, so that $k=1$ and $k'=w$. The formula gives $v_A(1)-H_1(\delta)-H_1(w\delta)=0+\frac{1}{2}\log p-\frac{1}{2}\log p=0$, as it must, since $\alpha^{-1}(1)=1$.

Lemma 7.8. *Let $f\in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$, $a\in k_S^\times$, $\delta=\delta_a$, and $\alpha=\mathrm{Ad}(\delta)$.*

(a) For $x\in\overline{G(\mathbb{Q}_S)}$ with Iwasawa decompositions $x=n\tilde{a}\tilde{k}$, $wx=n'\tilde{a}'\tilde{k}'$,

$$v_A\big(\alpha^{-1}(x)\big)=v_A(x)-H_1(k\delta)-H_1(k'\delta).$$

(b) For $\gamma\in\overline{A(\mathbb{Q}_S)}$ regular, $J_A(\gamma,f^\delta)=J_A(\alpha\gamma,f)+f_{B,\delta}(\alpha\gamma)+f_{B,\delta}(w\alpha\gamma w^{-1})$, and $J_G(\gamma,f^\delta)=J_G(\alpha\gamma,f)$.

(c) For $\pi\in\Pi_u(\overline{A(\mathbb{Q}_S)})$ and $\lambda\in i\mathbb{R}$, with $\pi'=\pi\circ\alpha^{-1}=\pi\kappa_a$: $\mathrm{tr}\,I(\pi_\lambda,f^\delta)=\mathrm{tr}\,I(\pi'_\lambda,f)$, and

$$\varphi_A(f^\delta,\pi_\lambda)=\varphi_A(f,\pi'_\lambda)+\widehat{f_{B,\delta}}(\pi'_\lambda)+\widehat{f_{B,\delta}}\big((\pi')^w_{-\lambda}\big)-d_a(\pi,\lambda)\,\mathrm{tr}\,I(\pi'_\lambda,f),$$

where $d_a(\pi,\lambda)=\sum_{v\in S_k}\big(\frac{r_v'}{r_v}(\pi_v\kappa_a,\lambda)-\frac{r_v'}{r_v}(\pi_v,\lambda)\big)$. The condition (N5) says that $d_a=0$.

(d) The products $r_v(\pi_v,\lambda)r_v(\pi_v^w,-\lambda)$ and $r_v(\pi_v\kappa_a,\lambda)r_v((\pi_v\kappa_a)^w,-\lambda)$ are equal up to a sign that does not depend on λ. So (N5) is compatible with (N2).

Proof. (a) The image of $\alpha^{-1}(x)$ in $G(\mathbb{Q}_S)$ is $\delta^{-1}g\delta$, $g=p(x)$. By the properties of H_1, $H(\delta^{-1}g\delta)=H_1(\delta^{-1})+H(g)+H_1(k\delta)$. Since $w\delta^{-1}w^{-1}=\mathrm{diag}(1,a^{-1})$, $H(w\delta^{-1}g\delta)=H_1(\mathrm{diag}(1,a^{-1}))+H(wg)+H_1(k'\delta)$, and $H_1(\mathrm{diag}(1,a^{-1}))=-H_1(\delta^{-1})$. Subtract from $v_A=-H(\cdot)-H(w\cdot)$.

(b) $x\mapsto\alpha(x)$ preserves the Haar measure of $G(\mathbb{Q}_S)$ and fixes $A(\mathbb{Q}_S)$ pointwise, so it preserves the measure on the quotient. Substituting $x=\alpha^{-1}(x')$ in (4.12),

$$J_A(\gamma,f^\delta)=|D(\gamma)|^{1/2}\int f\big(x'^{-1}\alpha(\gamma)x'\big)\,v_A\big(\alpha^{-1}(x')\big)\,dx'.$$

Insert (a) and argue as in the proof of Lemma 6.3(b), which uses only the form of the difference of the weights.

(c) Realize $I(\pi_\lambda)$ on the space of functions Φ on $\overline{G(\mathbb{Q}_S)}$ with $\Phi(n\tilde{a}x) = \pi(\tilde{a})e^{(\lambda+1)H_S(\tilde{a})}\Phi(x)$, by right translation, and let $(A(\lambda)\Phi)(x) = \Phi(\alpha^{-1}x)$. Since α preserves $N(\mathbb{Q}_S)$ and H_S on $\overline{A(\mathbb{Q}_S)}$, $A(\lambda)$ maps $I(\pi_\lambda)$ to $I(\pi'_\lambda)$, and $A(\lambda)^{-1}I(\pi'_\lambda, y)A(\lambda) = I(\pi_\lambda, \alpha^{-1}y)$; integrating against f,

$$A(\lambda)^{-1}\, I(\pi'_\lambda, f)A(\lambda) = I(\pi_\lambda, f^\delta), \tag{7.13}$$

which gives the first statement. In the compact picture $A(\lambda)$ depends on λ through $e^{(\lambda+1)H_S(\alpha^{-1}\tilde{k})}$, so $\frac{d}{d\lambda}A(\lambda) = \tilde{H}A(\lambda)$, where $\tilde{H}$ is multiplication by $\tilde{k} \mapsto H_S(\alpha^{-1}\tilde{k}) = H_1(\delta^{-1}) + H_1(k\delta)$. Next we compare intertwining operators. By Lemma 7.1(d), $\alpha^{-1}((w,1)^{-1}) = (w,1)^{-1}\tilde{b}$ with $\tilde{b} \in \overline{A(\mathbb{Q}_S)}$ over $a(a)$, and $\alpha^{-1}(n(u)) = n(a^{-1}u)$. Two changes of variable in the integral defining $M(\pi', \lambda)$ give

$$M(\pi', \lambda)A(\lambda) = c(\lambda)A(-\lambda)M(\pi, \lambda), \qquad c(\lambda) = \pi^w(\tilde{b})\,|a|_S^{-\lambda},$$

where $A(-\lambda)$ is formed with π^w. Hence, with $R = r^{-1}M$ and $\rho(\lambda) = r(\pi', \lambda)\, c(\lambda)^{-1} r(\pi, \lambda)^{-1}$,

$$A(-\lambda)\, R_S(\pi, \lambda) = \rho(\lambda)\, R_S(\pi', \lambda)A(\lambda), \qquad \frac{\dot{\rho}}{\rho} = \frac{\dot{r}}{r}(\pi', \lambda) - \frac{\dot{r}}{r}(\pi, \lambda) + \log|a|_S,$$

where a dot denotes $d/d\lambda$. (The scalar $r(\pi, \lambda)$ is the product of the $r_v(\pi_v, \lambda)$, $v \in S_k$.) Differentiate, and multiply on the left by $(\rho R_S(\pi', \lambda))^{-1} = A(\lambda)R_S(\pi, \lambda)^{-1}A(-\lambda)^{-1}$. Writing R for $R_S(\pi, \lambda)$ and $R^\natural$ for $R_S(\pi', \lambda)$,

$$(R^\natural)^{-1}\dot{R}^\natural = A(\lambda)\, R^{-1}\dot{R}A(\lambda)^{-1} - \frac{\dot{\rho}}{\rho} - \tilde{H} - (R^\natural)^{-1}\tilde{H}R^\natural.$$

Now $\tilde{H} = -\frac{1}{2}\log|a|_S - u'$, where u' is multiplication by $u'_B(\tilde{k}, \delta) = -H_1(k\delta)$, and $\dot{\rho}/\rho = d_a(\pi, \lambda) + \log|a|_S$. The scalars cancel:

$$-\frac{\dot{\rho}}{\rho} - \tilde{H} - (R^\natural)^{-1}\tilde{H}R^\natural = \underbrace{-\log|a|_S + \tfrac{1}{2}\log|a|_S + \tfrac{1}{2}\log|a|_S}_{=0} - d_a(\pi, \lambda) + u' + (R^\natural)^{-1}u'R^\natural.$$

Take the trace against $I(\pi'_\lambda, f)$, use (7.13), (N1), and Lemma 6.1(c) for the multiplication operator $-H_1(k\delta) = u'_B(\tilde{k}, \delta)$:

$$\varphi_A(f, \pi'_\lambda) = \varphi_A(f^\delta, \pi_\lambda) - \widehat{f_{B,\delta}}(\pi'_\lambda) - \widehat{f_{B,\delta}}\big((\pi')^w_{-\lambda}\big) + d_a(\pi, \lambda)\,\mathrm{tr}\, I(\pi'_\lambda, f).$$

(d) Apply the relation $M(\pi', \lambda)A(\lambda) = c(\lambda)A(-\lambda)M(\pi, \lambda)$ to π and to π^w at $-\lambda$; the second constant is $\pi(\tilde{b})|a|_S^{\lambda}$. So $M((\pi')^w, -\lambda)M(\pi', \lambda)$ is the conjugate by $A(\lambda)$ of $\pi(\tilde{b})\pi^w(\tilde{b}) \cdot M(\pi^w, -\lambda)M(\pi, \lambda)$. By (N2) the two compositions are the scalars $r(\pi', \lambda)r((\pi')^w, -\lambda)$ and $r(\pi, \lambda)r(\pi^w, -\lambda)$, and $\pi(\tilde{b})\pi^w(\tilde{b}) = \pi\big(\tilde{b}w\tilde{b}w^{-1}\big)$, where $\tilde{b}w\tilde{b}w^{-1}$ lies over 1, is ± 1. The argument is local at each $v \in S_k$. □

The next proposition shows that (N5) is not a restriction. Let v be a finite place, $F = k_v$, $q = q_v$, and for a genuine unitary character π of $\overline{A(F)}$ let $\rho(\pi, \lambda)$ be the scalar with

$$M_v(\pi^w, -\lambda)\, M_v(\pi, \lambda) = \rho(\pi, \lambda) \cdot 1;$$

the composition commutes with $I(\pi_\lambda)$, which is irreducible for λ outside a discrete set (as in the proof of Lemma 6.6(a); for unitary data see [Mo88]). Up to the normalization of the measure du, $\rho(\pi,\lambda)^{-1}$ is the Plancherel density of the genuine principal series. Let $\omega(\pi) = \pi\big((w,1)^2\big)$; the element $(w,1)^2$ lies over -1 and is central, and its square is 1 or $(1,-1)$, so $\omega(\pi) \in \mu_4$.

Proposition 7.8′. *Let v be a finite place of k, dyadic or not, and let ψ_v be arbitrary.*

(a) $\rho(\pi,\lambda)$ is a rational function of $z = q^{-\lambda}$, not identically zero, and

$$\rho(\pi^w,-\lambda) = \rho(\pi,\lambda), \qquad \rho(\pi_\nu,\lambda) = \rho(\pi,\lambda+\nu) \ \ (\nu \in i\mathbb{R}),$$
$$\rho(\pi\kappa_a,\lambda) = \pm\rho(\pi,\lambda) \ \ (a \in F^\times),$$

with a sign that does not depend on λ. For $\lambda \in i\mathbb{R}$ the adjoint of $M_v(\pi,\lambda)$ is $\omega(\pi)M_v(\pi^w,-\lambda)$, so that $\rho(\pi,\lambda) = \overline{\omega(\pi)}\,p(\pi,\lambda)$ with $p(\pi,\lambda) = |\rho(\pi,\lambda)| \geq 0$ on $i\mathbb{R}$.

(b) There is a unique rational function h_π of z, with no zero and no pole in the disc $|z| < 1$ and with $h_\pi(0) > 0$, such that $|h_\pi(z)|^2 = p(\pi,\lambda)$ for $|z| = 1$, $z = q^{-\lambda}$. Fix a map $c\colon \mu_4 \to \mu_8$ with $c(\zeta)^2 = \bar\zeta$ and $c(1) = 1$. The factors

$$r_v(\pi,\lambda) = c\big(\omega(\pi)\big)\,h_\pi(q^{-\lambda})$$

satisfy (N1)–(N5), and $r_v(\pi\kappa_a,\lambda)/r_v(\pi,\lambda) = c(\omega(\pi\kappa_a))/c(\omega(\pi)) \in \mu_8$. For v, ψ_v, π as in (N4) they are the factors of Lemma 6.6(a).

(c) (Szpruch [Sz09, Sz16].) *Write $\pi = \chi_{\psi,v}\cdot(\chi\circ p)$. Then*

$$p(\pi,\lambda) = C\,\frac{L_v(2\lambda,\chi^2)\,L_v(-2\lambda,\chi^{-2})}{L_v(1+2\lambda,\chi^2)\,L_v(1-2\lambda,\chi^{-2})}$$

with a constant $C = C(\chi,\psi_v) > 0$. Hence $h_\pi(q^{-\lambda}) = \sqrt{C}\,L_v(2\lambda,\chi^2)/L_v(2\lambda+1,\chi^2)$, and the factors of (b) are

$$r_v(\pi,\lambda) = c\big(\omega(\pi)\big)\sqrt{C}\,\frac{L_v(2\lambda,\chi^2)}{L_v(2\lambda+1,\chi^2)}$$

at every finite place: they are of the form $C_v(\chi)F_v(\chi^2,\lambda)$, with no exponential factor.

Proof. (a) On each $\overline{K_v}$-type, which is finite-dimensional and preserved by the intertwining operators, $M_v(\pi,\lambda)$ is a rational function of z [Li12a], and it is invertible for generic λ; so ρ is rational and not zero. If AB is the scalar ρ on a finite-dimensional space and $\rho \neq 0$, then $BA = \rho$; this is the first identity. The second holds because $M_v(\pi_\nu,\lambda) = M_v(\pi,\lambda+\nu)$, as in Lemma 6.6(c). The third is the first step of the proof of Lemma 7.8(d), which does not use normalizing factors: $M((\pi')^w,-\lambda)M(\pi',\lambda)$ is the conjugate by $A(\lambda)$ of $\pi(\tilde b)\pi^w(\tilde b)M(\pi^w,-\lambda)M(\pi,\lambda)$, and $\pi(\tilde b)\pi^w(\tilde b) = \pm 1$. For the adjoint, let $\varphi \in V(\pi)$, $\varphi' \in V(\pi^w)$ and $\tilde w = (w,1)$. For $\operatorname{Re}\lambda > 0$ the usual unfolding of the pairing $\langle\Phi,\Phi'\rangle = \int_{\overline{K_v}}\Phi\overline{\Phi'}$ between $V(\pi^w)_{-\lambda}$ and $V(\pi^w)_{\bar\lambda}$ gives

$$\big\langle M_v(\pi,\lambda)\varphi,\varphi'\big\rangle = \int_{\overline{A(F)\backslash SL(2,F)}} \varphi_\lambda(\tilde w^{-1}x)\,\overline{\varphi'_{\bar\lambda}(x)}\,dx,$$

the integrand being left invariant under $\overline{A(F)}$. Since $\tilde{w}^2$ is central and lies in $\overline{A(F)}$ with $H_v(\tilde{w}^2)=0$, $\varphi_\lambda(\tilde{w}x)=\omega(\pi)\varphi_\lambda(\tilde{w}^{-1}x)$; so the substitution $x\mapsto\tilde{w}x$ turns the integral into $\omega(\pi)^{-1}\int\varphi_\lambda(x)\overline{\varphi'_{\bar\lambda}(\tilde{w}^{-1}x)}\,dx=\omega(\pi)^{-1}\langle\varphi,M_v(\pi^w,\bar\lambda)\varphi'\rangle$. By continuation, $M_v(\pi,\lambda)^*=\omega(\pi)M_v(\pi^w,-\lambda)$ on $i\mathbb{R}$ (this is [Ar89, §1], with the scalar by which $(w,1)^2$ acts; compare the proof of Lemma 6.6(a), where $\omega=1$). Then $M^*M=\omega(\pi)\rho(\pi,\lambda)$ is a nonnegative scalar.

(b) *The function h_π.* Let $P(z)$ be the rational function with $P(q^{-\lambda})=p(\pi,\lambda)$; it is real and nonnegative on $|z|=1$ away from its poles. Hence $P(z)=\overline{P(1/\bar z)}$: the zeros and poles of P outside the circle come in pairs α, $1/\bar\alpha$ of equal order m_α, those on the circle have even order m_α, and the orders at 0 and ∞ are equal. Put

$$h(z)=\prod_{1<|\alpha|<\infty}(1-z/\alpha)^{m_\alpha}\prod_{|\alpha|=1}(1-z/\alpha)^{m_\alpha/2}.$$

Then $P(z)/\big(h(z)\overline{h(1/\bar z)}\big)$ has no zero or pole in $\mathbb{C}^\times$, so it is Cz^j, and it is positive on the circle, so $j=0$ and $C>0$; $h_\pi=\sqrt{C}h$. If h_1 is a second such function, h_1/h_π has no zero or pole in $|z|<1$ and absolute value 1 on the circle, hence by reflection no zero or pole at all; it is a positive constant of absolute value 1.

Symmetries. By uniqueness and (a): $h_{\pi_\nu}(z)=h_\pi(q^{-\nu}z)$; $h_{\pi\kappa_a}=h_\pi$; and, since $p(\pi^w,\lambda)=p(\pi,-\lambda)$, $h_{\pi^w}(z)=\overline{h_\pi(\bar z)}$. The last identity gives $h_{\pi^w}(z^{-1})=\overline{h_\pi(z)}$ on the circle, hence

$$h_{\pi^w}(z^{-1})\,h_\pi(z)=P(z)$$

on the circle and therefore identically. Also $\omega(\pi^w)=\omega(\pi_\nu)=\omega(\pi)$, because $(w,1)^2$ is central and H_ν vanishes on it.

The conditions. (N1) holds for M_ν. For (N2),

$$r_\nu(\pi^w,-\lambda)\,r_\nu(\pi,\lambda)=c(\omega(\pi))^2\,h_{\pi^w}(z^{-1})h_\pi(z)=\overline{\omega(\pi)}\,p(\pi,\lambda)=\rho(\pi,\lambda),$$

so $R_\nu(\pi^w,-\lambda)R_\nu(\pi,\lambda)=1$; and for $\lambda\in i\mathbb{R}$, $R_\nu^*R_\nu=M_\nu^*M_\nu/|h_\pi|^2=1$ wherever R_ν is defined, so R_ν is bounded near every point of $i\mathbb{R}$ and, being rational on each $\overline{K_\nu}$-type, has no pole there. (N3) is the rationality of M_ν and of h_π, and $r_\nu(\pi_\nu,\lambda)=r_\nu(\pi,\lambda+\nu)$ by the symmetries. (N5) holds because $h_{\pi\kappa_a}=h_\pi$. In the situation of (N4), $\omega(\pi)=1$ and, by Lemma 6.6(a) and its proof, $p(\pi,\lambda)=|g|^2$ on $i\mathbb{R}$ for $g=L_\nu(2\lambda,\chi^2)/L_\nu(2\lambda+1,\chi^2)=(1-q^{-1}\chi^2(\varpi)z^2)/(1-\chi^2(\varpi)z^2)$, which has its zeros on $|z|=q^{1/2}$, its poles on $|z|=1$ and the value 1 at $z=0$; so $h_\pi=g$, and (N4) holds.

(c) By [Sz09] (Theorem 8.1 of [Sz10]) the local coefficient of $I(\pi_\lambda)$ is $k_\chi q^{d_\chi\lambda}$ times a quotient of L-factors of χ and of χ^2, in every residue characteristic, and the product of the local coefficients of π_λ and $\pi^w_{-\lambda}$, which is $\rho(\pi,\lambda)^{-1}$ up to a nonzero constant, is a constant times $L_\nu(1+2\lambda,\chi^2)L_\nu(1-2\lambda,\chi^{-2})/L_\nu(2\lambda,\chi^2)L_\nu(-2\lambda,\chi^{-2})$; the factors involving χ and the exponential factors cancel. In [Sz16, (5.8)] the result is stated as a formula for the Plancherel measure of the double cover, with the constant $q^{e(\psi_2)-e(\chi^2)}$ for the measure self-dual with respect to ψ. The quotient g of L-factors satisfies $\overline{g(\bar\lambda)}=L_\nu(-2\lambda,\chi^{-2})/L_\nu(1-2\lambda,\chi^{-2})$ on $i\mathbb{R}$, it has no zero or pole in $|z|<1$, and $g=1$ at $z=0$; so $h_\pi=\sqrt{C}\,g$ by the uniqueness in (b). □

The proof of (a) and (b) uses nothing about the place beyond the rationality of M_ν; so the existence of factors with (N1)–(N5) does not rest on (c). Part (c) rests on the computations of Szpruch,

which we have not repeated (see Remark 2026.18 for what was read and what was checked numerically). At a real place the factors of Lemma 6.6(b) have the same shape: there $\omega(\pi) = e^{-i\pi m_0}$, the adjoint of $M_v(\pi,\lambda)$ on $i\mathbb{R}$ is $\omega(\pi)M_v(\pi^w,-\lambda)$ by (6.7), and $r_v = \sqrt{2}\,e^{i\pi m_0/2}L_v(2\lambda)/L_v(2\lambda+1)$ with $(e^{i\pi m_0/2})^2 = \overline{\omega(\pi)}$. *From here on the normalizing factors at the finite places are those of Proposition 7.8′.* With (c), the global factor of (6.10) is then

$$m(\mu,\lambda) = c_\mu \frac{\Lambda(2\lambda,\chi^2)}{\Lambda(2\lambda+1,\chi^2)}$$

with a nonzero constant c_μ and the completed Hecke L-function $\Lambda(s,\chi^2) = \prod_v L_v(s,\chi_v^2)$, so that m'/m is the logarithmic derivative of $\Lambda(2\lambda,\chi^2)/\Lambda(2\lambda+1,\chi^2)$ and the set $S_{\mathrm{bad}}(\mu)$ of Section 6.2 plays no role. (This is consistent with $m(\mu^w,-\lambda)m(\mu,\lambda) = 1$: by the functional equation the product is $c_\mu c_{\mu^w}\varepsilon(2\lambda,\chi^2)\varepsilon(-2\lambda,\chi^{-2})$, which does not depend on λ.)

Proposition 7.9. *Let the normalizing factors satisfy (N5), as those of Proposition 7.8′ do. Let $f \in C_c^\infty(\overline{G(\mathbb{Q}_S)})_-$ and $h \in G_1(k_S)$.*

(a) $I_A(x,f^h) = I_A(\mathrm{Ad}(h)x,f)$ *for $x \in \overline{A(\mathbb{Q}_S)}$ regular or central, whenever $\mathrm{Ad}(h)x \in \overline{A(\mathbb{Q}_S)}$.*

(b) For $h = \delta_a$ and x over $a(t)$: $I_A(x,f^{\delta_a}) = (a,t)_S I_A(x,f)$, with $(a,t)_S = \prod_{v\in S_k}(a_v,t_v)_v$; for a lift $\tilde z$ of -1, $I_A(\tilde z,f^{\delta_a}) = (a,-1)_S I_A(\tilde z,f)$.

(c) $\mathrm{tr}\,I(\pi,f^{\delta_a}) = \mathrm{tr}\,I(\pi\kappa_a,f)$, *and* $\mathrm{tr}\,I(\pi,f^{[h]}) = \mathrm{tr}\,I(\pi,f)$*: the characters of the principal series are stable. (This part does not use (N5). The function $f^{[h]}$ need not be smooth; the trace is then defined by the integral of orbital integrals in the proof.)*

Proof. (a) For $h \in SL(2,k_S)$ this is Theorem 6.9(b). Let $h = \delta = \delta_a$. By Lemma 7.8(c) and Fourier inversion on $\overline{A(\mathbb{Q}_S)}$, substituting $\pi \mapsto \pi'$ and using $\pi(\tilde a) = \pi'(\alpha\tilde a)$,

$$\Phi_A(\tilde a,f^\delta) = \Phi_A(\alpha\tilde a,f) + f_{B,\delta}(\alpha\tilde a) + f_{B,\delta}(w\,\alpha\tilde a\,w^{-1}).$$

Subtract from Lemma 7.8(b). For central x use the limit of Lemma 4.10(b), as for Lemma 6.3(c). (b) follows from (a), Lemma 7.1(d) and the fact that $I_A(\cdot,f)$ is anti-genuine. (c) The first statement is Lemma 7.8(c). By the proof of Theorem 6.9(a), $\mathrm{tr}\,I(\pi,f) = \int_{\overline{A(\mathbb{Q}_S)}} \pi(\tilde a)J_G(\tilde a,f)\,d\tilde a$, an integral of hyperbolic orbital integrals, and $J_G(\tilde a,f^{[h]}) = J_G(h\star\tilde a,f) = J_G(\tilde a,f)$ because $h\star\tilde a$ is conjugate to $\tilde a$. □

Remark 2026.18. *The condition (N5).* By the proof of Lemma 7.8(c), (N5) holds at v if and only if

$$R_v(\pi_v\kappa_a,\lambda) = C\,|a|_v^{-\lambda}\,A(-\lambda)\,R_v(\pi_v,\lambda)\,A(\lambda)^{-1}$$

with a constant C; the factor $|a|_v^{-\lambda}$ removes the scalar $|a|_v^{\lambda}$ contained in $A(-\lambda)A(\lambda)^{-1}$. It is natural: the inducing data π_v and $\pi_v\kappa_a$ have the same χ_v^2, hence the same parameter on the side of $SO(3)$ (Section 7.2, *The dual group*). Earlier drafts of this section had (N5) as a hypothesis at the dyadic and ramified places and formulated its existence as a problem. Proposition 7.8′ settles it, in two steps of different standing.

Existence (parts (a), (b)) is proved here. The point is that the scalar $\rho(\pi,\lambda)$ is the same for π and $\pi\kappa_a$ up to a sign (Lemma 7.8(d)), and that a normalizing factor can be read off from ρ alone,

as the factor of $|\rho|$ without zeros or poles in $\mathrm{Re}\,\lambda > 0$. One cannot instead transport an arbitrary r_v along the orbits of $k_v^\times/k_v^{\times 2}$ on the genuine characters: when $\pi_v^w = \pi_v\kappa_{a_0}$ with a_0 not a square, that is, when $\chi_v^2 = (a_0,\cdot)_v \neq 1$, the relation (N2) constrains the factor of π_v itself. The canonical choice meets this constraint because $h_{\pi^w}(z) = \overline{h_\pi(\bar z)}$. What is quoted in the proof is the rationality of M_v on the $\overline{K_v}$-types and generic irreducibility. We have not checked whether the factors of [Li12a] have property (N5); nothing here depends on it.

The explicit form (part (c)) is quoted. Sources, and how they were read (November 2026, in the arXiv versions; the journal version of [Sz09] was not opened): the local coefficient is Theorem 8.1 of [Sz10], which is the main theorem of [Sz09] and includes the 2-adic fields; the product formula is used in this form in the proof of Theorem 10.1 of [Sz10]; and [Sz16, (5.8)] states the Plancherel measure for $n = 2$ with its constant, deriving it from [Sz09]. (The general theorem of [Sz16] for covers of degree $n \not\equiv 0 \bmod 4$ is proved there for odd n; we use only the case $n = 2$.) The Plancherel measures of [GS16] are computed for residue characteristic prime to n and are not needed. As a check that does not depend on our reading, the script evaluates the Tate-type integrals $\int_{F^\times} \gamma_\psi(u)^{-1}\chi(u)\psi(u)|u|^s\,d^\times u$, which by [Sz09] are the inverse local coefficients, for $F = \mathbb{Q}_2$ and all characters χ of conductor dividing 32 modulo unramified twists, and for $F = \mathbb{Q}_3, \mathbb{Q}_5$ and all tamely ramified χ: the product of the integrals for (χ, s) and $(\chi^{-1}, -s)$ is a constant times $L(2s,\chi^2)L(-2s,\chi^{-2})/L(1+2s,\chi^2)L(1-2s,\chi^{-2})$ in every case. The relation between these integrals and ρ is Szpruch's and was not rederived.

For factors without (N5), every formula below that is stated under (N5) holds with the additional term coming from d_a in Lemma 7.8(c). The distributions $I_A(x,f)$ depend on the choice of the factors in any case (Remark 2026.15); with the factors of Proposition 7.8′ they are canonical.

So $I_A(x,\cdot)$ transforms under $G_1(k_S)$ by the character $h \mapsto (\det h, t)_S = \eta_x(\det h)$, as the orbital integral $J_G(x,\cdot)$ does. This does *not* say that $I_A(x,\cdot)$ is stable. Stability is invariance under $f \mapsto f^{[h]}$, where the sign $\eta_y(\det h)$ varies with the point y of the support of f, and $I_A(x,f)$ depends on f away from the class of x. At a real place, for instance, $\Phi_A(x,f)$ is built from the operators $R_v^{-1}R_v'$, whose poles are the points $\lambda = a_i, b_i$ at which $I(\pi_\lambda)$ has a single discrete series representation as a subquotient (Example 6.B), while a stable character involves two discrete series representations with different central characters [Ad98]. The stabilization of these terms in [Li21] expresses them through stable distributions on the endoscopic groups for both endoscopic data; it requires the weighted fundamental lemma [Li12b] and is not attempted here.

7.6 The central terms and the unipotent coefficients

The class $\mathcal{O}_z$ of $z = \pm 1$ contributes $J_{\mathrm{unip}}(f_z)$, which Proposition 4.8 expresses through the zeta integral $Z(f,s)$. Its second form in (4.9) is a sum over the rational unipotent classes $\xi \in k^\times/k^{\times 2}$; grouped according to the image of ξ in $k_S^\times/k_S^{\times 2}$ it diverges at $s = 1$, class by class; this is the difficulty noted in Remark 2026.10. The way out, as in [LL79], is to expand in the characters of the compact group $\mathbb{A}_k^\times/k^\times\mathbb{A}_k^{\times 2}$ instead. This computes the coefficients $a^G(S,u_\xi)$ of Theorem 4.11.

Let S be as in Theorem 5.16: a finite set of places of $\mathbb{Q}$ containing ∞, 2 and the places below which ψ is ramified. In this subsection we assume in addition that the ring $\mathcal{O}_S$ of elements of k that are integral outside S_k is a principal ideal domain, which holds once S is large enough (and always for $k = \mathbb{Q}$). Let $\mathfrak{X}_S$ be the group of quadratic characters κ of $k^\times\backslash\mathbb{A}_k^\times$ that are unramified outside S_k, κ_S the restriction of κ to $k_S^\times$, and $L^S(s,\kappa) = \prod_{v\notin S_k}(1-\kappa_v(\varpi_v)q_v^{-s})^{-1}$. For $\kappa = 1$ this is $\zeta_k^S(s)$;

write

$$\zeta_k^S(s) = \frac{\rho_S}{s-1} + c_S + O(s-1).$$

By Lemma 5.3(d) and its proof, $\mathfrak{X}_S = \{\kappa_b \mid b \in \mathscr{O}_S^\times / \mathscr{O}_S^{\times 2}\}$ with $\kappa_b(t) = \prod_v (b,t_v)_v$; it has $2^{|S_k|}$ elements, by the unit theorem. For $f_S = \bigotimes_{v\in S_k} f_v$, $z \in \{\pm 1\}$ and Φ_v as in Lemma 4.10, put

$$Z_v(\Phi_v, \kappa_v, s) = \int_{k_v^\times} \Phi_v(y)\kappa_v(y)|y|_v^s d^\times y, \qquad Z_S(\Phi_S, \kappa_S, s) = \prod_{v\in S_k} Z_v(\Phi_v, \kappa_v, s),$$

Tate's local zeta integrals [Ta67], holomorphic for $\operatorname{Re} s > 0$; by linearity Z_S is defined for all $f_S \in \mathscr{H}(\overline{G(\mathbb{Q}_S)})$. For $\xi \in k_S^\times$ let

$$J_G(zu_\xi, f_S) = |\xi|_S \int_{\{\pm 1\}N(\mathbb{Q}_S)\backslash G(\mathbb{Q}_S)} f_S\big(x^{-1} z n(\xi) x\big)\, dx = \prod_{v\in S_k} \frac{|\xi|_v}{2} \int_{k_v^\times} \Phi_v(\xi t^2)|t|_v^2 d^\times t, \tag{7.14}$$

the second expression by Lemma 4.10(c). The factor $|\xi|_S$ makes $J_G(zu_\xi, f_S)$ depend only on the class of ξ in $k_S^\times / k_S^{\times 2}$; it is the orbital integral for the measure on the centralizer of $n(\xi)$ transported from that of $n(1)$.

Proposition 7.10. *Let S be as above (in particular the ring $\mathscr{O}_S$ of S_k-integers is principal), $f_S \in \mathscr{H}(\overline{G(\mathbb{Q}_S)})$, $f = f_S \otimes e^S$ and $z \in \{\pm 1\}$.*

(a) For $\operatorname{Re} s > 1$,

$$Z(f_z, s) = \frac{1}{2} \sum_{\kappa \in \mathfrak{X}_S} L^S(s,\kappa) Z_S(\Phi_S, \kappa_S, s).$$

(b) For every character κ_S of $k_S^\times / k_S^{\times 2}$,

$$Z_S(\Phi_S, \kappa_S, 1) = \sum_{\xi \in k_S^\times / k_S^{\times 2}} \kappa_S(\xi) J_G(zu_\xi, f_S) =: J^\kappa_{\mathrm{unip}}(z, f_S),$$

and $\frac{d}{ds} Z_S(\Phi_S, 1, s)\big|_{s=1} = \rho_S^{-1} c_k J_A(z, f_S)$.

(c) Consequently

$$J_{\mathscr{O}_z}(f) = \operatorname{vol}\big(G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}\big) f_S(z) + \frac{c_k}{2} J_A(z, f_S) + \frac{c_S}{2} J^1_{\mathrm{unip}}(z, f_S) + \frac{1}{2} \sum_{\kappa \in \mathfrak{X}_S,\ \kappa \neq 1} L^S(1,\kappa) J^\kappa_{\mathrm{unip}}(z, f_S).$$

This is Theorem 4.11(a), with

$$a^G(S, u_\xi) = \frac{1}{2}\Big(c_S + \sum_{\kappa \in \mathfrak{X}_S,\ \kappa \neq 1} \kappa_S(\xi) L^S(1,\kappa)\Big), \qquad \xi \in k_S^\times / k_S^{\times 2}.$$

(d) For $a \in k_S^\times$,

$$J_G(zu_\xi, f_S^{\delta_a}) = (a,z)_S J_G(zu_{a\xi}, f_S), \qquad J^\kappa_{\mathrm{unip}}(z, f_S^{\delta_a}) = \kappa_S(a)\,(a,z)_S J^\kappa_{\mathrm{unip}}(z, f_S).$$

Proof. We may replace f by f_z and take $z=1$; then $\Phi_f = \Phi_S \otimes \bigotimes_{v\notin S_k} \mathbf{1}_{\mathscr{O}_v}$.

A local identity. For h integrable on $k_v^\times$ and $\xi \in k_v^\times$,

$$\int_{k_v^\times} h(\xi t^2)\, d^\times t = \frac{1}{2}\sum_{\kappa_v} \kappa_v(\xi) \int_{k_v^\times} h(y)\kappa_v(y)\, d^\times y, \tag{7.15}$$

the sum over the characters of $k_v^\times/k_v^{\times 2}$. Indeed $t\mapsto t^2$ is two-to-one onto the open subgroup $k_v^{\times 2}$ and multiplies $d^\times t$ by $|2|_v$, so the left side is $2|2|_v^{-1}\int_{\xi k_v^{\times 2}} h\, d^\times y$; and the characteristic function of $\xi k_v^{\times 2}$ is $[k_v^\times : k_v^{\times 2}]^{-1}\sum_{\kappa_v}\kappa_v(\xi)\kappa_v$. The index is $4|2|_v^{-1}$ at every place ($4|2|_v^{-1}$ for v finite, 2 for v real, 1 for v complex, with the normalized absolute values).

(a) Let $\Gamma = \mathscr{O}_S^\times$ and $Y_1 = k_S^\times \times \prod_{v\notin S_k}\mathscr{O}_v^\times$. Since $\mathscr{O}_S$ is principal, $\mathbb{A}_k^\times = k^\times Y_1$ and $k^\times \cap Y_1 = \Gamma$, so $\mathbb{A}_k^\times$ is the disjoint union of the cY_1, $c\in k^\times/\Gamma$. For $\eta\in k^\times$ let $J(\eta) = \int_{Y_1}\Phi_f(\eta t^2)|t|^{2s} d^\times t$. Then the integral in the second form of (4.9) is $\sum_{c\in k^\times/\Gamma} J(\xi c^2)$, and $(\xi, c)\mapsto \xi c^2$ is a bijection from $(k^\times/k^{\times 2})\times(k^\times/\Gamma)$ onto $k^\times/\Gamma^2$ (because $-1\in\Gamma$). So $Z(f,s) = \frac{1}{2}\sum_{\eta\in k^\times/\Gamma^2} J(\eta)$. Now $J(\eta)=0$ unless $\eta\in\mathscr{O}_S$, and by (7.15) at each $v\in S_k$, applied to $h = \Phi_v|\cdot|_v^s$,

$$J(\eta) = |\eta|_S^{-s} 2^{-|S_k|}\sum_{\kappa_S}\kappa_S(\eta) Z_S(\Phi_S,\kappa_S,s) \qquad (\eta\in\mathscr{O}_S\smallsetminus\{0\}).$$

Summing over η modulo Γ^2: the sum of κ_S over Γ/Γ^2 is $2^{|S_k|}$ if κ_S is trivial on Γ and 0 otherwise. The characters κ_S trivial on Γ are the restrictions of the $\kappa\in\mathfrak{X}_S$, and for such κ and $\eta\in\mathscr{O}_S$, $\kappa_S(\eta) = \prod_{v\notin S_k}\kappa_v(\varpi_v)^{\mathrm{ord}_v\eta}$ and $|\eta|_S^{-1} = \prod_{v\notin S_k} q_v^{\mathrm{ord}_v\eta}$, by the product formula. The sum over $\eta\in(\mathscr{O}_S\smallsetminus\{0\})/\Gamma$ runs over the nonzero ideals of $\mathscr{O}_S$ and gives $L^S(s,\kappa)$.

(b) By (7.15) with $h=\Phi_v|\cdot|_v$ and (7.14), $J_{G,v}(u_\xi, f_v) = \frac{1}{4}\sum_{\kappa_v}\kappa_v(\xi) Z_v(\Phi_v,\kappa_v,1)$; invert on $k_v^\times/k_v^{\times 2}$ and multiply over $v\in S_k$. The constant is $\prod_{v\in S_k} 4\,[k_v^\times : k_v^{\times 2}]^{-1} = \prod_{v\in S_k}|2|_v = 1$, because S_k contains all places at which $|2|_v\neq 1$. For the derivative, let $d^\times t = c_v' dt/|t|_v$ at $v\in S_k$, dt the self-dual measure, and $C' = \prod_{v\in S_k} c_v'$. Then $Z_S(\Phi_S,1,1) = C'\prod_v\hat\Phi_v(0)$ and

$$\frac{d}{ds} Z_S(\Phi_S,1,s)\Big|_{s=1} = C'\sum_{v\in S_k}\Big(\int_{k_v}\Phi_v(y)\log|y|_v\, dy\Big)\prod_{w\neq v}\hat\Phi_w(0) = C' J_A(1, f_S),$$

by Lemma 4.10(b), since $\sum_{v\in S_k}\log|2|_v = 0$. Comparing the residue of (a) at $s=1$, which is $\frac{1}{2}\rho_S C'\prod_v\hat\Phi_v(0)$, with the residue $\frac{1}{2}c_k\hat\Phi_f(0)$ of Proposition 4.8 gives $\rho_S C' = c_k$.

(c) By (a), the finite part of $Z(f,s)$ at $s=1$ is $\frac{1}{2}\big(c_S Z_S(\Phi_S,1,1) + \rho_S\frac{d}{ds}Z_S(\Phi_S,1,s)|_{s=1}\big) + \frac{1}{2}\sum_{\kappa\neq 1} L^S(1,\kappa) Z_S(\Phi_S,\kappa_S,1)$; for $\kappa\neq 1$, $L^S(s,\kappa)$ is holomorphic at $s=1$. Insert (b) and Proposition 4.8. The coefficient of $J_G(zu_\xi, f_S)$ is read off from the definition of J^κ_{unip}.

(d) $x\mapsto \delta_a x\delta_a^{-1}$ preserves the Haar measure of $G(\mathbb{Q}_S)$ and multiplies that of $N(\mathbb{Q}_S)$ by $|a|_S$; so $\int h(\delta_a x\delta_a^{-1})\, dx = |a|_S\int h(x)\, dx$ for the invariant measure on the quotient by $\{\pm 1\}N(\mathbb{Q}_S)$. With Lemma 7.1(d), this gives $\int f_S^{\delta_a}(x^{-1}zn(\xi)x)\, dx = (a,z)_S|a|_S\int f_S(x^{-1}zn(a\xi)x)\, dx$. Multiply by $|\xi|_S$, and sum against $\kappa_S(\xi)$. □

Part (c) proves the fine expansion at the central classes directly from the coarse one, for the sets S considered, and so removes the appeal to [Ar86, Li14a] in Theorem 4.11(a). The terms with $\kappa\neq 1$ are the analog of the terms of [LL79, §5] that involve $L(1,\kappa)$.

Example 7.G. Let $k=\mathbb{Q}$ and $S=\{\infty,2\}$. Then $\mathscr{O}_S=\mathbb{Z}[\frac12]$, $\mathfrak{X}_S=\{\kappa_b \mid b=\pm1,\pm2\}$, and $\zeta^S(s)=(1-2^{-s})\zeta(s)$, so $\rho_S=\frac12$ and $c_S=\frac12(\gamma_E+\log 2)$, with Euler's constant γ_E. The characters κ_{-1}, κ_2, κ_{-2} are the Dirichlet characters of conductor 4, 8, 8, and

$$L^S(1,\kappa_{-1})=\frac{\pi}{4},\qquad L^S(1,\kappa_2)=\frac{\log(1+\sqrt2)}{\sqrt2},\qquad L^S(1,\kappa_{-2})=\frac{\pi}{2\sqrt2}.$$

For the 16 classes $\xi\in\mathbb{Q}_S^\times/\mathbb{Q}_S^{\times2}$,

$$a^G(S,u_\xi)=\frac12\Big(\frac{\gamma_E+\log 2}{2}+\sum_{b\in\{-1,2,-2\}}(b,\xi_\infty)_\infty(b,\xi_2)_2\,L^S(1,\kappa_b)\Big);$$

for instance $a^G(S,u_\xi)\approx 1.57726$ for $\xi=1$, and $a^G(S,u_\xi)\approx -0.31886$ for the class with $\xi_\infty<0$ and $\xi_2=1$. The coefficients are not all positive, and they depend on ξ only through the pairings with $b\in\{\pm1,\pm2\}$, the same parameters as those of the ramified classes χ_{ψ_b} in Example 5.A. In particular $a^G(S,u_{\xi\varepsilon})=a^G(S,u_\xi)$ for $\varepsilon\in\mathscr{O}_S^\times$, since the $\kappa\in\mathfrak{X}_S$ are trivial on $\mathscr{O}_S^\times$: the classes of the rational elements ±1, ±2 all have the coefficient of $\xi=1$, and the 16 classes carry four distinct values (the invariance holds for every k and S as above). With $\varepsilon_b=(b,\xi_\infty)_\infty(b,\xi_2)_2$ they are

$(\varepsilon_{-1},\varepsilon_2,\varepsilon_{-2})$	a class $\xi=(\xi_\infty,\xi_2)$	$a^G(S,u_\xi)$
$(+,+,+)$	$(1,1)$	1.57726
$(-,-,+)$	$(1,3)$	0.16864
$(+,-,-)$	$(1,5)$	−0.15668
$(-,+,-)$	$(-1,1)$	−0.31886

Each value is taken on four classes, and the four values add up to $2c_S=\gamma_E+\log 2$. The script checks (a) numerically at $s=3$ for a test function with Φ_∞ not even and Φ_2 the characteristic function of $6+16\mathbb{Z}_2$.

By part (d), $J^\kappa_{\mathrm{unip}}(1,\cdot)$ transforms under $G_1(k_S)$ by $\kappa_S(\det h)$. Near the identity $\eta_x=1$ (because D^+ is close to 4), so there the twisted and the untwisted actions agree: $J^1_{\mathrm{unip}}(1,\cdot)$, the sum over all unipotent classes, is the stable unipotent distribution, and the $J^\kappa_{\mathrm{unip}}(1,\cdot)$ with $\kappa\neq1$ are the unstable ones. For $SL(2)$ the latter are the limits, as $\gamma\to1$ in the torus T_E with $\kappa_E=\kappa$, of the κ-orbital integrals, up to the factor $L(1,\kappa)$ [LL79, §§2, 5]. On the cover the same should hold near the identity, where the cover is canonically split on the topologically unipotent set, and by Lemma 7.6(c) the κ-orbital integrals near 1 are the stable orbital integrals of f_- near -1. So the terms $L^S(1,\kappa)J^\kappa_{\mathrm{unip}}(1,f_S)$, $\kappa\neq1$, should be counted with $S(f_-)$, and the corresponding terms at $z=-1$ with $S(f)$. We have not proved these limit formulas for the cover (Remark 2026.19).

7.7 The spectral side

On the spectral side the group $G_1(k)$ acts on the space of automorphic forms, and the pieces of the spectral decomposition are permuted. We record this, and then compute the one family of discrete terms whose stable structure is visible without endoscopy: the theta functions.

For $h\in G_1(k)$ and a function φ on $G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}$ put $(U_h\varphi)(x)=\varphi(\mathrm{Ad}(h)^{-1}x)$; this is well defined by Lemma 7.1(f). For $a\in k^\times$ let κ_a also denote the character $\tilde a\mapsto\prod_v(a,t_v)_v$ of $\overline{A(\mathbb{A})}$.

Lemma 7.11. *Let $h \in G_1(k)$ and $a = \det h$.*

(a) U_h is a unitary operator on L^2_{gen} with $U_h R(y) U_h^{-1} = R(\mathrm{Ad}(h)y)$ and $U_h R(f) U_h^{-1} = R(f^{h^{-1}})$. It preserves L^2_{cusp}, L^2_{res} and L^2_{cont}.

(b) U_h maps L^2_χ onto $L^2_{\chi^h}$, where $(G,\pi)^h = (G, \pi \circ \mathrm{Ad}(h)^{-1})$ and $(A,\mu)^h = (A, \mu\kappa_a)$. In particular $m(\pi \circ \mathrm{Ad}(h)) = m(\pi)$, $\chi_{\psi_b}\kappa_a = \chi_{\psi_{ab}}$, and $U_h V_b = V_{ab}$.

(c) Let $f = f_S \otimes e^S$ and $h_v \in Z(k_v)GL(2,\mathscr{O}_v)$ for all $v \notin S_k$. Then $\mathrm{tr}(R(f^h)|_{L^2_{\chi,\mathrm{disc}}}) = \mathrm{tr}(R(f)|_{L^2_{\chi^h,\mathrm{disc}}})$ for every χ, and if (N5) holds, $I_\chi(f^h) = I_{\chi^h}(f)$. Hence $\sum_\chi I_\chi(f^h) = \sum_\chi I_\chi(f)$.

Proof. (a) $\mathrm{Ad}(h)$ preserves $G(\mathbb{Q})$ and the Tamagawa measure, which gives unitarity and the two formulas. It preserves $N(\mathbb{A})$ and its Haar measure up to the factor $|a|_{\mathbb{A}_k} = 1$, so $(U_h\varphi)_B = U_h(\varphi_B)$ for the constant term along B; hence cusp forms go to cusp forms, and the orthogonal complement, the discrete and the continuous parts of it, are preserved. (b) By Lemma 7.1(d), $\mathrm{Ad}(h)^{-1}$ acts on $\overline{A(\mathbb{A})}$ by the character κ_a (for $h = \delta_a$, and conjugation by $SL(2,k)$ does not change the class of a cuspidal datum). So U_h maps $\mathscr{H}_B(\mu)$ to $\mathscr{H}_B(\mu\kappa_a)$ and Eisenstein series, their wave packets and their residues to the corresponding objects for $\mu\kappa_a$. The formula for $\chi_{\psi_b}\kappa_a$ is Lemma 3.3(d). (c) By Lemma 7.1(e), $f^h = f_S^{h_S} \otimes e^S$. The first statement follows from (a) and (b): $\mathrm{tr}(R(f^h)|_L) = \mathrm{tr}(R(f)|_{U_h L})$ for an invariant subspace L. This gives $I_\chi(f^h) = I_{\chi^h}(f)$ for $\chi \in \mathscr{X}_G$ and for the residual terms. For $\mu \in \Pi_{\mathrm{aut}}(X_A)$, $\mathrm{tr}\, I(\mu_{it}, f^h) = \mathrm{tr}\, I((\mu\kappa_a)_{it}, f)$ by Lemma 7.8(c), and $\frac{m'}{m}(\mu\kappa_a,\lambda) = \frac{m'}{m}(\mu,\lambda)$: in (6.10) the partial L-functions depend only on χ^2, and the local factors change by constants, by (N5). So the terms of Theorem 6.11(e),(f) for χ and f^h are those for χ^h and f. □

The geometric counterpart of (c) is Theorem 7.13(c) below.

Now fix $b \in k^\times$ and put $\psi' = \psi_{-b}$, as in Theorem 5.8(c). For a pattern of signs $\varepsilon = (\varepsilon_v)_v$, with $\varepsilon_v = +$ for almost all v, let $r^\varepsilon_{\psi'} = \bigotimes'_v r^{\varepsilon_v}_{\psi'_v}$, an irreducible genuine unitary representation of $\overline{G(\mathbb{A})}$ on the functions in $\mathscr{S}(\mathbb{A}_k)$ of parity ε_v at each v. Let

$$\Theta_b = \left\{ \theta^{\psi'}_\phi \;\middle|\; \phi \in \mathscr{S}(\mathbb{A}_k) \right\}^- \subset L^2_{\mathrm{gen}},$$

the closure of the space of all theta functions (5.6), even or not; and for $f = \bigotimes f_v$ let

$$\Theta^{\mathrm{st}}_b(f) = \prod_v \left(\mathrm{tr}\, r^+_{\psi'_v}(f_v) - \mathrm{tr}\, r^-_{\psi'_v}(f_v) \right) = \prod_v \int f_v(x) \frac{\Delta_{\psi'_v}(x)}{|D^+(x)|_v^{1/2}}\, dx,$$

the second expression by Proposition 7.5. Almost all factors are 1. Both Θ_b and Θ^{st}_b depend only on the class of b in $k^\times/k^{\times 2}$: the substitution $\phi(x) \mapsto \phi(cx)$, $c \in k^\times$, carries $\omega_{\psi'}$ to $\omega_{\psi'_{c^{-2}}}$, preserves parities, and does not change $\sum_{\xi \in k}$.

Proposition 7.12. *(a) $\phi \mapsto \theta^{\psi'}_\phi$ is zero on $r^\varepsilon_{\psi'}$ if $\prod_v \varepsilon_v = -1$ and injective if $\prod_v \varepsilon_v = +1$. So*

$$\Theta_b \cong \bigoplus_{\prod_v \varepsilon_v = +1} r^\varepsilon_{\psi'},$$

a subspace of L^2_{disc}; the summand with all $\varepsilon_v = +$ is V_b, and the others are cuspidal.

(b) For $f=\bigotimes_v f_v\in\mathscr{H}(\overline{G(\mathbb{A})})$,

$$\operatorname{tr}\big(R(f)\big|_{\Theta_b}\big)=\tfrac12\,\Theta_b^{\mathrm{st}}(f)+\tfrac12\,\Theta_b^{\mathrm{st}}(f_-).$$

(c) Θ_b^{st} is a stable distribution: $\Theta_b^{\mathrm{st}}(f^{[h]})=\Theta_b^{\mathrm{st}}(f)$ for $h\in G_1(\mathbb{A}_k)$. Moreover $U_h\Theta_b=\Theta_{ab}$ and $\Theta_b^{\mathrm{st}}(f^h)=\Theta_{ab}^{\mathrm{st}}(f)$ for $h\in G_1(k)$, $a=\det h$.

Proof. (a) $\theta_{\omega(y)\phi}(x)=\theta_\phi(xy)$, so $\phi\mapsto\theta_\phi$ intertwines $\omega_{\psi'}$ with R. The parity operator $P=\bigotimes_v P_v$ satisfies $\theta_{P\phi}=\theta_\phi$ (replace ξ by $-\xi$ in (5.6); P commutes with ω). If $\prod\varepsilon_v=-1$, then $P\phi=-\phi$ on r^ε and $\theta_\phi=0$. If $\prod\varepsilon_v=+1$, let S_0 be a finite set of places containing the archimedean ones and those with $\varepsilon_v=-$, and $\xi_0=1$. Put $\phi_v=\mathbf{1}_{\mathscr{O}_v}$ for $v\notin S_0$, and $\phi_v=h_v+\varepsilon_vP_vh_v$ for $v\in S_0$, where h_v is supported in a neighborhood U_v of ξ_0 with $U_v\cap(-U_v)=\emptyset$ and $h_v(\xi_0)=1$. Then $\phi=\bigotimes\phi_v\in r^\varepsilon$. Since k is discrete in $\mathbb{A}_k$, for U_v small enough the only $\xi\in k$ that are integral outside S_0 and lie in $U_v\cup(-U_v)$ for every $v\in S_0$ are $\pm\xi_0$, and $\theta_\phi(1)=\phi(\xi_0)+\phi(-\xi_0)=1+\prod_{v\in S_0}\varepsilon_v=2$. So the map is not zero on the irreducible r^ε, hence injective. The constant term of θ_ϕ is $x\mapsto(\omega(x)\phi)(0)$ (Section 5.2), which vanishes when some ϕ_v is odd; so those θ_ϕ are cuspidal, and the even ones span V_b by Theorem 5.8(c). That theta functions are square integrable, and that $\phi\mapsto\theta_\phi$ is continuous for the L^2 norms on each r^ε, is standard [We64, Ge76, GPS80]: on a Siegel set $\theta_\phi(n\tilde{a}\tilde{k})=O(e^{H(\tilde{a})/2})$. The representations r^ε are pairwise inequivalent, because $r^+_{\psi'_v}$ and $r^-_{\psi'_v}$ have different central characters; so the sum is direct and orthogonal.

(b) By (a), $\operatorname{tr}(R(f)|_{\Theta_b})=\sum_{\prod\varepsilon_v=1}\prod_v\operatorname{tr}r^{\varepsilon_v}_{\psi'_v}(f_v)$, a finite sum because $\operatorname{tr}r^-_{\psi'_v}(e_v)=0$ at the unramified places. With $t_v^\pm=\operatorname{tr}r^\pm_{\psi'_v}(f_v)$ this is $\frac12\prod_v(t_v^+-t_v^-)+\frac12\prod_v(t_v^++t_v^-)$. The first product is $\Theta_b^{\mathrm{st}}(f)$, and the second is $\operatorname{tr}\omega_{\psi'}(f)$. By Lemma 5.7, $\omega_{\psi'_v}(-1,1)=\gamma'_v(1)^2P_v$, and $\prod_v\gamma'_v(1)=1$ (Lemma 3.3(b)); so $\omega_{\psi'}(s(-1))=P$. Therefore $\operatorname{tr}\omega_{\psi'}(f)=\operatorname{tr}\big(\omega_{\psi'}(s(-1))\,\omega_{\psi'}(f)\,P\big)=\operatorname{tr}\big(\omega_{\psi'}(f_-)P\big)=\Theta_b^{\mathrm{st}}(f_-)$, since $\omega(z)\omega(f)=\omega(f_{z^{-1}})$.

(c) Stability is Proposition 7.5 at each place. By Lemma 7.4, $\omega_{\psi'}\circ\mathrm{Ad}(\delta_a)=\omega_{\psi'_a}$ and $\psi'_a=\psi_{-ab}$; for $a\in k^\times$ the identity map of $\mathscr{S}(\mathbb{A}_k)$ therefore carries $\theta_\phi^{\psi'}\circ\mathrm{Ad}(\delta_a)$ to $\theta_\phi^{\psi'_a}$, which gives both statements (replace a by a^{-1}, which is in the same square class). □

Part (b) has the same two-term form as Theorem 7.7: a stable distribution evaluated at f and at f_-, each with coefficient $\frac12$. The multiplicity of $r^\varepsilon_{\psi'}$ in Θ_b is $\frac12(1+\prod_v\varepsilon_v)$. Waldspurger [Wa80, Wa91] proved that this is its multiplicity in L^2_{disc}, and that the rest of the discrete spectrum has a similar structure: it is a multiplicity-free sum over the cuspidal automorphic representations σ of $PGL(2,\mathbb{A}_k)$, each contributing the representations $\bigotimes_v\pi_v^{\varepsilon_v}$ of a packet with $\prod_v\varepsilon_v=\varepsilon(\frac12,\sigma)$, where the local packet is $\{\pi_v^+\}$ or $\{\pi_v^+,\pi_v^-\}$. Here the packets are those of the theta correspondence for the additive character ψ': π_v^+ is the lift of σ_v from $PGL(2,k_v)=SO(2,1)$, and π_v^-, when it exists, is the lift of the Jacquet–Langlands transfer of σ_v from the anisotropic $SO(3)$; the labels depend on ψ'. See [GS12] for the local correspondence and its epsilon dichotomy, [GI18] for the double cover of $Sp(2n)$, and [Li24]. In these terms Θ_b is the packet attached to the character κ_b of $PGL(2)$. One difference should be noted. For the theta functions the stable combination is the difference $\operatorname{tr}r^+-\operatorname{tr}r^-$ (Proposition 7.5). For the packets of cuspidal σ it is the sum $\operatorname{tr}\pi_v^++\operatorname{tr}\pi_v^-$, and the difference is unstable. The reason is the character identity of the endoscopic transfer from $SO(3)$: on the regular set,

$$\Theta_{\pi_v^+}(x)+\Theta_{\pi_v^-}(x)=\big(\Theta^+_{\psi'}(x)-\Theta^-_{\psi'}(x)\big)\cdot\Theta_{\sigma_v}\big(t(x)\big),\tag{7.16}$$

where $t(x)$ is the conjugacy class in $PGL(2,k_v) = SO(2,1)$ whose image in $SO(3)$ has the eigenvalues $\lambda, 1, \lambda^{-1}$ if $p(x)$ has the eigenvalues λ, λ^{-1}, and the left side is $\Theta_{\pi_v^+}$ alone when the packet has one element. For $k_v = \mathbb{R}$ this is the lifting of Adams [Ad98], whose last section treats the pair $(SO(3), \widetilde{SL}(2,\mathbb{R}))$; for k_v non-archimedean it is the theorem of Schultz [Sc98, Theorem 5.35] (see also [Ho10]); and for tempered parameters of $Mp(2n)$ over any local field of characteristic 0 it is the case $s = 1$ of the endoscopic character identities of Luo [Lu20], stated there through the transfer of [Li11] and for the packets of [GS12]. *Given* (7.16)*, the sum is stable in the sense of this section*: the first factor on the right is invariant under $\star$ by Proposition 7.5; the second depends only on the eigenvalues of $p(x)$, and $p(h \star x) = hp(x)h^{-1}$; and the characters are locally integrable functions [Li12a, §4], so that the distribution $\operatorname{tr}\pi_v^+ + \operatorname{tr}\pi_v^-$ is invariant under $f \mapsto f^{[h]}$. We quote (7.16) and do not prove it; we read [Sc98] and [Lu20] in their statements only, and did not match their normalizations of the transfer factor and of the correspondence t with ours beyond Proposition 7.5. (In [Sc98] the identification of the packets with Waldspurger's is stated without proof; in [Lu20] the packets are those of the theta correspondence by definition.) The case $s = -1$ of [Lu20] gives the difference $\Theta_{\pi_v^+} - \Theta_{\pi_v^-}$, up to the sign $\varepsilon(\frac{1}{2}, \sigma_v, \psi')$, as the transfer for the other endoscopic datum, whose transfer factor is built from $\Theta^+_{\psi'} + \Theta^-_{\psi'}$.

Two cases can be checked directly, without (7.16). At a finite place, the packet of the Steinberg representation is $\{\mathrm{sp}, r^-_{\psi'}\}$ ([GS12, §5], the example after Theorem 5.2 there), where sp is the other constituent of the principal series that has $r^+_{\psi'}$ as a quotient (it is the lift of the trivial representation of the anisotropic $SO(3)$, and $r^-_{\psi'}$ is the lift of the Steinberg representation); the sum of their characters is the character of that principal series minus $\Theta^+_{\psi'} - \Theta^-_{\psi'}$, a difference of two stable characters (Propositions 7.9(c) and 7.5). At a real place a packet consists of the discrete series representations with weights $m, m+2, \dots$ and $-m, -m-2, \dots$, $m \in \frac{1}{2} + \mathbb{Z}$, $m > 1$ (see [Ad98] for the stable discrete series characters of $Mp(2,\mathbb{R})$); on $\tilde{k}_\theta$ the sum of their characters is $-\sin((m-1)\theta)/\sin\theta$, which is even in θ, and the difference is $i\cos((m-1)\theta)/\sin\theta$, which is odd, while $\tilde{k}_\theta$ and $\tilde{k}_{-\theta}$ are stably conjugate (Example 7.B). In both kinds of packet the two members have opposite central characters ([GS12, §5] in general), so the unstable combination for f is, up to a constant, the stable one for f_-; this is how the pattern $S(f) + S(f_-)$ arises on the spectral side. We do not derive any of this from the trace formula. The local statements above are Theorems 5.1–5.3 of [GS12], where they are attributed to Waldspurger; the global multiplicity statement is the case $n = 1$ of [GI18, Theorem 1.4]. We have checked these two sources, and not the original papers [Wa80, Wa91], which we could not obtain.

By Proposition 7.9(c) the terms of Corollary 6.14 that come from the continuous spectrum, $-\frac{1}{4}\operatorname{tr} I(\chi_{\psi_b,0}, f)$ and $-\frac{1}{4\pi}\int \frac{m'}{m}(\mu, it)\operatorname{tr} I(\mu_{it}, f)\,dt$, are stable distributions. They are unchanged when f is replaced by f_-, because $I(\mu_\lambda, s(-1)) = \mu(s(-1)) = 1$ for $\mu \in \Pi_{\mathrm{aut}}(X_A)$.

7.8 The stable(r) form

We collect the results. For $f = f_S \otimes e^S$ as in Theorem 6.11, with S as in Section 7.6, write $I(f) = \sum_{\mathscr{O}} I_{\mathscr{O}}(f) = \sum_\chi I_\chi(f)$.

Not everything in the theorem depends on (N5). The decomposition of the regular elliptic terms, the unipotent coefficients, the trace on the theta functions and the stability of the principal series characters are proved without it. It is used for the equivariance of the normalized weighted distributions I_A (Proposition 7.9(a), (b)), and through it in the statement of part (c) about I_A and in

the invariance of the individual $I_{\mathscr{O}}$ and I_χ under $G_1(k)$. Parts (a) and (b) as displayed do not use it. By Proposition 7.8′ the condition is met by a canonical choice of the factors, so part (c) is not conditional; it is a statement about the distributions I_A defined with factors of that kind.

Theorem 7.13. *Let S, f_S and $f = f_S \otimes e^S$ be as above. In part (c) let the normalizing factors satisfy (N5) (for instance those of Proposition 7.8′).*

(a) (Geometric side.)

$$I(f) = S_{\mathrm{ell}}(f) + S_{\mathrm{ell}}(f_-) + c_k \sum_{\{t,t^{-1}\}} I_A\big(a(t)_S, f_S\big) + \sum_{z=\pm 1} \Big(\mathrm{vol}\big(G(\mathbb{Q})\backslash \overline{G(\mathbb{A})}\big) f_S(z) + \frac{c_k}{2} I_A(z, f_S) + \frac{1}{2} \sum_{\kappa \in \mathfrak{X}_S} \ell_S(\kappa) J^\kappa_{\mathrm{unip}}(z, f_S) \Big),$$

where S_{ell} is the stable distribution of Theorem 7.7, t runs over $\mathscr{O}_S^\times \smallsetminus \{\pm 1\}$ modulo inversion, $\ell_S(1) = c_S$ and $\ell_S(\kappa) = L^S(1,\kappa)$ for $\kappa \neq 1$.

(b) (Spectral side.) $I(f)$ is the right side of Corollary 6.14. The part of the discrete sum carried by the theta functions is $\frac{1}{2}\sum_b \big(\Theta_b^{\mathrm{st}}(f) + \Theta_b^{\mathrm{st}}(f_-)\big)$, the sum over the classes $b \in k^\times/k^{\times 2}$ with Θ_b pairwise orthogonal; the terms of the continuous spectrum are stable.

(c) (Equivariance.) The individually indexed distributions occurring in (a) transform under $G_1(k_S)$, acting by $f_S \mapsto f_S^h$, by the following characters of $\det h = a$: η_γ for $J_G^{\mathrm{st}}(\gamma,\cdot)$ and $\eta_\gamma \kappa_E$ for $J_G^\kappa(\gamma,\cdot)$; $(\cdot,t)_S$ for $I_A(a(t)_S,\cdot)$; $(\cdot,z)_S$ for $f_S(z)$ and $I_A(z,\cdot)$; $\kappa_S \cdot (\cdot,z)_S$ for $J^\kappa_{\mathrm{unip}}(z,\cdot)$. The characters differ from one index to the next, so the sums displayed in (a) are in general not eigendistributions; in particular S_{ell} is stable for the twisted action and is not an eigendistribution for the ordinary one. Both sides are invariant under $h \in G_1(k) \cap \prod_{v \notin S_k} Z(k_v) GL(2,\mathscr{O}_v)$.

Proof. (a) Theorem 6.11(a)–(c), with Theorem 7.7 for the elliptic classes and Proposition 7.10(c) for the central ones; in (c) of Theorem 6.11 the class of $a(t)$ contributes only if t is a unit outside S_k. (b) Proposition 7.12 and the remark after it; the spaces Θ_b for inequivalent b are orthogonal because their irreducible constituents are pairwise inequivalent (the constituents of $\omega_{\psi'_v}$ and $\omega_{\psi''_v}$ are inequivalent when $\psi'' \neq \psi'_{c^2}$, by their restrictions to N). (c) Lemma 7.6(b), Proposition 7.9(b), Lemma 7.1(d) and Proposition 7.10(d). For the last statement, such an h acts on $f = f_S \otimes e^S$ through its component in $G_1(k_S)$, by Lemma 7.1(e), and Lemma 7.11(a) applies. □

What the theorem does not say. A stable trace formula for the cover would be an identity

$$I(f) = S(f) + S(f_-)$$

with S a stable distribution, and then an identification of $S(f)$ with $\frac{1}{2}$ of the stable trace formula of $SO(3)$ evaluated at a transfer of f; this is Li's theorem [Li21]. Parts (a) and (b) exhibit this form for the regular elliptic terms and for the theta functions. What is missing, in increasing order of difficulty, is:

(i) the limit formulas that attach the terms $J^\kappa_{\mathrm{unip}}(z,\cdot)$, $\kappa \neq 1$, to the singular elliptic points of the tori, and so distribute them between $S(f)$ and $S(f_-)$ (Section 7.6);

(ii) the stabilization of the weighted terms $I_A(a(t),\cdot)$ and $I_A(z,\cdot)$, which are equivariant by Proposition 7.9 but not stable;

(iii) the transfer $f \mapsto f'$ to $SO(3)$ and the corresponding fundamental lemma [Li11, Li12b];

(iv) on the spectral side, the description of the cuspidal spectrum by packets [Wa80, Wa91, GI18].

The equivariance in (c) is weaker than stability, and the part of $I(f)$ that is invariant under all of $G_1(k_S)$ is small: by Lemma 7.6(b) and Proposition 7.9(b), a regular elliptic or hyperbolic γ contributes to it only if, at each place of S_k, $D^+(\gamma)$ is a square or θ times a square; for instance if γ is a square in its torus.

7.9 Status of the results

This subsection lists the statements of Section 7 that rest on citation or on numerical checks.

Remark 2026.19. *What rests on citation.* Proved here, given the earlier sections: Lemmas 7.1 and 7.2 (with the cocycle (2.13) on $GL(2)$ from [Ku69]; part (c) of Lemma 7.1 is also checked directly); Lemma 7.3; Lemma 7.4; Lemma 7.6; Theorem 7.7, with the Hasse norm theorem and the index of the global norm group from [CF67]; Lemma 7.8; Proposition 7.9, parts (a) and (b) under (N5); Proposition 7.8′(a), (b), the existence of normalizing factors with (N1)–(N5) (given the rationality of the local intertwining operators [Li12a]); Proposition 7.10; Proposition 7.12(b), (c) given (a); and Proposition 7.5 on the regular set, with Lemma 7.5′. Taken from the literature: at the archimedean places, the standard fact that an operator with a Schwartz kernel is of trace class with trace the integral over the diagonal; and at all places, that the character of the Weil representation is a locally integrable function on the whole group [Ma99, Li12a], which is used to call $\Theta_\psi^+ - \Theta_\psi^-$ a stable character and in the second formula for Θ_b^{st} in Section 7.7. Formula (7.7) agrees with [Th08, Theorem 1C] up to the sign left open there, so its sign rests on our computation alone; we have not compared it with the formulas of [Ma99, Th13] or with [Th08, Theorem 2A] in detail. Also from the literature: the explicit form of the normalizing factors at the dyadic and ramified places, Proposition 7.8′(c), which is Szpruch's computation [Sz09, Sz16]; the convergence and square-integrability of theta functions in Proposition 7.12(a); Tate's local zeta integrals [Ta67]; and the statements about Waldspurger's packets after Proposition 7.12, including the character identity (7.16) of [Ad98, Sc98, Lu20], which are not used in any proof. Which of the cited statements were checked against their sources, and which were not, is said where they are cited: checked (in the arXiv or publisher versions, at the level of statements) are [GS12, §5], [GI18, Theorem 1.4], [Th08, Theorem 1C, Theorem 2A, §12.2], [Sz10, Sz16], [Sc98], [Lu20]; not opened are [Wa80, Wa91], the journal version of [Sz09], and [Ad98] beyond its review and the account of it in [Sc98].

What is not done. The condition (N5) is no longer a hypothesis (Proposition 7.8′); we have not determined whether the factors of [Li12a] satisfy it, nor rederived Szpruch's formula for the local coefficients at the dyadic places (Remark 2026.18). The character identity (7.16) is quoted, not proved. The statements in the text about the relation with the endoscopy of [Li11, Li15, Li21] are comparisons of shape: we have not identified J_G^{st} with Li's stable orbital integral beyond Proposition 7.5, nor Δ_ψ with his transfer factor, nor checked the normalizations in $\iota(\tilde{G},H) = \frac{1}{2}$. The limit formulas of Section 7.6 at the singular elliptic points, the stabilization of the weighted terms, the

transfer to $SO(3)$ and the description of the cuspidal spectrum are not treated. Sections 2–6 and this section were refereed independently (see the Acknowledgments). An unpublished stabilization of the trace formula for covers of $SL(2)$ by Hiraga and Ikeda is mentioned in [Li15]; see also [HI13].

Remark 2026.20. The script `checks/sec7_n2_check.py` verifies, in exact arithmetic at $v = \infty, 2, 3, 5, 7$ on random elements of $SL(2,\mathbb{Q})$ and $GL(2,\mathbb{Q})$: Lemma 7.1(c),(d), both as a homomorphism for β_v and as conjugation in the cover of $GL(2)$; Lemma 7.2 on the tori $T_\theta(\mathbb{Q})$ and on the split torus, and Example 7.A(d); the conjugation invariance of Δ_ψ, Lemma 7.3(b),(c), and Example 7.B; the identity between Weil indices behind (7.5); the consistency of (7.6) with (7.7) under translation by $(-1,1)$, and the agreement of (7.7) with [Th08, Theorem 1C] up to sign; identity (7.9); and Lemma 7.8(a). Numerically: Example 7.C for $q = 3, 5, 7, 11$; Proposition 7.10(a) for $k = \mathbb{Q}$, $S = \{\infty, 2\}$ at $s = 3$, the local identity (7.15) at 2 and 3, and the constants of Example 7.G, including the invariance of the coefficients under $\mathscr{O}_S^\times$; Lemma 7.5′at 2, 3 and 5; for Proposition 7.8′, the form of the product of the Tate-type integrals of Remark 2026.18 at 2, 3 and 5, and at the real place the adjoint relation and the identity $c^2 = \bar{\omega}$ for the factors of Lemma 6.6(b); the two facts used in the proof of Proposition 7.12(b); and the parity in θ of the sum and the difference of the two discrete series characters in the paragraph after that proposition. The script `checks/examples_n2_check.py` checks Example 7.F. Example 7.D is derived from Lemma 7.6(d) and its proof, and has no numerical check.

8 Summary of results and open questions

This section collects the main results of the paper, with the status that the text gives them, and the questions that the paper leaves open. The status of each statement in detail is in the subsections "Status of the results" at the end of Sections 2–7.

Results

1. *The cover* (Section 2). The double cover of $G(\mathbb{A})$ is built from Kubota's cocycle, with its topology fixed intrinsically, and it splits over $G(\mathbb{Q})$ (Lemmas 2.1 and 2.6). The cocycle and the existence of the topological cover are taken from [Ku67, Mo68]. The rest is proved in the text.

2. *Rank* 0 (Section 3). For the cover of a torus the trace formula is Poisson summation, twisted by a genuine automorphic character (Theorem 3.4 and Lemma 3.5). For the split torus Weil's character χ_ψ gives a canonical choice (Lemma 3.3, whose parts (a) and (b) are Weil's).

3. *The geometric side* (Section 4). The coarse geometric expansion is proved directly from the height inequality, with truncation parameter $T \geq 0$ (Lemma 4.3 and Theorem 4.5). Each class contributes a polynomial in T of degree at most one, written down in Propositions 4.6, 4.7 and 4.8. The fine expansion, Theorem 4.11, is quoted from [Ar86, Li14a]. Its part (a) is derived again in Proposition 7.10 when the ring of S_k-integers is principal.

4. *The spectral side* (Section 5). The characters fixed by the Weyl group are the Weil characters χ_{ψ_b} (Lemma 5.3). The normalizing factor of the intertwining operator is $L^S(2\lambda, \chi^2)/L^S(2\lambda+1, \chi^2)$ (Lemma 5.6). The residual spectrum is the sum of the even Weil

representations $r^+_{\psi_{-b}}$ (Theorem 5.8). Parts (a) and (b) of that theorem, and the matching of parameters in (c), are proved here. The rest of (c) is taken from [Ge76, GPS80, Wa80]. The trace formula is Theorem 5.16. The spectral decomposition, the Eisenstein series and the polynomial behavior of the spectral terms are taken from [La76, MW95, Ar80, Li13].

5. *The invariant form* (Section 6). The invariant distributions are built by Fourier inversion on the abelian cover of A (Theorem 6.9). The invariant trace formula is Theorem 6.11, with Corollary 6.14, and $M(w,0) = -1$ for every character fixed by the Weyl group (Proposition 6.13). These are proved here, given the existence of normalizing factors, Theorem 6.5, which is taken from [Ar89, Li12a].

6. *The stable(r) form* (Section 7). Conjugation by $GL(2)$ can exchange the two lifts of an element (Lemma 7.2). Stable conjugacy is conjugation by $GL(2)$ twisted by the sign $(\det h, \det(1+\gamma))_v$, and its classes are described by the function Δ_ψ (Lemma 7.3). The character of $r^+_\psi - r^-_\psi$ is computed on the regular set (Proposition 7.5). The regular elliptic terms take the form $S_{\mathrm{ell}}(f) + S_{\mathrm{ell}}(f_-)$ (Theorem 7.7). The unipotent coefficients are $a^G(S,u_\xi) = \frac{1}{2}\big(c_S + \sum_{\kappa\neq 1}\kappa_S(\xi)L^S(1,\kappa)\big)$ (Proposition 7.10). The theta functions contribute to the spectral side in the same form as the elliptic terms (Proposition 7.12). The results are collected in Theorem 7.13. Its statements about the weighted terms use the property (N5) of the normalizing factors. Factors with this property exist at every place (Proposition 7.8′(a), (b)). Their explicit form at the dyadic and ramified places, Proposition 7.8′(c), is taken from Szpruch [Sz09, Sz16].

Open questions

The following questions are named as open in Sections 1 and 7. They are not answered here.

1. The stabilization of the weighted terms $I_A(a(t),\cdot)$ and $I_A(z,\cdot)$, which are equivariant under $GL(2)$ but not stable (Proposition 7.9 and Section 7.8).

2. The transfer $f \mapsto f'$ to $SO(3)$ and the corresponding fundamental lemma [Li11, Li12b], and with them the identification of Δ_ψ with Li's transfer factor and of J^{st}_G with his stable orbital integral (Remark 2026.19).

3. The description of the cuspidal spectrum by packets [Wa80, Wa91, GI18]. The character identity (7.16) used for the packets is quoted, not proved.

4. Whether the normalizing factors of [Li12a] have the property (N5), and an independent derivation of Szpruch's formula for the local coefficients at the dyadic places (Remark 2026.18).

5. Whether $T \geq 0$ suffices in Theorem 5.12, as it does on the geometric side (Remark 2026.13).

6. The limit formulas of Section 7.6 at the singular elliptic points.

7. The n-fold cover for $n > 2$, which the 1995 notes treated and this revision does not (Remarks 2026.3 and 2026.6).

Glossary

Notation

Symbol	Meaning	Defined in
Groups		
k, d, $\mathbb{A}_k$, $\mathbb{A}$	number field, its degree $[k:\mathbb{Q}]$, its adeles, the adeles of $\mathbb{Q}$	§2
G	$\mathrm{Res}_{k/\mathbb{Q}} SL(2)$; $G(\mathbb{Q}) = SL(2,k)$	§2
B, N, A	upper triangular Borel subgroup, its unipotent radical, the diagonal torus	§2.1
$a(t)$, $n(u)$, w	$\mathrm{diag}(t,t^{-1})$, the upper unipotent matrix with entry u, the Weyl element $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$	§2.1, §4
T_θ, T_μ	anisotropic torus attached to $k(\sqrt{\theta})$; centralizer of an elliptic μ	§2.1, Lemma 4.1
K_v, K_m, K_0, K_f	maximal compact subgroup at v; principal congruence subgroup of level ϖ^m; $\prod_v K_v$; an open compact subgroup of $G(\mathbb{A}_f)$	§2.1, §2.2
$A(\mathbb{A})^1$, $A_B(\mathbb{R})^0$	kernel of H on $A(\mathbb{A})$; the split component $\{a(u) \mid u > 0\}$	§3.1, §4.3
The cover		
μ_2	$\{\pm 1\}$, the kernel of the covering	§2
$(a,b)_v$	quadratic Hilbert symbol of k_v	§2
β_v	Kubota's cocycle on $SL(2,k_v)$; Borel, not continuous	(2.1), Lemma 2.1
κ_v, $\kappa_{\mathbb{A}}$	Kubota's correction on K_v (v finite, non-dyadic); its product over v, defined on $G(\mathbb{Q})$	(2.4), (2.12)
σ_v, $\sigma_{\mathbb{A}}$	β_v times the coboundary of κ_v; the adelic cocycle $\prod_v \sigma_v$	(2.10), (2.11)
$\overline{SL(2,k_v)}$, $\overline{G(\mathbb{A})}$	the local and the adelic double cover; $\overline{H} = p^{-1}(H)$ for a subgroup H	Lemma 2.1, §2.2
(g,ζ)	Borel coordinates on a cover; $g \mapsto (g,1)$ is not continuous	(2.2), Example 2.A
K_v^*, K_f^*	the image of K_v under $k \mapsto (k,\kappa_v(k))$, a compact open subgroup of the cover; the image of K_f	Lemma 2.2(b), Lemma 2.6(c)
$\overline{G(\mathbb{Q}_S)}$, $\overline{A(\mathbb{Q}_S)}$	the covers of $\prod_{v\in S_k} SL(2,k_v)$ and of its diagonal torus	§4.4, §6
Heights, measures, functions		
$H = H_B$, H_v, H_S	the height $\log\lvert t\rvert$ of the A-component in $x = nak$	(2.5), Lemma 2.8
δ_B	modulus character, e^{2H}	§2.1.5
$\mathfrak{a}$, α, $\alpha^\vee$	$\mathrm{Hom}(X(A)_{\mathbb{Q}},\mathbb{R}) \cong \mathbb{R}$; the simple root $2/d$; the coroot d	§2.1.3, Remark 2026.9
$\tau_B = \hat{\tau}_B$	characteristic function of $(0,\infty)$	(2.6)
$C_c^\infty(\cdot)_-$	smooth, compactly supported, anti-genuine functions on a cover	§2.1.4
e_v, e^S	unit of the spherical Hecke algebra at v; $\bigotimes_{v\notin S} e_v$	(2.8), §4.4
$\mathscr{H}(\overline{G(\mathbb{A})})$, $\mathscr{H}(\overline{G(\mathbb{Q}_S)})$	Hecke algebras of test functions; $f = f_S \otimes e^S$	§2.2, §4.4
f^y	$x \mapsto f(yxy^{-1})$	§6
c_k	$\mathrm{vol}(A(\mathbb{Q})\backslash A(\mathbb{A})^1)$	§4.3
Rank 0		
ψ, ψ_b	a nontrivial character of $\mathbb{A}_k/k$; $x \mapsto \psi(bx)$	§3.1

Symbol	Meaning	Defined in
$\gamma_v(a)$	Weil index of $x \mapsto \psi_v(ax^2)$, an eighth root of unity	(3.2)
χ_ψ, χ_{ψ_b}	Weil's genuine character of $\overline{A(\mathbb{A})}$, $(a(t),\zeta) \mapsto \zeta \prod_v \gamma_v(t_v)/\gamma_v(1)$	Lemma 3.3
X_T, $\Pi_{\mathrm{aut}}(X_T)$	$T(\mathbb{Q})\backslash\overline{T(\mathbb{A})^1}$; its genuine characters	§3.1
$I^A_{\mathscr{O}}$, $I^A_{\tilde{\chi}}$	the two kinds of terms in the trace formula for the cover of A	(3.4)
Geometric side		
$\mathscr{C}$, $\mathscr{O}$, $\mathscr{O}_z$	the set of classes in $G(\mathbb{Q})$; a class; the central class of $z = \pm 1$	Lemma 4.1
K_P, $K_{P,\mathscr{O}}$, $J_{P,\mathscr{O}}$	kernels attached to $P \in \{B, G\}$	(4.1), (4.2)
k^T, $k^T_{\mathscr{O}}$, $j^T_{\mathscr{O}}$	truncated kernels	(4.4)
$J^T_{\mathscr{O}}(f)$, $J_{\mathscr{O}}(f)$	integral of $k^T_{\mathscr{O}}$, a polynomial in T; its value at $T = 0$	Theorem 4.5
v_A, $v_{A,v}$	the weight $-H(x) - H(wx)$	(4.3), Lemma 4.9
Φ_f, $Z(f,s)$	the function $y \mapsto \int_K f(k^{-1}n(y)k)\,dk$ and its zeta integral (not to be confused with Φ_A)	(4.8), (4.9)
$J_G(\gamma, f_S)$	orbital integral on $\overline{G(\mathbb{Q}_S)}$	(4.11)
$J_A(\gamma, f_S)$	weighted orbital integral, weight v_A; not invariant	(4.12)
$J_A(z, f_S)$	its regularized value at $z = \pm 1$	Lemma 4.10
u_ξ, $a^M(S,\gamma)$	the unipotent element $n(\xi)$; the global coefficients of the fine expansion	Theorem 4.11
Spectral side		
$\mathscr{X}$, $\mathscr{X}_G$, $\mathscr{X}_A$	classes of cuspidal data; those of (G,π); those of (A,μ)	§5.1
μ, μ^w, μ_λ	a genuine automorphic character of the cover of A; its Weyl conjugate; $\mu\, e^{\lambda H}$	§5.1
χ_b	the ramified class of (A, χ_{ψ_b})	Lemma 5.3
$\mathscr{H}_B(\mu)$, $I(\mu_\lambda)$	the space of the principal series and the induced representation	(5.1)
$E(x,\varphi,\lambda)$	Eisenstein series	(5.2)
$M(w,\lambda)$, M_v	global and local intertwining operators	(5.3), Lemma 5.6
$m^S(\mu,\lambda)$	$L^S(2\lambda,\chi^2)/L^S(2\lambda+1,\chi^2)$	(5.4)
$\omega_{\psi'}$, $r^{\pm}_{\psi'}$	Weil representation and its even and odd parts	(5.5)
V_b	residual representation of the class χ_b, isomorphic to $r^+_{\psi_{-b}}$	Theorem 5.8
Λ^T	truncation operator	(5.9)
$K_{P,\chi}$, k^T_χ, J^T_χ, J_χ	spectral kernels, their truncation and its integral	§5.3
$M^T_B(\mu,\lambda)$	$2T - M(w,\lambda)^{-1}M'(w,\lambda)$	(5.12)
Invariant form		
$u'_B(k,y)$, $f_{B,y}$	the weight $-H(ky)$; the function on the cover of A that measures $J(f^y) - J(f)$	(6.2), (6.3)
$V(\pi)$, $I(\pi_\lambda)$	local principal series of $\overline{G(\mathbb{Q}_S)}$	§6
$r_v(\pi_v,\lambda)$, R_v, R_S	local normalizing factor; normalized operator $r_v^{-1}M_v$; $\bigotimes_{v\in S_k} R_v$	Theorem 6.5
$S_{\mathrm{bad}}(\mu)$	the places where r_v is not explicit in Section 6 (see Proposition 7.8′)	§6.2
$\rho(\pi,\lambda)$, $\omega(\pi)$, h_π	the scalar $M_v(\pi^w,-\lambda)M_v(\pi,\lambda)$; the value $\pi((w,1)^2) \in \mu_4$; the factor of $\lvert\rho\rvert$ without zeros or poles in $\lvert q^{-\lambda}\rvert < 1$	Proposition 7.8′
$m(\mu,\lambda)$	global normalizing factor, $M(w,\lambda) = m \cdot R_S$	(6.10)

Symbol	Meaning	Defined in
$\varphi_A(f,\pi)$	$-\operatorname{tr}(R_S^{-1}R_S' I(\pi,f))$, a function of the *character* π of $\overline{A(\mathbb{Q}_S)}$	(6.11)
$\Phi_A(\tilde{a},f)$	the inverse Fourier transform of $\varphi_A(f,\cdot)$, a function of the *element* $\tilde{a}$ of $\overline{A(\mathbb{Q}_S)}$	(6.12)
$I_A(\gamma,f)$, $I_A(z,f)$	$J_A-\Phi_A$, the invariant distributions	(6.13)
$I_{\mathcal{O}}(f)$, $I_\chi(f)$	the terms of the invariant trace formula	(6.14)
Section 7		
G_1, δ_a, $\mathrm{Ad}(h)$	$GL(2)$; $\mathrm{diag}(a,1)$; the automorphism of the cover lifting conjugation by $h\in G_1$	Lemma 2.10, Lemma 7.1
f^h, $f^{[h]}$, f_z, f_-	$f\circ\mathrm{Ad}(h)$; $f(h\star\cdot)$; $f(z\cdot)$ for central z; f_{-1}	§7, §7.2
$D^+(\gamma)$, η_γ	$\det(1+\gamma)=2+\operatorname{tr}\gamma$; the character $a\mapsto(a,D^+(\gamma))_v$	(7.2)
$h\star x$	$\eta_x(\det h)\,\mathrm{Ad}(h)x$, the action whose orbits are the stable classes	(7.3)
Δ_ψ	genuine function with values in μ_4, constant on stable classes; $\lvert D^+\rvert^{1/2}$ times the character of $r_\psi^+-r_\psi^-$	Lemma 7.3, Prop. 7.5
κ_E, $NE^\times$	the quadratic character of E/F and its kernel, the norms	§7
J_G^{st}, J_G^κ	stable and κ-orbital integrals	(7.8), (7.11)
$S_{\mathrm{ell}}(f)$	the stable elliptic regular distribution	Theorem 7.7
H_1, κ_a	the height on $GL(2)$; the character $a(t)\mapsto(a,t)$ of the cover of A	§7.5
$\mathfrak{X}_S$, $L^S(s,\kappa)$, ρ_S, c_S	quadratic Hecke characters unramified outside S; their partial L-functions; the residue and constant term of ζ_k^S at 1	§7.6
$J_G(zu_\xi,f_S)$, $J^\kappa_{\mathrm{unip}}(z,f_S)$	normalized unipotent orbital integral; its κ-sum over ξ	(7.14), Prop. 7.10
U_h, Θ_b, Θ_b^{st}	the action of $G_1(k)$ on automorphic forms; the span of the theta functions of ψ_{-b}; its stable character	§7.7

Terms

Anti-genuine function. A function f on a cover with $f(x\zeta)=\zeta^{-1}f(x)$ for $\zeta\in\mu_2$. All test functions are anti-genuine. For the double cover this is the same condition as *genuine*, $f(x\zeta)=\zeta f(x)$.

Borel coordinates. The coordinates (g,ζ) in which a cover is described by a cocycle. They give the group law and the Borel structure, but not the topology (Lemma 2.1).

Class (geometric). A set of elements of $G(\mathbb{Q})$ whose semisimple parts are conjugate: central, hyperbolic or elliptic (Lemma 4.1). A class is *unramified* if it consists of semisimple elements.

Cuspidal datum. A pair (G,π) with π cuspidal, or a pair (A,μ) with μ a genuine automorphic character of the cover of A; (A,μ) and (A,μ^w) are equivalent. A class χ of data (A,μ) is *ramified* if $\mu^w=\mu$, and then $\mu=\chi_{\psi_b}$.

Dyadic place. A place of k above 2. The explicit splitting κ_v and the explicit normalizing factors are not available there.

Genuine representation. A representation of a cover in which μ_2 acts by the identity character.

(G,S)**-equivalence.** Arthur's equivalence relation on $G(\mathbb{Q})$, for a finite set S of places: equal semisimple parts up to conjugacy, and unipotent parts conjugate in the centralizer over $\mathbb{Q}_S$. It indexes the terms of the fine expansion (Theorem 4.11).

Endoscopic group. For the double cover of $SL(2)$ the elliptic endoscopic groups are two copies of $SO(3)$, attached to the central elements ± 1 of the dual group $SL(2,\mathbb{C})$ [Li11]. They are discussed but not used.

Invariant distribution. A linear form I on test functions with $I(f^y) = I(f)$ for all y in the group.

Normalizing factor. A scalar function $r_v(\pi_v, \lambda)$ such that $r_v^{-1} M_v$ has the properties (N1)–(N4) of Theorem 6.5. Its existence, an explicit formula for it, and its role in the global factor m are three different matters (Section 6.2).

Residual spectrum. The part of the discrete spectrum that is not cuspidal. For this group it is spanned by the residues of Eisenstein series at $\lambda = \frac{1}{2}$ and consists of even Weil representations.

Stable conjugacy. On $SL(2,F)$: conjugacy under $GL(2,F)$. On the cover: the relation whose classes are the orbits of $h \star x$; a stable class lies over a stable class of $SL(2,F)$ and contains one of the two lifts of each element (Section 7.2). Conjugation by $GL(2,F)$ without the twist is coarser.

Stable distribution. An invariant distribution given by a class function that is constant on stable classes, or a stable orbital integral J_G^{st}; equivalently, one that is invariant under $f \mapsto f^{[h]}$.

Stable(r) form. The trace formula with its terms organized by stable classes, before any transfer to $SO(3)$ (Theorem 7.13).

Truncation. Subtracting from a kernel, or from an automorphic form, its constant term in the cusp $\{H(\delta x) > T\}$.

Weighted orbital integral. An orbital integral with the weight v_A inserted, (4.12). It arises from truncation and is not invariant.

Weil index. The eighth root of unity $\gamma_v(a)$ attached to the character of second degree $x \mapsto \psi_v(ax^2)$, (3.2).

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ChatGPT (OpenAI), an AI model, acted as an independent referee. It refereed the November 2026 draft of Sections 2–6 in one report and Section 7 in a second report. Several definitions, proofs and examples of the present draft were added or corrected in response to these reports.

Review by AI systems is not an independent certification of the results. The errors that remain are the authors'. Further details on the AI usage, including the referee reports, the responses to them and a list of every change, can be found in the revision log (`revision_log.md`) that accompanies the source.

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